

IARPA SuperTools Deliverable

ColdFlux RSFQ Logic Cell Library for MIT-LL SFQ Process

Submitted by

ColdFlux Team
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Version 3.0

Version History

Version 3.0

This is the version of the cell library for Phase 3 release to the Test & Evaluation teams. The key changes for the RSFQ cell library are as follows:

- Complete migration of schematic/symbol/netlist data from previous Xic formats to the gEDA gschem formats.
- Re-designed cells with PTL interconnects to use 160 μA input junctions and 250 μA output junctions.
- Updated junction parasitic inductance to use 0.5 pH instead of 0.2 pH.
- Complete re-design of NOT, NOTT and XNOR cells.
- Compacted cell layouts and changed the biasing structure of cells with PTL interconnects to bring in the biasing on M5 at the top and bottom of the cells from external bias lines .

Version 2.1

This is an updated version of the cell library for Phase 2B after taking into account feedback from the Test & Evaluation teams. The key changes for the RSFQ cell library are as follows:

- Minor updates to cell layouts to adhere to latest DRC rules.
- LEF files, extracted using qPALACE, are included in the library.

Version 2.1

This version is the ColdFlux cell library deliverable for SuperTools Phase 2B. The key changes for the RSFQ cell library are as follows:

- RSFQ cells were redesigned from first principles using phase-based equations.
- Cell versions both with and without integrated PTL transmitters and receivers are included in version 2.1.
- Base circuit netlists are included to show how the base cell is designed. An optimized circuit netlist represents the optimized circuit, as extracted from the layout using InductEx.

- A testbench for each cell is included for easy user verification of cell functionality.
- The mesh file is also included in version 2.1 to allow the user to view the 3D layout as generated by TTH/InductEx.
- An XNOR and XNORT cell was developed.
- Additional Always0 (both synchronous and asynchronous) and Always0T cells were developed.
- Parameterized cell layouts are in development, exposing parameters such as inductor widths and track pitch. This allows scaling and regeneration of layouts while maintaining inductance.

Version 2.0

Updated the RSFQ logic cell libraries for SuperTools Phase 2A.

Version 1.1

Previous release of RSFQ Logic cell library document.

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1. Introduction and Setup

1.1 Introduction

This RSFQ cell library is developed under the IARPA SuperTools/ColdFlux contract via the U.S. Army Research Office grant W911NF-17-1-0120. The aim is to create a generic and open-source cell library with RSFQ logic [1] as part of the IARPA SuperTools Program [2], [3]. The cell library is continually updated and the latest version of the library can soon be found at: <https://github.com/sunmagnetics/RSFQlib>.

The free and open-source tools *gschem* [4], *JoSIM* [5], [6], *JoSIM-tools* [7], *KLayout* [8] and *TimEx* [9], [10] are used to develop and test the RSFQ cells. The circuit schematics are drawn using *gschem*. *JoSIM* is used as the SPICE engine for simulating the cells, while *JoSIM-tools* is used for operating margin analysis as well as cell parameter optimization. *KLayout* is used to construct the cell layouts. *TimEx* is used to extract the characteristics of the cell to generate the Mealy Finite State Machine diagram and Verilog files. Icarus Verilog [11] and *GTKWave* [12] can be used to simulate and view the verilog files for each cell. Additionally, *InductEx* [13], [14] is used for impedance extraction during cell layout design and *InductEx-LVS* is used for Layout Versus Schematic checking. A free version of InductEx is available, but has limited capacity.

Version 3.0 of the RSFQ cell library includes two versions of each cell: one with standard connections designed to be connected directly with other cells, and a version designed to be connected to Passive Transmission Lines (PTLs). The version of the cell designed to be connected to PTLs includes integrated PTL transmitter and receiver cells. To indicate the integration of PTL transmitters and receivers within a cell, the letter ‘T’ is added at the end of a cell name, for example the DFF with integrated PTL transmitters and receivers will be referred to as DFFT. The cell library is designed for PTLs with characteristic impedance $5\ \Omega$. Version 3.0 of the cell library does not include an XNOR cell with integrated PTL transmitter and receiver.

The following core cells are included in the RSFQ cell library:

- Interconnects: JTL, JTLT, SPLIT, SPLITT, MERGE, MERGET, PTLTX, PTLRX, Always0 (synchronous and asynchronous) and Always0T (synchronous and asynchronous).
- Logic cells: AND2, AND2T, OR2, OR2T, XOR, XORT, NOT, NOTT, XNOR.
- Buffers: DFF, DFFT, NDRO, NDROT, BUFF and BUFFT.
- Interfacing cells: DCSFQ, DCSFQ-PTLTX, PTLRX-SFQDC and SFQDC.

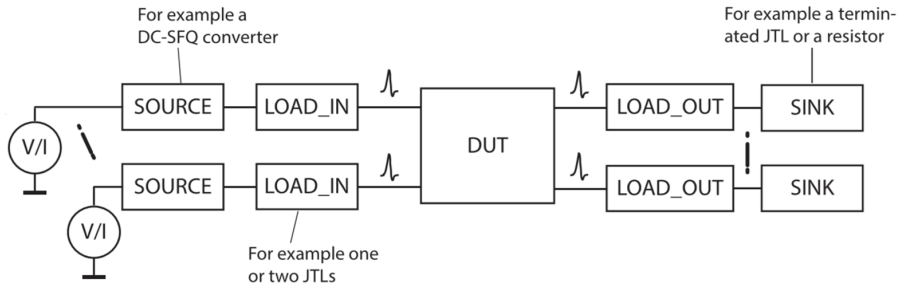


Figure 1.1: Testbench for functionality verification of the RSFQ cell library.

More complex functions can be constructed through connecting several core cells. The cells are currently optimized to run at a maximum clock frequency of 50 GHz.

A testbench for each cell is included within the RSFQ cell library. The testbench circuit is constructed through the circuit in Fig. 1.1 with the DUT representing the device-under-test. The SOURCE cell represents either the DCSFQ or DCSFQ-PTLTX cell, depending on whether a cell with integrated PTL transmitters and receivers are under test. The LOAD_IN and LOAD_OUT cells represent either a JTL or PTL and the SINK cell represents either a resistor or a JTLT terminated through a resistor. The functionality of the DUT is only confirmed if it is able to drive the LOAD_OUT cell (or SINK_OUT cell in the case of PTL cells) successfully. Future versions of the cell library will also require the DUT to drive the LOAD_OUT cell within a specified time. This will constrain the maximum throughput delay of the DUT to minimize timing violations when the cell is used to synthesize larger circuits.

Each delivered cell is documented in 5 parts:

1. **Schematic:** The schematic of a cell is constructed using *gschem* and is delivered in the native circuit schematic format.
2. **Layout:** The physical layout of the cells is constructed using *KLayout* and is delivered in standard GDSII format.
3. **Analog model:**
 - (a) **Netlist:** The netlist presents the device-level construction of a circuit. Each cell is delivered with a “base” and “extracted” circuit netlist files. The base netlist shows how the cell is designed from first principles using phase-based equations. The cells are then optimized before the physical layout is done. The extracted circuit netlists provides the back-annotated optimized netlist extracted through *InductEx*. The extracted netlist is the one included in this document.
 - (b) **Pin list:** The pin labels and function of each pin is listed.
 - (c) **Simulation results:** JoSIM is used for all circuit simulations. The simulation uses the cell testbench to verify cell functionality.
4. **Digital model:**

- (a) **Verilog model:** The behavior-level model of a cell with timing specifications included within the model. The verilog models are delivered in two parts – a basic verilog model accompanied by a SDF file to include timing delays, and a self-contained verilog model which contains the timing delays of the cell within the verilog model itself. All verilog models are extracted using *TimEx* and is delivered in standard HDL Verilog format. The verilog model included within this document is the self-contained verilog model.
 - (b) **Simulation results:** The digital simulation testbench is generated through *TimEx* and is run using *Icarus Verilog* and wave viewer *GTKWave*. Each edge event indicates an SFQ pulse.
 - (c) **Mealy finite state machine diagram:** The state machine diagram is extracted using *TimEx* and is delivered in standard PDF format.
5. **Power consumption:** The power consumption of each cell is calculated in terms of static and dynamic power consumption. As a rough estimate, it is assumed that each junction switches with every clock signal. For asynchronous cells, the power consumption is calculated through assuming that an input pulse train is applied at the same frequency as the specified clock frequency. Following [15], dynamic power consumption can be calculated as $P_d = f\Phi_0 I_c$ and static power consumption can be calculated as $P_s = I_b V_b$.

1.2 Setup

The latest version of the RSFQ cell library can be found at: <https://github.com/sunmagnetics/RSFQlib>. The RSFQ cell library is simulated and tested using several free and/or open-source tools:

- *gschem* is part of the *gEDA* project and can be found at <http://www.geda-project.org/>.
- *JoSIM* can be found at <https://github.com/JoeyDelp/JoSIM/>.
- *JoSIM-tools* can be found at <https://github.com/pleroux0/josim-tools>.
- *TimEx* can be found at <https://github.com/sunmagnetics/TimEx>.
- *KLayout* can be found at <https://www.klayout.de/>.
- *InductEx* can be found at <https://www.inductex.info>. *InductEx-LVS* is included with *InductEx*.
- *Icarus Verilog* can be found at <http://iverilog.icarus.com/>.
- *GTKWave* can be found at <http://gtkwave.sourceforge.net/>.

No additional setup is required to use the RSFQ cell library.

1.3 License

The generic RSFQ cell library is free to distribute and/or modify under the terms of the MIT license.

2. RSFQ Cell Library: Standard Connections

2.1 Interconnects

2.1.1 JTL

The JTL, Josephson transmission line, cell is commonly used to re-establish and propagate SFQ pulses. The cell is not designed to be directly connected to a PTL.

Schematic

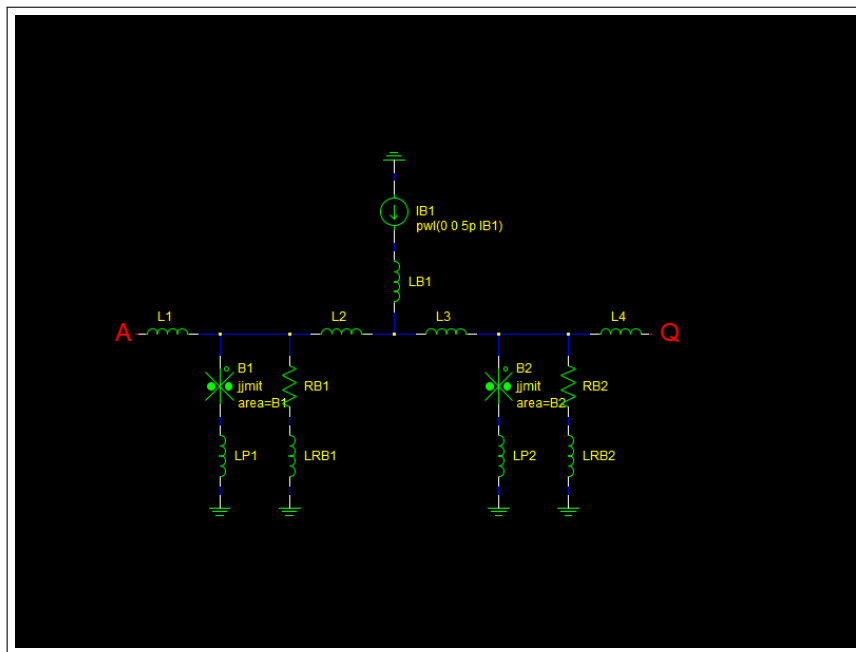


Figure 2.1: Schematic of RSFQ JTL.

Layout

The physical layout of the RSFQ JTL is shown in Fig. 2.2. The layout height is $70\ \mu m$ and the width is $20\ \mu m$. The cell includes an integrated bias line on M0 within the top row of track blocks. A bias pillar from M0 to M6 connects the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

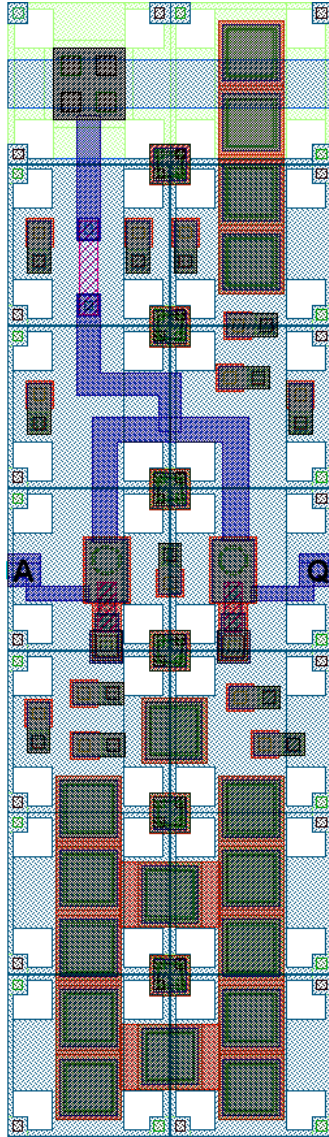


Figure 2.2: RSFQ JTL Layout

Analog model

```

1  * Back-annotated simulation file written by InductEx v.6.1.52 on 2022/07/25.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 21 July 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      q
8  .subckt THmit11_JTL a q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap=0.07pF, r0=160, rn=16, icrit=0.1mA)
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=IC
23 .param B2=IC
24
25 .param IB1=(B1+B2)*Ic0*BiasCoef
26
27 .param LB1=LB
28
29 .param L1=Phi0/(4*B1*Ic0)
30 .param L2=Phi0/(4*B1*Ic0)
31 .param L3=Phi0/(4*B2*Ic0)
32 .param L4=Phi0/(4*B2*Ic0)
33
34 .param LP1=LP
35 .param LP2=LP
36
37 .param RB1=B0Rs/B1
38 .param RB2=B0Rs/B2
39
40 .param LRB1=(RB1/Rsheet)*Lsheet+LP
41 .param LRB2=(RB2/Rsheet)*Lsheet+LP
42
43 B1 1 2 jjmit area=B1
44 B2 5 6 jjmit area=B2
45
46 IB1 0 4 pwl(0 0 5p IB1)
47
48 LB1 4 3 2.336E-012
49
50 L1 a 1 2.07E-012
51 L2 1 3 2.088E-012
52 L3 3 5 2.082E-012
53 L4 5 q 2.072E-012
54
55 LP1 2 0 3.137E-013
56 LP2 6 0 3.123E-013
57
58 RB1 1 101 RB1
59 LRB1 101 0 LRB1
60 RB2 5 105 RB2
61 LRB2 105 0 LRB2
62 .ends

```

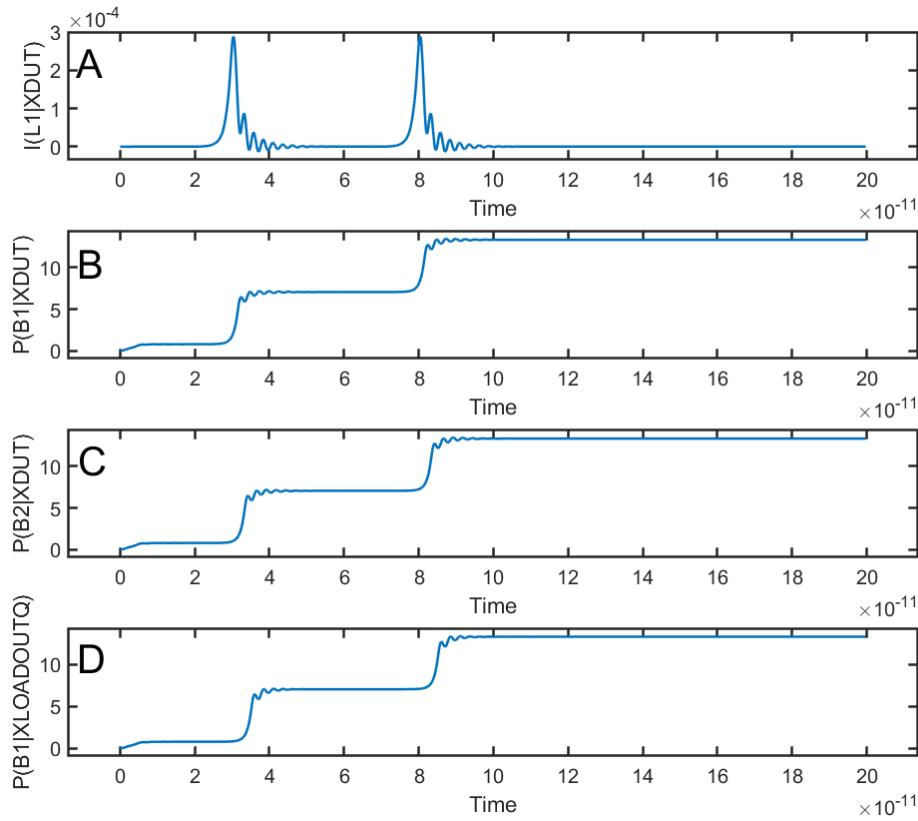
Listing 2.1: RSFQ JTL JoSIM netlist.

Table 2.1: RSFQ JTL pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ JTL using JoSIM is shown in Fig. 2.3. The testbench is included within the cell library for user verification. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the phase over the output JJ of pin **q**, and
- (d) the phase over the input JJ of the load circuit.

**Figure 2.3:** RSFQ JTL analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 25 July 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_JTL_v3p0_extracted (a, q);
9
10 input
11     a;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_a_q = 3.5,
21     ct_state0_a_a = 5.2;
22
23 reg
24     errorsignal_a;
25
26 integer
27     outfile,
28     cell_state; // internal state of the cell
29
30 initial
31     begin
32         errorsignal_a = 0;
33         cell_state = 0; // Startup state
34         q = 0; // All outputs start at 0
35     end
36
37 always @(posedge a or negedge a) // execute at positive and negative edges of input
38     begin
39         if ($time>4) // arbitrary steady-state time)
40             begin
41                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
42                     begin
43                         outfile = $fopen("errors.txt", "a");
44                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
45                             ↪ ", $stime);
46                         $fclose(outfile);
47                         q <= 1'bX; // Set all outputs to unknown
48                     end
49                 if (errorsignal_a == 0)
50                     begin
51                         case (cell_state)
52                             0: begin
53                                 q <= #(delay_state0_a_q) !q;
54                                 errorsignal_a = 1; // Critical timing on this input; assign
55                                     ↪ immediately
56                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
57                                     ↪ after critical timing expires
58                             end
59                         endcase
60                     end
61             end
62     end
63 endmodule

```

Listing 2.2: RSFQ JTL verilog model with self-contained timing.

The digital simulation results for the RSFQ JTL is shown in Fig. 2.4 and the Mealy finite state machine diagram, extracted using *TimEx*, is shown in Fig. 2.5.

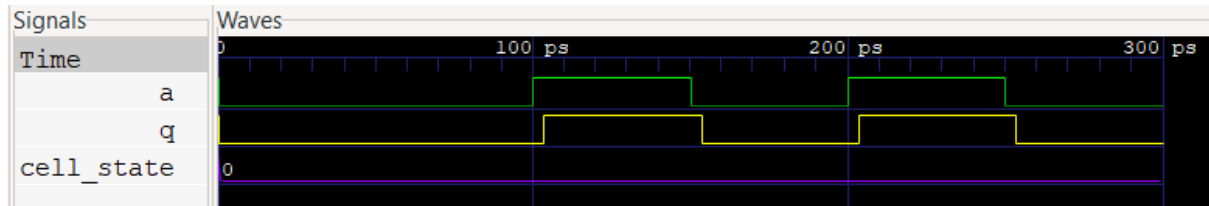


Figure 2.4: RSFQ JTL digital simulation results.

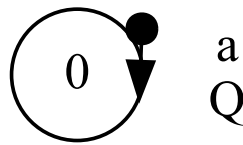


Figure 2.5: RSFQ JTL Mealy finite state machine diagram.

Power Consumption

Table 2.2: RSFQ JTL power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	910	1.03
2	910	2.07
5	910	5.17
10	910	10.3
20	910	20.7
50	910	51.7

2.1.2 SPLIT

The SPLIT cell is used to split a single pulse signal line into two duplicate output pulse signal lines. The cell is not designed to be directly connected to a PTL.

Schematic

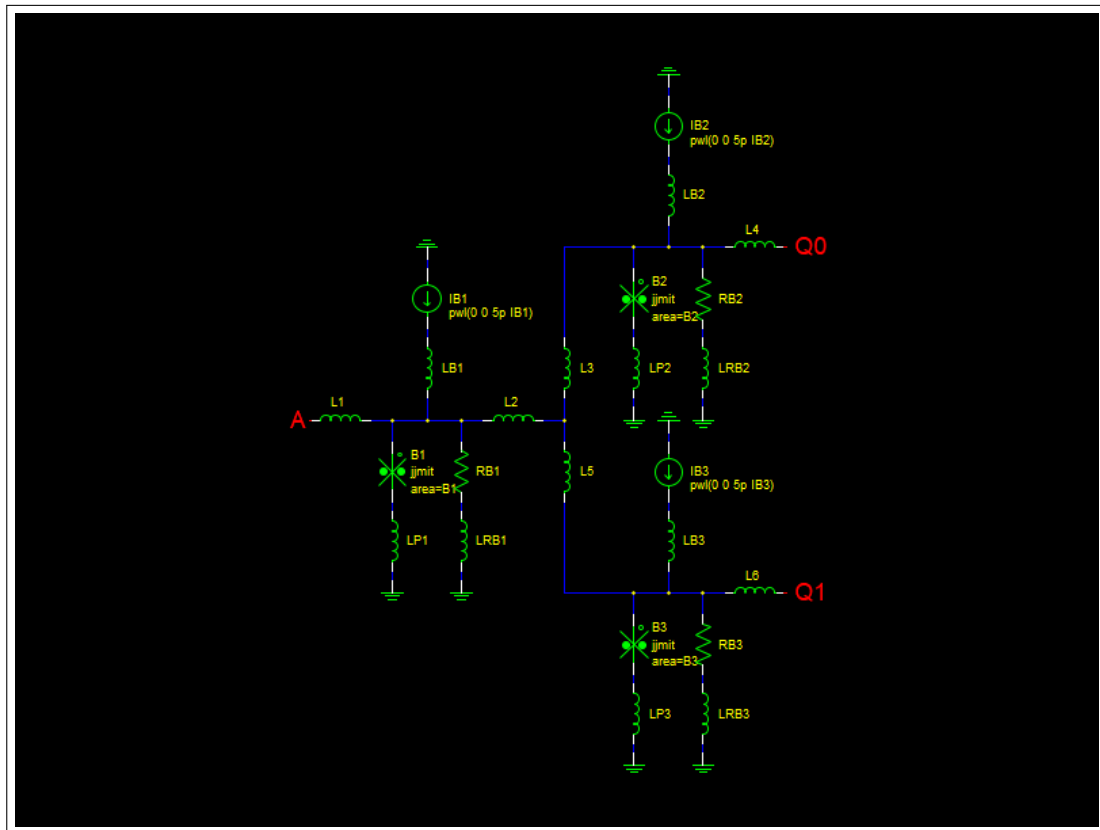


Figure 2.6: Schematic of RSFQ SPLIT.

Layout

The physical layout of the RSFQ SPLIT is shown in Fig. 2.7. The layout height is $70\ \mu\text{m}$ and the width is $20\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

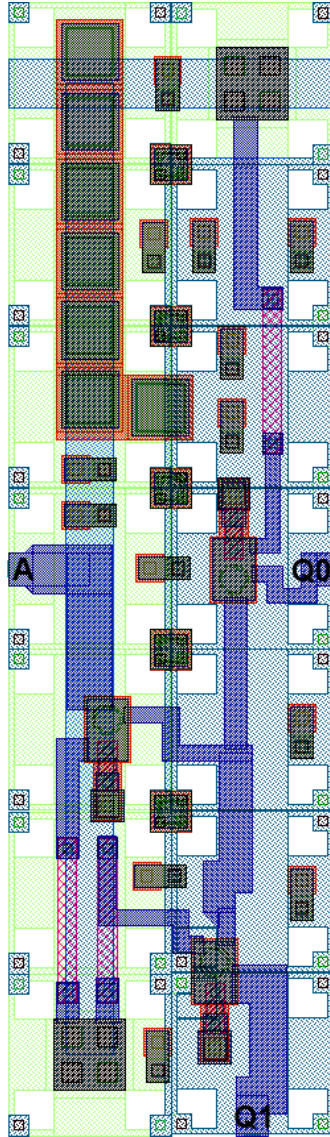


Figure 2.7: RSFQ SPLIT layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/07/25.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 25 July 2022
5  * Last modification by: T. Hall
6
7  *$Ports a      q0      q1
8  .subckt THmit11_SPLIT a q0 q1
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=IC
23 .param B2=IC
24 .param B3=IC
25
26 .param IB1=BiasCoef*Ic0*B1
27 .param IB2=BiasCoef*Ic0*B2
28 .param IB3=IB2
29
30 .param LB1=LB
31 .param LB2=LB
32 .param LB3=LB
33
34 .param L1=Phi0/(4*IC*Ic0)
35 .param L2=(Phi0/(2*B1*Ic0))/2
36 .param L3=L2
37 .param L4=Phi0/(4*IC*Ic0)
38 .param L5=L3
39 .param L6=L4
40
41 .param LP1=LP
42 .param LP2=LP
43 .param LP3=LP
44
45 .param RB1=B0Rs/B1
46 .param RB2=B0Rs/B2
47 .param RB3=B0Rs/B3
48
49 .param LRB1=(RB1/Rsheet)*Lsheet+LP
50 .param LRB2=(RB2/Rsheet)*Lsheet+LP
51 .param LRB3=(RB3/Rsheet)*Lsheet+LP
52
53 B1 1 2 jjmit area=B1
54 B2 5 6 jjmit area=B2
55 B3 8 9 jjmit area=B3
56
57 IB1 0 3 pw1(0 0 5p IB1)
58 IB2 0 7 pw1(0 0 5p IB2)
59 IB3 0 10 pw1(0 0 5p IB3)
60
61 LB1 3 1 1.712E-012
62 LB2 7 5 2.279E-012
63 LB3 10 8 2.858E-012
64
65 L1 a 1 2.062E-012
66 L2 1 4 2.06E-012
67 L3 5 4 2.079E-012
68 L4 5 q0 2.081E-012
69 L5 4 8 2.076E-012
70 L6 8 q1 2.086E-012
71
72 LP1 2 0 4.263E-013
73 LP2 6 0 3.75E-013
74 LP3 9 0 4.312E-013
75
76 RB1 1 101 RB1
77 LRB1 101 0 LRB1
78 RB2 5 105 RB2
79 LRB2 105 0 LRB2
80 RB3 8 108 RB3
81 LRB3 108 0 LRB3
82 .ends

```

Listing 2.3: RSFQ SPLIT JoSIM netlist.

Table 2.3: RSFQ SPLIT pin list.

Pin	Description
a	Data input
q0	Data output
q1	Data output

The JoSIM simulation results for the RSFQ SPLIT are shown in Fig. 2.8. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the phase over the output JJ of pin **q0**,
- (d) the phase over the output JJ of pin **q1**,
- (e) the phase over the input JJ of the load cell connected to pin q0, and
- (f) the phase over the input JJ of the load cell connected to pin q1.

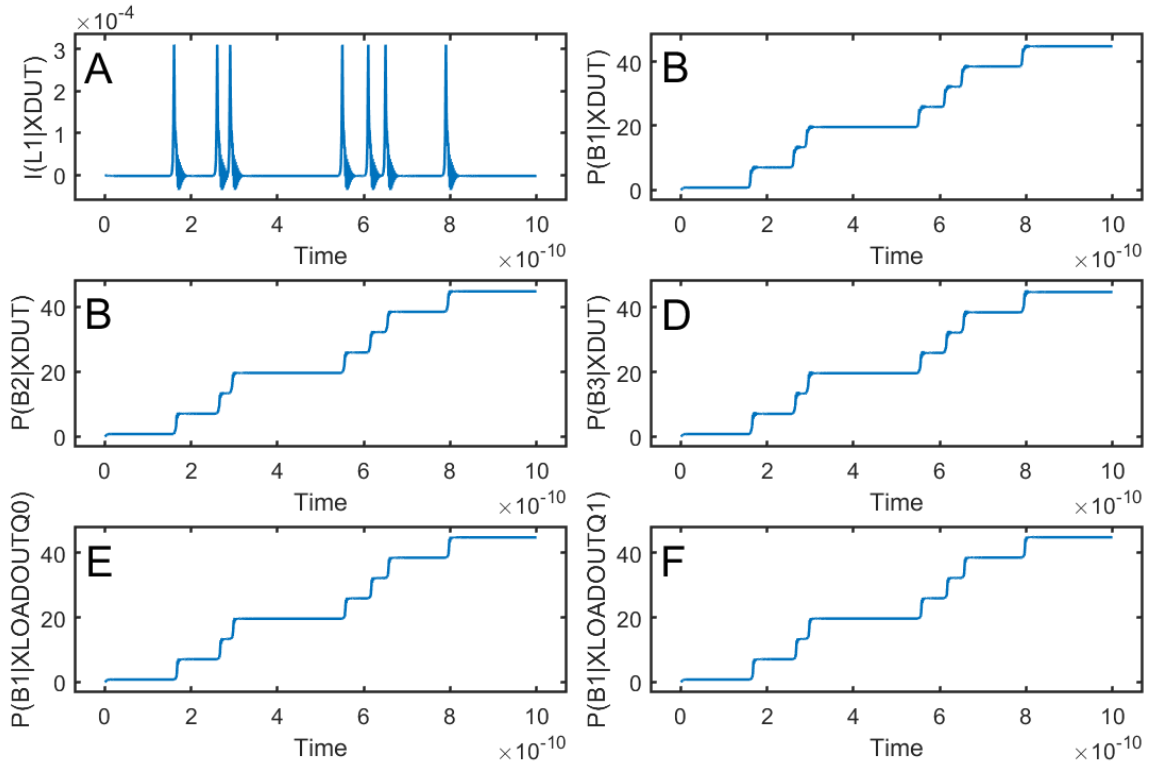


Figure 2.8: RSFQ SPLIT analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 25 July 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_SPLIT_v3p0_extracted (a, q0, q1);
9
10 input
11     a;
12
13 output
14     q0, q1;
15
16 reg
17     q0, q1;
18
19 real
20     delay_state0_a_q0 = 6.3,
21     delay_state0_a_q1 = 6.3,
22     ct_state0_a_a = 7.0;
23
24 reg
25     errorsignal_a;
26 integer
27     outfile,
28     cell_state; // internal state of the cell
29
30 initial
31     begin
32         errorsignal_a = 0;
33         cell_state = 0; // Startup state
34         q0 = 0; // All outputs start at 0
35         q1 = 0; // All outputs start at 0
36     end
37
38 always @(posedge a or negedge a) // execute at positive and negative edges of input
39     begin
40         if ($time>4) // arbitrary steady-state time)
41             begin
42                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
43                     begin
44                         outfile = $fopen("errors.txt", "a");
45                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0dps.\n
46                             ↪ ", $stime);
47                         $fclose(outfile);
48                         q0 <= 1'bX; // Set all outputs to unknown
49                         q1 <= 1'bX; // Set all outputs to unknown
50                     end
51                 if (errorsignal_a == 0)
52                     begin
53                         case (cell_state)
54                             0: begin
55                                 q0 <= #(delay_state0_a_q0) !q0;
56                                 q1 <= #(delay_state0_a_q1) !q1;
57                                 errorsignal_a = 1; // Critical timing on this input; assign
58                                     ↪ immediately
59                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
60                                     ↪ after critical timing expires
61                             end
62                         endcase
63                     end
64             end
65     end
66 endmodule

```

Listing 2.4: RSFQ SPLIT verilog model.

The digital simulation results for the RSFQ SPLIT is shown in Fig. 2.9 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.10.

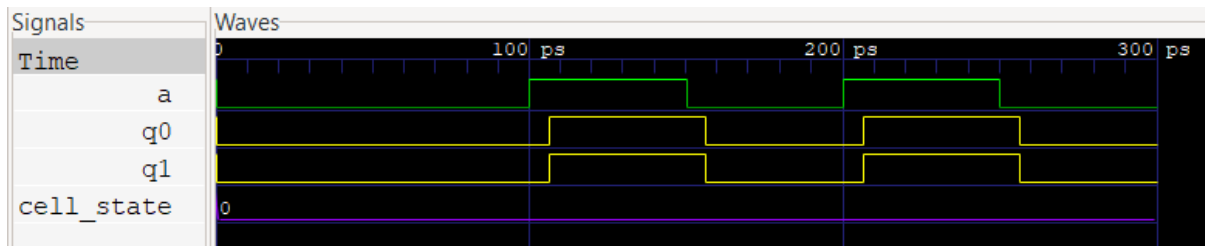


Figure 2.9: RSFQ SPLIT digital simulation results.

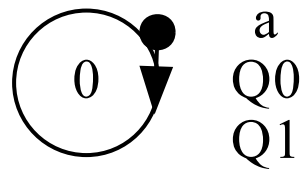


Figure 2.10: RSFQ SPLIT Mealy finite state diagram.

Power consumption

Table 2.4: RSFQ SPLIT power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	1370	1.55
2	1370	3.10
5	1370	7.75
10	1370	15.5
20	1370	31.0
50	1370	77.5

2.1.3 MERGE

The MERGE joins two input pulse signal lines and provides a single output pulse signal line. If there is a pulse on either input line, the MERGE will generate a pulse on the output signal line. The cell is not designed to be directly connected to a PTL.

Schematic

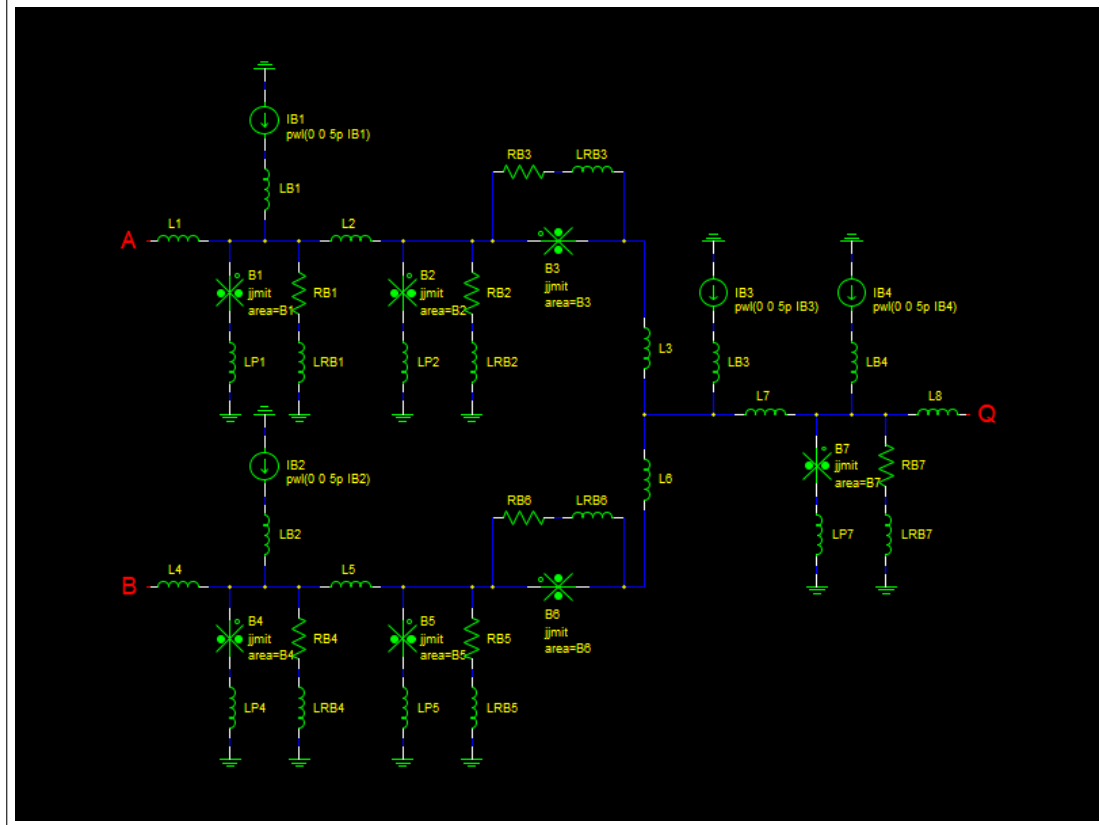


Figure 2.11: Schematic of RSFQ MERGE.

Layout

The physical layout of the RSFQ MERGE is shown in Fig. 2.12. The height of the layout is $70\ \mu\text{m}$ and the width is $30\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

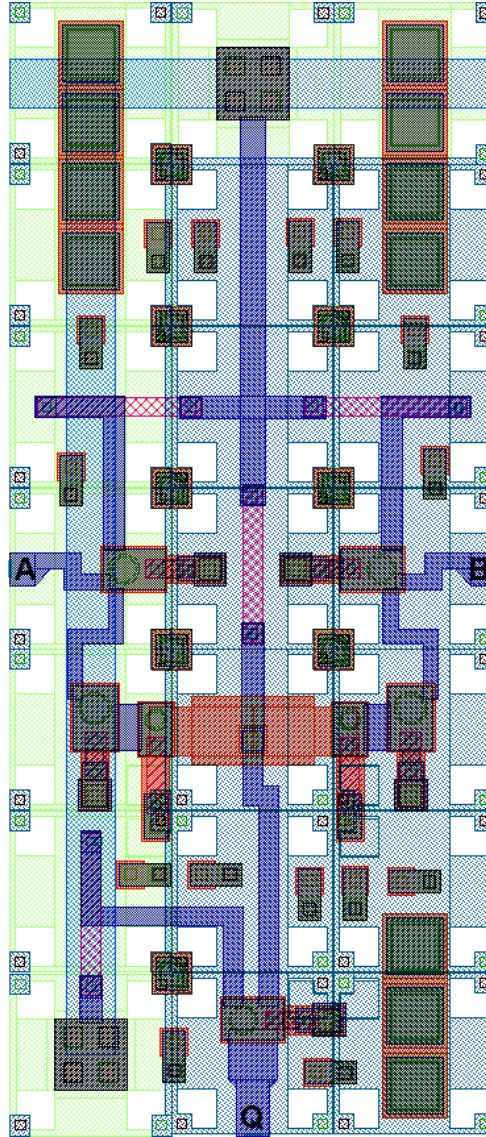


Figure 2.12: RSFQ MERGE layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/07/29.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 29 July 2022
5  * Last modification by: T. Hall
6
7  *$Ports a      b      q
8  .subckt THmit11_MERGE a b q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7'
21
22 .param B1=2.5
23 .param B2=3.01
24 .param B3=1.18
25 .param B4=2.5
26 .param B5=3.01
27 .param B6=1.18
28 .param B7=2.5
29
30 .param IB1=175u
31 .param IB2=175u
32 .param IB3=183u
33 .param IB4=175u
34
35 .param L1=2.0465p
36 .param L2=2.4953p
37 .param L3=0.9488p
38 .param L4=2.0465p
39 .param L5=2.4953p
40 .param L6=0.9488p
41 .param L7=3.6821p
42 .param L8=1.3761p
43
44 .param LB1=LB
45 .param LB2=LB
46 .param LB3=LB
47 .param LB4=LB
48
49 .param LP1=LP
50 .param LP2=LP
51 .param LP4=LP
52 .param LP5=LP
53 .param LP7=LP
54
55 .param RB1=B0Rs/B1
56 .param RB2=B0Rs/B2
57 .param RB3=B0Rs/B3
58 .param RB4=B0Rs/B4
59 .param RB5=B0Rs/B5
60 .param RB6=B0Rs/B6
61 .param RB7=B0Rs/B7
62
63 .param LRB1=(RB1/Rsheet)*Lsheet+LP
64 .param LRB2=(RB2/Rsheet)*Lsheet+LP
65 .param LRB3=(RB3/Rsheet)*Lsheet
66 .param LRB4=(RB4/Rsheet)*Lsheet+LP
67 .param LRB5=(RB5/Rsheet)*Lsheet+LP
68 .param LRB6=(RB6/Rsheet)*Lsheet
69 .param LRB7=(RB7/Rsheet)*Lsheet+LP
70
71 B1 1 2 jjmit area=B1
72 B2 4 5 jjmit area=B2
73 B3 4 6 jjmit area=B3
74 B4 8 9 jjmit area=B4
75 B5 11 12 jjmit area=B5
76 B6 11 13 jjmit area=B6
77 B7 15 16 jjmit area=B7
78
79 IB1 0 3 pwl(0 0 5p IB1)
80 IB2 0 10 pwl(0 0 5p IB2)
81 IB3 0 14 pwl(0 0 5p IB3)
82 IB4 0 17 pwl(0 0 5p IB4)
83
84 LB1 3 1 LB1
85 LB2 10 8 LB2
86 LB3 14 7 LB3
87 LB4 17 15 LB4
88
89 L1 a 1 2.048E-012
90 L2 1 4 2.485E-012
91 L3 6 7 9.486E-013
92 L4 b 8 2.059E-012
93 L5 8 11 2.473E-012
94 L6 7 13 9.53E-013
95 L7 7 15 3.697E-012
96 L8 15 q 1.377E-012
97
98 LP1 2 0 4.88E-013
99 LP2 5 0 3.854E-013
100 LP4 9 0 4.857E-013
101 LP5 12 0 3.834E-013
102 LP7 16 0 3.61E-013
103
104 RB1 1 101 RB1
105 LRB1 101 0 LRB1
106 RB2 4 104 RB2
107 LRB2 104 0 LRB2
108 RB3 4 106 RB3
109 LRB3 106 6 LRB3
110 RB4 8 108 RB4
111 LRB4 108 0 LRB4
112 RB5 11 111 RB5
113 LRB5 111 0 LRB5
114 RB6 11 113 RB6
115 LRB6 113 13 LRB6
116 RB7 15 115 RB7
117 LRB7 115 0 LRB7
118 .ends

```

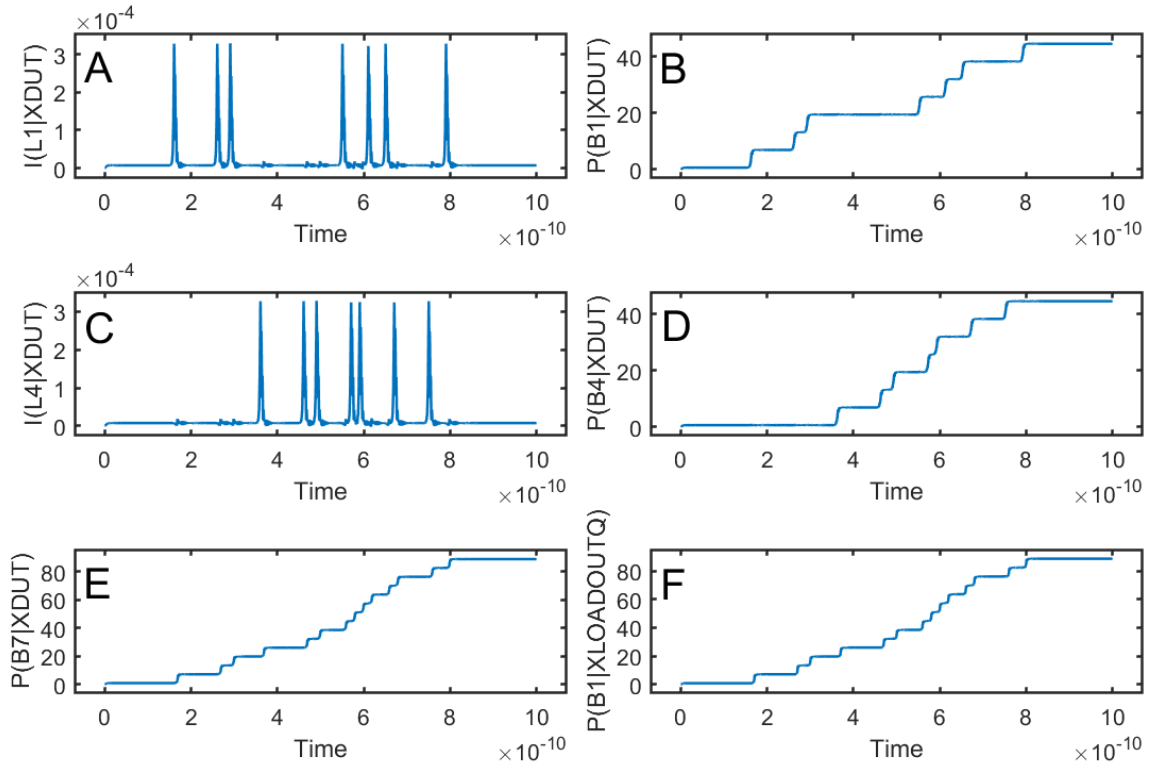
Listing 2.5: RSFQ MERGE JoSIM netlist.

Table 2.5: RSFQ MERGE pin list.

Pin	Description
a	Data input
b	Data input
q	Data output

The JoSIM simulation results for the RSFQ MERGE are shown in Fig. 2.13. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b**,
- (d) the phase over the input JJ of pin **b**,
- (e) the phase over the output JJ of pin **q**, and
- (f) the phase over the input JJ of the load cell connected to pin **q**.

**Figure 2.13:** RSFQ MERGE analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 29 July 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_MERGE_v3p0_extracted (a, b, q);
9
10 input
11     a, b;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_a_q = 9.0,
21     delay_state0_b_q = 9.0,
22     ct_state0_a_a = 10.2,
23     ct_state0_a_b = 2.3,
24     ct_state0_b_a = 2.2,
25     ct_state0_b_b = 10.2;
26
27 reg
28     errorsignal_a,
29     errorsignal_b;
30
31 integer
32     outfile,
33     cell_state; // internal state of the cell
34
35 initial
36     begin
37         errorsignal_a = 0;
38         errorsignal_b = 0;
39         cell_state = 0; // Startup state
40         q = 0; // All outputs start at 0
41     end
42
43 always @(posedge a or negedge a) // execute at positive and negative edges of input
44     begin
45         if ($time>4) // arbitrary steady-state time)
46             begin
47                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
48                     begin
49                         outfile = $fopen("errors.txt", "a");
50                         $fdisplay(outfile, "Violation of critical timing in module %m; %0d ps.\n
51                             ↳ ", $stime);
52                         $fclose(outfile);
53                         q <= 1'bX; // Set all outputs to unknown
54                     end
55                 if (errorsignal_a == 0)
56                     begin
57                         case (cell_state)
58                             0: begin
59                                 q <= #(delay_state0_a_q) !q;
60                                 errorsignal_a = 1; // Critical timing on this input; assign
61                                     ↳ immediately
62                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
63                                     ↳ after critical timing expires
64                                 errorsignal_b = 1; // Critical timing on this input; assign
65                                     ↳ immediately
66                                 errorsignal_b <= #(ct_state0_a_b) 0; // Clear error signal
67                                     ↳ after critical timing expires

```

```

63         end
64     endcase
65 end
66 end
67 end
68
69 always @(posedge b or negedge b) // execute at positive and negative edges of input
70 begin
71     if ($time>4) // arbitrary steady-state time)
72     begin
73         if (errorsignal_b == 1'b1) // A critical timing is active for this input
74         begin
75             outfile = $fopen("errors.txt", "a");
76             $fdisplay(outfile, "Violation of critical timing in module %m; %0d ps.\n
77                 ↪ ", $stime);
78             $fclose(outfile);
79             q <= 1'bX; // Set all outputs to unknown
80         end
81         if (errorsignal_b == 0)
82         begin
83             case (cell_state)
84             0: begin
85                 q <= #(delay_state0_b_q) !q;
86                 errorsignal_a = 1; // Critical timing on this input; assign
87                     ↪ immediately
88                 errorsignal_a <= #(ct_state0_b_a) 0; // Clear error signal
89                     ↪ after critical timing expires
90                 errorsignal_b = 1; // Critical timing on this input; assign
91                     ↪ immediately
92                 errorsignal_b <= #(ct_state0_b_b) 0; // Clear error signal
93                     ↪ after critical timing expires
94             end
95             endcase
96         end
97     end
98 end
99 endmodule

```

Listing 2.6: RSFQ MERGE verilog model.

The digital simulation results for the RSFQ MERGE is shown in Fig. 2.14 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.15.

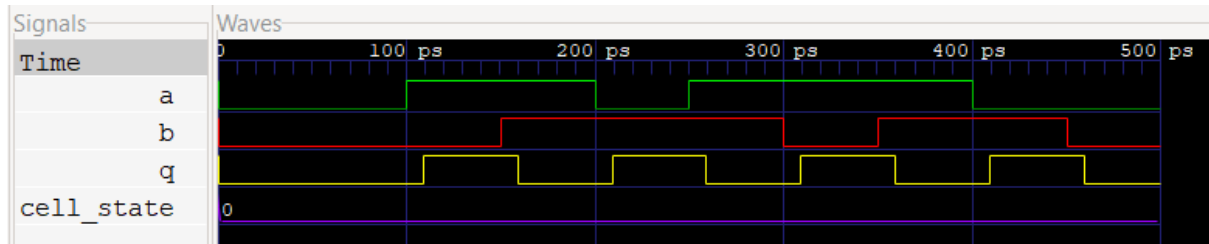


Figure 2.14: RSFQ MERGE digital simulation results.

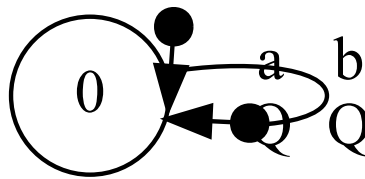


Figure 2.15: RSFQ MERGE Mealy finite state diagram.

Power consumption

Table 2.6: RSFQ MERGE power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	1840	3.28
2	1840	6.57
5	1840	16.4
10	1840	32.8
20	1840	65.7
50	1840	164

2.1.4 PTLTX

The RSFQ PTLTX is a cell which transmits a pulse signal over a PTL. It is connected to a cell which is not designed to connect to PTLs, if a PTL connection is required.

Schematic

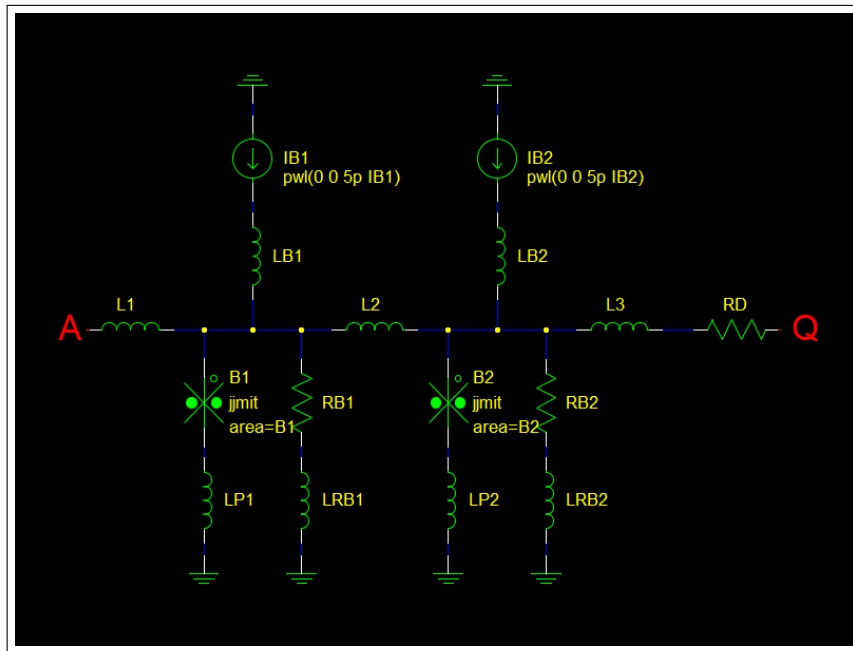


Figure 2.16: Schematic of RSFQ PTLTX.

Layout

The physical layout of the RSFQ PTLTX is shown in Fig. 2.17. The layout height is $70\ \mu\text{m}$ and the width is $20\ \mu\text{m}$. The biasing is brought in on M5 from an external bias line at the top of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

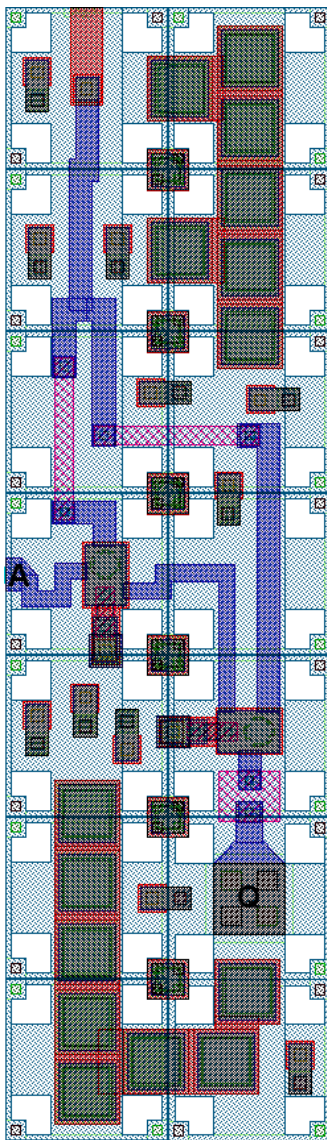


Figure 2.17: RSFQ PTLTX Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/15.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 4 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      q
8  .subckt THmit11_PTLTX a q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICtrans=2.5
20 .param LB=2p
21 .param BiasCoef=0.7
22 .param Lptl=2p
23 .param RD=1.36
24
25 .param B1=IC
26 .param B2=ICtrans
27
28 .param IB1=BiasCoef*Ic0*B1
29 .param IB2=BiasCoef*Ic0*B2
30
31 .param LB1=LB
32 .param LB2=LB
33
34 .param L1=Phi0/(4*B1*Ic0)
35 .param L2=Phi0/(2*B1*Ic0)
36 .param L3=Lptl
37
38 .param LP1=LP
39 .param LP2=LP
40
41 .param RB1=B0Rs/B1
42 .param RB2=B0Rs/B2
43
44 .param LRB1=(RB1/Rsheet)*Lsheet+LP
45 .param LRB2=(RB2/Rsheet)*Lsheet+LP
46
47 B1 1 2 jjmit area=B1
48 B2 4 5 jjmit area=B2
49
50 IB1 0 3 pwl(0 0 5p IB1)
51 IB2 0 6 pwl(0 0 5p IB2)
52
53 LB1 3 1 1.003E-012
54 LB2 6 4 3.702E-012
55
56 L1 a 1 2.061E-012
57 L2 1 4 4.1E-012
58 L3 4 7 6.098E-013
59
60 LP1 2 0 4.298E-013
61 LP2 5 0 3.112E-013
62
63 RD 7 q RD
64
65 RB1 1 101 RB1
66 LRB1 101 0 LRB1
67 RB2 4 104 RB2
68 LRB2 104 0 LRB2
69 .ends

```

Listing 2.7: RSFQ PTLTX JoSIM netlist.

Table 2.7: RSFQ PTLTX pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ PTLTX using JoSIM is shown in Fig. 2.18. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the phase over the output JJ of pin **q**, and
- (d) the phase over the input JJ of the load circuit connected via a PTL to the PTLTX.

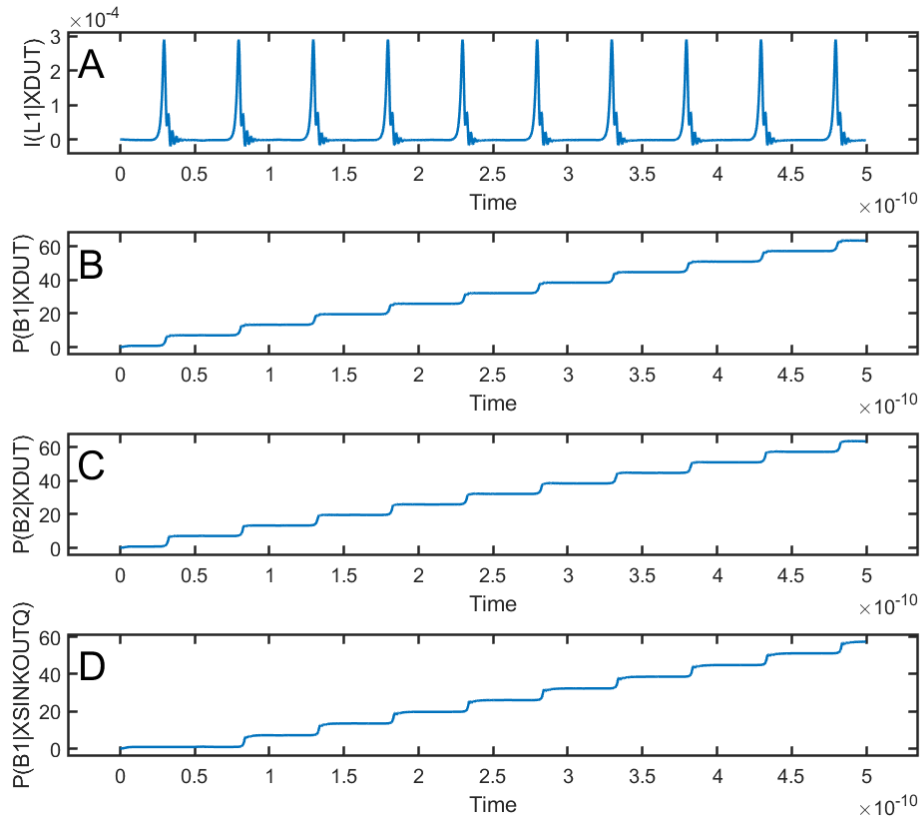


Figure 2.18: RSFQ PTLTX analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 15 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_PTLTX_v3p0_extracted (a, q);
9
10 input
11     a;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_a_q = 3.3,
21     ct_state0_a_a = 5.2;
22
23 reg
24     errorsignal_a;
25
26 integer
27     outfile,
28     cell_state; // internal state of the cell
29
30 initial
31     begin
32         errorsignal_a = 0;
33         cell_state = 0; // Startup state
34         q = 0; // All outputs start at 0
35     end
36
37 always @(posedge a or negedge a) // execute at positive and negative edges of input
38     begin
39         if ($time>4) // arbitrary steady-state time)
40             begin
41                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
42                     begin
43                         outfile = $fopen("errors.txt", "a");
44                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
45                             ↪ ", $stime);
46                         $fclose(outfile);
47                         q <= 1'bX; // Set all outputs to unknown
48                     end
49                 if (errorsignal_a == 0)
50                     begin
51                         case (cell_state)
52                             0: begin
53                                 q <= #(delay_state0_a_q) !q;
54                                 errorsignal_a = 1; // Critical timing on this input; assign
55                                     ↪ immediately
56                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
57                                     ↪ after critical timing expires
58                             end
59                         endcase
60                     end
61             end
62     end
63 endmodule

```

Listing 2.8: RSFQ PTLTX verilog model.

The digital simulation results for the RSFQ PTLTX is shown in Fig. 2.19 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.20.

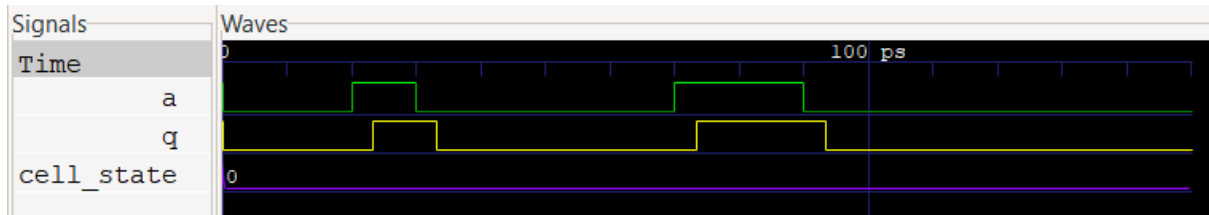


Figure 2.19: RSFQ PTLTX digital simulation results.

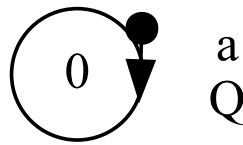


Figure 2.20: RSFQ PTLTX Mealy finite state machine diagram.

Power Consumption

Table 2.8: RSFQ PTLTX power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	910	1.03
2	910	2.07
5	910	5.17
10	910	10.3
20	910	20.7
50	910	51.7

2.1.5 PTLRX

The PTLRX is a receiver cell which receives a pulse signal from a PTL. It is connected to cells that are not designed to connect to PTLs when a PTL connection is required.

Schematic

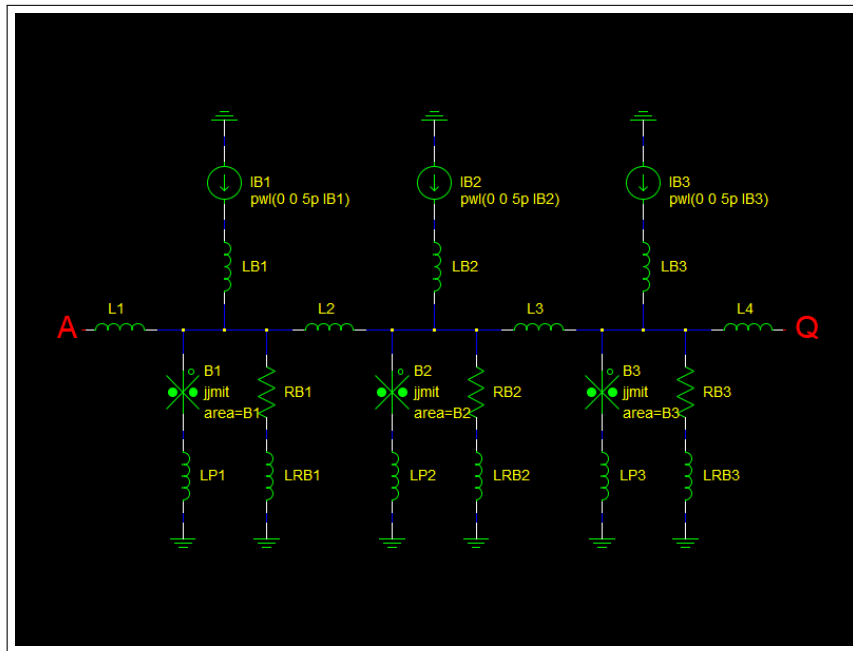


Figure 2.21: Schematic of RSFQ PTLRX.

Layout

The physical layout of the RSFQ PTLRX is shown in Fig. 2.22. The layout height is $70\ \mu\text{m}$ and the width is $20\ \mu\text{m}$. The biasing is brought in on M5 from an external bias line at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

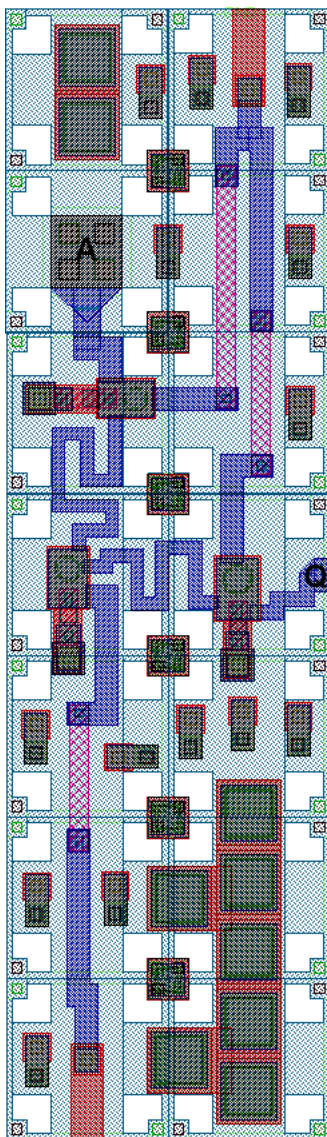


Figure 2.22: RSFQ PTLRX Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/23.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 23 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      q
8  .subckt THmit11_PTLRX a q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param LB=2p
21 .param BiasCoef=0.7
22 .param Lptl=2p
23
24 .param B1=ICreceive
25 .param B2=IC/1.25
26 .param B3=IC
27
28 .param IB1=BiasCoef*Ic0*B1
29 .param IB2=Ic0*B2
30 .param IB3=BiasCoef*Ic0*B3
31
32 .param LB1=LB
33 .param LB2=LB
34 .param LB3=LB
35
36 .param L1=Lptl
37 .param L2=Phi0/(2*B1*Ic0)
38 .param L3=Phi0/(2*B2*Ic0)
39 .param L4=Phi0/(4*IC*Ic0)
40
41 .param LP1=LP
42 .param LP2=LP
43 .param LP3=LP
44
45 .param RB1=B0Rs/B1
46 .param RB2=B0Rs/B2
47 .param RB3=B0Rs/B3
48
49 .param LRB1=(RB1/Rsheet)*Lsheet+LP
50 .param LRB2=(RB2/Rsheet)*Lsheet+LP
51 .param LRB3=(RB3/Rsheet)*Lsheet+LP
52
53 B1 1 2 jjmit area=B1
54 B2 4 5 jjmit area=B2
55 B3 7 8 jjmit area=B3
56
57 IB1 0 3 pwl(0 0 5p IB1)
58 IB2 0 6 pwl(0 0 5p IB2)
59 IB3 0 9 pwl(0 0 5p IB3)
60
61 LB1 3 1 1.265E-012
62 LB2 6 4 2.061E-012
63 LB3 9 7 1.659E-012
64
65 L1 a 1 1.414E-012
66 L2 1 4 6.505E-012
67 L3 4 7 5.179E-012
68 L4 7 q 2.06E-012
69
70 LP1 2 0 3.679E-013
71 LP2 5 0 4.598E-013
72 LP3 8 0 3.89E-013
73
74 RB1 1 101 RB1
75 LRB1 101 0 LRB1
76 RB2 4 104 RB2
77 LRB2 104 0 LRB2
78 RB3 7 107 RB3
79 LRB3 107 0 LRB3
80 .ends

```

Listing 2.9: RSFQ PTLRX JoSIM netlist.

Table 2.9: RSFQ PTLRX pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ PTLRX using JoSIM is shown in Fig. 2.23. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the phase over the output JJ of pin **q**, and
- (d) the phase over the input JJ of the load circuit connected to pin **q**.

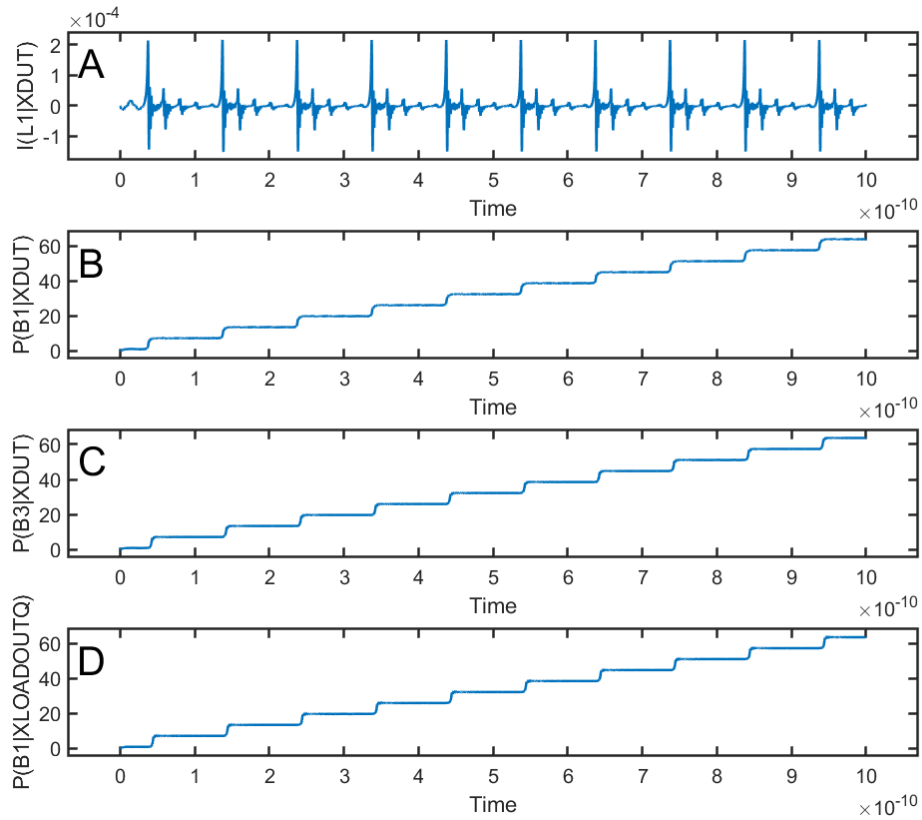


Figure 2.23: RSFQ PTLRX analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 23 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_PTLRX_v3p0_extracted (a, q);
9
10 input
11     a;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_a_q = 5.3,
21     ct_state0_a_a = 5.4;
22
23 reg
24     errorsignal_a;
25
26 integer
27     outfile,
28     cell_state; // internal state of the cell
29
30 initial
31     begin
32         errorsignal_a = 0;
33         cell_state = 0; // Startup state
34         q = 0; // All outputs start at 0
35     end
36
37 always @(posedge a or negedge a) // execute at positive and negative edges of input
38     begin
39         if ($time>4) // arbitrary steady-state time)
40             begin
41                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
42                     begin
43                         outfile = $fopen("errors.txt", "a");
44                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
45                             ↪ ", $stime);
46                         $fclose(outfile);
47                         q <= 1'bX; // Set all outputs to unknown
48                     end
49                 if (errorsignal_a == 0)
50                     begin
51                         case (cell_state)
52                             0: begin
53                                 q <= #(delay_state0_a_q) !q;
54                                 errorsignal_a = 1; // Critical timing on this input; assign
55                                     ↪ immediately
56                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
57                                     ↪ after critical timing expires
58                             end
59                         endcase
60                     end
61             end
62     end
63 endmodule

```

Listing 2.10: RSFQ PTLRX verilog model.

The digital simulation results for the RSFQ PTLRX is shown in Fig. 2.24 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.25.

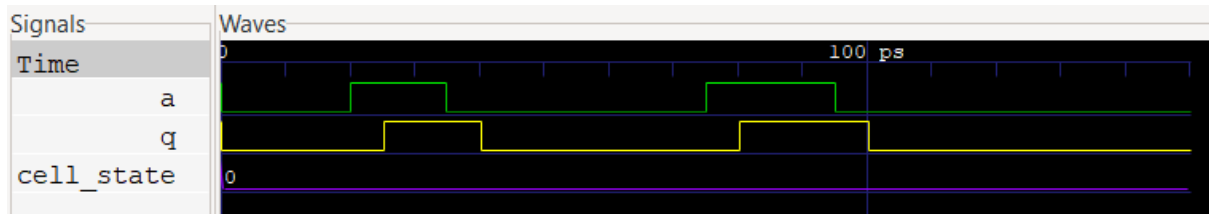


Figure 2.24: RSFQ PTLRX digital simulation results.

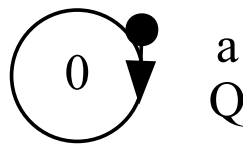


Figure 2.25: RSFQ PTLRX Mealy finite state machine diagram.

Power Consumption

Table 2.10: RSFQ PTLRX power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	1270	1.26
2	1270	2.52
5	1270	6.31
10	1270	12.6
20	1270	25.2
50	1270	63.1

2.1.6 Always0 Asynchronous

The Always0 Asynchronous cell provides an output which is always zero. Two versions of the cell exist: One with an **a** input port, as seen in Fig. 2.26, and one without an **a** input port, as seen in Fig. 2.27. The cell is not designed to be directly connected to a PTL.

Schematic

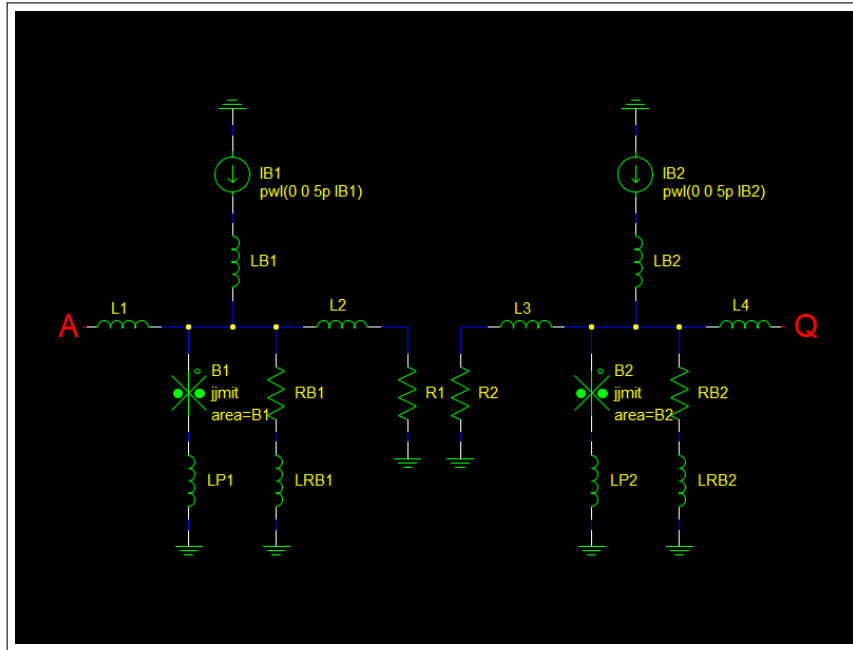


Figure 2.26: Schematic of RSFQ Always0 Asynchronous.

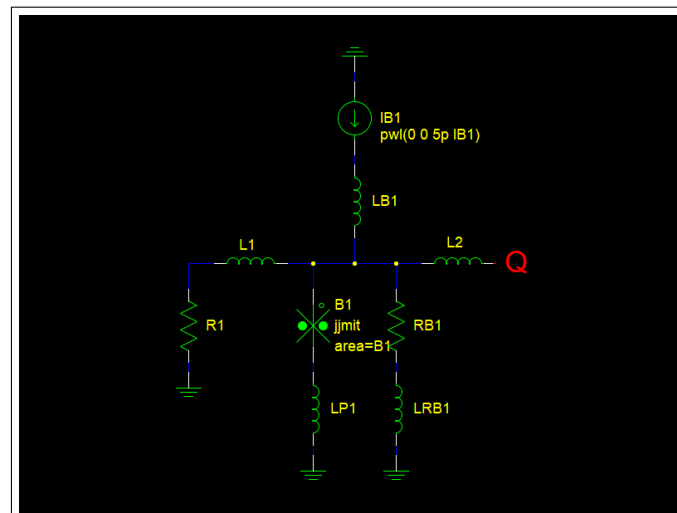


Figure 2.27: Schematic of RSFQ Always0 Asynchronous without an **a** input port.

Layout

The physical layouts of the two RSFQ Always0 Asynchronous cell versions are shown in Fig. 2.28a and 2.28b respectively. The height of the Always0 Asynchronous layout is $70\ \mu\text{m}$ and the width is $20\ \mu\text{m}$. For the version without the **a** input port, the width of the layout is $10\ \mu\text{m}$. The cells include an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

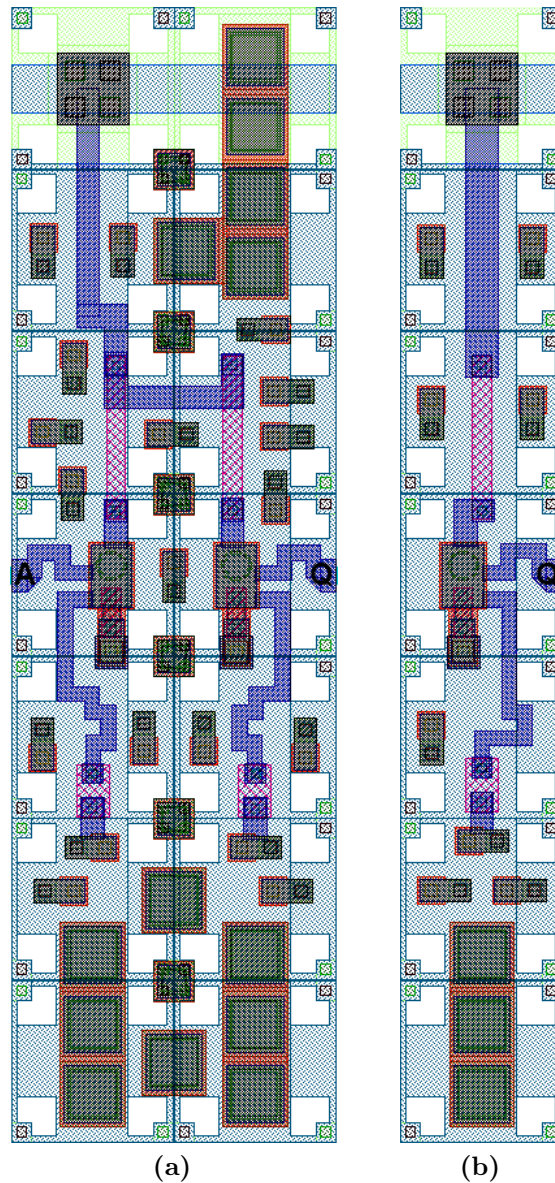


Figure 2.28: The physical layouts for (a) the RSFQ Always0 Asynchronous and (b) the RSFQ Always0 Asynchronous without an **a** input port.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/23.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 3 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      q
8  .subckt THmit11_ALWAYS0_ASYNC a q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=IC
23 .param B2=IC
24
25 .param IB1=B1*Ic0*BiasCoef
26 .param IB2=B2*Ic0*BiasCoef
27
28 .param LB1=LB
29 .param LB2=LB
30
31 .param L1=Phi0/(4*B1*Ic0)
32 .param L2=Phi0/(2*B1*Ic0)
33 .param L3=Phi0/(2*B2*Ic0)
34 .param L4=Phi0/(4*B2*Ic0)
35
36 .param R1=2
37 .param R2=2
38
39 .param LP1=LP
40 .param LP2=LP
41
42 .param RB1=B0Rs/B1
43 .param RB2=B0Rs/B2
44
45 .param LRB1=(RB1/Rsheet)*Lsheet+LP
46 .param LRB2=(RB2/Rsheet)*Lsheet+LP
47
48 B1 1 2 jjmit area=B1
49 B2 6 7 jjmit area=B2
50
51 IB1 0 3 pwl(0 0 5p IB1)
52 IB2 0 8 pwl(0 0 5p IB2)
53
54 LB1 3 1 7.025E-013
55 LB2 8 6 6.963E-013
56
57 L1 a 1 2.08E-012
58 L2 1 4 4.162E-012
59 L3 5 6 4.149E-012
60 L4 6 q 2.083E-012
61
62 R1 4 0 R1
63 R2 5 0 R2
64
65 LP1 2 0 3.835E-013
66 LP2 7 0 3.846E-013
67
68 RB1 1 101 RB1
69 LRB1 101 0 LRB1
70 RB2 6 106 RB2
71 LRB2 106 0 LRB2
72 .ends

```

Listing 2.11: RSFQ Always0 Asynchronous JoSIM netlist.

Table 2.11: RSFQ Always0 Asynchronous pin list.

Pin	Description
a	Data input
q	Data output

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/23.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 3 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports          q
8  .subckt THmit11_ALWAYS0_ASYNC_NOA q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=IC
23
24 .param IB1=B1*Ic0*BiasCoef
25
26 .param LB1=LB
27
28 .param L1=Phi0/(2*B1*Ic0)
29 .param L2=Phi0/(4*B1*Ic0)
30
31 .param R1=2
32
33 .param LP1=LP
34
35 .param RB1=B0Rs/B1
36
37 .param LRB1=(RB1/Rsheet)*Lsheet+LP
38
39 B1 2 3 jjmit area=B1
40
41 IB1 0 4 pwl(0 0 5p IB1)
42
43 LB1 4 2 7.52E-013
44
45 LP1 3 0 4.232E-013
46
47 L1 1 2 4.162E-012
48 L2 2 q 2.078E-012
49
50 R1 1 0 R1
51
52 RB1 2 102 RB1
53 LRB1 102 0 LRB1
54 .ends

```

Listing 2.12: RSFQ Always0 Asynchronous, without an **a** input port, JoSIM netlist.

Table 2.12: RSFQ Always0 Asynchronous, without an **a** input port, pin list.

Pin	Description
q	Data output

Digital model

```

1 // -----
2 // Author: L. Schindler
3 // Version: 3.0
4 // Last modification date: 23 August 2022
5 // Last modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_ALWAYS0_ASYNC (a, q);
9
10 input
11     a;
12
13 output
14     q;
15
16 reg
17     q;
18
19 initial
20     begin
21         q = 0; // Output always 0
22     end
23
24 always
25     begin
26         #10 q = 0; // Output always 0
27     end
28 endmodule

```

Listing 2.13: RSFQ Always0 Asynchronous verilog model.

```

1 // -----
2 // Author: L. Schindler
3 // Version: 3.0
4 // Last modification date: 23 August 2022
5 // Last modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_ALWAYS0_ASYNC_NOA (q);
9
10
11 output
12     q;
13
14 reg
15     q;
16
17 initial
18     begin
19         q = 0; // Output always 0
20     end
21
22 always
23     begin
24         #10 q = 0; // Output always 0
25     end
26 endmodule

```

Listing 2.14: RSFQ Always0 Asynchronous, without an **a** input port, verilog model.

Power consumption

Table 2.13: RSFQ Always0 Asynchronous power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	910	0.52
2	910	1.03
5	910	2.58
10	910	5.17
20	910	10.3
50	910	25.8

Table 2.14: RSFQ Always0 Asynchronous, without an **a** input port, power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	455	N/A
2	455	N/A
5	455	N/A
10	455	N/A
20	455	N/A
50	455	N/A

2.1.7 Always0 Synchronous

The Always0 Synchronous cell provides an output which is always zero synchronous to a clock signal. Two versions of the cell exist: One with an **a** input port, as seen in Fig. 2.29, and one without an **a** input port, as seen in Fig. 2.30. The Always0 Synchronous is not designed to be directly connected to a PTL.

Schematic

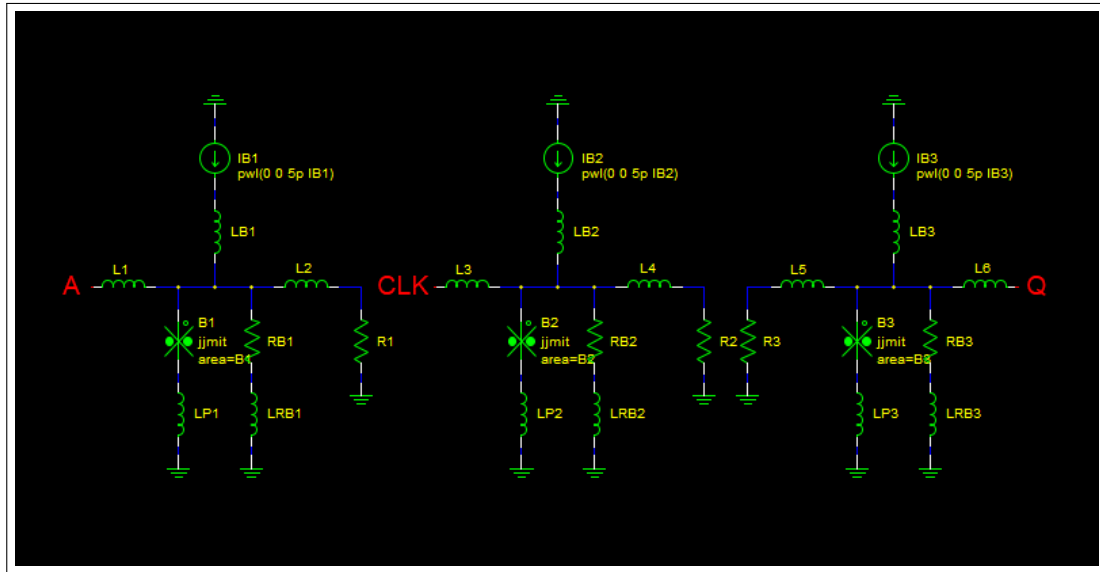


Figure 2.29: Schematic of RSFQ Always0 Synchronous.

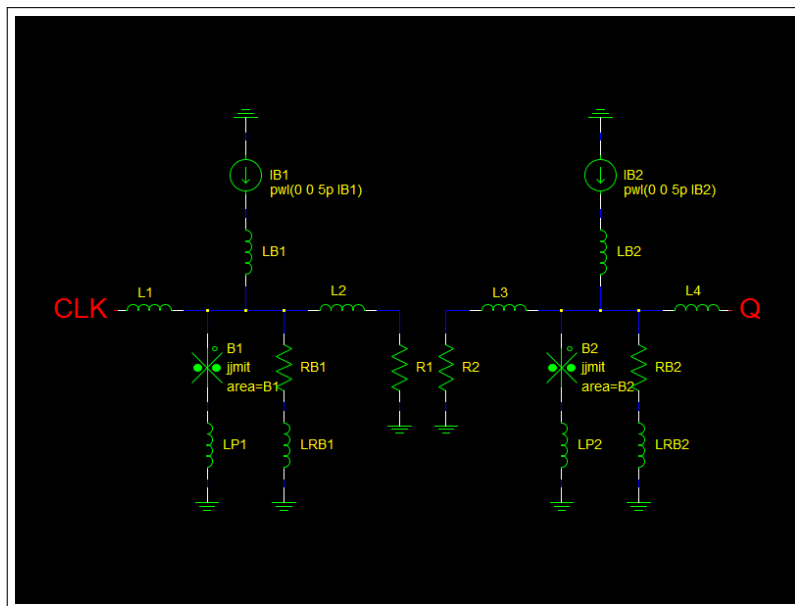


Figure 2.30: Schematic of RSFQ Always0 Synchronous without an **a** input port.

Layout

The physical layouts of the two RSFQ Always0 Synchronous cell versions are shown in Fig. 2.31a and 2.31b respectively. The height of the Always0 Synchronous layout is $70\ \mu\text{m}$ and the width is $20\ \mu\text{m}$. For the version without the **a** input port, the width of the layout is $20\ \mu\text{m}$. The cells include an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

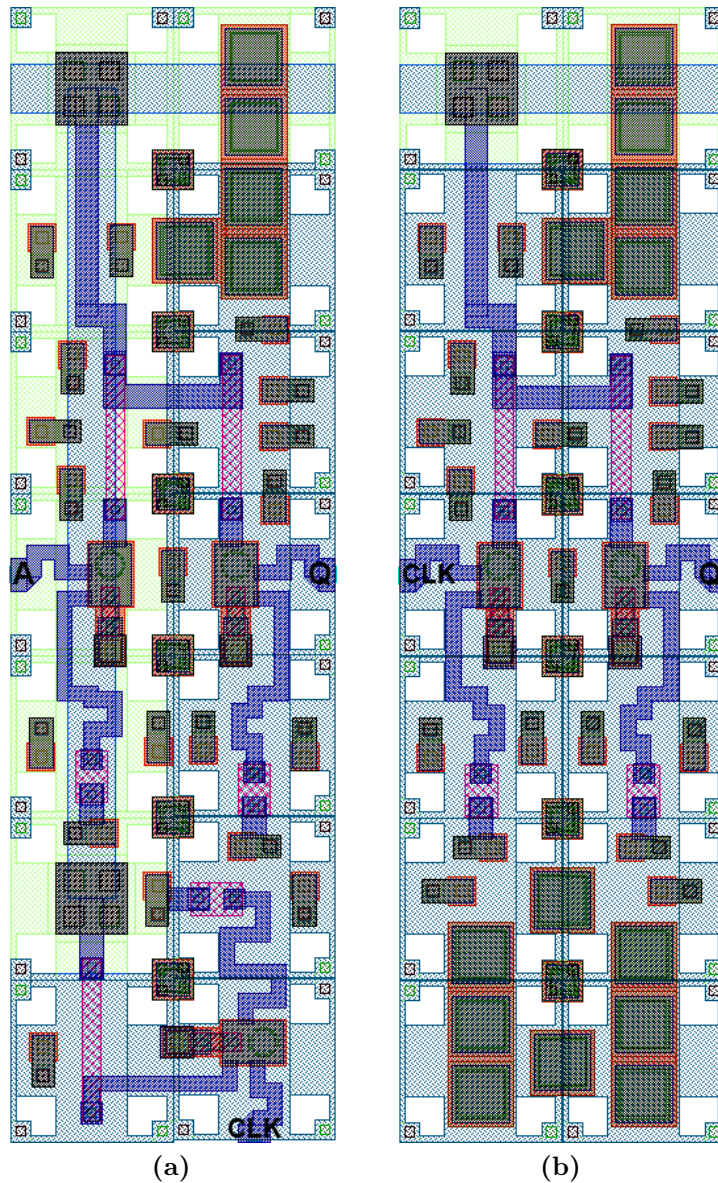


Figure 2.31: The physical layouts for (a) the RSFQ Always0 Synchronous and (b) the RSFQ Always0 Synchronous without an **a** input port.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/23.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 3 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      clk      q
8  .subckt THmit11_ALWAYS0_SYNC a clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=IC
23 .param B2=IC
24 .param B3=IC
25
26 .param IB1=B1*Ic0*BiasCoef
27 .param IB2=B2*Ic0*BiasCoef
28 .param IB3=B3*Ic0*BiasCoef
29
30 .param LB1=LB
31 .param LB2=LB
32 .param LB3=LB
33
34 .param L1=Phi0/(4*B1*Ic0)
35 .param L2=Phi0/(2*B1*Ic0)
36 .param L3=Phi0/(4*B2*Ic0)
37 .param L4=Phi0/(2*B2*Ic0)
38 .param L5=Phi0/(2*B3*Ic0)
39 .param L6=Phi0/(4*B3*Ic0)
40
41 .param R1=2
42 .param R2=2
43 .param R3=2
44
45 .param LP1=LP
46 .param LP2=LP
47 .param LP3=LP
48
49 .param RB1=B0Rs/B1
50 .param RB2=B0Rs/B2
51 .param RB3=B0Rs/B3
52
53 .param LRB1=(RB1/Rsheet)*Lsheet+LP
54 .param LRB2=(RB2/Rsheet)*Lsheet+LP
55 .param LRB3=(RB3/Rsheet)*Lsheet+LP
56
57 B1 1 2 jjmit area=B1
58 B2 5 6 jjmit area=B2
59 B3 10 11 jjmit area=B3
60
61 IB1 0 3 pw1(0 0 5p IB1)
62 IB2 0 7 pw1(0 0 5p IB2)
63 IB3 0 12 pw1(0 0 5p IB3)
64
65 LB1 3 1 7.02E-013
66 LB2 7 5 3.529E-012
67 LB3 12 10 6.962E-013
68
69 LP1 2 0 3.842E-013
70 LP2 6 0 3.469E-013
71 LP3 11 0 3.849E-013
72
73 L1 a 1 2.08E-012
74 L2 1 4 4.129E-012
75 L3 clk 5 2.057E-012
76 L4 5 8 4.16E-012
77 L5 9 10 4.148E-012
78 L6 10 q 2.082E-012
79
80 R1 4 0 R1
81 R2 8 0 R2
82 R3 9 0 R3
83
84 RB1 1 101 RB1
85 LRB1 101 0 LRB1
86 RB2 5 105 RB2
87 LRB2 105 0 LRB2
88 RB3 10 110 RB3
89 LRB3 110 0 LRB3
90 .ends

```

Listing 2.15: RSFQ Always0 Synchronous JoSIM netlist.

Table 2.15: RSFQ Always0 Synchronous pin list.

Pin	Description
a	Data input
clk	Clock input
q	Data output


```

1  * Back-annotated simulation file written
   ↳ by InductEx v.6.1.52 on 2022/08/23.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 23 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports          clk      q
8  .subckt THmit11_ALWAYS0_SYNC_NOA clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
   ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
   ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=IC
23 .param B2=IC
24
25 .param IB1=B1*Ic0*BiasCoef
26 .param IB2=B2*Ic0*BiasCoef
27
28 .param LB1=LB
29 .param LB2=LB
30
31 .param L1=Phi0/(4*B1*Ic0)
32 .param L2=Phi0/(2*B1*Ic0)
33 .param L3=Phi0/(2*B2*Ic0)
34 .param L4=Phi0/(4*B2*Ic0)
35
36 .param R1=2
37 .param R2=2
38
39 .param LP1=LP
40 .param LP2=LP
41
42 .param RB1=B0Rs/B1
43 .param RB2=B0Rs/B2
44
45 .param LRB1=(RB1/Rsheet)*Lsheet+LP
46 .param LRB2=(RB2/Rsheet)*Lsheet+LP
47
48 B1 1 2 jjmit area=B1
49 B2 6 7 jjmit area=B2
50
51 IB1 0 3 pwl(0 0 5p IB1)
52 IB2 0 8 pwl(0 0 5p IB2)
53
54 LB1 3 1 7.025E-013
55 LB2 8 6 6.963E-013
56
57 LP1 2 0 3.835E-013
58 LP2 7 0 3.846E-013
59
60 L1 clk 1 2.08E-012
61 L2 1 4 4.162E-012
62 L3 5 6 4.149E-012
63 L4 6 q 2.083E-012
64
65 R1 4 0 R1
66 R2 5 0 R2
67
68 RB1 1 101 RB1
69 LRB1 101 0 LRB1
70 RB2 6 106 RB2
71 LRB2 106 0 LRB2
72 .ends

```

Listing 2.16: RSFQ Always0 Synchronous, without an **a** input port, JoSIM netlist.

Table 2.16: RSFQ Always0 Synchronous, without an **a** input port, pin list.

Pin	Description
clk	Clock input
q	Data output

Digital model

```

1 // -----
2 // Author: L. Schindler
3 // Version: 3.0
4 // Last modification date: 23 August 2022
5 // Last modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_ALWAYS0_SYNC (a, clk, q);
9
10 input
11     a;
12     clk;
13
14 output
15     q;
16
17 reg
18     q;
19
20 initial
21     begin
22         q = 0; // Output always 0
23     end
24
25 always
26     begin
27         #10 q = 0; // Output always 0
28     end
29 endmodule

```

Listing 2.17: RSFQ Always0 Synchronous verilog model.

```

1 // -----
2 // Author: L. Schindler
3 // Version: 3.0
4 // Last modification date: 23 August 2022
5 // Last modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_ALWAYS0_SYNC_NOA (clk, q);
9
10 input
11     clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 initial
20     begin
21         q = 0; // Output always 0
22     end
23
24 always
25     begin
26         #10 q = 0; // Output always 0
27     end
28 endmodule

```

Listing 2.18: RSFQ Always0 Synchronous, without an **a** input port, verilog model.

Power consumption

Table 2.17: RSFQ Always0 Synchronous power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	1365	1.03
2	1365	2.07
5	1365	5.17
10	1365	10.3
20	1365	20.7
50	1365	51.7

Table 2.18: RSFQ Always0 Synchronous, without an **a** input port, power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	910	0.52
2	910	1.03
5	910	2.58
10	910	5.17
20	910	10.3
50	910	25.8

2.2 Logic Cells

2.2.1 AND2

The RSFQ AND2 cell generates an output pulse if pulses from both input signal lines were received before the clock signal. The cell is not designed to be directly connected to a PTL.

Schematic

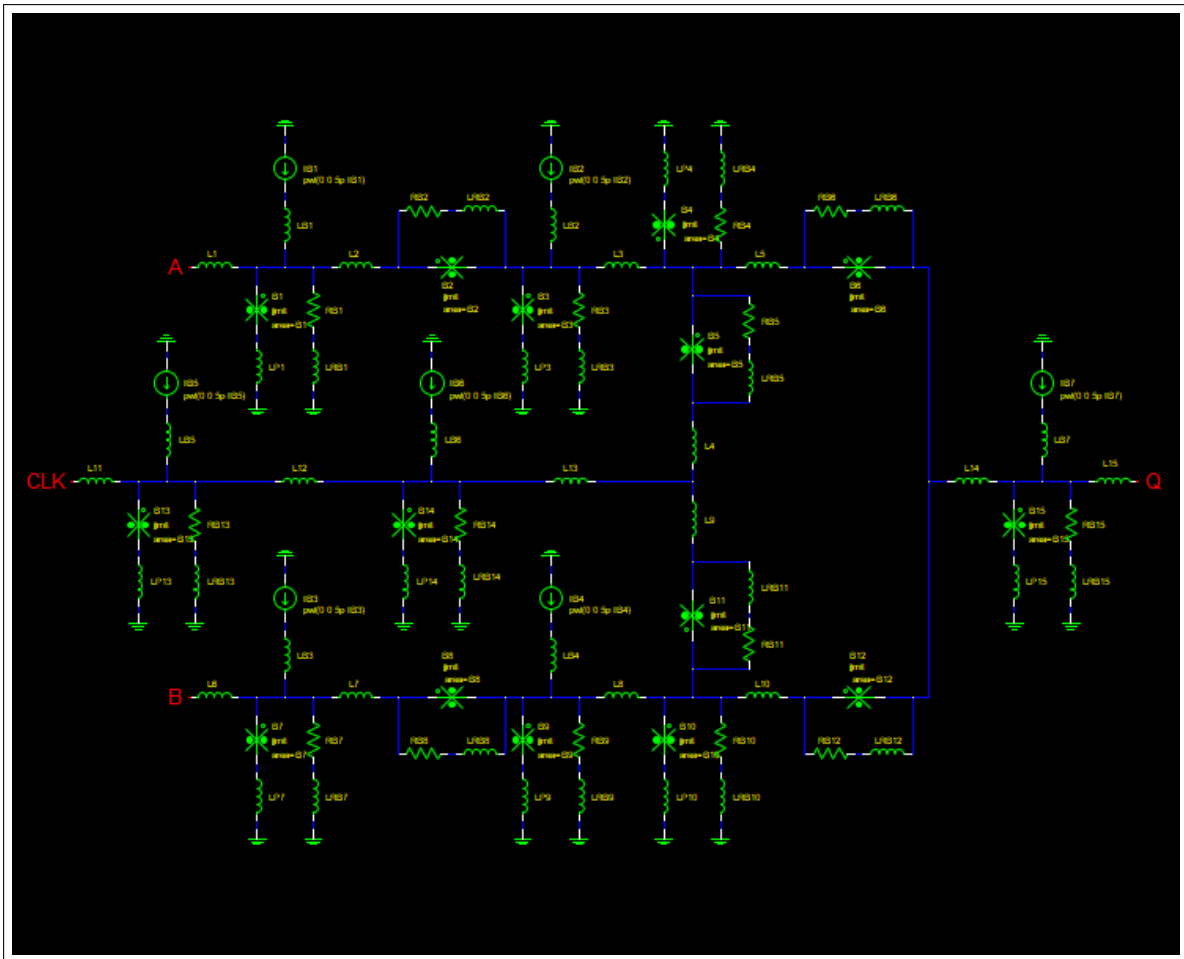


Figure 2.32: Schematic of RSFQ AND2.

Layout

The physical layout of the RSFQ AND2 is shown in Fig. 2.33. The layout height is $70\ \mu\text{m}$ and the width is $50\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

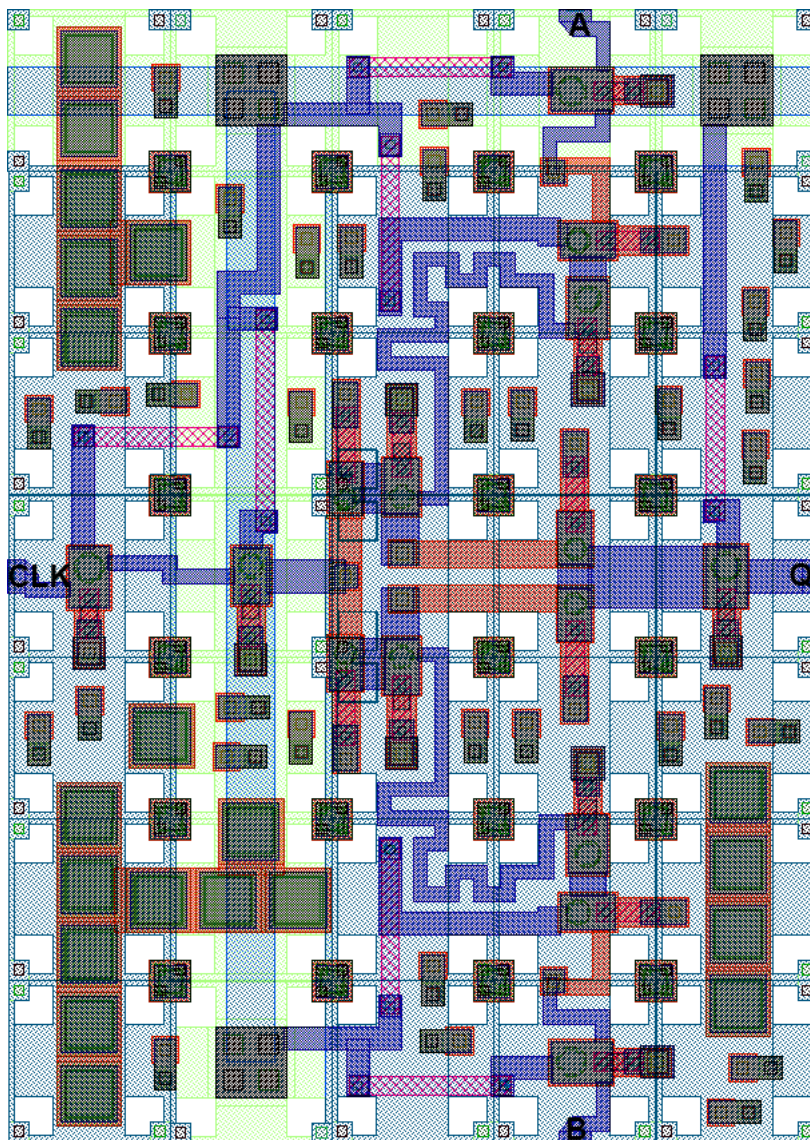


Figure 2.33: RSFQ AND2 Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/09.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 4 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      b      clk      q
8  .subckt THmit11_AND2 a b clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=2.5
23 .param B2=1.60
24 .param B3=1.94
25 .param B4=1.57
26 .param B5=1.25
27 .param B6=1.25
28 .param B7=2.5
29 .param B8=1.60
30 .param B9=1.94
31 .param B10=1.57
32 .param B11=1.25
33 .param B12=1.25
34 .param B13=2.5
35 .param B14=1.71
36 .param B15=2.5
37
38 .param IB1=175u
39 .param IB2=162u
40 .param IB3=175u
41 .param IB4=162u
42 .param IB5=175u
43 .param IB6=124u
44 .param IB7=175u
45
46 .param LB1=LB
47 .param LB2=LB
48 .param LB3=LB
49 .param LB4=LB
50 .param LB5=LB
51 .param LB6=LB
52 .param LB7=LB
53
54 .param L1=1.8843p
55 .param L2=4.2266p
56 .param L3=9.6820p
57 .param L4=0.9913p
58 .param L5=2.5745p
59 .param L6=1.8843p
60 .param L7=4.2266p
61 .param L8=9.6820p
62 .param L9=0.9913p
63 .param L10=2.5745p
64 .param L11=1.5293p
65 .param L12=2.6230p
66 .param L13=0.7965p
67 .param L14=0.7671p
68 .param L15=1.2144p
69
70 .param LP1=LP
71 .param LP3=LP
72 .param LP4=LP
73 .param LP7=LP
74 .param LP9=LP
75 .param LP10=LP
76 .param LP13=LP
77 .param LP14=LP
78 .param LP15=LP
79
80 .param RB1=B0Rs/B1
81 .param RB2=B0Rs/B2
82 .param RB3=B0Rs/B3
83 .param RB4=B0Rs/B4
84 .param RB5=B0Rs/B5
85 .param RB6=B0Rs/B6
86 .param RB7=B0Rs/B7
87 .param RB8=B0Rs/B8
88 .param RB9=B0Rs/B9
89 .param RB10=B0Rs/B10
90 .param RB11=B0Rs/B11
91 .param RB12=B0Rs/B12
92 .param RB13=B0Rs/B13
93 .param RB14=B0Rs/B14
94 .param RB15=B0Rs/B15
95
96 .param LRB1=(RB1/Rsheet)*Lsheet+LP
97 .param LRB2=(RB2/Rsheet)*Lsheet
98 .param LRB3=(RB3/Rsheet)*Lsheet+LP
99 .param LRB4=(RB4/Rsheet)*Lsheet+LP
100 .param LRB5=(RB5/Rsheet)*Lsheet
101 .param LRB6=(RB6/Rsheet)*Lsheet
102 .param LRB7=(RB7/Rsheet)*Lsheet+LP
103 .param LRB8=(RB8/Rsheet)*Lsheet
104 .param LRB9=(RB9/Rsheet)*Lsheet+LP
105 .param LRB10=(RB10/Rsheet)*Lsheet+LP
106 .param LRB11=(RB11/Rsheet)*Lsheet
107 .param LRB12=(RB12/Rsheet)*Lsheet
108 .param LRB13=(RB13/Rsheet)*Lsheet+LP
109 .param LRB14=(RB14/Rsheet)*Lsheet+LP
110 .param LRB15=(RB15/Rsheet)*Lsheet+LP
111
112 B1 1 2 jjmit area=B1
113 B2 4 5 jjmit area=B2
114 B3 5 6 jjmit area=B3
115 B4 8 9 jjmit area=B4
116 B5 8 10 jjmit area=B5
117 B6 12 13 jjmit area=B6
118 B7 14 15 jjmit area=B7
119 B8 17 18 jjmit area=B8
120 B9 18 19 jjmit area=B9
121 B10 21 22 jjmit area=B10
122 B11 21 23 jjmit area=B11
123 B12 24 13 jjmit area=B12
124 B13 25 26 jjmit area=B13
125 B14 28 29 jjmit area=B14
126 B15 31 32 jjmit area=B15
127
128 IB1 0 3 pwl(0 0 5p IB1)
129 IB2 0 7 pwl(0 0 5p IB2)

```

```

130 IB3 0 16 pwl(0 0 5p IB3)
131 IB4 0 20 pwl(0 0 5p IB4)
132 IB5 0 27 pwl(0 0 5p IB5)
133 IB6 0 30 pwl(0 0 5p IB6)
134 IB7 0 33 pwl(0 0 5p IB7)
135
136 LB1 3 1 1.032E-012
137 LB2 7 5 3.135E-012
138 LB3 16 14 1.016E-012
139 LB4 20 18 3.19E-012
140 LB5 27 25 1.708E-012
141 LB6 30 28 6.363E-013
142 LB7 33 31 7.55E-013
143
144 L1 a 1 1.874E-012
145 L2 1 4 4.249E-012
146 L3 5 8 9.672E-012
147 L4 10 11 9.976E-013
148 L5 8 12 2.561E-012
149 L6 b 14 1.875E-012
150 L7 14 17 4.233E-012
151 L8 18 21 9.691E-012
152 L9 11 23 9.928E-013
153 L10 21 24 2.556E-012
154 L11 clk 25 1.531E-012
155 L12 25 28 2.626E-012
156 L13 28 11 7.973E-013
157 L14 13 31 7.691E-013
158 L15 31 q 1.217E-012
159
160 LP1 2 0 3.464E-013
161 LP3 6 0 6.827E-013
162 LP4 0 9 4.723E-013
163 LP7 15 0 3.484E-013
164 LP9 19 0 6.81E-013
165 LP10 22 0 4.756E-013
166 LP13 26 0 3.392E-013

167 LP14 29 0 3.972E-013
168 LP15 32 0 3.699E-013
169
170 RB1 1 101 RB1
171 LRB1 101 0 LRB1
172 RB2 4 104 RB2
173 LRB2 104 5 LRB2
174 RB3 5 105 RB3
175 LRB3 105 0 LRB3
176 RB4 108 8 RB4
177 LRB4 0 108 LRB4
178 RB5 8 110 RB5
179 LRB5 110 10 LRB5
180 RB6 12 112 RB6
181 LRB6 112 13 LRB6
182 RB7 14 114 RB7
183 LRB7 114 0 LRB7
184 RB8 17 117 RB8
185 LRB8 117 18 LRB8
186 RB9 18 118 RB9
187 LRB9 118 0 LRB9
188 RB10 21 121 RB10
189 LRB10 121 0 LRB10
190 RB11 123 21 RB11
191 LRB11 23 123 LRB11
192 RB12 24 124 RB12
193 LRB12 124 13 LRB12
194 RB13 25 125 RB13
195 LRB13 125 0 LRB13
196 RB14 28 128 RB14
197 LRB14 128 0 LRB14
198 RB15 31 131 RB15
199 LRB15 131 0 LRB15
200 .ends

```

Listing 2.19: RSFQ AND2 JoSIM netlist.**Table 2.19:** RSFQ AND2 pin list.

Pin	Description
a	Data input
b	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ AND2 using JoSIM is shown in Fig. 2.34. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b**,
- (d) the phase over the input JJ of pin **b**,
- (e) the current through the input inductor connected to pin **clk**,
- (f) the phase over the input JJ of pin **clk**,
- (g) the phase over the output JJ of pin **q**, and
- (h) the phase over the input JJ of the load circuit connected to pin **q**.

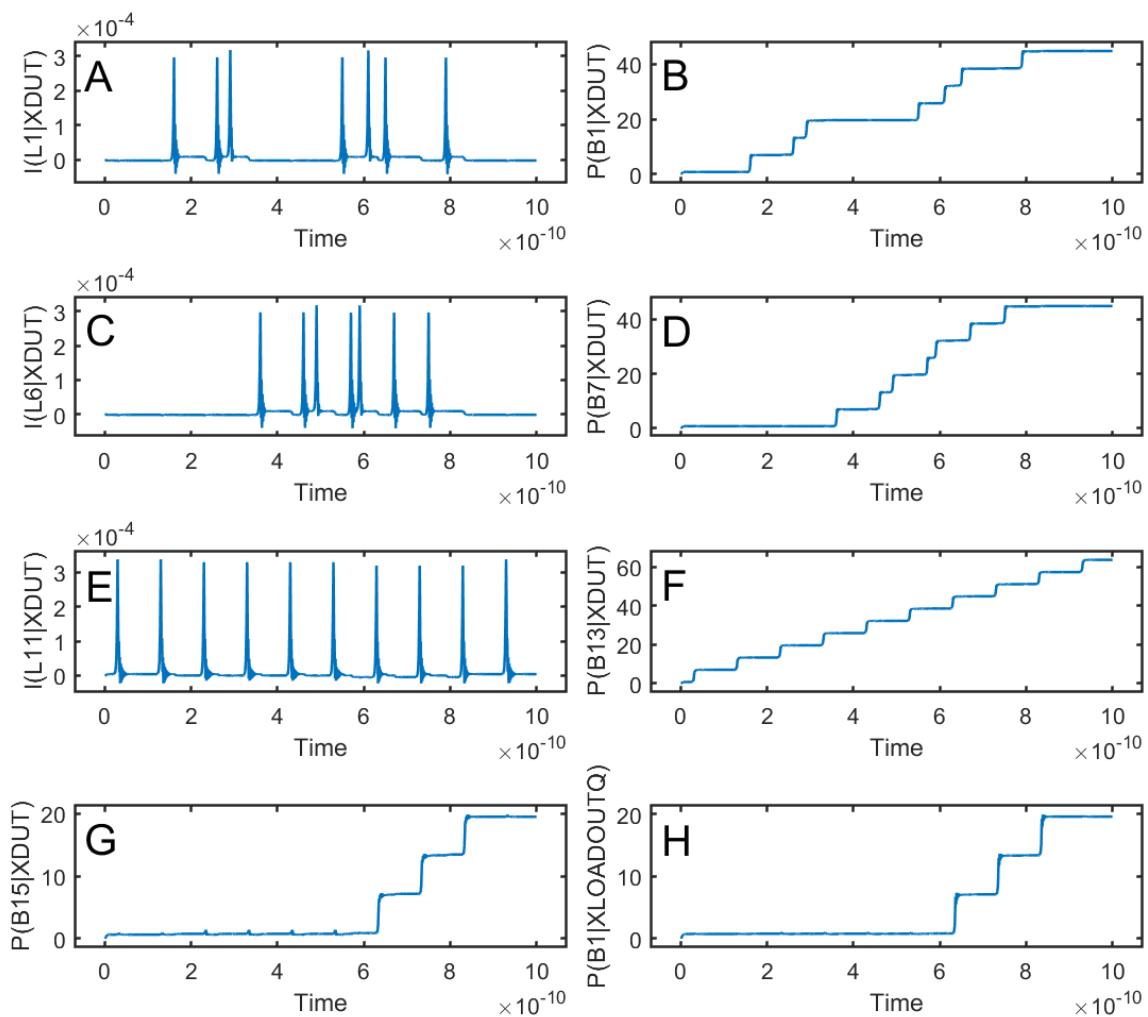


Figure 2.34: RSFQ AND2 analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 9 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit1l_AND2_v3p0_extracted (a, b, clk, q);
9
10 input
11     a, b, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state3_clk_q = 5.0,
21     ct_state0_clk_a = 1.2,
22     ct_state0_clk_b = 1.2,
23     ct_state1_clk_a = 1.6,
24     ct_state1_clk_b = 1.0,
25     ct_state2_clk_a = 1.0,
26     ct_state2_clk_b = 1.6,
27     ct_state3_clk_a = 0.7,
28     ct_state3_clk_b = 0.7;
29
30 reg
31     errorsignal_a,
32     errorsignal_b,
33     errorsignal_clk;
34
35 integer
36     outfile,
37     cell_state; // internal state of the cell
38
39 initial
40     begin
41         errorsignal_a = 0;
42         errorsignal_b = 0;
43         errorsignal_clk = 0;
44         cell_state = 0; // Startup state
45         q = 0; // All outputs start at 0
46     end
47
48 always @(posedge a or negedge a) // execute at positive and negative edges of input
49     begin
50         if ($time>4) // arbitrary steady-state time)
51             begin
52                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
53                     begin
54                         outfile = $fopen("errors.txt", "a");
55                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
56                         $fclose(outfile);
57                         q <= 1'bX; // Set all outputs to unknown
58                     end
59                 if (errorsignal_a == 0)
60                     begin
61                         case (cell_state)
62                             0: begin
63                                 cell_state = 1; // Blocking statement -- immediately
64                             end
65                             1: begin
66

```

```

67         2: begin
68             cell_state = 3; // Blocking statement -- immediately
69         end
70         3: begin
71             end
72         endcase
73     end
74 end
75 end
76
77 always @(posedge b or negedge b) // execute at positive and negative edges of input
78 begin
79     if ($time>4) // arbitrary steady-state time)
80     begin
81         if (errorsignal_b == 1'b1) // A critical timing is active for this input
82         begin
83             outfile = $fopen("errors.txt", "a");
84             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
            ↪ ", $stime);
85             $fclose(outfile);
86             q <= 1'bX; // Set all outputs to unknown
87         end
88         if (errorsignal_b == 0)
89         begin
90             case (cell_state)
91             0: begin
92                 cell_state = 2; // Blocking statement -- immediately
93             end
94             1: begin
95                 cell_state = 3; // Blocking statement -- immediately
96             end
97             2: begin
98             end
99             3: begin
100            end
101            endcase
102        end
103    end
104 end
105
106 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
107 begin
108     if ($time>4) // arbitrary steady-state time)
109     begin
110         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
111         begin
112             outfile = $fopen("errors.txt", "a");
113             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
            ↪ ", $stime);
114             $fclose(outfile);
115             q <= 1'bX; // Set all outputs to unknown
116         end
117         if (errorsignal_clk == 0)
118         begin
119             case (cell_state)
120             0: begin
121                 errorsignal_a = 1; // Critical timing on this input; assign
122                 ↪ immediately
123                 errorsignal_a <= #(ct_state0_clk_a) 0; // Clear error signal
124                 ↪ after critical timing expires
125                 errorsignal_b = 1; // Critical timing on this input; assign
126                 ↪ immediately
127                 errorsignal_b <= #(ct_state0_clk_b) 0; // Clear error signal
128                 ↪ after critical timing expires
129             end
130             1: begin
131                 cell_state = 0; // Blocking statement -- immediately
132                 errorsignal_a = 1; // Critical timing on this input; assign
133                 ↪ immediately
134                 errorsignal_a <= #(ct_state1_clk_a) 0; // Clear error signal

```

```

130         ↪ after critical timing expires
        errorsignal_b = 1; // Critical timing on this input; assign
131         ↪ immediately
        errorsignal_b <= #(ct_state1_clk_b) 0; // Clear error signal
        ↪ after critical timing expires
132     end
133 2: begin
134     cell_state = 0; // Blocking statement -- immediately
135     errorsignal_a = 1; // Critical timing on this input; assign
        ↪ immediately
136     errorsignal_a <= #(ct_state2_clk_a) 0; // Clear error signal
        ↪ after critical timing expires
137     errorsignal_b = 1; // Critical timing on this input; assign
        ↪ immediately
138     errorsignal_b <= #(ct_state2_clk_b) 0; // Clear error signal
        ↪ after critical timing expires
139     end
140 3: begin
141     q <= #(delay_state3_clk_q) !q;
142     cell_state = 0; // Blocking statement -- immediately
143     errorsignal_a = 1; // Critical timing on this input; assign
        ↪ immediately
144     errorsignal_a <= #(ct_state3_clk_a) 0; // Clear error signal
        ↪ after critical timing expires
145     errorsignal_b = 1; // Critical timing on this input; assign
        ↪ immediately
146     errorsignal_b <= #(ct_state3_clk_b) 0; // Clear error signal
        ↪ after critical timing expires
147     end
148     endcase
149     end
150 end
151 end
152
153 endmodule

```

Listing 2.20: RSFQ AND2 verilog model.

The digital simulation results for the RSFQ AND2 is shown in Fig. 2.35 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.36.

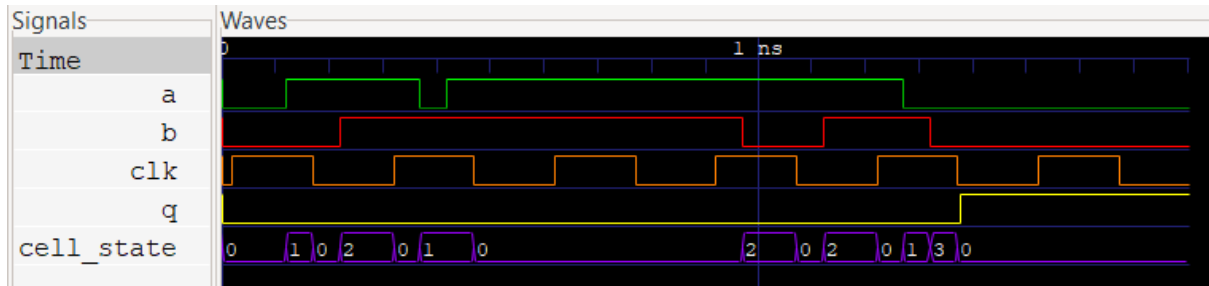


Figure 2.35: RSFQ AND2 digital simulation results.

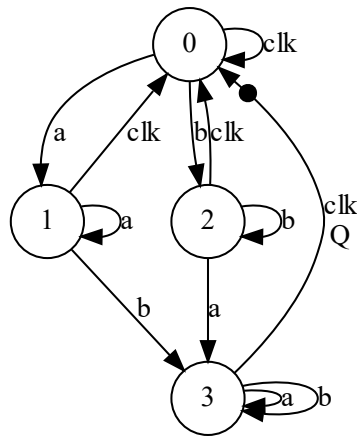


Figure 2.36: RSFQ AND2 Mealy finite state machine diagram.

Power Consumption

Table 2.20: RSFQ AND2 power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	2980	5.57
2	2980	11.1
5	2980	27.8
10	2980	55.7
20	2980	111
50	2980	278

2.2.2 OR2

The RSFQ OR2 cell generates an output pulse if an input pulse from either input line was received before the clock signal. The cell is not designed to be directly connected to a PTL.

Schematic

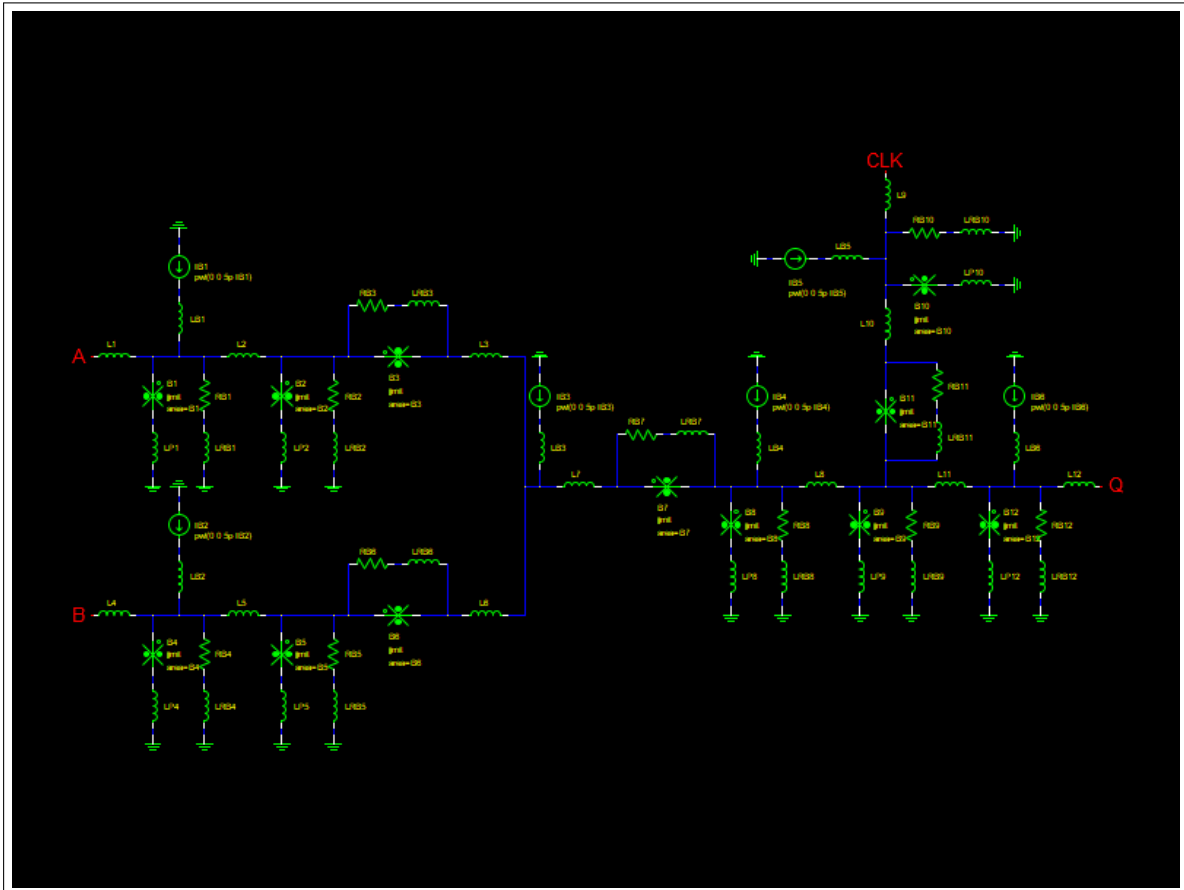


Figure 2.37: Schematic of RSFQ OR2.

Layout

The physical layout for the RSFQ OR2 is shown in Fig. 2.38. The layout height is $70\ \mu\text{m}$ and the width is $40\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

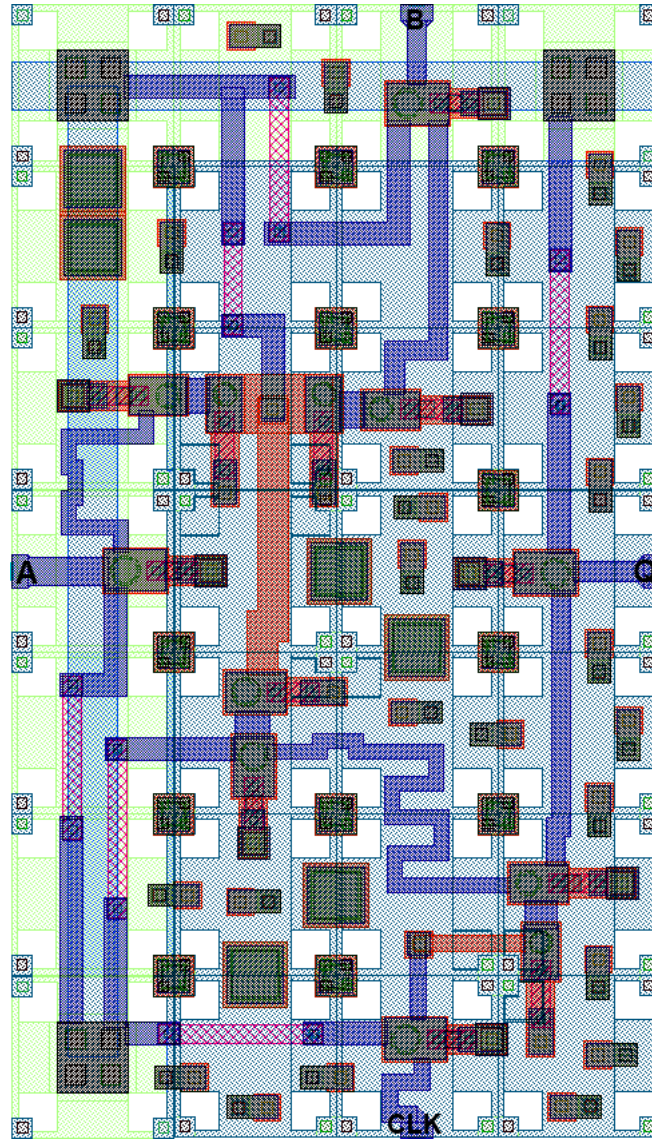


Figure 2.38: RSFQ OR2 Layout

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/07.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 4 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      b      clk      q
8  .subckt THmit11_OR2 a b clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=2.5
23 .param B2=1.67
24 .param B3=1.48
25 .param B4=2.5
26 .param B5=1.67
27 .param B6=1.48
28 .param B7=2.27
29 .param B8=2.36
30 .param B9=1.76
31 .param B10=2.5
32 .param B11=1.44
33 .param B12=2.5
34
35 .param IB1=175u
36 .param IB2=175u
37 .param IB3=283u
38 .param IB4=158u
39 .param IB5=175u
40 .param IB6=175u
41
42 .param LB1=LB
43 .param LB2=LB
44 .param LB3=LB
45 .param LB4=LB
46 .param LB5=LB
47 .param LB6=LB
48
49 .param L1=1.6049p
50 .param L2=4.4173p
51 .param L3=0.7163p
52 .param L4=1.6049p
53 .param L5=4.4173p
54 .param L6=0.7163p
55 .param L7=3.2947p
56 .param L8=8.7190p
57 .param L9=2.1422p
58 .param L10=3.7843p
59 .param L11=3.9628p
60 .param L12=1.6906p
61
62 .param LP1=LP
63 .param LP2=LP
64 .param LP4=LP
65 .param LP5=LP
66 .param LP8=LP
67 .param LP9=LP
68 .param LP10=LP
69 .param LP12=LP
70
71 .param RB1=B0Rs/B1
72 .param RB2=B0Rs/B2
73 .param RB3=B0Rs/B3
74 .param RB4=B0Rs/B4
75 .param RB5=B0Rs/B5
76 .param RB6=B0Rs/B6
77 .param RB7=B0Rs/B7
78 .param RB8=B0Rs/B8
79 .param RB9=B0Rs/B9
80 .param RB10=B0Rs/B10
81 .param RB11=B0Rs/B11
82 .param RB12=B0Rs/B12
83
84 .param LRB1=(RB1/Rsheet)*Lsheet+LP
85 .param LRB2=(RB2/Rsheet)*Lsheet+LP
86 .param LRB3=(RB3/Rsheet)*Lsheet
87 .param LRB4=(RB4/Rsheet)*Lsheet+LP
88 .param LRB5=(RB5/Rsheet)*Lsheet+LP
89 .param LRB6=(RB6/Rsheet)*Lsheet
90 .param LRB7=(RB7/Rsheet)*Lsheet
91 .param LRB8=(RB8/Rsheet)*Lsheet+LP
92 .param LRB9=(RB9/Rsheet)*Lsheet+LP
93 .param LRB10=(RB10/Rsheet)*Lsheet+LP
94 .param LRB11=(RB11/Rsheet)*Lsheet
95 .param LRB12=(RB12/Rsheet)*Lsheet+LP
96
97 B1 1 2 jjmit area=B1
98 B2 4 5 jjmit area=B2
99 B3 4 6 jjmit area=B3
100 B4 8 9 jjmit area=B4
101 B5 11 12 jjmit area=B5
102 B6 11 13 jjmit area=B6
103 B7 15 16 jjmit area=B7
104 B8 16 17 jjmit area=B8
105 B9 19 20 jjmit area=B9
106 B10 21 22 jjmit area=B10
107 B11 24 19 jjmit area=B11
108 B12 25 26 jjmit area=B12
109
110 IB1 0 3 pw1(0 0 5p IB1)
111 IB2 0 10 pw1(0 0 5p IB2)
112 IB3 0 14 pw1(0 0 5p IB3)
113 IB4 0 18 pw1(0 0 5p IB4)
114 IB5 0 23 pw1(0 0 5p IB5)
115 IB6 0 27 pw1(0 0 5p IB6)
116
117 LB1 3 1 LB1
118 LB2 10 8 LB2
119 LB3 14 7 LB3
120 LB4 18 16 LB4
121 LB5 23 21 LB5
122 LB6 27 25 LB6
123
124 L1 a 1 1.592E-012
125 L2 1 4 4.395E-012
126 L3 6 7 7.134E-013
127 L4 b 8 1.61E-012
128 L5 8 11 4.378E-012
129 L6 13 7 7.212E-013

```

```

130 | L7 7 15 3.29E-012
131 | L8 16 19 8.747E-012
132 | L9 clk 21 2.161E-012
133 | L10 21 24 3.766E-012
134 | L11 19 25 3.926E-012
135 | L12 25 q 1.675E-012
136 |
137 | LP1 2 0 4.683E-013
138 | LP2 5 0 4.125E-013
139 | LP4 9 0 3.472E-013
140 | LP5 12 0 4.43E-013
141 | LP8 17 0 4.666E-013
142 | LP9 20 0 4.384E-013
143 | LP10 22 0 3.63E-013
144 | LP12 26 0 4.421E-013
145 |
146 | RB1 1 101 RB1
147 | LRB1 101 0 LRB1
148 | RB2 4 104 RB2
149 | LRB2 104 0 LRB2
150 | RB3 4 106 RB3
151 | LRB3 106 6 LRB3

152 | RB4 8 108 RB4
153 | LRB4 108 0 LRB4
154 | RB5 11 111 RB5
155 | LRB5 111 0 LRB5
156 | RB6 11 113 RB6
157 | LRB6 113 13 LRB6
158 | RB7 15 115 RB7
159 | LRB7 115 16 LRB7
160 | RB8 16 116 RB8
161 | LRB8 116 0 LRB8
162 | RB9 19 119 RB9
163 | LRB9 119 0 LRB9
164 | RB10 21 121 RB10
165 | LRB10 121 0 LRB10
166 | RB11 24 124 RB11
167 | LRB11 124 19 LRB11
168 | RB12 25 125 RB12
169 | LRB12 125 0 LRB12
170 | .ends

```

Listing 2.21: RSFQ OR2 JoSIM netlist.**Table 2.21:** RSFQ OR2 pin list.

Pin	Description
a	Data input
b	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ OR2 using JoSIM is shown in Fig. 2.39. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b**,
- (d) the phase over the input JJ of pin **b**,
- (e) the current through the input inductor connected to pin **clk**,
- (f) the phase over the input JJ of pin **clk**,
- (g) the phase over the output JJ of pin **q**, and
- (h) the phase over the input JJ of the load circuit connected to pin **q**.

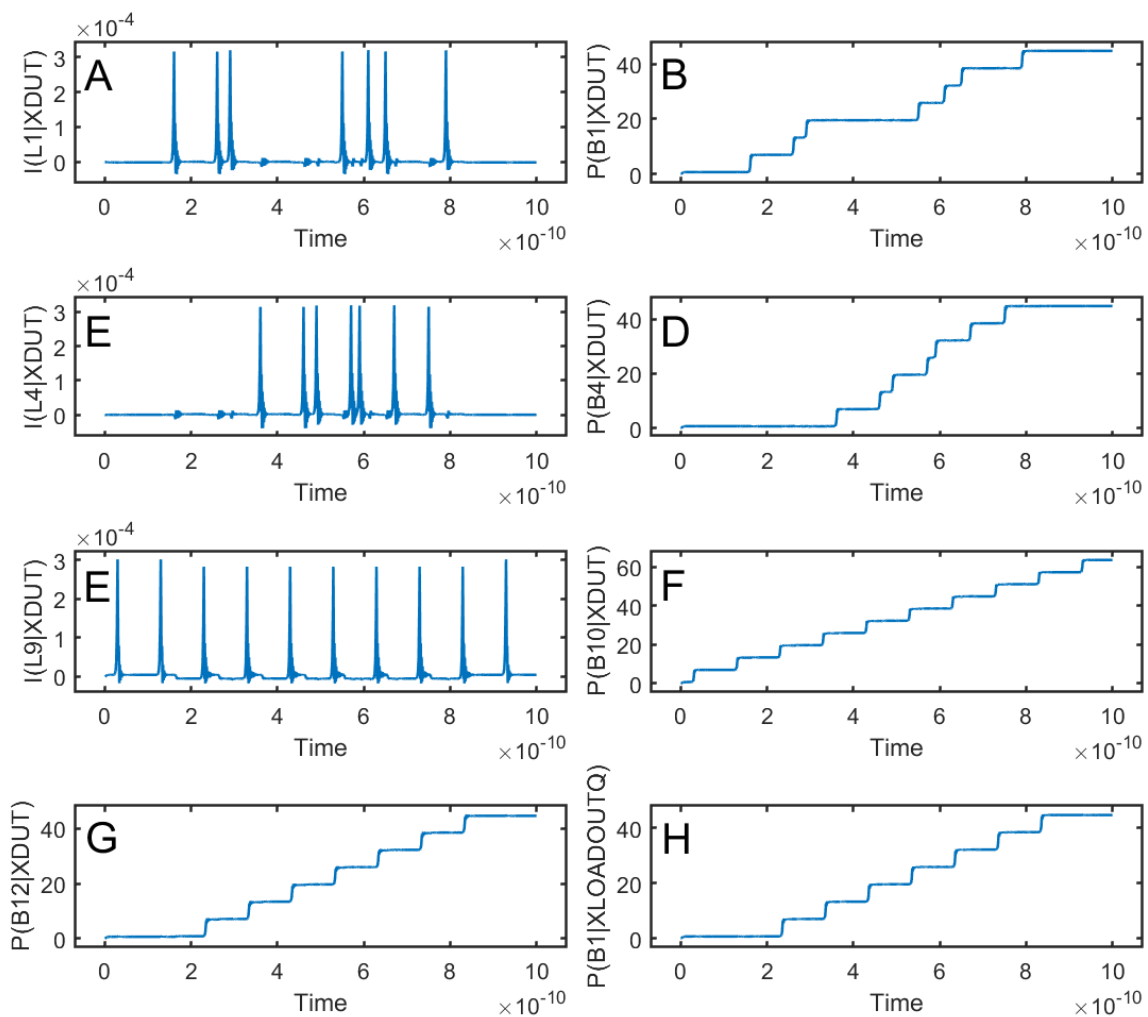


Figure 2.39: RSFQ OR2 analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 7 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit1l_OR2_v3p0_extracted (a, b, clk, q);
9
10 input
11     a, b, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state1_clk_q = 5.5,
21     ct_state0_a_clk = 2.9,
22     ct_state0_b_clk = 2.9,
23     ct_state1_a_clk = 3.8,
24     ct_state1_b_clk = 3.7;
25
26 reg
27     errorsignal_a,
28     errorsignal_b,
29     errorsignal_clk;
30
31 integer
32     outfile,
33     cell_state; // internal state of the cell
34
35 initial
36     begin
37         errorsignal_a = 0;
38         errorsignal_b = 0;
39         errorsignal_clk = 0;
40         cell_state = 0; // Startup state
41         q = 0; // All outputs start at 0
42     end
43
44 always @(posedge a or negedge a) // execute at positive and negative edges of input
45     begin
46         if ($time>4) // arbitrary steady-state time)
47             begin
48                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
49                     begin
50                         outfile = $fopen("errors.txt", "a");
51                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n",
52                             ↪ ", $time);
53                         $fclose(outfile);
54                         q <= 1'bX; // Set all outputs to unknown
55                     end
56                 if (errorsignal_a == 0)
57                     begin
58                         case (cell_state)
59                             0: begin
60                                 cell_state = 1; // Blocking statement -- immediately
61                                 errorsignal_clk = 1; // Critical timing on this input; assign
62                                     ↪ immediately
63                                 errorsignal_clk <= #(ct_state0_a_clk) 0; // Clear error signal
64                                     ↪ after critical timing expires
65                             end
66                             1: begin
67                                 errorsignal_clk = 1; // Critical timing on this input; assign

```

```

65         ↪ immediately
        errorsignal_clk <= #(ct_state1_a_clk) 0; // Clear error signal
66         ↪ after critical timing expires
        end
67     endcase
68 end
69 end
70 end
71
72 always @(posedge b or negedge b) // execute at positive and negative edges of input
73 begin
74     if ($time>4) // arbitrary steady-state time)
75     begin
76         if (errorsignal_b == 1'b1) // A critical timing is active for this input
77         begin
78             outfile = $fopen("errors.txt", "a");
79             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
            ↪ ", $time);
80             $fclose(outfile);
81             q <= 1'bX; // Set all outputs to unknown
82         end
83         if (errorsignal_b == 0)
84         begin
85             case (cell_state)
86             0: begin
87                 cell_state = 1; // Blocking statement -- immediately
88                 errorsignal_clk = 1; // Critical timing on this input; assign
89                 ↪ immediately
89                 errorsignal_clk <= #(ct_state0_b_clk) 0; // Clear error signal
90                 ↪ after critical timing expires
90             end
91             1: begin
92                 errorsignal_clk = 1; // Critical timing on this input; assign
93                 ↪ immediately
93                 errorsignal_clk <= #(ct_state1_b_clk) 0; // Clear error signal
94                 ↪ after critical timing expires
94             end
95             endcase
96         end
97     end
98 end
99
100 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
101 begin
102     if ($time>4) // arbitrary steady-state time)
103     begin
104         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
105         begin
106             outfile = $fopen("errors.txt", "a");
107             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
            ↪ ", $time);
108             $fclose(outfile);
109             q <= 1'bX; // Set all outputs to unknown
110         end
111         if (errorsignal_clk == 0)
112         begin
113             case (cell_state)
114             0: begin
115                 end
116             1: begin
117                 q <= #(delay_state1_clk_q) !q;
118                 cell_state = 0; // Blocking statement -- immediately
119             end
120             endcase
121         end
122     end
123 end
124 endmodule

```

Listing 2.22: RSFQ OR2 verilog model.

The digital simulation results for the RSFQ OR2 is shown in Fig. 2.40 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.41.

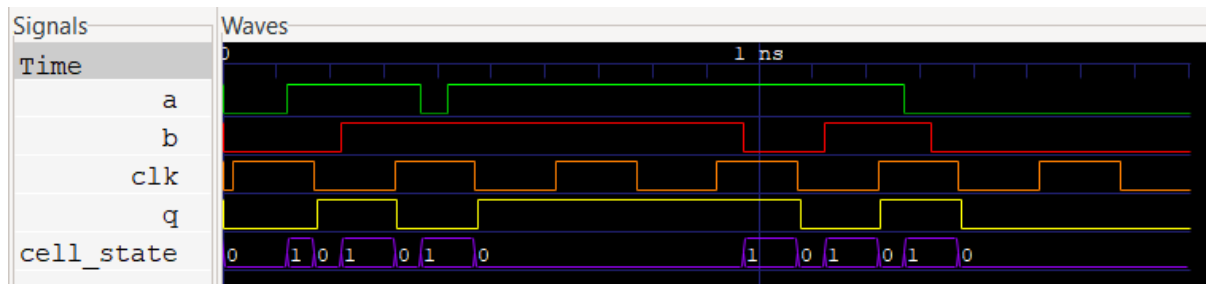


Figure 2.40: RSFQ OR2 digital simulation results.

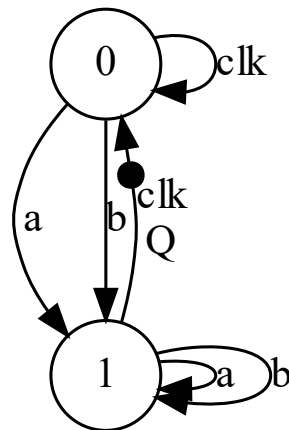


Figure 2.41: RSFQ OR2 Mealy finite state machine diagram.

Power Consumption

Table 2.22: RSFQ OR2 power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	2970	4.99
2	2970	9.98
5	2970	24.9
10	2970	49.9
20	2970	99.8
50	2970	249

2.2.3 XOR

The RSFQ XOR cell generates an output pulse exclusively if a pulse from a single input line was received before the clock signal. The cell is not designed to be directly connected to a PTL.

Schematic

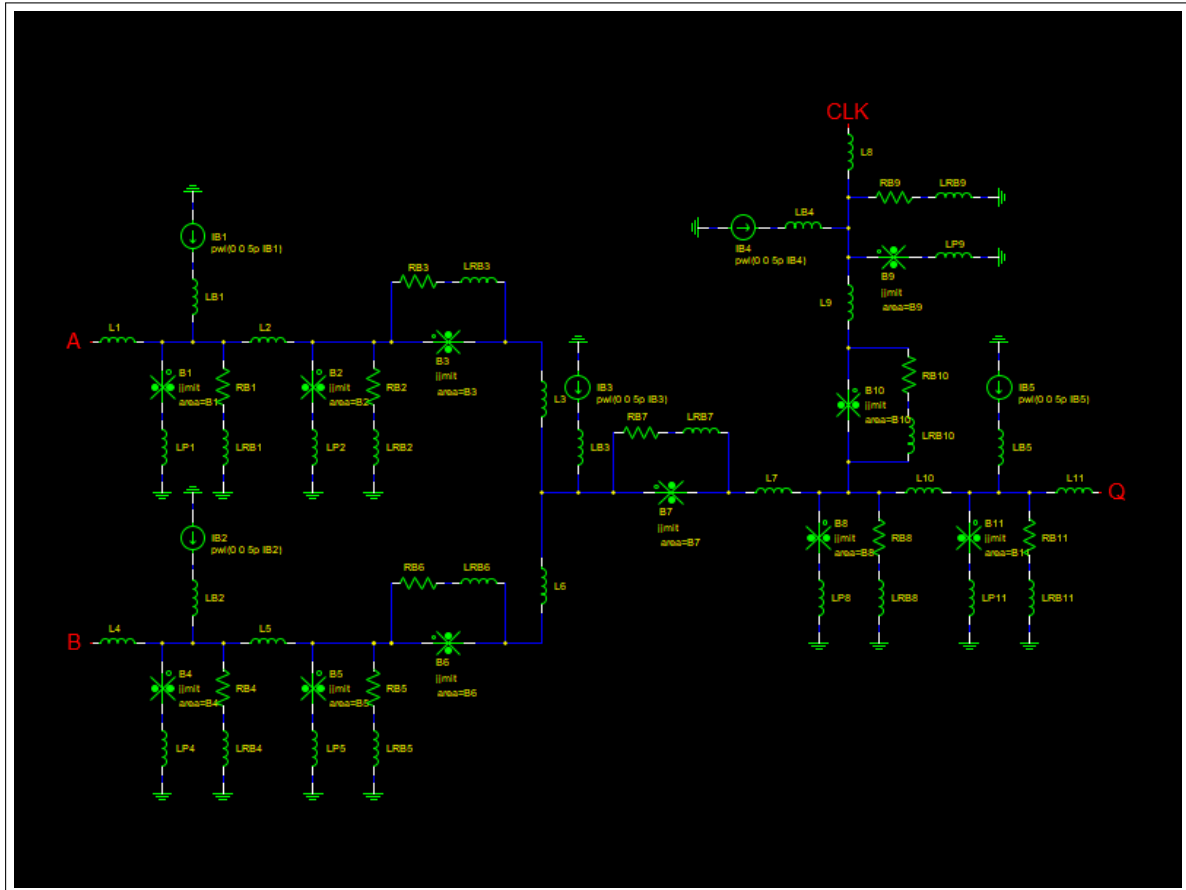


Figure 2.42: Schematic of RSFQ XOR.

Layout

The physical layout for the RSFQ XOR is shown in Fig. 2.43. The layout height is $70\ \mu\text{m}$ and the width is $40\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

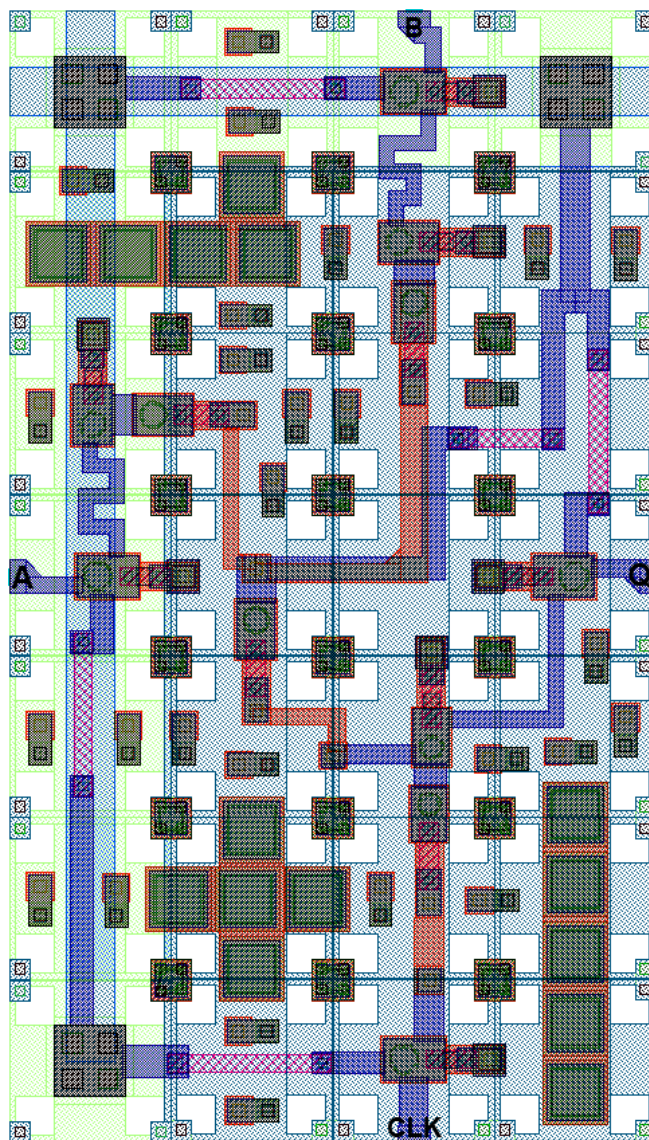


Figure 2.43: RSFQ XOR Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/29.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 28 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      b      clk      q
8  .subckt THmit11_XOR a b clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=2.5
23 .param B2=2.0
24 .param B3=2.02
25 .param B4=2.5
26 .param B5=2.0
27 .param B6=2.02
28 .param B7=1.96
29 .param B8=1.65
30 .param B9=2.5
31 .param B10=1.46
32 .param B11=2.5
33
34 .param IB1=175u
35 .param IB2=175u
36 .param IB3=273u
37 .param IB4=175u
38 .param IB5=175u
39
40 .param LB1=LB
41 .param LB2=LB
42 .param LB3=LB
43 .param LB4=LB
44 .param LB5=LB
45
46 .param L1=1.5740p
47 .param L2=3.1407p
48 .param L3=4.7381p
49 .param L4=1.5740p
50 .param L5=3.1407p
51 .param L6=4.7381p
52 .param L7=4.7751p
53 .param L8=1.1983p
54 .param L9=3.2191p
55 .param L10=3.7711p
56 .param L11=1.4703p
57
58 .param LP1=LP
59 .param LP2=LP
60 .param LP4=LP
61 .param LP5=LP
62 .param LP8=LP
63 .param LP9=LP
64 .param LP11=LP
65
66 .param RB1=B0Rs/B1
67 .param RB2=B0Rs/B2
68 .param RB3=B0Rs/B3
69 .param RB4=B0Rs/B4
70 .param RB5=B0Rs/B5
71 .param RB6=B0Rs/B6
72 .param RB7=B0Rs/B7
73 .param RB8=B0Rs/B8
74 .param RB9=B0Rs/B9
75 .param RB10=B0Rs/B10
76 .param RB11=B0Rs/B11
77
78 .param LRB1=(RB1/Rsheet)*Lsheet+LP
79 .param LRB2=(RB2/Rsheet)*Lsheet+LP
80 .param LRB3=(RB3/Rsheet)*Lsheet
81 .param LRB4=(RB4/Rsheet)*Lsheet+LP
82 .param LRB5=(RB5/Rsheet)*Lsheet+LP
83 .param LRB6=(RB6/Rsheet)*Lsheet
84 .param LRB7=(RB7/Rsheet)*Lsheet
85 .param LRB8=(RB8/Rsheet)*Lsheet+LP
86 .param LRB9=(RB9/Rsheet)*Lsheet+LP
87 .param LRB10=(RB10/Rsheet)*Lsheet
88 .param LRB11=(RB11/Rsheet)*Lsheet+LP
89
90 B1 1 2 jjmit area=B1
91 B2 4 5 jjmit area=B2
92 B3 4 6 jjmit area=B3
93 B4 8 9 jjmit area=B4
94 B5 11 12 jjmit area=B5
95 B6 11 13 jjmit area=B6
96 B7 7 15 jjmit area=B7
97 B8 16 17 jjmit area=B8
98 B9 18 19 jjmit area=B9
99 B10 21 16 jjmit area=B10
100 B11 22 23 jjmit area=B11
101
102 IB1 0 3 pwl(0 0 5p IB1)
103 IB2 0 10 pwl(0 0 5p IB2)
104 IB3 0 14 pwl(0 0 5p IB3)
105 IB4 0 20 pwl(0 0 5p IB4)
106 IB5 0 24 pwl(0 0 5p IB5)
107
108 LB1 3 1 8.314E-013
109 LB2 10 8 9.179E-013
110 LB3 14 7 3.234E-012
111 LB4 20 18 1.047E-012
112 LB5 24 22 1.004E-012
113
114 L1 a 1 1.57E-012
115 L2 1 4 3.128E-012
116 L3 6 7 4.746E-012
117 L4 b 8 1.578E-012
118 L5 8 11 3.145E-012
119 L6 7 13 4.763E-012
120 L7 15 16 4.74E-012
121 L8 clk 18 1.196E-012
122 L9 18 21 3.224E-012
123 L10 16 22 3.775E-012
124 L11 22 q 1.459E-012
125
126 LP1 2 0 4.215E-013
127 LP2 5 0 4.657E-013
128 LP4 9 0 3.32E-013
129 LP5 12 0 3.87E-013

```

```

130 | LP8 17 0 4.698E-013
131 | LP9 19 0 3.611E-013
132 | LP11 23 0 4.394E-013
133 |
134 | RB1 1 101 RB1
135 | LRB1 101 0 LRB1
136 | RB2 4 104 RB2
137 | LRB2 104 0 LRB2
138 | RB3 4 106 RB3
139 | LRB3 106 6 LRB3
140 | RB4 8 108 RB4
141 | LRB4 108 0 LRB4
142 | RB5 11 111 RB5
143 | LRB5 111 0 LRB5
144 | RB6 11 113 RB6
145 | LRB6 113 13 LRB6
146 | RB7 7 115 RB7
147 | LRB7 115 15 LRB7
148 | RB8 16 116 RB8
149 | LRB8 116 0 LRB8
150 | RB9 18 118 RB9
151 | LRB9 118 0 LRB9
152 | RB10 21 121 RB10
153 | LRB10 121 16 LRB10
154 | RB11 22 122 RB11
155 | LRB11 122 0 LRB11
156 | .ends

```

Listing 2.23: RSFQ XOR JoSIM netlist.**Table 2.23:** RSFQ XOR pin list.

Pin	Description
a	Data input
b	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ XOR using JoSIM is shown in Fig. 2.44. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b**,
- (d) the phase over the input JJ of pin **b**,
- (e) the current through the input inductor connected to pin **clk**,
- (f) the phase over the input JJ of pin **clk**,
- (g) the phase over the output JJ of pin **q**, and
- (h) the phase over the input JJ of the load circuit connected to pin **q**.

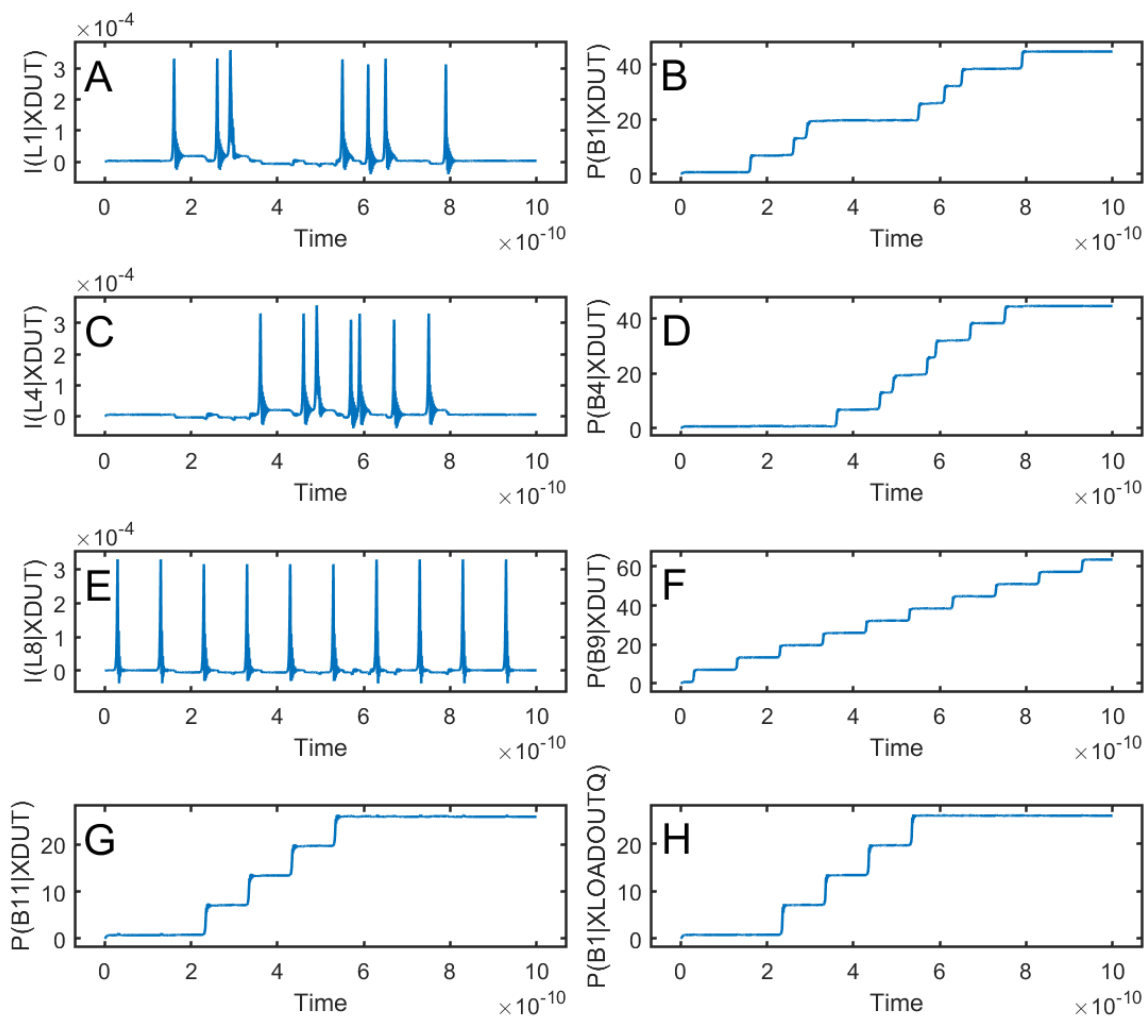


Figure 2.44: RSFQ XOR analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 29 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit1l_XOR_v3p0_extracted (a, b, clk, q);
9
10 input
11     a, b, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state1_clk_q = 5.0,
21     delay_state2_clk_q = 5.0,
22     ct_state0_a_clk = 0.5,
23     ct_state0_b_clk = 0.4,
24     ct_state1_a_b = 8.0,
25     ct_state1_a_clk = 7.3,
26     ct_state1_b_b = 5.2,
27     ct_state1_clk_b = 5.9,
28     ct_state2_a_a = 5.2,
29     ct_state2_b_a = 7.7,
30     ct_state2_b_clk = 7.0,
31     ct_state2_clk_a = 6.1;
32
33 reg
34     errorsignal_a,
35     errorsignal_b,
36     errorsignal_clk;
37
38 integer
39     outfile,
40     cell_state; // internal state of the cell
41
42 initial
43     begin
44         errorsignal_a = 0;
45         errorsignal_b = 0;
46         errorsignal_clk = 0;
47         cell_state = 0; // Startup state
48         q = 0; // All outputs start at 0
49     end
50
51 always @(posedge a or negedge a) // execute at positive and negative edges of input
52     begin
53         if ($time>4) // arbitrary steady-state time)
54             begin
55                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
56                     begin
57                         outfile = $fopen("errors.txt", "a");
58                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
59                         $fclose(outfile);
60                         q <= 1'bX; // Set all outputs to unknown
61                     end
62                 if (errorsignal_a == 0)
63                     begin
64                         case (cell_state)
65                             0: begin
66                                 cell_state = 1; // Blocking statement -- immediately

```

```

67         errorsignal_clk = 1; // Critical timing on this input; assign
68         ↪ immediately
69         errorsignal_clk <= #(ct_state0_a_clk) 0; // Clear error signal
70         ↪ after critical timing expires
71     end
72 1: begin
73     errorsignal_b = 1; // Critical timing on this input; assign
74     ↪ immediately
75     errorsignal_b <= #(ct_state1_a_b) 0; // Clear error signal
76     ↪ after critical timing expires
77     errorsignal_clk = 1; // Critical timing on this input; assign
78     ↪ immediately
79     errorsignal_clk <= #(ct_state1_a_clk) 0; // Clear error signal
80     ↪ after critical timing expires
81     end
82 2: begin
83     cell_state = 0; // Blocking statement -- immediately
84     errorsignal_a = 1; // Critical timing on this input; assign
85     ↪ immediately
86     errorsignal_a <= #(ct_state2_a_a) 0; // Clear error signal
87     ↪ after critical timing expires
88     end
89 endcase
90 end
91 end
92
93 always @(posedge b or negedge b) // execute at positive and negative edges of input
94 begin
95     if ($time>4) // arbitrary steady-state time)
96     begin
97         if (errorsignal_b == 1'b1) // A critical timing is active for this input
98         begin
99             outfile = $fopen("errors.txt", "a");
100             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0dps.\n", $time);
101             ↪ ", $time);
102             $fclose(outfile);
103             q <= 1'bX; // Set all outputs to unknown
104         end
105         if (errorsignal_b == 0)
106         begin
107             case (cell_state)
108             0: begin
109                 cell_state = 2; // Blocking statement -- immediately
110                 errorsignal_clk = 1; // Critical timing on this input; assign
111                 ↪ immediately
112                 errorsignal_clk <= #(ct_state0_b_clk) 0; // Clear error signal
113                 ↪ after critical timing expires
114             end
115             1: begin
116                 cell_state = 0; // Blocking statement -- immediately
117                 errorsignal_b = 1; // Critical timing on this input; assign
118                 ↪ immediately
119                 errorsignal_b <= #(ct_state1_b_b) 0; // Clear error signal
120                 ↪ after critical timing expires
121             end
122             2: begin
123                 errorsignal_a = 1; // Critical timing on this input; assign
124                 ↪ immediately
125                 errorsignal_a <= #(ct_state2_b_a) 0; // Clear error signal
126                 ↪ after critical timing expires
127                 errorsignal_clk = 1; // Critical timing on this input; assign
128                 ↪ immediately
129                 errorsignal_clk <= #(ct_state2_b_clk) 0; // Clear error signal
130                 ↪ after critical timing expires
131             end
132             endcase
133         end
134     end
135 end

```

```

120
121 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
122     begin
123         if ($time>4) // arbitrary steady-state time)
124             begin
125                 if (errorsignal_clk == 1'b1) // A critical timing is active for this input
126                     begin
127                         outfile = $fopen("errors.txt", "a");
128                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n",
129                             ↪ ", $time);
130                         $fclose(outfile);
131                         q <= 1'bX; // Set all outputs to unknown
132                     end
133                 if (errorsignal_clk == 0)
134                     begin
135                         case (cell_state)
136                             0: begin
137                                 end
138                             1: begin
139                                 q <= #(delay_state1_clk_q) !q;
140                                 cell_state = 0; // Blocking statement -- immediately
141                                 errorsignal_b = 1; // Critical timing on this input; assign
142                                 ↪ immediately
143                                 errorsignal_b <= #(ct_state1_clk_b) 0; // Clear error signal
144                                 ↪ after critical timing expires
145                             end
146                             2: begin
147                                 q <= #(delay_state2_clk_q) !q;
148                                 cell_state = 0; // Blocking statement -- immediately
149                                 errorsignal_a = 1; // Critical timing on this input; assign
150                                 ↪ immediately
151                                 errorsignal_a <= #(ct_state2_clk_a) 0; // Clear error signal
152                                 ↪ after critical timing expires
153                             end
154                         endcase
155                     end
156             end
157     end
158 endmodule

```

Listing 2.24: RSFQ XOR verilog model.

The digital simulation results for the RSFQ XOR is shown in Fig. 2.45 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.46.

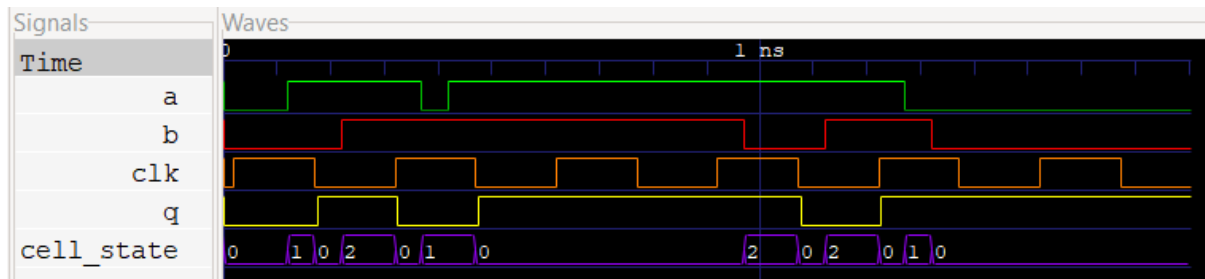


Figure 2.45: RSFQ XOR digital simulation results.

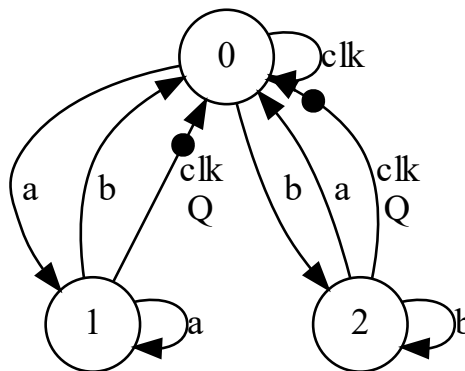


Figure 2.46: RSFQ XOR Mealy finite state machine diagram.

Power Consumption

Table 2.24: RSFQ XOR power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	2530	4.78
2	2530	9.56
5	2530	23.9
10	2530	47.8
20	2530	95.6
50	2530	239

2.2.4 NOT

The RSFQ NOT cell is a signal inverting cell driven by a clock pulse signal line. The cell is not designed to be directly connected to a PTL.

Schematic

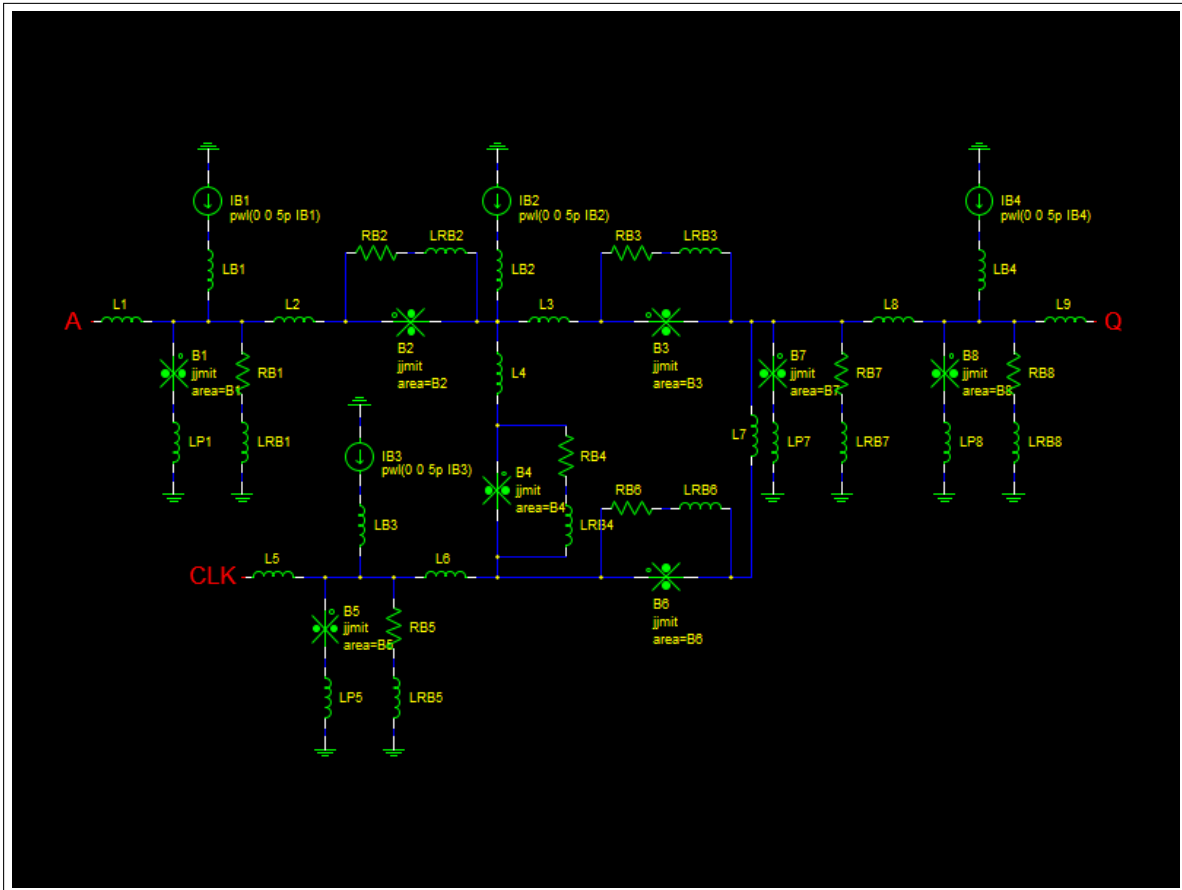


Figure 2.47: Schematic of RSFQ NOT.

Layout

The physical layout of the RSFQ NOT is shown in Fig. 2.48. The layout height is $70\ \mu\text{m}$ and the width is $40\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a $2.6\ \text{mV}$ voltage source.

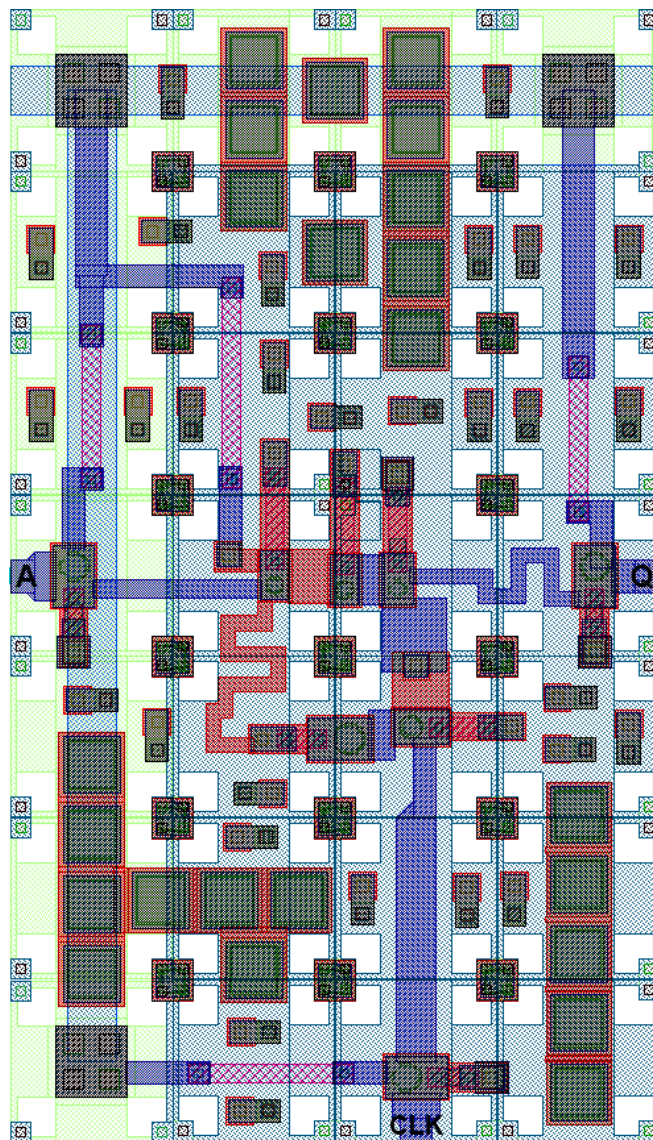


Figure 2.48: RSFQ NOT Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/25.
2  * Author: T. Hall
3  * Version: 3.0
4  * Last modification date: 24 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      clk      q
8  .subckt THmit11_NOT a clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=2.5
23 .param B2=0.78
24 .param B3=0.85
25 .param B4=2.69
26 .param B5=2.5
27 .param B6=1.42
28 .param B7=1.01
29 .param B8=2.5
30
31 .param IB1=175u
32 .param IB2=131u
33 .param IB3=175u
34 .param IB4=175u
35
36 .param LB1=LB
37 .param LB2=LB
38 .param LB3=LB
39 .param LB4=LB
40
41 .param L1=0.8555p
42 .param L2=2.7537p
43 .param L3=0.7553p
44 .param L4=7.3609p
45 .param L5=0.8585p
46 .param L6=3.0349p
47 .param L7=0.9373p
48 .param L8=4.2782p
49 .param L9=0.9209p
50
51 .param LP1=LP
52 .param LP5=LP
53 .param LP7=LP
54 .param LP8=LP
55
56 .param RB1=B0Rs/B1
57 .param RB2=B0Rs/B2
58 .param RB3=B0Rs/B3
59 .param RB4=B0Rs/B4
60 .param RB5=B0Rs/B5
61 .param RB6=B0Rs/B6
62 .param RB7=B0Rs/B7
63 .param RB8=B0Rs/B8
64
65 .param LRB1=(RB1/Rsheet)*Lsheet+LP
66 .param LRB2=(RB2/Rsheet)*Lsheet
67 .param LRB3=(RB3/Rsheet)*Lsheet
68 .param LRB4=(RB4/Rsheet)*Lsheet
69 .param LRB5=(RB5/Rsheet)*Lsheet+LP
70 .param LRB6=(RB6/Rsheet)*Lsheet
71 .param LRB7=(RB7/Rsheet)*Lsheet+LP
72 .param LRB8=(RB8/Rsheet)*Lsheet+LP
73
74 B1 1 2 jjmit area=B1
75 B2 4 5 jjmit area=B2
76 B3 7 8 jjmit area=B3
77 B4 9 10 jjmit area=B4
78 B5 11 12 jjmit area=B5
79 B6 10 14 jjmit area=B6
80 B7 8 15 jjmit area=B7
81 B8 16 17 jjmit area=B8
82
83 IB1 0 3 pwl(0 0 5p IB1)
84 IB2 0 6 pwl(0 0 5p IB2)
85 IB3 0 13 pwl(0 0 5p IB3)
86 IB4 0 18 pwl(0 0 5p IB4)
87
88 LB1 3 1 1.246E-012
89 LB2 6 5 1.707E-012
90 LB3 13 11 8.002E-013
91 LB4 18 16 8.582E-013
92
93 L1 a 1 8.521E-013
94 L2 1 4 2.756E-012
95 L3 5 7 7.55E-013
96 L4 5 9 7.358E-012
97 L5 clk 11 8.595E-013
98 L6 11 10 3.051E-012
99 L7 8 14 9.435E-013
100 L8 8 16 4.287E-012
101 L9 16 q 9.3E-013
102
103 LP1 2 0 3.288E-013
104 LP5 12 0 3.499E-013
105 LP7 15 0 4.831E-013
106 LP8 17 0 3.407E-013
107
108 RB1 1 101 RB1
109 LRB1 101 0 LRB1
110 RB2 4 104 RB2
111 LRB2 104 5 LRB2
112 RB3 7 107 RB3
113 LRB3 107 8 LRB3
114 RB4 9 109 RB4
115 LRB4 109 10 LRB4
116 RB5 11 111 RB5
117 LRB5 111 0 LRB5
118 RB6 10 110 RB6
119 LRB6 110 14 LRB6
120 RB7 8 108 RB7
121 LRB7 108 0 LRB7
122 RB8 16 116 RB8
123 LRB8 116 0 LRB8
124 .ends

```

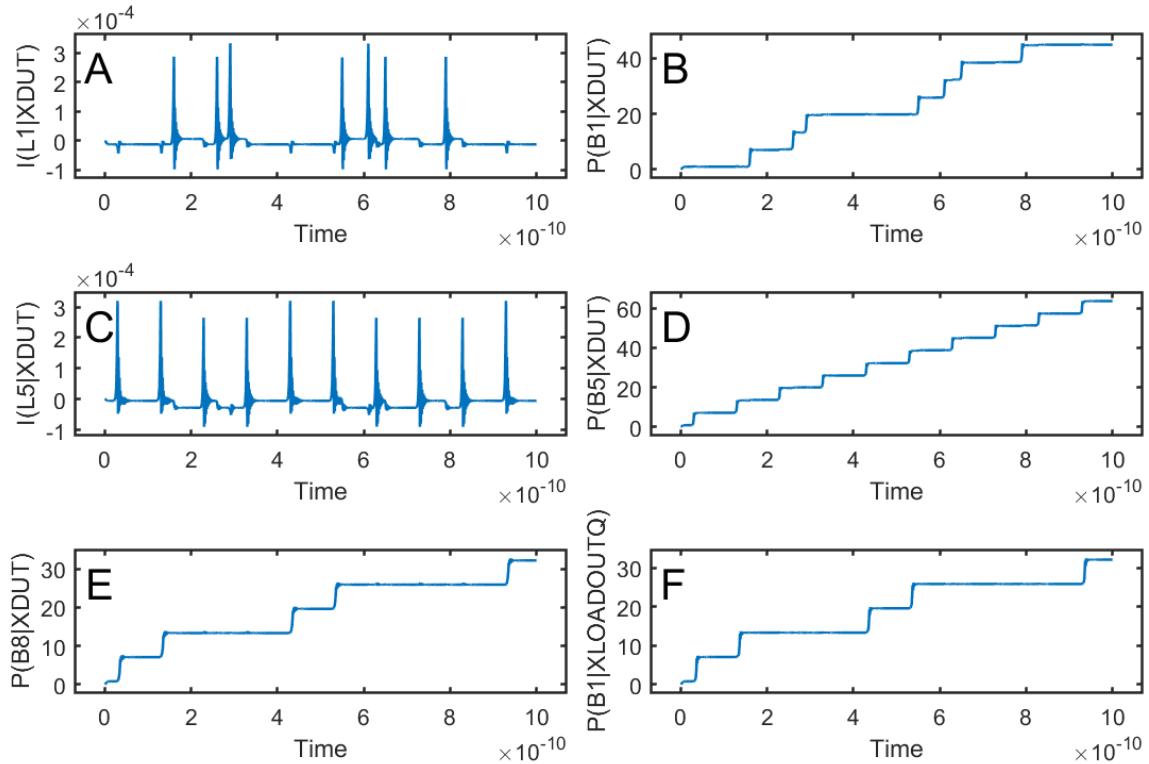
Listing 2.25: RSFQ NOT JoSIM netlist.

Table 2.25: RSFQ NOT pin list.

Pin	Description
a	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ NOT using JoSIM is shown in Fig. 2.49. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **clk**,
- (d) the phase over the input JJ of pin **clk**,
- (e) the phase over the output JJ of pin **q**, and
- (f) the phase over the input JJ of the load circuit connected to pin **q**.

**Figure 2.49:** RSFQ NOT analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 25 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_NOT_v3p0_extracted (a, clk, q);
9
10 input
11     a, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_clk_q = 5.5,
21     ct_state0_a_clk = 0.8,
22     ct_state0_clk_a = 4.5,
23     ct_state0_clk_clk = 5.2,
24     ct_state1_a_clk = 2.1;
25
26 reg
27     errorsignal_a,
28     errorsignal_clk;
29
30 integer
31     outfile,
32     cell_state; // internal state of the cell
33
34 initial
35     begin
36         errorsignal_a = 0;
37         errorsignal_clk = 0;
38         cell_state = 0; // Startup state
39         q = 0; // All outputs start at 0
40     end
41
42 always @(posedge a or negedge a) // execute at positive and negative edges of input
43     begin
44         if ($time>4) // arbitrary steady-state time)
45             begin
46                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
47                     begin
48                         outfile = $fopen("errors.txt", "a");
49                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
50                             ↪ ", $stime);
51                         $fclose(outfile);
52                         q <= 1'bX; // Set all outputs to unknown
53                     end
54                 if (errorsignal_a == 0)
55                     begin
56                         case (cell_state)
57                             0: begin
58                                 cell_state = 1; // Blocking statement -- immediately
59                                 errorsignal_clk = 1; // Critical timing on this input; assign
60                                     ↪ immediately
61                                 errorsignal_clk <= #(ct_state0_a_clk) 0; // Clear error signal
62                                     ↪ after critical timing expires
63                             end
64                             1: begin
65                                 errorsignal_clk = 1; // Critical timing on this input; assign
66                                     ↪ immediately
67                                 errorsignal_clk <= #(ct_state1_a_clk) 0; // Clear error signal

```

```

64                                     ↪ after critical timing expires
65                                     end
66                               endcase
67                               end
68       end
69
70   always @(posedge clk or negedge clk) // execute at positive and negative edges of input
71   begin
72       if ($time>4) // arbitrary steady-state time)
73       begin
74           if (errorsignal_clk == 1'b1) // A critical timing is active for this input
75           begin
76               outfile = $fopen("errors.txt", "a");
77               $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
78               $fclose(outfile);
79               q <= 1'bX; // Set all outputs to unknown
80           end
81       if (errorsignal_clk == 0)
82       begin
83           case (cell_state)
84           0: begin
85               q <= #(delay_state0_clk_q) !q;
86               errorsignal_a = 1; // Critical timing on this input; assign
87                               ↪ immediately
88               errorsignal_a <= #(ct_state0_clk_a) 0; // Clear error signal
89                               ↪ after critical timing expires
90               errorsignal_clk = 1; // Critical timing on this input; assign
91                               ↪ immediately
92               errorsignal_clk <= #(ct_state0_clk_clk) 0; // Clear error
93                               ↪ signal after critical timing expires
94           end
95           1: begin
96               cell_state = 0; // Blocking statement -- immediately
97           end
98       endcase
99   end
100 end
101 endmodule

```

Listing 2.26: RSFQ NOT verilog model.

The digital simulation results for the RSFQ NOT is shown in Fig. 2.50 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.51.

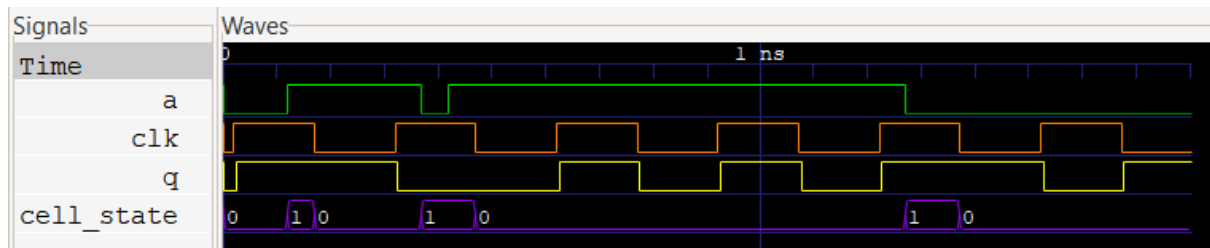


Figure 2.50: RSFQ NOT digital simulation results.

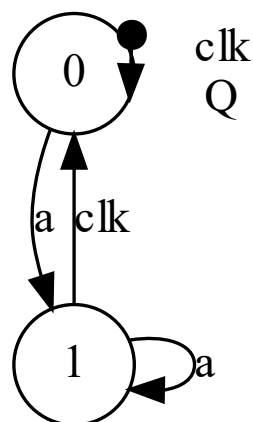


Figure 2.51: RSFQ NOT Mealy finite state machine diagram.

Power Consumption

Table 2.26: RSFQ NOT power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	1710	2.95
2	1710	5.89
5	1710	14.7
10	1710	29.5
20	1710	58.9
50	1710	147

2.2.5 XNOR

The RSFQ XNOR cell is a combination of a NOT and XOR cell. It generates an output pulse exclusively if no input pulse was received or if a pulse from both input lines was received before the clock signal. The cell is not designed to be directly connected to a PTL.

Schematic

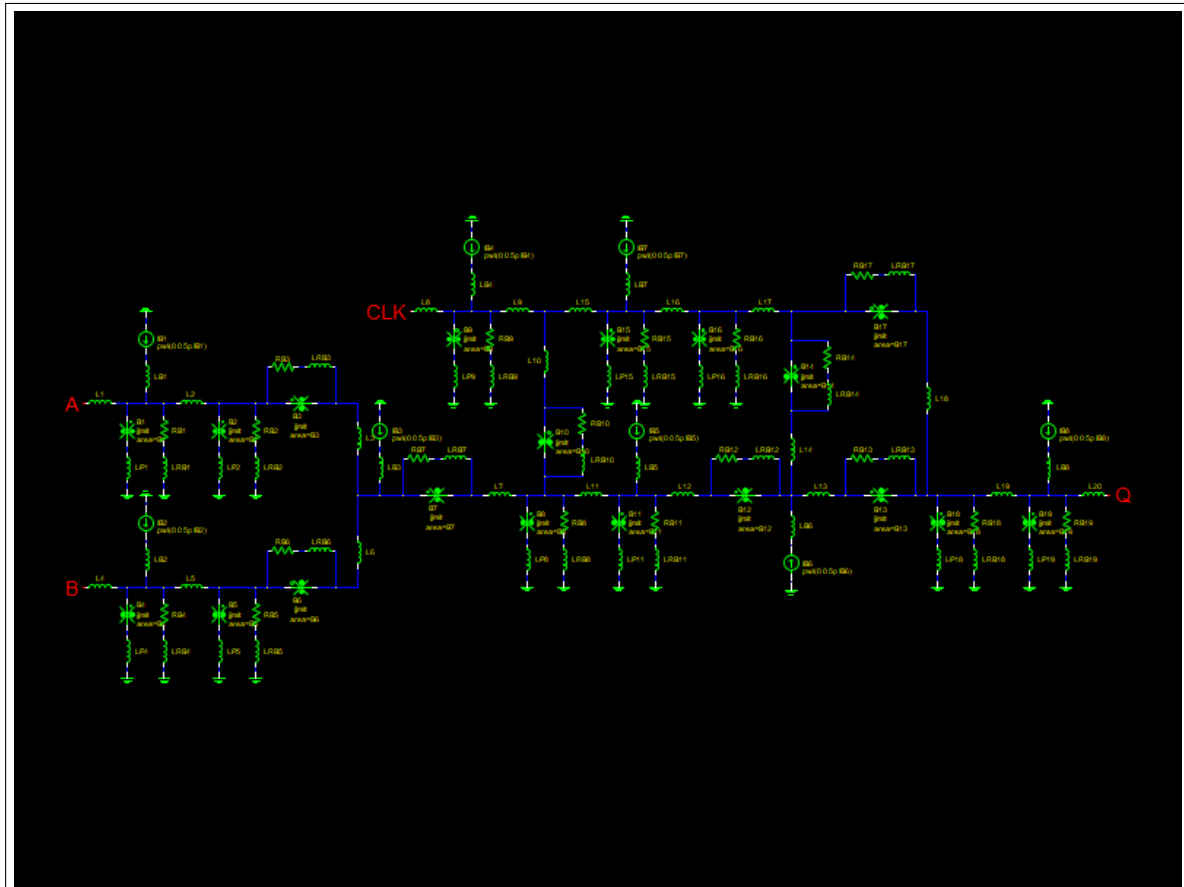


Figure 2.52: Schematic of RSFQ XNOR.

Layout

The physical layout for the RSFQ XNOR is shown in Fig. 2.53. The layout height is $70\ \mu\text{m}$ and the width is $60\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

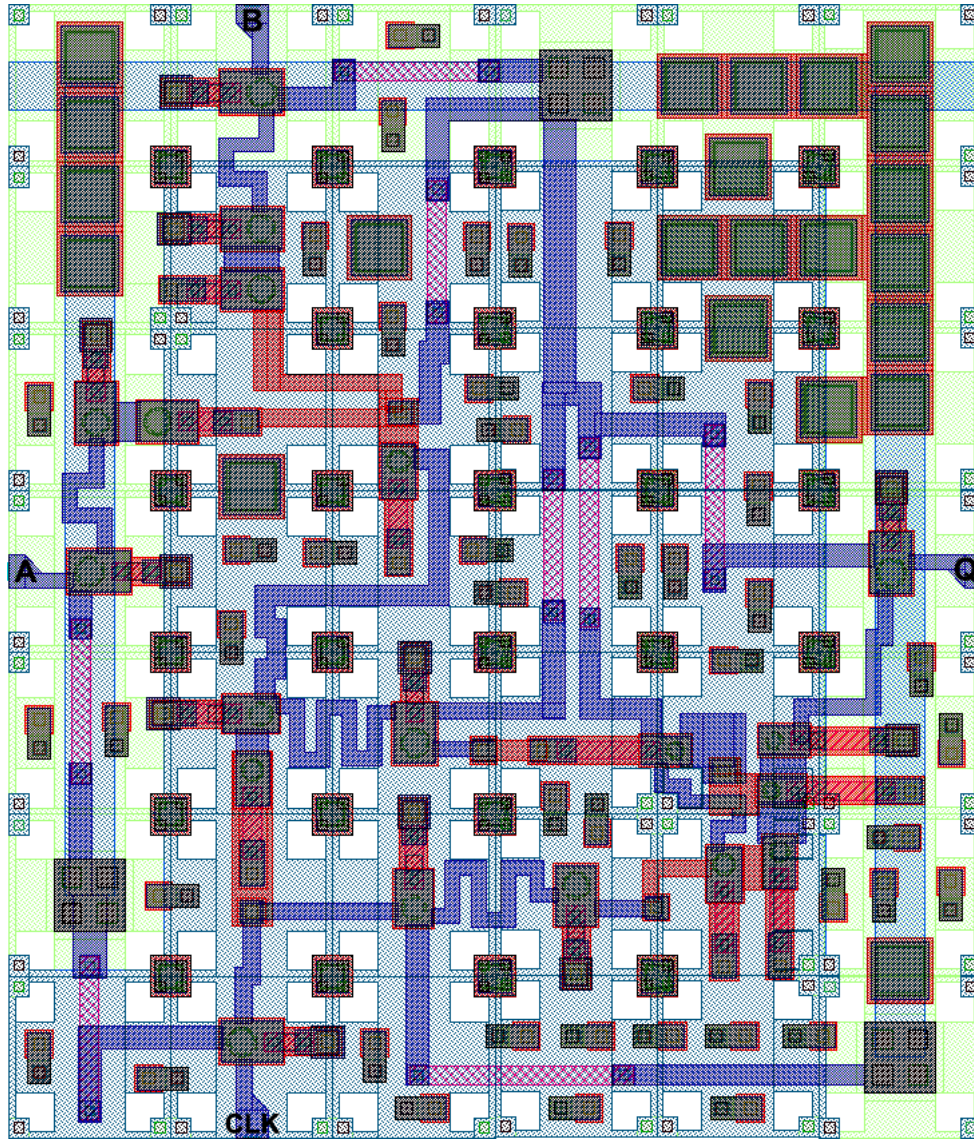


Figure 2.53: RSFQ XNOR Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/09/07.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 6 September 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      b      clk      q
8  .subckt THmit11_XNOR a b clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=2.5
23 .param B2=2.22
24 .param B3=2.12
25 .param B4=2.5
26 .param B5=2.22
27 .param B6=2.12
28 .param B7=1.39
29 .param B8=1.57
30 .param B9=2.5
31 .param B10=1.28
32 .param B11=2.43
33 .param B12=0.77
34 .param B13=0.72
35 .param B14=1.34
36 .param B15=1.63
37 .param B16=2.15
38 .param B17=1.05
39 .param B18=0.83
40 .param B19=2.5
41
42 .param IB1=175u
43 .param IB2=175u
44 .param IB3=207u
45 .param IB4=175u
46 .param IB5=193u
47 .param IB6=147u
48 .param IB7=122u
49 .param IB8=175u
50
51 .param LB1=LB
52 .param LB2=LB
53 .param LB3=LB
54 .param LB4=LB
55 .param LB5=LB
56 .param LB6=LB
57 .param LB7=LB
58 .param LB8=LB
59
60 .param L1=1.4696p
61 .param L2=2.7700p
62 .param L3=3.6981p
63 .param L4=1.4696p
64 .param L5=2.7700p
65 .param L6=3.6981p
66 .param L7=6.0060p
67 .param L8=1.5890p
68 .param L9=1.8231p
69 .param L10=1.6294p
70 .param L11=4.5825p
71 .param L12=3.3983p
72 .param L13=1.1575p
73 .param L14=3.0580p
74 .param L15=2.1812p
75 .param L16=4.5604p
76 .param L17=2.9745p
77 .param L18=0.8577p
78 .param L19=3.5636p
79 .param L20=1.5943p
80
81 .param LP1=LP
82 .param LP2=LP
83 .param LP4=LP
84 .param LP5=LP
85 .param LP8=LP
86 .param LP9=LP
87 .param LP11=LP
88 .param LP15=LP
89 .param LP16=LP
90 .param LP18=LP
91 .param LP19=LP
92
93 .param RB1=B0Rs/B1
94 .param RB2=B0Rs/B2
95 .param RB3=B0Rs/B3
96 .param RB4=B0Rs/B4
97 .param RB5=B0Rs/B5
98 .param RB6=B0Rs/B6
99 .param RB7=B0Rs/B7
100 .param RB8=B0Rs/B8
101 .param RB9=B0Rs/B9
102 .param RB10=B0Rs/B10
103 .param RB11=B0Rs/B11
104 .param RB12=B0Rs/B12
105 .param RB13=B0Rs/B13
106 .param RB14=B0Rs/B14
107 .param RB15=B0Rs/B15
108 .param RB16=B0Rs/B16
109 .param RB17=B0Rs/B17
110 .param RB18=B0Rs/B18
111 .param RB19=B0Rs/B19
112 .param LRB1=(RB1/Rsheet)*Lsheet+LP
113 .param LRB2=(RB2/Rsheet)*Lsheet+LP
114 .param LRB3=(RB3/Rsheet)*Lsheet
115 .param LRB4=(RB4/Rsheet)*Lsheet+LP
116 .param LRB5=(RB5/Rsheet)*Lsheet+LP
117 .param LRB6=(RB6/Rsheet)*Lsheet
118 .param LRB7=(RB7/Rsheet)*Lsheet
119 .param LRB8=(RB8/Rsheet)*Lsheet+LP
120 .param LRB9=(RB9/Rsheet)*Lsheet+LP
121 .param LRB10=(RB10/Rsheet)*Lsheet
122 .param LRB11=(RB11/Rsheet)*Lsheet+LP
123 .param LRB12=(RB12/Rsheet)*Lsheet
124 .param LRB13=(RB13/Rsheet)*Lsheet
125 .param LRB14=(RB14/Rsheet)*Lsheet
126 .param LRB15=(RB15/Rsheet)*Lsheet+LP
127 .param LRB16=(RB16/Rsheet)*Lsheet+LP
128 .param LRB17=(RB17/Rsheet)*Lsheet
129 .param LRB18=(RB18/Rsheet)*Lsheet+LP

```

```

130 .param LRB19=(RB19/Rsheet)*Lsheet+LP
131
132 B1 1 2 jjmit area=B1
133 B2 4 5 jjmit area=B2
134 B3 4 6 jjmit area=B3
135 B4 8 9 jjmit area=B4
136 B5 11 12 jjmit area=B5
137 B6 11 13 jjmit area=B6
138 B7 7 15 jjmit area=B7
139 B8 16 17 jjmit area=B8
140 B9 18 19 jjmit area=B9
141 B10 22 16 jjmit area=B10
142 B11 23 24 jjmit area=B11
143 B12 26 27 jjmit area=B12
144 B13 29 30 jjmit area=B13
145 B14 32 31 jjmit area=B14
146 B15 33 34 jjmit area=B15
147 B16 36 37 jjmit area=B16
148 B17 32 38 jjmit area=B17
149 B18 30 39 jjmit area=B18
150 B19 40 41 jjmit area=B19
151
152 IB1 0 3 pwl(0 0 5p IB1)
153 IB2 0 10 pwl(0 0 5p IB2)
154 IB3 0 14 pwl(0 0 5p IB3)
155 IB4 0 20 pwl(0 0 5p IB4)
156 IB5 0 25 pwl(0 0 5p IB5)
157 IB6 0 28 pwl(0 0 5p IB6)
158 IB7 0 35 pwl(0 0 5p IB7)
159 IB8 0 42 pwl(0 0 5p IB8)
160
161 LB1 3 1 5.929E-013
162 LB2 10 8 1.279E-012
163 LB3 14 7 1.737E-012
164 LB4 20 18 2.846E-012
165 LB5 25 23 3.286E-012
166 LB6 27 28 2.4E-012
167 LB7 35 33 1.996E-012
168 LB8 42 40 2.415E-012
169
170 L1 a 1 1.471E-012
171 L2 1 4 2.745E-012
172 L3 6 7 3.693E-012
173 L4 b 8 1.476E-012
174 L5 8 11 2.749E-012
175 L6 7 13 3.671E-012
176 L7 15 16 5.95E-012
177 L8 clk 18 1.58E-012
178 L9 18 21 1.818E-012
179 L10 21 22 1.634E-012
180 L11 16 23 4.569E-012
181 L12 23 26 3.384E-012
182 L13 27 29 1.168E-012
183 L14 31 27 3.042E-012
184 L15 21 33 2.198E-012
185 L16 33 36 4.555E-012
186 L17 36 32 2.964E-012
187 L18 38 30 8.576E-013
188 L19 30 40 3.531E-012
189 L20 40 q 1.583E-012
190
191 LP1 2 0 4.616E-013
192 LP2 5 0 4.744E-013
193 LP4 9 0 3.882E-013
194 LP5 12 0 3.984E-013
195 LP8 17 0 4.778E-013
196 LP9 19 0 3.834E-013
197 LP11 24 0 4.104E-013
198 LP15 34 0 5.341E-013
199 LP16 37 0 3.884E-013
200 LP18 39 0 4.944E-013
201 LP19 41 0 3.606E-013
202
203 RB1 1 101 RB1
204 LRB1 101 0 LRB1
205 RB2 4 104 RB2
206 LRB2 104 0 LRB2
207 RB3 4 106 RB3
208 LRB3 106 6 LRB3
209 RB4 8 108 RB4
210 LRB4 108 0 LRB4
211 RB5 11 111 RB5
212 LRB5 111 0 LRB5
213 RB6 11 113 RB6
214 LRB6 113 13 LRB6
215 RB7 7 115 RB7
216 LRB7 115 15 LRB7
217 RB8 16 116 RB8
218 LRB8 116 0 LRB8
219 RB9 18 118 RB9
220 LRB9 118 0 LRB9
221 RB10 22 122 RB10
222 LRB10 122 16 LRB10
223 RB11 23 123 RB11
224 LRB11 123 0 LRB11
225 RB12 26 126 RB12
226 LRB12 126 27 LRB12
227 RB13 29 129 RB13
228 LRB13 129 30 LRB13
229 RB14 32 131 RB14
230 LRB14 131 31 LRB14
231 RB15 33 133 RB15
232 LRB15 133 0 LRB15
233 RB16 36 136 RB16
234 LRB16 136 0 LRB16
235 RB17 32 138 RB17
236 LRB17 138 38 LRB17
237 RB18 30 130 RB18
238 LRB18 130 0 LRB18
239 RB19 40 140 RB19
240 LRB19 140 0 LRB19
241 .ends

```

Listing 2.27: RSFQ XNOR JoSIM netlist.

Table 2.27: RSFQ XNOR pin list.

Pin	Description
a	Data input
b	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ XNOR using JoSIM is shown in Fig. 2.54. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b**,
- (d) the phase over the input JJ of pin **b**,
- (e) the current through the input inductor connected to pin **clk**,
- (f) the phase over the input JJ of pin **clk**,
- (g) the phase over the output JJ of pin **q**, and
- (h) the phase over the input JJ of the load circuit connected to pin **q**.

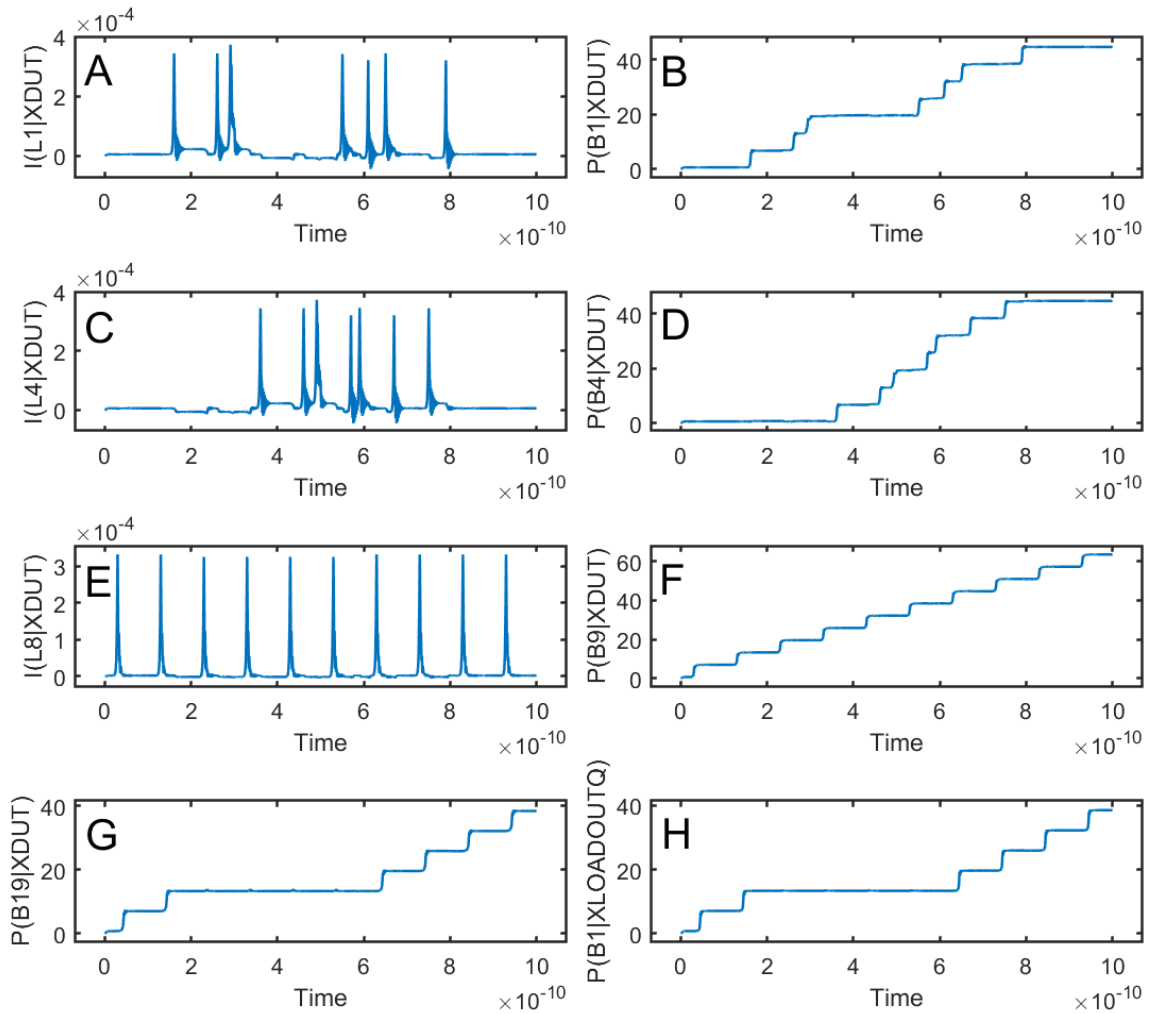


Figure 2.54: RSFQ XNOR analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 7 September 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_XNOR_v3p0_extracted (a, b, clk, q);
9
10 input
11     a, b, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_clk_q = 14.3,
21     ct_state0_clk_a = 1.3,
22     ct_state0_clk_b = 1.2,
23     ct_state0_clk_clk = 10.5,
24     ct_state1_a_b = 9.2,
25     ct_state1_a_clk = 7.3,
26     ct_state1_b_b = 5.2,
27     ct_state1_clk_b = 7.7,
28     ct_state2_a_a = 5.2,
29     ct_state2_b_a = 9.2,
30     ct_state2_b_clk = 7.4,
31     ct_state2_clk_a = 7.7;
32
33 reg
34     errorsignal_a,
35     errorsignal_b,
36     errorsignal_clk;
37
38 integer
39     outfile,
40     cell_state; // internal state of the cell
41
42 initial
43     begin
44         errorsignal_a = 0;
45         errorsignal_b = 0;
46         errorsignal_clk = 0;
47         cell_state = 0; // Startup state
48         q = 0; // All outputs start at 0
49     end
50
51 always @(posedge a or negedge a) // execute at positive and negative edges of input
52     begin
53         if ($time>4) // arbitrary steady-state time)
54             begin
55                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
56                     begin
57                         outfile = $fopen("errors.txt", "a");
58                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
59                         $fclose(outfile);
60                         q <= 1'bX; // Set all outputs to unknown
61                     end
62                 if (errorsignal_a == 0)
63                     begin
64                         case (cell_state)
65                             0: begin
66                                 cell_state = 1; // Blocking statement -- immediately

```

```

67         end
68     1: begin
69         errorsignal_b = 1; // Critical timing on this input; assign
70             ↪ immediately
71         errorsignal_b <= #(ct_state1_a_b) 0; // Clear error signal
72             ↪ after critical timing expires
73         errorsignal_clk = 1; // Critical timing on this input; assign
74             ↪ immediately
75         errorsignal_clk <= #(ct_state1_a_clk) 0; // Clear error signal
76             ↪ after critical timing expires
77     end
78     2: begin
79         cell_state = 0; // Blocking statement -- immediately
80         errorsignal_a = 1; // Critical timing on this input; assign
81             ↪ immediately
82         errorsignal_a <= #(ct_state2_a_a) 0; // Clear error signal
83             ↪ after critical timing expires
84     end
85 endcase
86 end
87 end
88
89 always @(posedge b or negedge b) // execute at positive and negative edges of input
90 begin
91     if ($time>4) // arbitrary steady-state time)
92     begin
93         if (errorsignal_b == 1'b1) // A critical timing is active for this input
94         begin
95             outfile = $fopen("errors.txt", "a");
96             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d.ps.\n", $time);
97             ↪ ", $time);
98             $fclose(outfile);
99             q <= 1'bX; // Set all outputs to unknown
100         end
101         if (errorsignal_b == 0)
102         begin
103             case (cell_state)
104             0: begin
105                 cell_state = 2; // Blocking statement -- immediately
106             end
107             1: begin
108                 cell_state = 0; // Blocking statement -- immediately
109                 errorsignal_b = 1; // Critical timing on this input; assign
110                     ↪ immediately
111                 errorsignal_b <= #(ct_state1_b_b) 0; // Clear error signal
112                     ↪ after critical timing expires
113             end
114             2: begin
115                 errorsignal_a = 1; // Critical timing on this input; assign
116                     ↪ immediately
117                 errorsignal_a <= #(ct_state2_b_a) 0; // Clear error signal
118                     ↪ after critical timing expires
119                 errorsignal_clk = 1; // Critical timing on this input; assign
120                     ↪ immediately
121                 errorsignal_clk <= #(ct_state2_b_clk) 0; // Clear error signal
122                     ↪ after critical timing expires
123             end
124         endcase
125     end
126 end
127
128 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
129 begin
130     if ($time>4) // arbitrary steady-state time)
131     begin
132         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
133         begin
134             outfile = $fopen("errors.txt", "a");

```

```

124         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d.ps.\n
           ↳ ", $stime);
125     $fclose(outfile);
126     q <= 1'bX; // Set all outputs to unknown
127 end
128 if (errorsignal_clk == 0)
129     begin
130         case (cell_state)
131             0: begin
132                 q <= #(delay_state0_clk_q) !q;
133                 errorsignal_a = 1; // Critical timing on this input; assign
134                     ↳ immediately
135                 errorsignal_a <= #(ct_state0_clk_a) 0; // Clear error signal
136                     ↳ after critical timing expires
137                 errorsignal_b = 1; // Critical timing on this input; assign
138                     ↳ immediately
139                 errorsignal_b <= #(ct_state0_clk_b) 0; // Clear error signal
140                     ↳ after critical timing expires
141                 errorsignal_clk = 1; // Critical timing on this input; assign
142                     ↳ immediately
143                 errorsignal_clk <= #(ct_state0_clk_clk) 0; // Clear error
144                     ↳ signal after critical timing expires
145             end
146             1: begin
147                 cell_state = 0; // Blocking statement -- immediately
148                 errorsignal_b = 1; // Critical timing on this input; assign
149                     ↳ immediately
150                 errorsignal_b <= #(ct_state1_clk_b) 0; // Clear error signal
151                     ↳ after critical timing expires
152             end
153             2: begin
154                 cell_state = 0; // Blocking statement -- immediately
155                 errorsignal_a = 1; // Critical timing on this input; assign
156                     ↳ immediately
157                 errorsignal_a <= #(ct_state2_clk_a) 0; // Clear error signal
158                     ↳ after critical timing expires
159             end
160         endcase
161     end
162 end
163 end
164 endmodule

```

Listing 2.28: RSFQ XNOR verilog model.

The digital simulation results for the RSFQ XNOR is shown in Fig. 2.55 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.56.

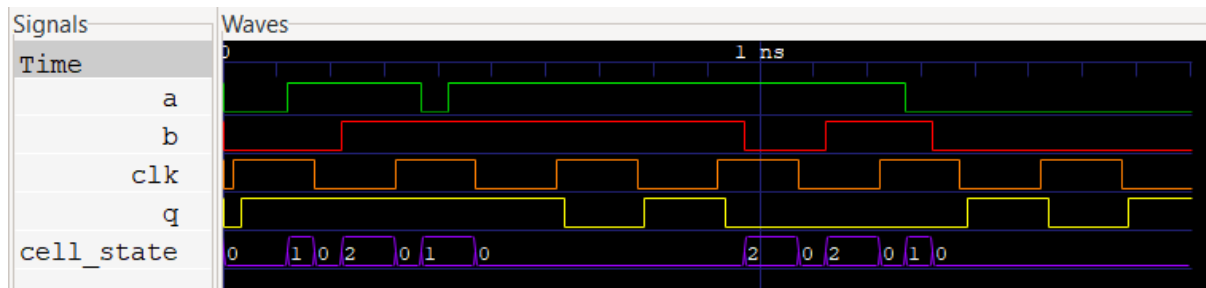


Figure 2.55: RSFQ XNOR digital simulation results.

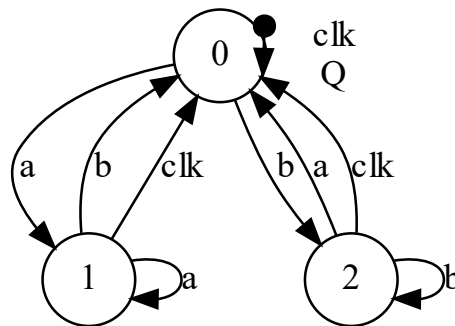


Figure 2.56: RSFQ XNOR Mealy finite state machine diagram.

Power Consumption

Table 2.28: RSFQ XNOR power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	3560	7.00
2	3560	14.0
5	3560	35.0
10	3560	70.0
20	3560	140
50	3560	350

2.3 Buffers

2.3.1 DFF

The RSFQ DFF, D flip-flop, is a multi-state device used to transmit an input set pulse synchronised with a reset (typically clock) signal. The cell is not designed to be directly connected to a PTL.

Schematic

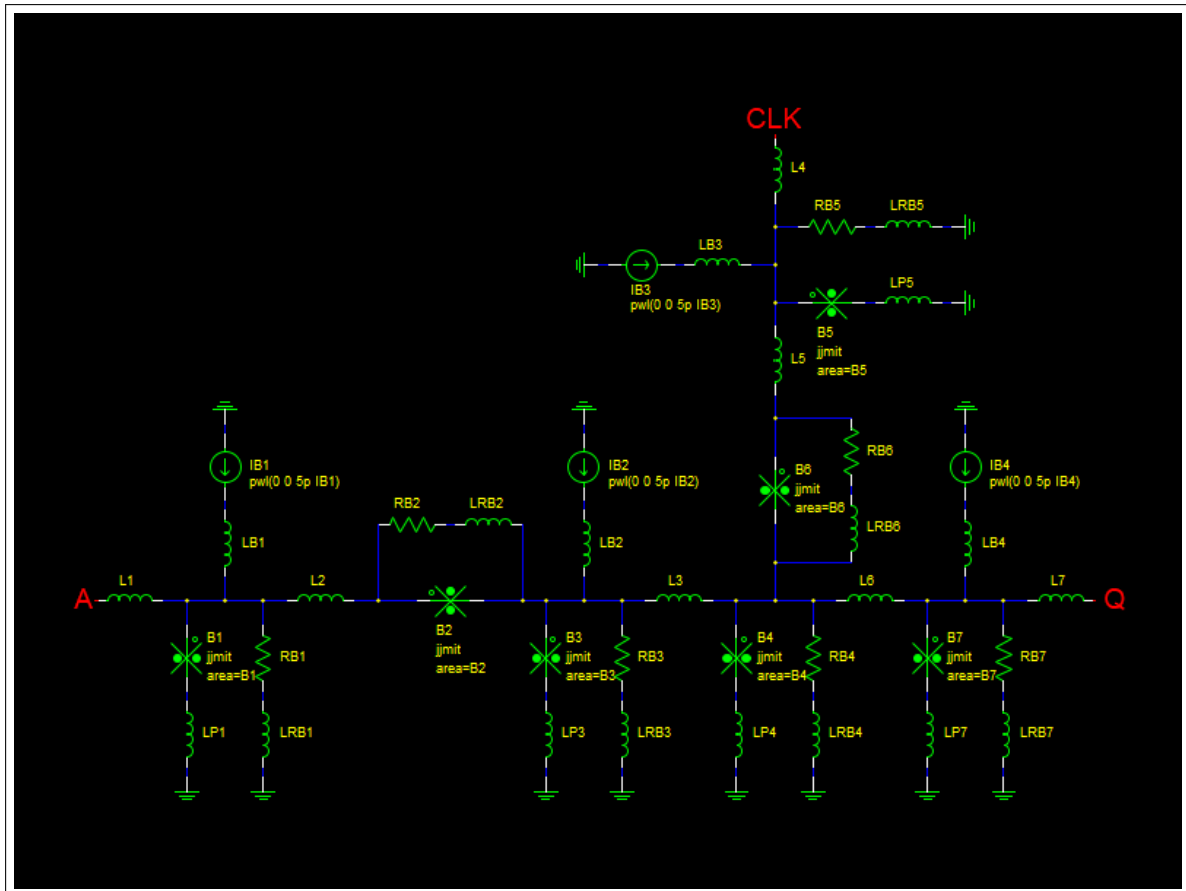


Figure 2.57: Schematic of RSFQ DFF.

Layout

The physical layout for the RSFQ DFF is shown in Fig. 2.58. The layout height is $70\ \mu m$ and the width is $30\ \mu m$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

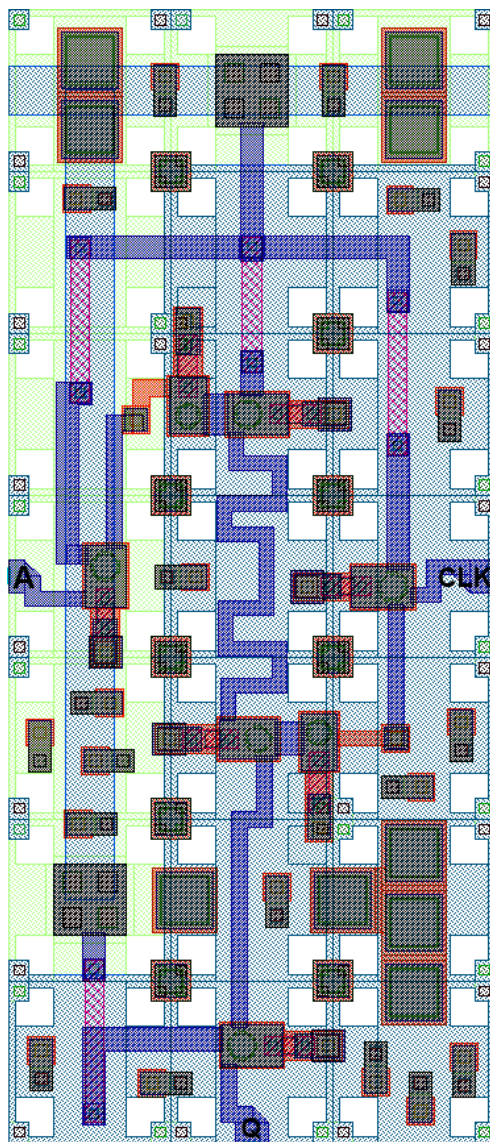


Figure 2.58: RSFQ DFF Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/02.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 1 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      clk      q
8  .subckt THmit11_DFF a clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=2.5
23 .param B2=1.67
24 .param B3=2.32
25 .param B4=2.02
26 .param B5=2.5
27 .param B6=1.69
28 .param B7=2.5
29
30 .param IB1=175u
31 .param IB2=222u
32 .param IB3=175u
33 .param IB4=175u
34
35 .param LB1=LB
36 .param LB2=LB
37 .param LB3=LB
38 .param LB4=LB
39
40 .param L1=1.9553p
41 .param L2=3.8899p
42 .param L3=8.6023p
43 .param L4=2.0461p
44 .param L5=3.8124p
45 .param L6=4.6626p
46 .param L7=1.8248p
47
48 .param LP1=LP
49 .param LP3=LP
50 .param LP4=LP
51 .param LP5=LP
52 .param LP7=LP
53
54 .param RB1=B0Rs/B1
55 .param RB2=B0Rs/B2
56 .param RB3=B0Rs/B3
57 .param RB4=B0Rs/B4
58 .param RB5=B0Rs/B5
59 .param RB6=B0Rs/B6
60 .param RB7=B0Rs/B7
61
62 .param LRB1=(RB1/Rsheet)*Lsheet+LP
63 .param LRB2=(RB2/Rsheet)*Lsheet
64 .param LRB3=(RB3/Rsheet)*Lsheet+LP
65 .param LRB4=(RB4/Rsheet)*Lsheet+LP
66 .param LRB5=(RB5/Rsheet)*Lsheet+LP
67 .param LRB6=(RB6/Rsheet)*Lsheet
68 .param LRB7=(RB7/Rsheet)*Lsheet+LP
69
70 B1 1 2 jjmit area=B1
71 B2 4 5 jjmit area=B2
72 B3 5 6 jjmit area=B3
73 B4 8 9 jjmit area=B4
74 B5 10 11 jjmit area=B5
75 B6 13 8 jjmit area=B6
76 B7 14 15 jjmit area=B7
77
78 IB1 0 3 pwl(0 0 5p IB1)
79 IB2 0 7 pwl(0 0 5p IB2)
80 IB3 0 12 pwl(0 0 5p IB3)
81 IB4 0 16 pwl(0 0 5p IB4)
82
83 LB1 3 1 LB1
84 LB2 7 5 LB2
85 LB3 12 10 LB3
86 LB4 16 14 LB4
87
88 L1 a 1 1.948E-012
89 L2 1 4 3.879E-012
90 L3 5 8 8.641E-012
91 L4 clk 10 2.035E-012
92 L5 10 13 3.799E-012
93 L6 8 14 4.675E-012
94 L7 14 q 1.809E-012
95
96 LP1 2 0 3.511E-013
97 LP3 6 0 3.717E-013
98 LP4 9 0 4.668E-013
99 LP5 11 0 4.339E-013
100 LP7 15 0 4.147E-013
101
102 RB1 1 101 RB1
103 LRB1 101 0 LRB1
104 RB2 4 104 RB2
105 LRB2 104 5 LRB2
106 RB3 5 105 RB3
107 LRB3 105 0 LRB3
108 RB4 8 108 RB4
109 LRB4 108 0 LRB4
110 RB5 10 110 RB5
111 LRB5 110 0 LRB5
112 RB6 13 113 RB6
113 LRB6 113 8 LRB6
114 RB7 14 114 RB7
115 LRB7 114 0 LRB7
116 .ends

```

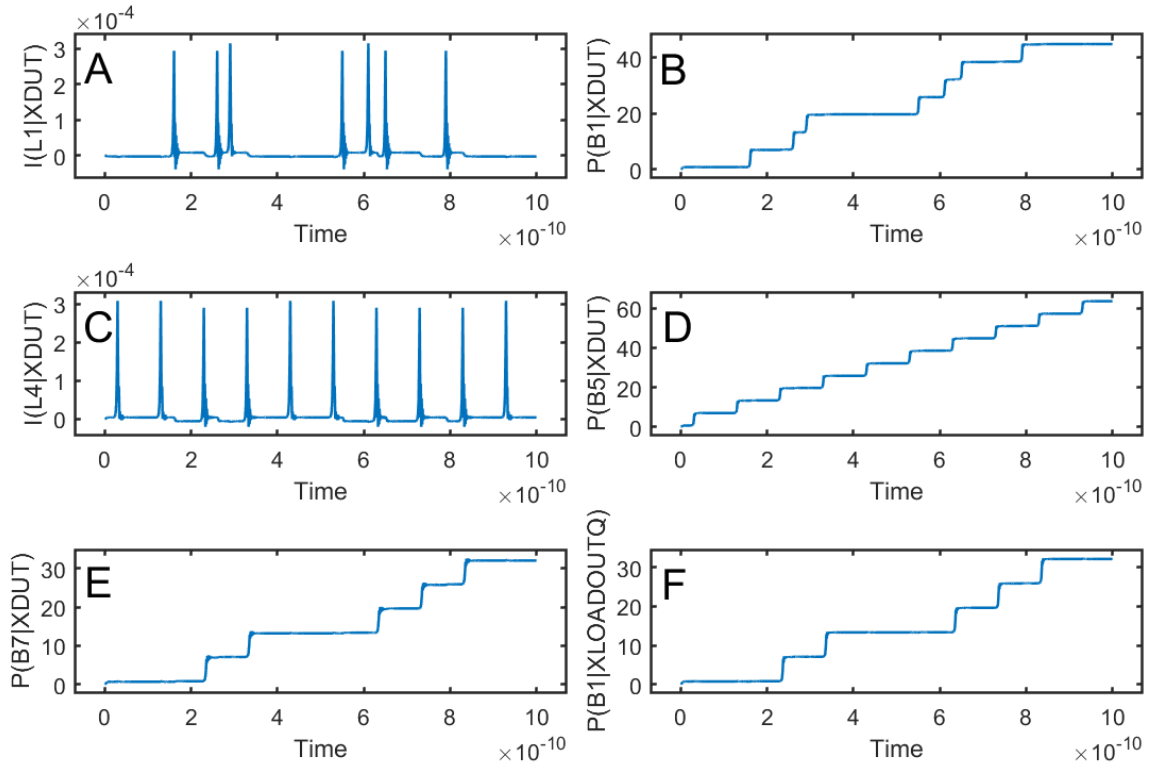
Listing 2.29: RSFQ DFF JoSIM netlist.

Table 2.29: RSFQ DFF pin list.

Pin	Description
a	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ DFF using JoSIM is shown in Fig. 2.59. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **clk**,
- (d) the phase over the input JJ of pin **clk**,
- (e) the phase over the output JJ of pin **q**, and
- (f) the phase over the input JJ of the load circuit connected to pin **q**.

**Figure 2.59:** RSFQ DFF analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 2 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_DFF_v3p0_extracted (a, clk, q);
9
10 input
11     a, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state1_clk_q = 6.3,
21     ct_state0_clk_a = 0.4;
22
23 reg
24     errorsignal_a,
25     errorsignal_clk;
26
27 integer
28     outfile,
29     cell_state; // internal state of the cell
30
31 initial
32     begin
33         errorsignal_a = 0;
34         errorsignal_clk = 0;
35         cell_state = 0; // Startup state
36         q = 0; // All outputs start at 0
37     end
38
39 always @(posedge a or negedge a) // execute at positive and negative edges of input
40     begin
41         if ($time>4) // arbitrary steady-state time)
42             begin
43                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
44                     begin
45                         outfile = $fopen("errors.txt", "a");
46                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
47                             ↪ ", $time);
48                         $fclose(outfile);
49                         q <= 1'bX; // Set all outputs to unknown
50                     end
51                 if (errorsignal_a == 0)
52                     begin
53                         case (cell_state)
54                             0: begin
55                                 cell_state = 1; // Blocking statement -- immediately
56                                 end
57                             1: begin
58                                 end
59                         endcase
60                     end
61             end
62     end
63
64 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
65     begin
66         if ($time>4) // arbitrary steady-state time)
67             begin

```

```

67         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
68             begin
69                 outfile = $fopen("errors.txt", "a");
70                 $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
71                 $fclose(outfile);
72                 q <= 1'bX; // Set all outputs to unknown
73             end
74         if (errorsignal_clk == 0)
75             begin
76                 case (cell_state)
77                     0: begin
78                         errorsignal_a = 1; // Critical timing on this input; assign
79                             ↪ immediately
80                         errorsignal_a <= #(ct_state0_clk_a) 0; // Clear error signal
81                             ↪ after critical timing expires
82                     end
83                     1: begin
84                         q <= #(delay_state1_clk_q) !q;
85                         cell_state = 0; // Blocking statement -- immediately
86                     end
87                 endcase
88             end
89         end
90     endmodule

```

Listing 2.30: RSFQ DFF verilog model.

The digital simulation results for the RSFQ DFF is shown in Fig. 2.60 and the Mealy finite state machine diagram, extracted using *TimEx*, is shown in Fig. 2.61.

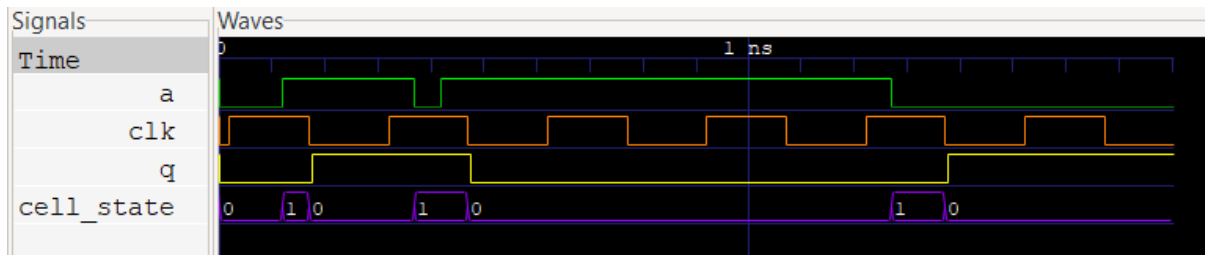


Figure 2.60: RSFQ DFF digital simulation results.

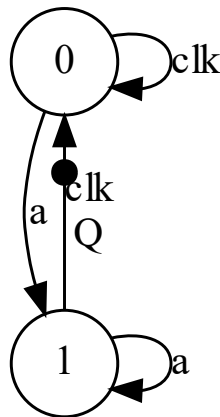


Figure 2.61: RSFQ DFF Mealy finite state machine diagram.

Power Consumption

Table 2.30: RSFQ DFF power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	1940	3.14
2	1940	6.29
5	1940	15.7
10	1940	31.4
20	1940	62.9
50	1940	157

2.3.2 NDRO

The NDRO, non-destructive readout, cell is a memory device controlled by a set, reset and clock input signal. When an input set signal is received, the NDRO will generate an output pulse after each clock signal until an input reset signal is received. The cell is not designed to be directly connected to a PTL.

Schematic

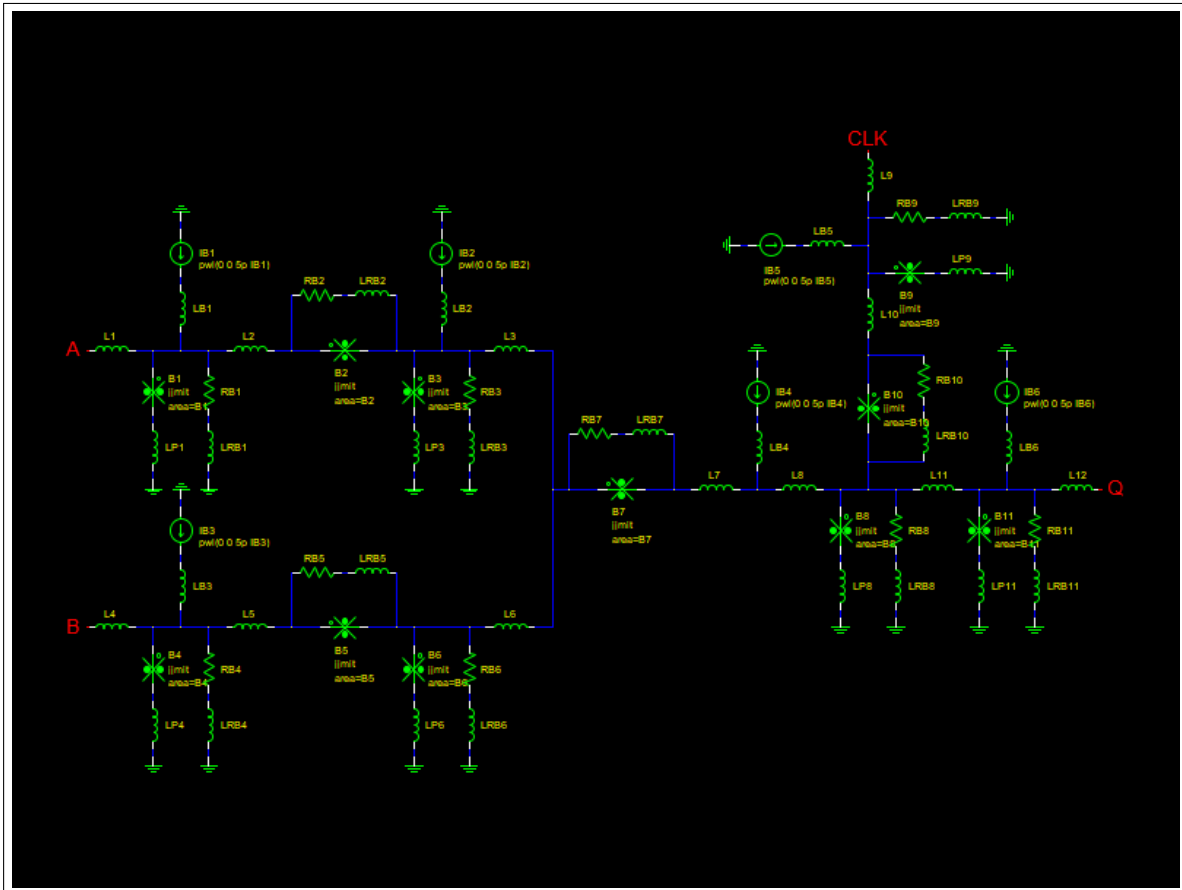


Figure 2.62: Schematic of RSFQ NDRO.

Layout

The physical layout of the RSFQ NDRO is shown in Fig. 2.63. The layout height is $70\ \mu m$ and the width is $40\ \mu m$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

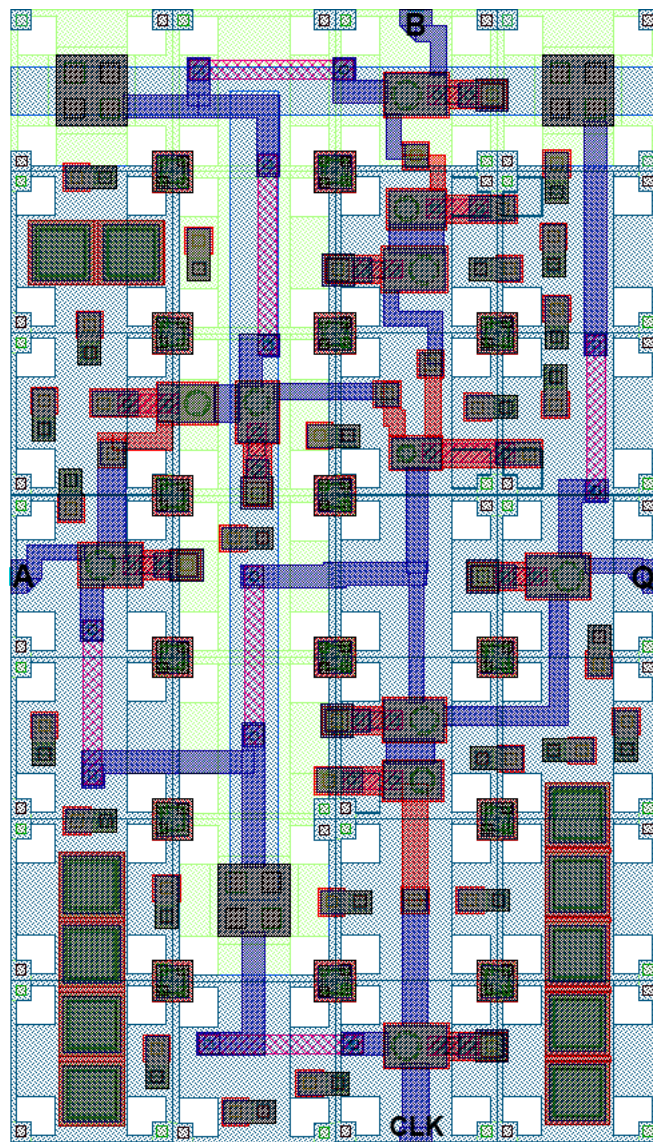


Figure 2.63: RSFQ NDRO Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/09/01.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 30 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      b      clk      q
8  .subckt THmit11_NDR0 a b clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=2.5
23 .param B2=1.80
24 .param B3=2.01
25 .param B4=2.5
26 .param B5=1.64
27 .param B6=2.58
28 .param B7=0.90
29 .param B8=2.30
30 .param B9=2.5
31 .param B10=1.77
32 .param B11=2.5
33
34 .param IB1=175u
35 .param IB2=139u
36 .param IB3=175u
37 .param IB4=158u
38 .param IB5=175u
39 .param IB6=175u
40
41 .param LB1=LB
42 .param LB2=LB
43 .param LB3=LB
44 .param LB4=LB
45 .param LB5=LB
46 .param LB6=LB
47
48 .param L1=1.8663p
49 .param L2=3.1100p
50 .param L3=2.8212p
51 .param L4=1.7117p
52 .param L5=2.7767p
53 .param L6=3.6623p
54 .param L7=1.4501p
55 .param L8=2.0153p
56 .param L9=1.3783p
57 .param L10=3.2508p
58 .param L11=3.3341p
59 .param L12=1.6989p
60
61 .param LP1=LP
62 .param LP3=LP
63 .param LP4=LP
64 .param LP6=LP
65 .param LP8=LP
66 .param LP9=LP
67 .param LP11=LP
68
69 .param RB1=B0Rs/B1
70 .param RB2=B0Rs/B2
71 .param RB3=B0Rs/B3
72 .param RB4=B0Rs/B4
73 .param RB5=B0Rs/B5
74 .param RB6=B0Rs/B6
75 .param RB7=B0Rs/B7
76 .param RB8=B0Rs/B8
77 .param RB9=B0Rs/B9
78 .param RB10=B0Rs/B10
79 .param RB11=B0Rs/B11
80
81 .param LRB1=(RB1/Rsheet)*Lsheet+LP
82 .param LRB2=(RB2/Rsheet)*Lsheet
83 .param LRB3=(RB3/Rsheet)*Lsheet+LP
84 .param LRB4=(RB4/Rsheet)*Lsheet+LP
85 .param LRB5=(RB5/Rsheet)*Lsheet
86 .param LRB6=(RB6/Rsheet)*Lsheet+LP
87 .param LRB7=(RB7/Rsheet)*Lsheet
88 .param LRB8=(RB8/Rsheet)*Lsheet+LP
89 .param LRB9=(RB9/Rsheet)*Lsheet+LP
90 .param LRB10=(RB10/Rsheet)*Lsheet
91 .param LRB11=(RB11/Rsheet)*Lsheet+LP
92
93 B1 1 2 jjmit area=B1
94 B2 4 5 jjmit area=B2
95 B3 5 6 jjmit area=B3
96 B4 9 10 jjmit area=B4
97 B5 12 13 jjmit area=B5
98 B6 13 14 jjmit area=B6
99 B7 8 15 jjmit area=B7
100 B8 18 19 jjmit area=B8
101 B9 20 21 jjmit area=B9
102 B10 23 18 jjmit area=B10
103 B11 24 25 jjmit area=B11
104
105 IB1 0 3 pw1(0 0 5p IB1)
106 IB2 0 7 pw1(0 0 5p IB2)
107 IB3 0 11 pw1(0 0 5p IB3)
108 IB4 0 17 pw1(0 0 5p IB4)
109 IB5 0 22 pw1(0 0 5p IB5)
110 IB6 0 26 pw1(0 0 5p IB6)
111
112 LB1 3 1 7.35E-013
113 LB2 7 5 7.798E-013
114 LB3 11 9 1.023E-012
115 LB4 17 16 1.981E-012
116 LB5 22 20 6.382E-013
117 LB6 26 24 1.181E-012
118
119 L1 a 1 1.861E-012
120 L2 1 4 3.103E-012
121 L3 5 8 2.808E-012
122 L4 b 9 1.712E-012
123 L5 9 12 2.775E-012
124 L6 13 8 3.652E-012
125 L7 15 16 1.455E-012
126 L8 16 18 2.018E-012
127 L9 clk 20 1.37E-012
128 L10 20 23 3.239E-012
129 L11 18 24 3.321E-012

```

```

130 | L12 24 q 1.698E-012
131 |
132 | LP1 2 0 4.356E-013
133 | LP3 6 0 4.322E-013
134 | LP4 10 0 3.473E-013
135 | LP6 14 0 3.698E-013
136 | LP8 19 0 4.794E-013
137 | LP9 21 0 3.788E-013
138 | LP11 25 0 4.491E-013
139 |
140 | RB1 1 101 RB1
141 | LRB1 101 0 LRB1
142 | RB2 4 104 RB2
143 | LRB2 104 5 LRB2
144 | RB3 5 105 RB3
145 | LRB3 105 0 LRB3
146 | RB4 9 109 RB4
147 | LRB4 109 0 LRB4

148 | RB5 12 112 RB5
149 | LRB5 112 13 LRB5
150 | RB6 13 113 RB6
151 | LRB6 113 0 LRB6
152 | RB7 8 108 RB7
153 | LRB7 108 15 LRB7
154 | RB8 18 118 RB8
155 | LRB8 118 0 LRB8
156 | RB9 20 120 RB9
157 | LRB9 120 0 LRB9
158 | RB10 23 123 RB10
159 | LRB10 123 18 LRB10
160 | RB11 24 124 RB11
161 | LRB11 124 0 LRB11
162 | .ends

```

Listing 2.31: RSFQ NDRO JoSIM netlist.**Table 2.31:** RSFQ NDRO pin list.

Pin	Description
a	Data input (set signal)
b	Data input (reset signal)
clk	Clock input
q	Data output

The simulation results for the RSFQ NDRO using JoSIM is shown in Fig. 2.64. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a** (set signal),
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b** (reset signal),
- (d) the phase over the input JJ of pin **b**,
- (e) the current through the input inductor connected to pin **clk**,
- (f) the phase over the input JJ of pin **clk**,
- (g) the phase over the output JJ of pin **q**, and
- (h) the phase over the input JJ of the load circuit connected to pin **q**.

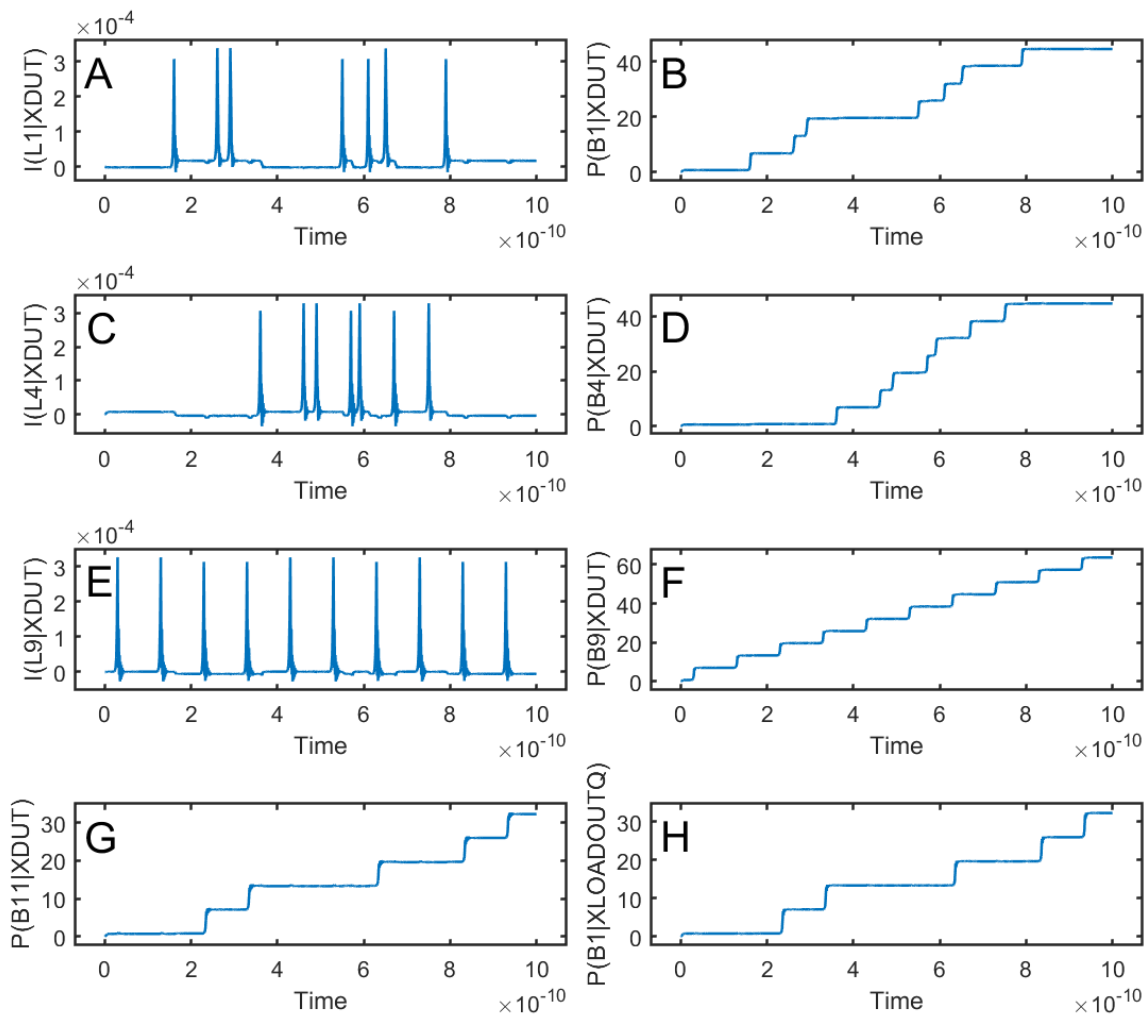


Figure 2.64: RSFQ NDRO analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 1 September 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_NDRO_v3p0_extracted (a, b, clk, q);
9
10 input
11     a, b, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state1_clk_q = 5.5,
21     ct_state0_a_b = 0.9,
22     ct_state1_b_a = 1.9,
23     ct_state1_clk_clk = 9.1;
24
25 reg
26     errorsignal_a,
27     errorsignal_b,
28     errorsignal_clk;
29
30 integer
31     outfile,
32     cell_state; // internal state of the cell
33
34 initial
35     begin
36         errorsignal_a = 0;
37         errorsignal_b = 0;
38         errorsignal_clk = 0;
39         cell_state = 0; // Startup state
40         q = 0; // All outputs start at 0
41     end
42
43 always @(posedge a or negedge a) // execute at positive and negative edges of input
44     begin
45         if ($time>4) // arbitrary steady-state time)
46             begin
47                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
48                     begin
49                         outfile = $fopen("errors.txt", "a");
50                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
51                         $fclose(outfile);
52                         q <= 1'bX; // Set all outputs to unknown
53                     end
54                 if (errorsignal_a == 0)
55                     begin
56                         case (cell_state)
57                             0: begin
58                                 cell_state = 1; // Blocking statement -- immediately
59                                 errorsignal_b = 1; // Critical timing on this input; assign
60                                     immediately
61                                 errorsignal_b <= #(ct_state0_a_b) 0; // Clear error signal
62                                     after critical timing expires
63                             end
64                             1: begin
65                                 end
66                         endcase
67                     end
68             end
69     end

```

```

65         end
66     end
67 end
68
69 always @(posedge b or negedge b) // execute at positive and negative edges of input
70 begin
71     if ($time>4) // arbitrary steady-state time)
72     begin
73         if (errorsignal_b == 1'b1) // A critical timing is active for this input
74         begin
75             outfile = $fopen("errors.txt", "a");
76             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
77                 ↳ ", $time);
78             $fclose(outfile);
79             q <= 1'bX; // Set all outputs to unknown
80         end
81         if (errorsignal_b == 0)
82         begin
83             case (cell_state)
84             0: begin
85                 end
86             1: begin
87                 cell_state = 0; // Blocking statement -- immediately
88                 errorsignal_a = 1; // Critical timing on this input; assign
89                 ↳ immediately
90                 errorsignal_a <= #(ct_state1_b_a) 0; // Clear error signal
91                 ↳ after critical timing expires
92             end
93             endcase
94         end
95     end
96 end
97
98 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
99 begin
100     if ($time>4) // arbitrary steady-state time)
101     begin
102         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
103         begin
104             outfile = $fopen("errors.txt", "a");
105             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
106                 ↳ ", $time);
107             $fclose(outfile);
108             q <= 1'bX; // Set all outputs to unknown
109         end
110         if (errorsignal_clk == 0)
111         begin
112             case (cell_state)
113             0: begin
114                 end
115             1: begin
116                 q <= #(delay_state1_clk_q) !q;
117                 errorsignal_clk = 1; // Critical timing on this input; assign
118                 ↳ immediately
119                 errorsignal_clk <= #(ct_state1_clk_clk) 0; // Clear error
120                 ↳ signal after critical timing expires
121             end
122             endcase
123         end
124     end
125 end
126 endmodule

```

Listing 2.32: RSFQ NDRO verilog model.

The digital simulation results for the RSFQ NDRO is shown in Fig. 2.65 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 2.66.

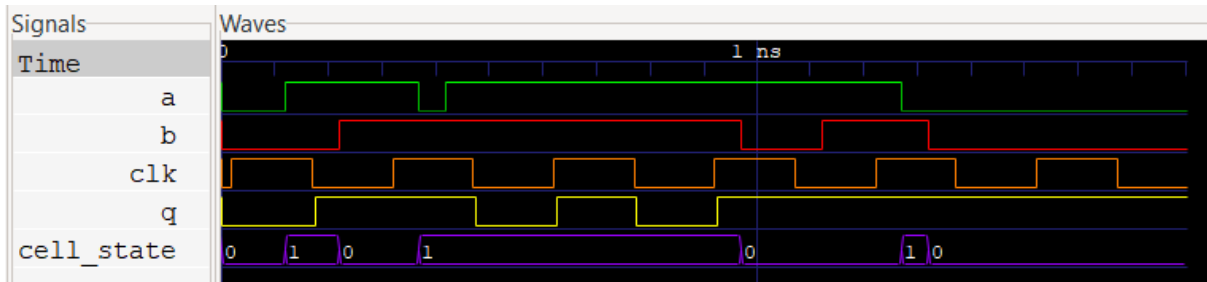


Figure 2.65: RSFQ NDRO digital simulation results.

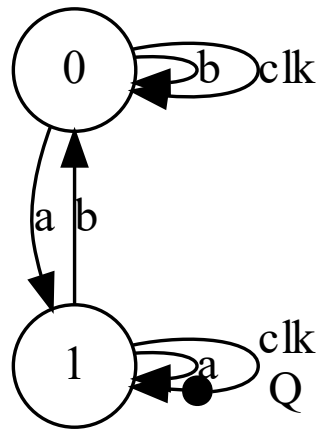


Figure 2.66: RSFQ NDRO Mealy finite state machine diagram.

Power Consumption

Table 2.32: RSFQ NDRO power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	2590	4.76
2	2590	9.51
5	2590	23.8
10	2590	47.6
20	2590	95.1
50	2590	238

2.3.3 BUFF

The RSFQ BUFF cell is a buffer cell intended for clock balancing. It is designed to have the same a-to-q delay as the SPLIT cell. The cell is not designed to be directly connected to a PTL.

Schematic

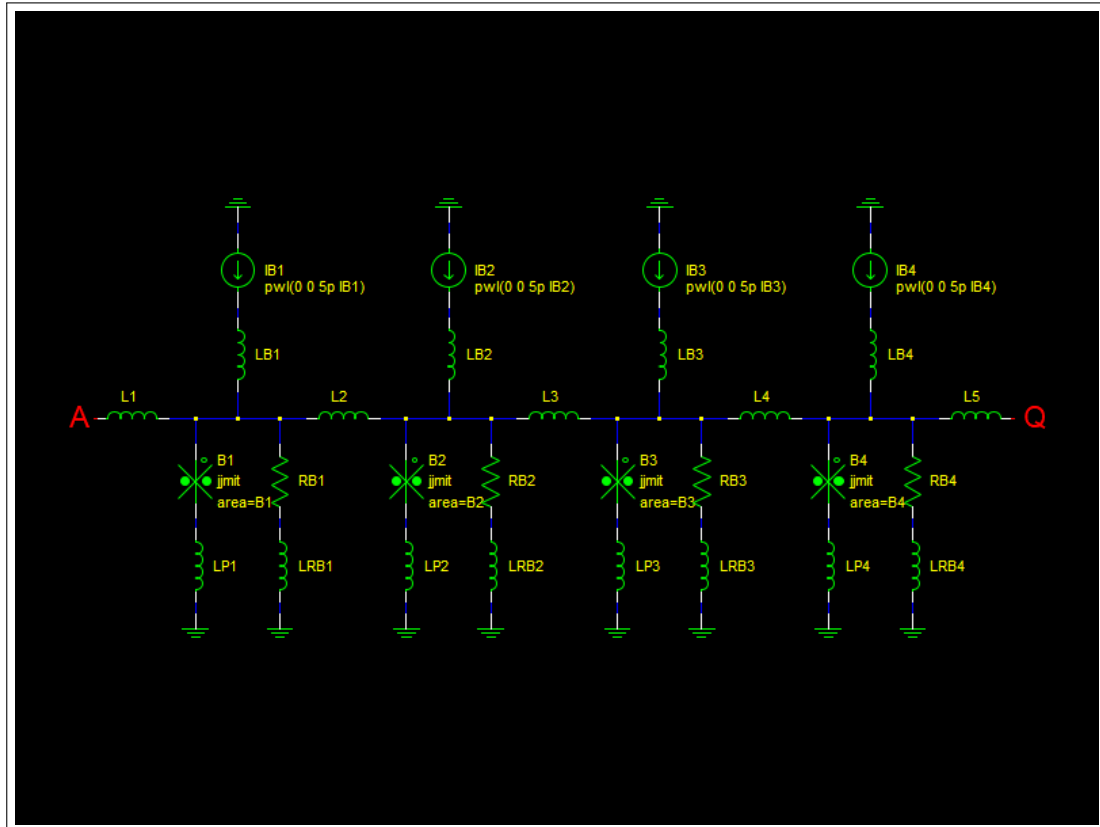


Figure 2.67: Schematic of RSFQ BUFF.

Layout

The physical layout of the RSFQ BUFF is shown in Fig. 2.68. The layout height is $70\ \mu\text{m}$ and the width is $30\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. Bias pillars from M0 to M6 connect the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

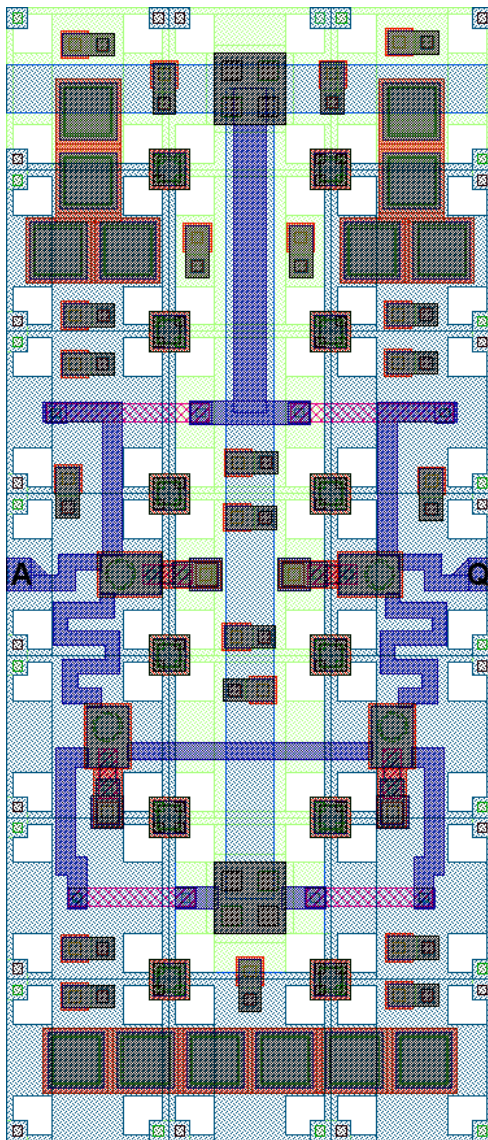


Figure 2.68: RSFQ BUFF Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/07/29.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 29 July 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      q
8  .subckt THmit11_BUFF a q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=IC
23 .param B2=IC
24 .param B3=IC
25 .param B4=IC
26
27 .param IB1=B1*Ic0*BiasCoef
28 .param IB2=B2*Ic0*0.95
29 .param IB3=B3*Ic0*0.95
30 .param IB4=B4*Ic0*BiasCoef
31
32 .param LB1=LB
33 .param LB2=LB
34 .param LB3=LB
35 .param LB4=LB
36
37 .param L1=Phi0/(4*B1*Ic0)
38 .param L2=Phi0/(2*B1*Ic0)
39 .param L3=Phi0/(2*B2*Ic0)
40 .param L4=Phi0/(2*B3*Ic0)
41 .param L5=Phi0/(4*B4*Ic0)
42
43 .param LP1=LP
44 .param LP2=LP
45 .param LP3=LP
46 .param LP4=LP
47
48 .param RB1=B0Rs/B1
49 .param RB2=B0Rs/B2
50 .param RB3=B0Rs/B3
51 .param RB4=B0Rs/B4
52
53 .param LRB1=(RB1/Rsheet)*Lsheet+LP
54 .param LRB2=(RB2/Rsheet)*Lsheet+LP
55 .param LRB3=(RB3/Rsheet)*Lsheet+LP
56 .param LRB4=(RB4/Rsheet)*Lsheet+LP
57
58 B1 1 2 jjmit area=B1
59 B2 4 5 jjmit area=B2
60 B3 7 8 jjmit area=B3
61 B4 10 11 jjmit area=B4
62
63 IB1 0 3 pwl(0 0 5p IB1)
64 IB2 0 6 pwl(0 0 5p IB2)
65 IB3 0 9 pwl(0 0 5p IB3)
66 IB4 0 12 pwl(0 0 5p IB4)
67
68 LB1 3 1 LB1
69 LB2 6 4 LB2
70 LB3 9 7 LB3
71 LB4 12 10 LB4
72
73 L1 a 1 2.06E-012
74 L2 1 4 4.141E-012
75 L3 4 7 4.153E-012
76 L4 7 10 4.142E-012
77 L5 10 q 2.078E-012
78
79 LP1 2 0 4.709E-013
80 LP2 5 0 4.152E-013
81 LP3 8 0 4.425E-013
82 LP4 11 0 4.583E-013
83
84 RB1 1 101 RB1
85 LRB1 101 0 LRB1
86 RB2 4 104 RB2
87 LRB2 104 0 LRB2
88 RB3 7 107 RB3
89 LRB3 107 0 LRB3
90 RB4 10 110 RB4
91 LRB4 110 0 LRB4
92 .ends

```

Listing 2.33: RSFQ BUFF JoSIM netlist.

Table 2.33: RSFQ BUFF pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ BUFF using JoSIM is shown in Fig. 2.69. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the phase over the output JJ of pin **q**, and
- (d) the phase over the input JJ of the load circuit connected to pin **q**.

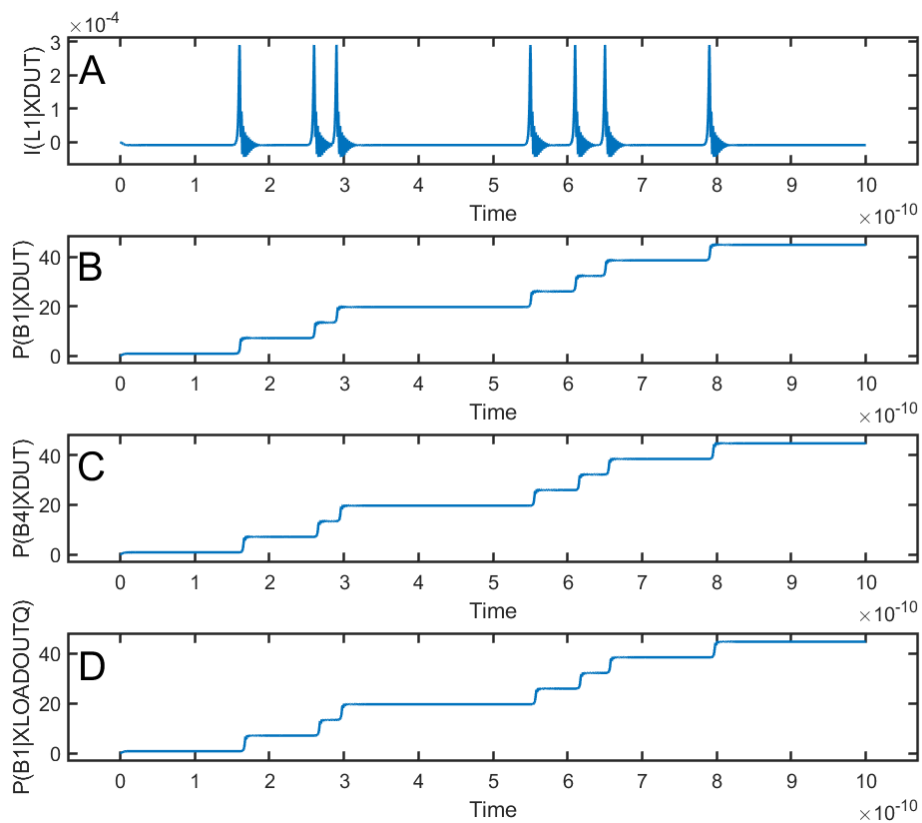


Figure 2.69: RSFQ BUFF analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 29 July 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_BUFF_v3p0_extracted (a, q);
9
10 input
11     a;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_a_q = 6.3,
21     ct_state0_a_a = 5.2;
22
23 reg
24     errorsignal_a;
25
26 integer
27     outfile,
28     cell_state; // internal state of the cell
29
30 initial
31     begin
32         errorsignal_a = 0;
33         cell_state = 0; // Startup state
34         q = 0; // All outputs start at 0
35     end
36
37 always @(posedge a or negedge a) // execute at positive and negative edges of input
38     begin
39         if ($time>4) // arbitrary steady-state time)
40             begin
41                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
42                     begin
43                         outfile = $fopen("errors.txt", "a");
44                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
45                             ↪ ", $stime);
46                         $fclose(outfile);
47                         q <= 1'bX; // Set all outputs to unknown
48                     end
49                 if (errorsignal_a == 0)
50                     begin
51                         case (cell_state)
52                             0: begin
53                                 q <= #(delay_state0_a_q) !q;
54                                 errorsignal_a = 1; // Critical timing on this input; assign
55                                     ↪ immediately
56                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
57                                     ↪ after critical timing expires
58                             end
59                         endcase
60                     end
61             end
62     end
63 endmodule

```

Listing 2.34: RSFQ BUFF verilog model.

The digital simulation results for the RSFQ BUFF is shown in Fig. 2.70 and the Mealy finite state machine diagram, extracted using *TimEx*, is shown in Fig. 2.71.

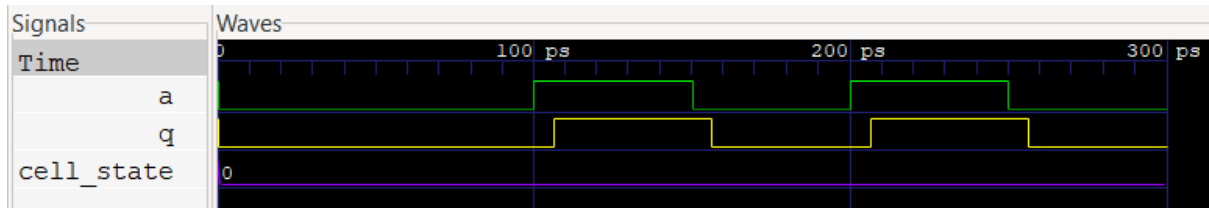


Figure 2.70: RSFQ BUFF digital simulation results.

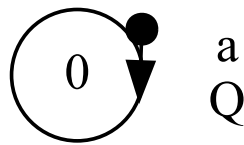


Figure 2.71: RSFQ BUFF Mealy finite state machine diagram.

Power Consumption

Table 2.34: RSFQ BUFF power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	2150	2.07
2	2150	4.14
5	2150	10.3
10	2150	20.7
20	2150	41.4
50	2150	103

2.4 Interface Cells

2.4.1 DCSFQ

The RSFQ DCSFQ is an interface cell designed to convert input voltage pulses into SFQ pulses. The DCSFQ does not have an integrated PTL transmitter and is not intended to connect directly to a PTL output.

Schematic

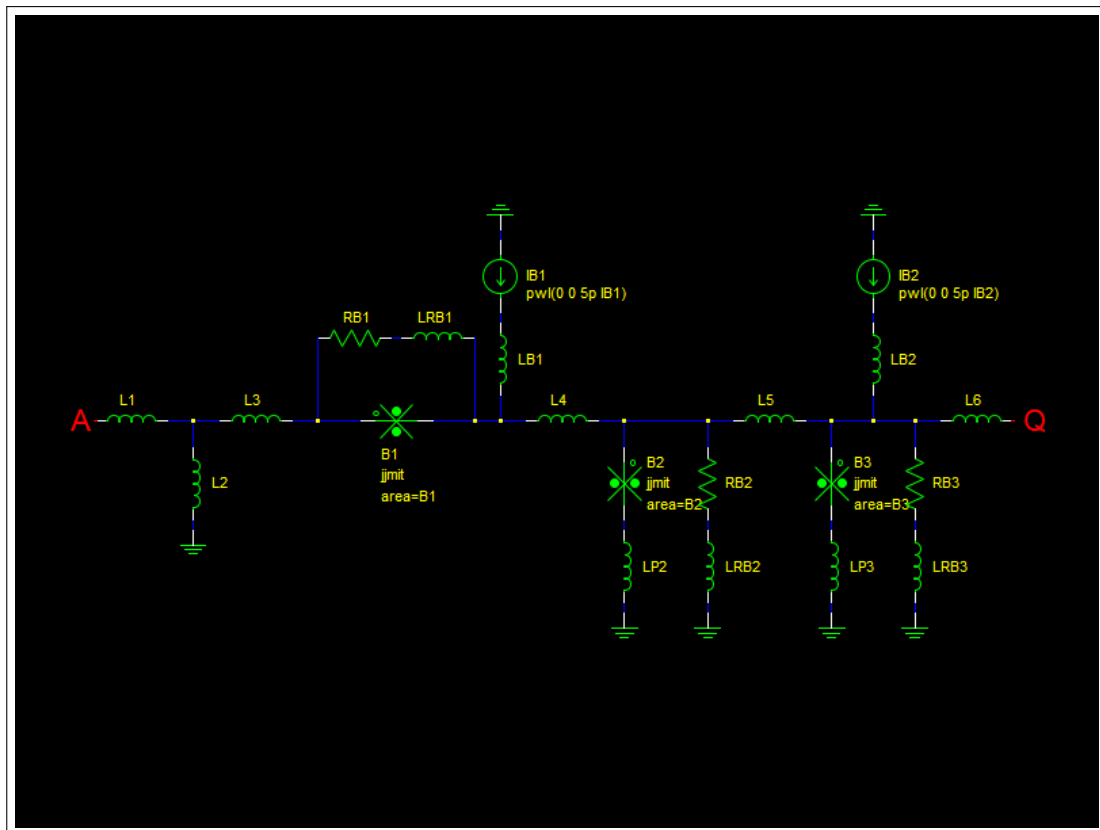


Figure 2.72: Schematic of RSFQ DCSFQ.

Layout

The physical layout of the RSFQ DCSFQ is shown in Fig. 2.73. The layout height is $70\ \mu\text{m}$ and the width is $20\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. A bias pillar from M0 to M6 connects the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

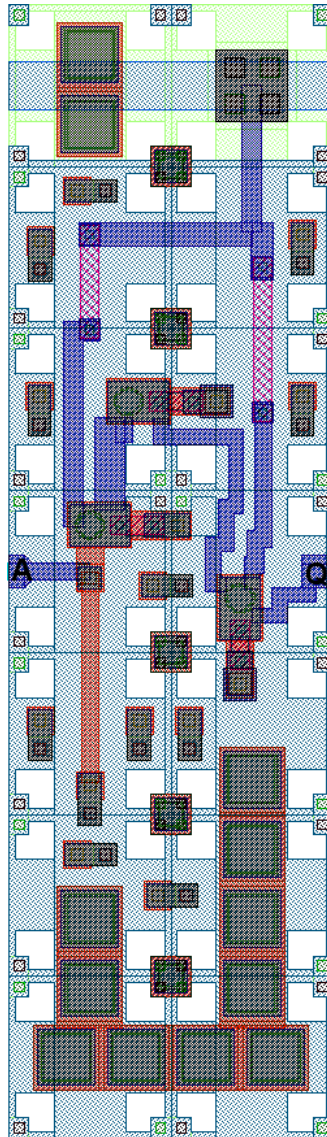


Figure 2.73: RSFQ DCSFQ Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/07/25.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 21 July 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      q
8  .subckt THmit11_DCSFQ a q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param LB=2p
20 .param BiasCoef=0.7
21
22 .param B1=2.25
23 .param B2=2.25
24 .param B3=IC
25
26 .param IB1=275u
27 .param IB2=B3*Ic0*BiasCoef
28
29 .param LB1=LB
30 .param LB2=LB
31
32 .param L1=1p
33 .param L2=3.9p
34 .param L3=0.6p
35 .param L4=1.1p
36 .param L5=4.5p
37 .param L6=Phi0/(4*IC*Ic0)
38
39 .param LP2=LP
40 .param LP3=LP
41
42 .param RB1=B0Rs/B1
43 .param RB2=B0Rs/B2
44 .param RB3=B0Rs/B3
45
46 .param LRB1=(RB1/Rsheet)*Lsheet
47 .param LRB2=(RB2/Rsheet)*Lsheet+LP
48 .param LRB3=(RB3/Rsheet)*Lsheet+LP
49
50 B1 2 3 jjmit area=B1
51 B2 5 6 jjmit area=B2
52 B3 7 8 jjmit area=B3
53
54 IB1 0 4 pwl(0 0 5p IB1)
55 IB2 0 9 pwl(0 0 5p IB2)
56
57 LB1 4 3 2.825E-012
58 LB2 9 7 2.942E-012
59
60 L1 a 1 1.672E-012
61 L2 1 0 3.901E-012
62 L3 1 2 5.953E-013
63 L4 3 5 1.1E-012
64 L5 5 7 4.542E-012
65 L6 7 q 2.012E-012
66
67 LP2 6 0 3.924E-013
68 LP3 8 0 3.841E-013
69
70 RB1 2 102 RB1
71 LRB1 102 3 LRB1
72 RB2 5 105 RB2
73 LRB2 105 0 LRB2
74 RB3 7 107 RB3
75 LRB3 107 0 LRB3
76 .ends

```

Listing 2.35: RSFQ DCSFQ JoSIM netlist.

Table 2.35: RSFQ DCSFQ pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ DCSFQ using JoSIM is shown in Fig. 2.74. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the current through the output inductor connected to pin **q**,
- (c) the phase over the output JJ of pin **q**, and
- (d) the phase over the input JJ of the load circuit connected to pin **q**.

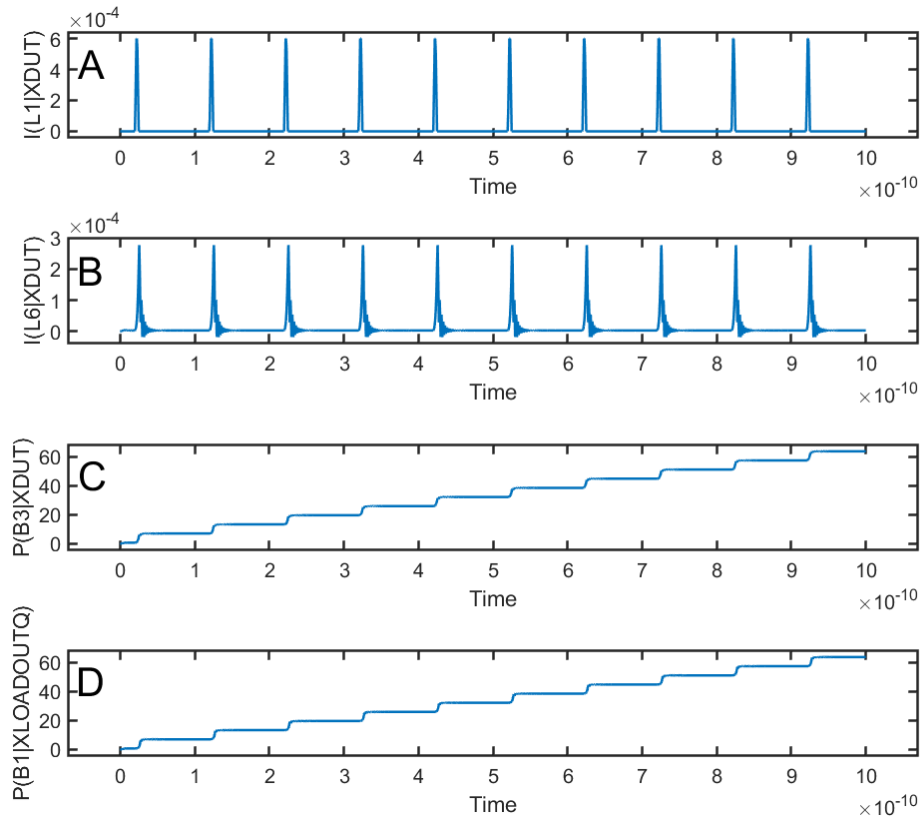


Figure 2.74: RSFQ DCSFQ analog simulation results.

2.4.2 SFQDC

The RSFQ SFQDC is an interface cell designed to convert SFQ pulses to an output voltage level which can be measured by standard equipment. The SFQDC does not have an integrated PTL receiver and is not intended to be connected directly to a PTL input.

Schematic

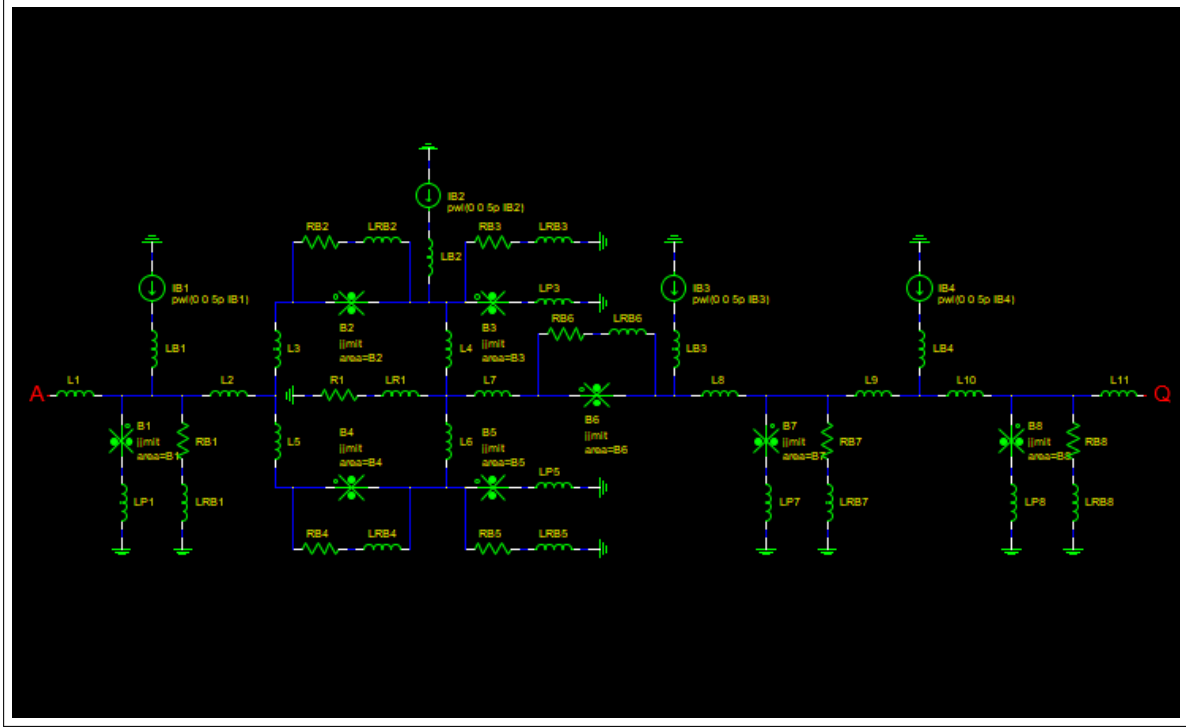


Figure 2.75: Schematic of RSFQ SFQDC.

Layout

The physical layout of the RSFQ SFQDC is shown in Fig. 2.76. The layout height is $70\ \mu\text{m}$ and the width is $40\ \mu\text{m}$. The cell includes an integrated bias line on M0 within the top row of track blocks. A bias pillar from M0 to M6 connects the circuit to the integrated bias line. All bias resistors are designed to be connected to a 2.6 mV voltage source.

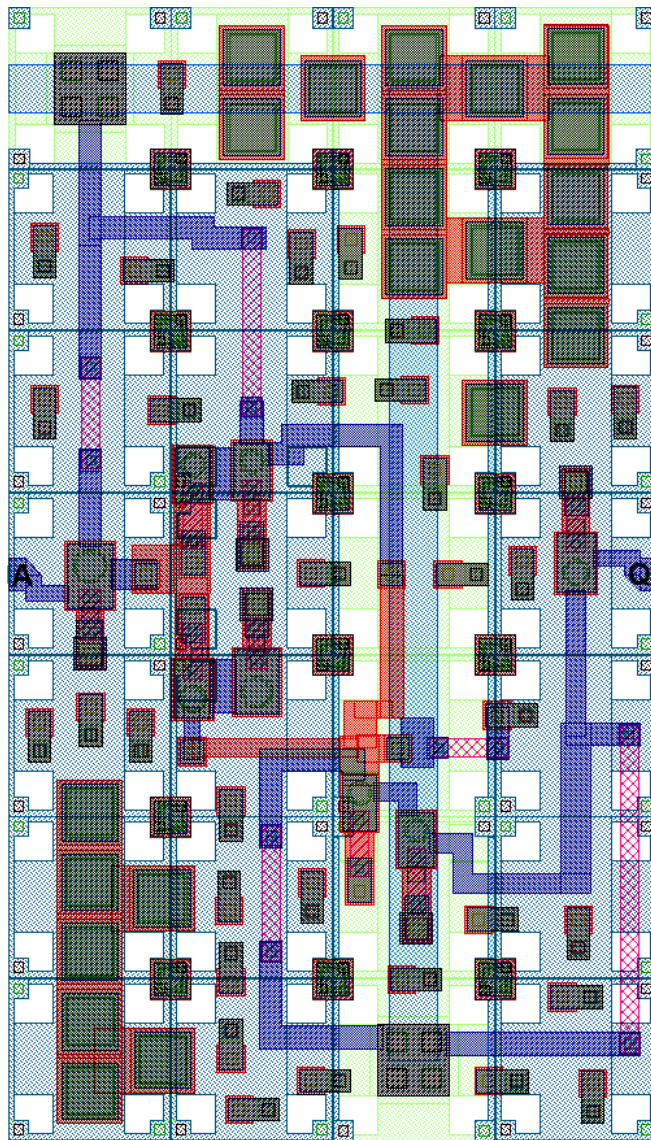


Figure 2.76: RSFQ SFQDC Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/08.
2  * Adapted from Fluxonics SDQDC_v5
3  * Author: L. Schindler
4  * Version: 3.0
5  * Last modification date: 4 August 2022
6  * Last modification by: T. Hall
7
8  *$Ports      a      q
9  .subckt THmit11_SFQDC a q
10 .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
11 .param Phi0=2.067833848E-15
12 .param B0=1
13 .param Ic0=0.0001
14 .param IcRs=100u*6.859904418
15 .param B0Rs=IcRs/Ic0*B0
16 .param Rsheet=2
17 .param Lsheet=1.13e-12
18 .param LP=0.5p
19 .param IC=2.5
20 .param LB=2p
21 .param BiasCoef=0.7
22
23 .param B1=3.25
24 .param B2=1.50
25 .param B3=1.75
26 .param B4=2.00
27 .param B5=3.00
28 .param B6=1.50
29 .param B7=1.50
30 .param B8=2.00
31
32 .param IB1=280u
33 .param IB2=150u
34 .param IB3=220u
35 .param IB4=80u
36
37 .param LB1=LB
38 .param LB2=LB
39 .param LB3=LB
40 .param LB4=LB
41
42 .param L1=1.522p
43 .param L2=0.827p
44 .param L3=1.12884p
45 .param L4=5.94p
46 .param L5=1.11098p
47 .param L6=3.216p
48 .param L7=0.215p
49 .param L8=0.954p
50 .param L9=3.699p
51 .param L10=2.010p
52 .param L11=1.510p
53
54 .param LR1=0.91p
55 .param R1=0.375
56
57 .param LP1=LP
58 .param LP3=LP
59 .param LP5=LP
60 .param LP7=LP
61 .param LP8=LP
62
63 .param RB1=B0Rs/B1
64 .param RB2=B0Rs/B2
65 .param RB3=B0Rs/B3
66 .param RB4=B0Rs/B4
67 .param RB5=B0Rs/B5
68 .param RB6=B0Rs/B6
69 .param RB7=B0Rs/B7
70 .param RB8=B0Rs/B8
71
72 .param LRB1=(RB1/Rsheet)*Lsheet+LP
73 .param LRB2=(RB2/Rsheet)*Lsheet
74 .param LRB3=(RB3/Rsheet)*Lsheet+LP
75 .param LRB4=(RB4/Rsheet)*Lsheet
76 .param LRB5=(RB5/Rsheet)*Lsheet+LP
77 .param LRB6=(RB6/Rsheet)*Lsheet
78 .param LRB7=(RB7/Rsheet)*Lsheet+LP
79 .param LRB8=(RB8/Rsheet)*Lsheet+LP
80
81 B1 1 2 jjmit area=B1
82 B2 5 6 jjmit area=B2
83 B3 6 8 jjmit area=B3
84 B4 10 11 jjmit area=B4
85 B5 11 12 jjmit area=B5
86 B6 14 15 jjmit area=B6
87 B7 17 18 jjmit area=B7
88 B8 21 22 jjmit area=B8
89
90 IB1 0 3 pwl(0 0 5p IB1)
91 IB2 0 7 pwl(0 0 5p IB2)
92 IB3 0 16 pwl(0 0 5p IB3)
93 IB4 0 20 pwl(0 0 5p IB4)
94
95 LB1 3 1 1.401E-012
96 LB2 7 6 6.064E-013
97 LB3 16 15 1.989E-012
98 LB4 20 19 8.87E-013
99
100 L1 a 1 1.529E-012
101 L2 1 4 8.239E-013
102 L3 5 4 1.124E-012
103 L4 6 9 5.898E-012
104 L5 4 10 1.111E-012
105 L6 9 11 3.227E-012
106 L7 9 14 2.138E-013
107 L8 15 17 9.503E-013
108 L9 17 19 3.696E-012
109 L10 19 21 2.001E-012
110 L11 21 q 1.496E-012
111
112 R1 0 13 R1
113 LR1 13 9 9.11E-013
114
115 LP1 2 0 3.162E-013
116 LP3 8 0 4.507E-013
117 LP5 12 0 6.57E-013
118 LP7 18 0 4.343E-013
119 LP8 22 0 3.726E-013
120
121 RB1 1 101 RB1
122 LRB1 101 0 LRB1
123 RB2 5 105 RB2
124 LRB2 105 6 LRB2
125 RB3 6 106 RB3
126 LRB3 106 0 LRB3
127 RB4 10 110 RB4
128 LRB4 110 11 LRB4
129 RB5 11 111 RB5

```

```

130 | LRB5 111 0 LRB5
131 | RB6 14 114 RB6
132 | LRB6 114 15 LRB6
133 | RB7 17 117 RB7
134 | LRB7 117 0 LRB7

```

```

135 | RB8 21 121 RB8
136 | LRB8 121 0 LRB8
137 | .ends

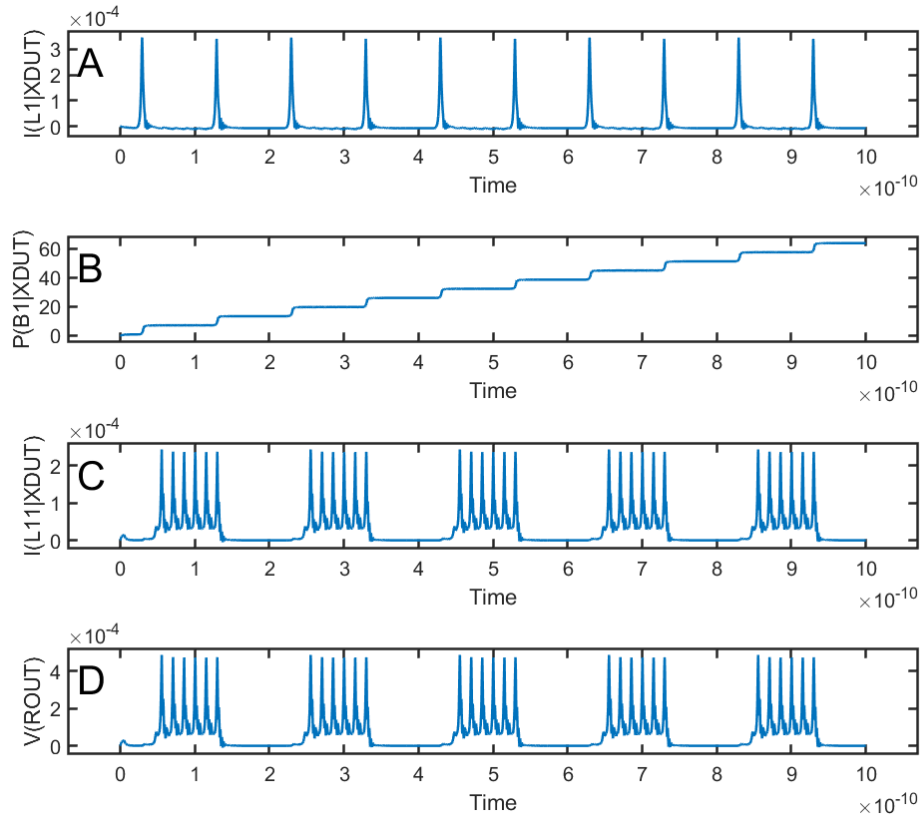
```

Listing 2.36: RSFQ SFQDC JoSIM netlist.**Table 2.36:** RSFQ SFQDC pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ SFQDC using JoSIM is shown in Fig. 2.77. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the output inductor connected to pin **q**, and
- (c) the voltage over the load resistor connected to pin **q**.

**Figure 2.77:** RSFQ SFQDC analog simulation results.

3. RSFQ Cell Library: PTL Connections

3.1 Interconnects

3.1.1 JTLT

The JTLT, Josephson transmission line, cell is commonly used to reestablish and propagate RSFQ pulses when long PTL connections are required. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

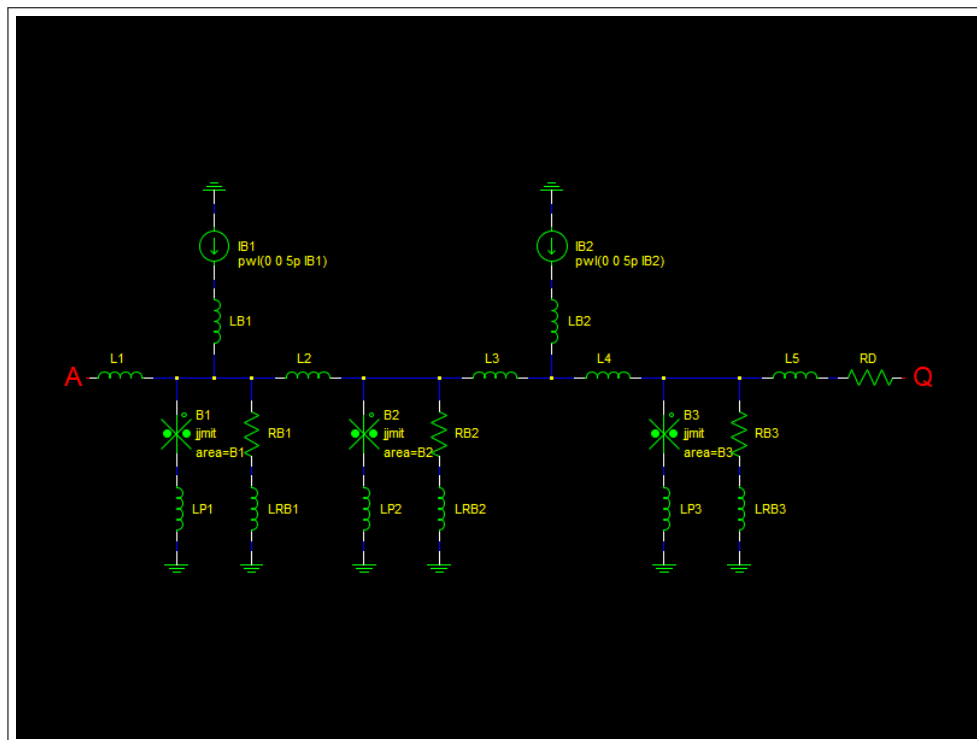


Figure 3.1: Schematic of RSFQ JTLT.

Layout

The physical layout of the RSFQ JTTL is shown in Fig. 3.2. The layout height is $70\ \mu\text{m}$ and the width is $20\ \mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

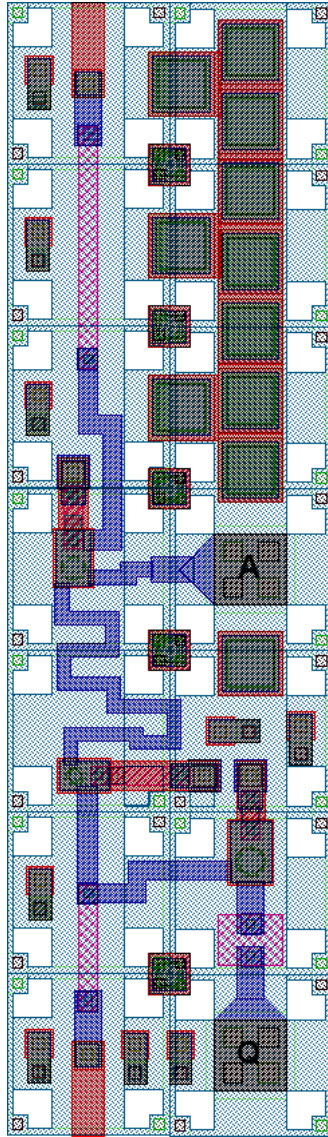


Figure 3.2: RSFQ JTTL Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↪ by InductEx v.6.1.52 on 2022/08/10.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 9 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      q
8  .subckt THmit11_JTTL a q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↪ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↪ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=1.6
27 .param B2=0.81
28 .param B3=2.5
29
30 .param IB1=112u
31 .param IB2=239u
32
33 .param LB1=LB
34 .param LB2=LB
35
36 .param L1=Lptl
37 .param L2=8.2192p
38 .param L3=1.4265p
39 .param L4=2.2384p
40 .param L5=Lptl
41
42 .param LP1=LP
43 .param LP2=LP
44 .param LP3=LP
45
46 .param RB1=B0Rs/B1
47 .param RB2=B0Rs/B2
48 .param RB3=B0Rs/B3
49
50 .param LRB1=(RB1/Rsheet)*Lsheet+LP
51 .param LRB2=(RB2/Rsheet)*Lsheet+LP
52 .param LRB3=(RB3/Rsheet)*Lsheet+LP
53
54 B1 1 2 jjmit area=B1
55 B2 4 5 jjmit area=B2
56 B3 8 9 jjmit area=B3
57
58 IB1 0 3 pwl(0 0 5p IB1)
59 IB2 0 7 pwl(0 0 5p IB2)
60
61 LB1 3 1 3.202E-012
62 LB2 7 6 2.142E-013
63
64 L1 a 1 1.986E-012
65 L2 1 4 8.188E-012
66 L3 4 6 1.431E-012
67 L4 6 8 2.254E-012
68 L5 8 10 7.542E-013
69
70 RD 10 q RD
71
72 LP1 2 0 4.596E-013
73 LP2 5 0 4.684E-013
74 LP3 9 0 4.113E-013
75
76 RB1 1 101 RB1
77 LRB1 101 0 LRB1
78 RB2 4 104 RB2
79 LRB2 104 0 LRB2
80 RB3 8 108 RB3
81 LRB3 108 0 LRB3
82
83 .ends

```

Listing 3.1: RSFQ JTTL JoSIM netlist.

Table 3.1: RSFQ JTTL pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ JTTL using JoSIM is shown in Fig. 3.3. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the phase over the output JJ of pin **q**, and
- (d) the phase over the input JJ of the load circuit connected via a PTL to the JTTL.

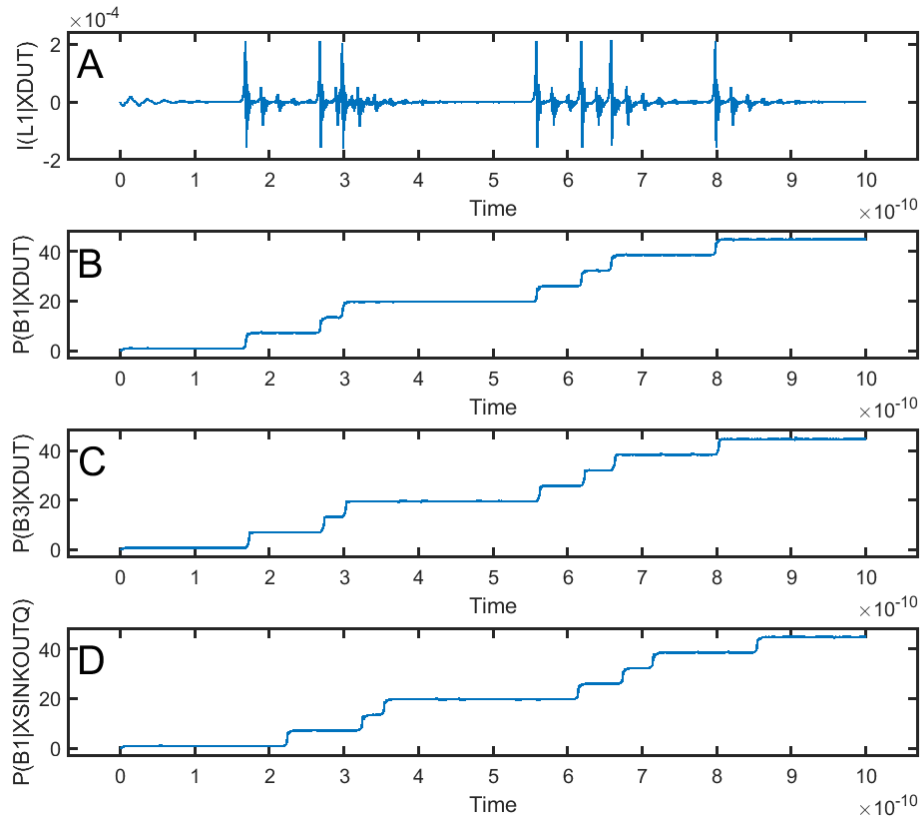


Figure 3.3: RSFQ JTTL analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 10 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_JTLT_v3p0_extracted (a, q);
9
10 input
11     a;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_a_q = 4.5,
21     ct_state0_a_a = 20.1;
22
23 reg
24     errorsignal_a;
25
26 integer
27     outfile,
28     cell_state; // internal state of the cell
29
30 initial
31     begin
32         errorsignal_a = 0;
33         cell_state = 0; // Startup state
34         q = 0; // All outputs start at 0
35     end
36
37 always @(posedge a or negedge a) // execute at positive and negative edges of input
38     begin
39         if ($time>4) // arbitrary steady-state time)
40             begin
41                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
42                     begin
43                         outfile = $fopen("errors.txt", "a");
44                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
45                             ↪ ", $stime);
46                         $fclose(outfile);
47                         q <= 1'bX; // Set all outputs to unknown
48                     end
49                 if (errorsignal_a == 0)
50                     begin
51                         case (cell_state)
52                             0: begin
53                                 q <= #(delay_state0_a_q) !q;
54                                 errorsignal_a = 1; // Critical timing on this input; assign
55                                     ↪ immediately
56                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
57                                     ↪ after critical timing expires
58                             end
59                         endcase
60                     end
61             end
62     end
63 endmodule

```

Listing 3.2: RSFQ JTLT verilog model.

The digital simulation results for the RSFQ JTTL is shown in Fig. 3.4 and the Mealy finite state machine diagram, extracted using *TimEx*, is shown in Fig. 3.5.

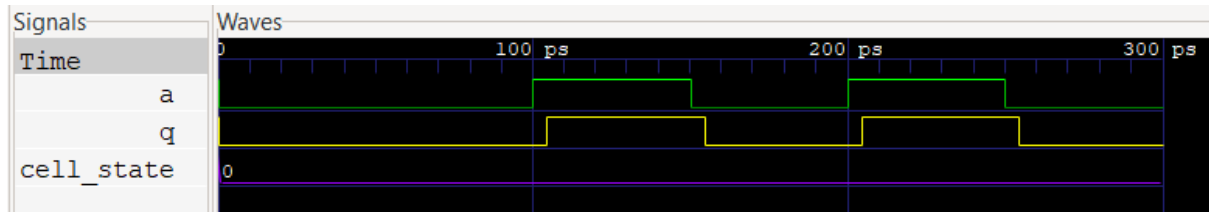


Figure 3.4: RSFQ JTTL digital simulation results.

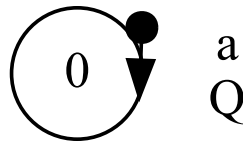


Figure 3.5: RSFQ JTTL Mealy finite state machine diagram.

Power Consumption

Table 3.2: RSFQ JTTL power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	913	1.02
2	913	2.03
5	913	5.08
10	913	10.2
20	913	20.3
50	913	50.8

3.1.2 SPLITT

The SPLITT cell is used to split a single pulse signal line into two duplicate output pulse signal lines. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

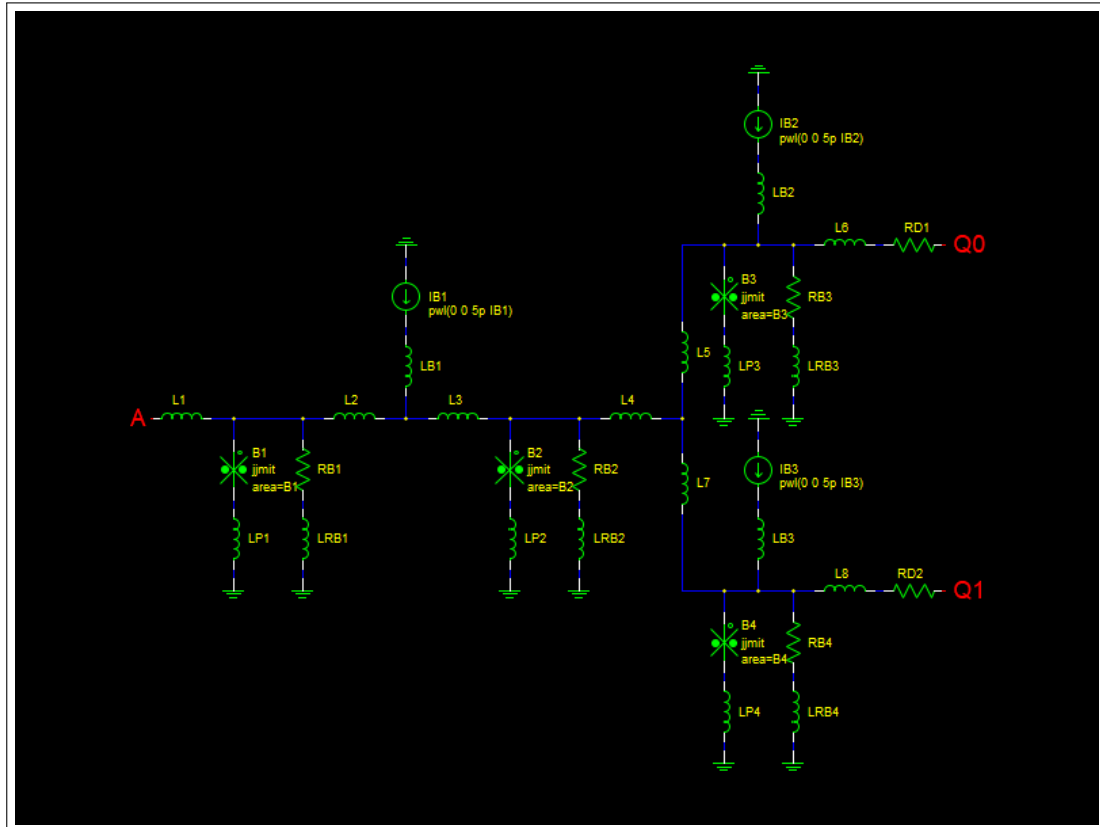


Figure 3.6: Schematic of RSFQ SPLITT.

Layout

The physical layout of the RSFQ SPLITT is shown in Fig. 3.7. The layout height is $70\ \mu\text{m}$ and the width is $30\ \mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

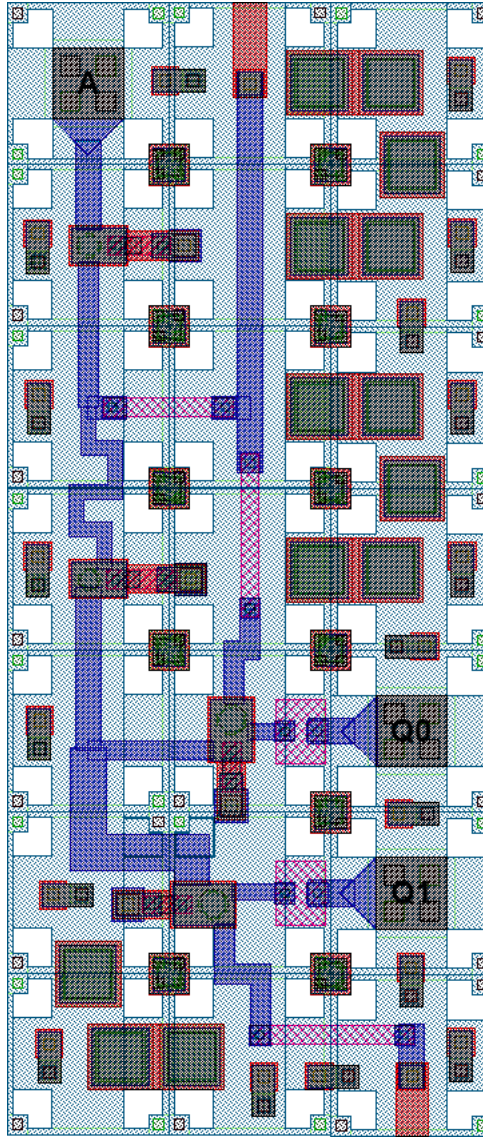


Figure 3.7: RSFQ SPLITT layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/10.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 10 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      q0      q1
8  .subckt THmit11_SPLITT a q0 q1
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=1.6
27 .param B2=1.36
28 .param B3=2.5
29 .param B4=2.5
30
31 .param IB1=231u
32 .param IB2=175u
33 .param IB3=175u
34
35 .param LB1=LB
36 .param LB2=LB
37 .param LB3=LB
38
39 .param L1=Lptl
40 .param L2=2.1010p
41 .param L3=3.6142p
42 .param L4=1.7400p
43 .param L5=1.8162p
44 .param L6=Lptl
45 .param L7=1.8162p
46 .param L8=Lptl
47
48 .param RD1=RD
49 .param RD2=RD
50
51 .param LP1=LP
52 .param LP2=LP
53 .param LP3=LP
54 .param LP4=LP
55
56 .param RB1=B0Rs/B1
57 .param RB2=B0Rs/B2
58 .param RB3=B0Rs/B3
59 .param RB4=B0Rs/B4
60
61 .param LRB1=(RB1/Rsheet)*Lsheet+LP
62 .param LRB2=(RB2/Rsheet)*Lsheet+LP
63 .param LRB3=(RB3/Rsheet)*Lsheet+LP
64 .param LRB4=(RB4/Rsheet)*Lsheet+LP
65
66 B1 1 2 jjmit area=B1
67 B2 5 6 jjmit area=B2
68 B3 8 9 jjmit area=B3
69 B4 12 13 jjmit area=B4
70
71 IB1 0 4 pw1(0 0 5p IB1)
72 IB2 0 10 pw1(0 0 5p IB2)
73 IB3 0 14 pw1(0 0 5p IB3)
74
75 LB1 4 3 4.087E-013
76 LB2 10 8 1.617E-012
77 LB3 14 12 2.099E-012
78
79 L1 a 1 1.432E-012
80 L2 1 3 2.097E-012
81 L3 3 5 3.589E-012
82 L4 5 7 1.755E-012
83 L5 8 7 1.82E-012
84 L6 8 11 7.505E-013
85 L7 7 12 1.805E-012
86 L8 12 15 1.137E-012
87
88 RD1 11 q0 RD1
89 RD2 15 q1 RD2
90
91 LP1 2 0 3.952E-013
92 LP2 6 0 4.754E-013
93 LP3 9 0 3.421E-013
94 LP4 13 0 3.856E-013
95
96 RB1 1 101 RB1
97 LRB1 101 0 LRB1
98 RB2 5 105 RB2
99 LRB2 105 0 LRB2
100 RB3 8 108 RB3
101 LRB3 108 0 LRB3
102 RB4 12 112 RB4
103 LRB4 112 0 LRB4
104
105 .ends

```

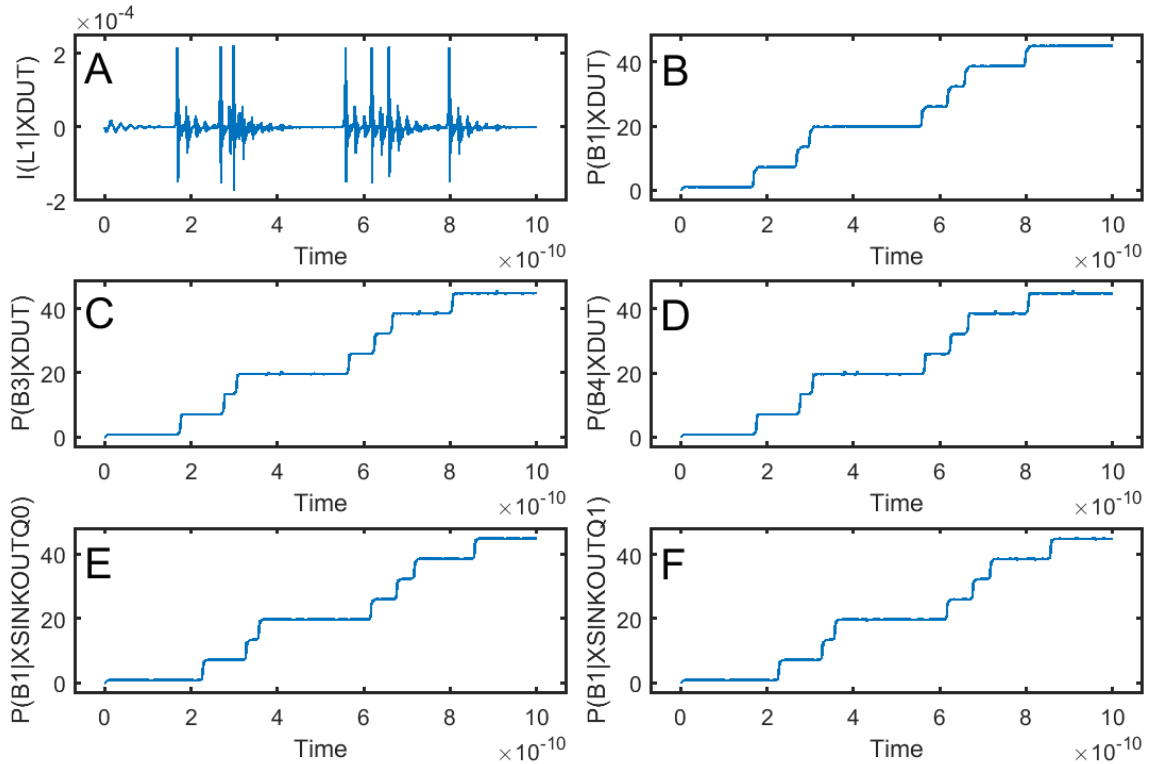
Listing 3.3: RSFQ SPLITT JoSIM netlist.

Table 3.3: RSFQ SPLITT pin list.

Pin	Description
a	Data input
q0	Data output
q1	Data output

The JoSIM simulation results for the RSFQ SPLITT are shown in Fig. 3.8. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the phase over the output JJ of pin **q0**,
- (d) the phase over the output JJ of pin **q1**,
- (e) the phase over the input JJ of the load cell connected to pin q0 through a PTL, and
- (f) the phase over the input JJ of the load cell connected to pin q1 through a PTL.

**Figure 3.8:** RSFQ SPLITT analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 10 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_SPLITT_v3p0_extracted (a, q0, q1);
9
10 input
11     a;
12
13 output
14     q0, q1;
15
16 reg
17     q0, q1;
18
19 real
20     delay_state0_a_q0 = 7.3,
21     delay_state0_a_q1 = 7.3,
22     ct_state0_a_a = 11.1;
23
24 reg
25     errorsignal_a;
26
27 integer
28     outfile,
29     cell_state; // internal state of the cell
30
31 initial
32     begin
33         errorsignal_a = 0;
34         cell_state = 0; // Startup state
35         q0 = 0; // All outputs start at 0
36         q1 = 0; // All outputs start at 0
37     end
38
39 always @(posedge a or negedge a) // execute at positive and negative edges of input
40     begin
41         if ($time>4) // arbitrary steady-state time)
42             begin
43                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
44                     begin
45                         outfile = $fopen("errors.txt", "a");
46                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0dps.\n", $time);
47                         $fclose(outfile);
48                         q0 <= 1'bX; // Set all outputs to unknown
49                         q1 <= 1'bX; // Set all outputs to unknown
50                     end
51                 if (errorsignal_a == 0)
52                     begin
53                         case (cell_state)
54                             0: begin
55                                 q0 <= #(delay_state0_a_q0) !q0;
56                                 q1 <= #(delay_state0_a_q1) !q1;
57                                 errorsignal_a = 1; // Critical timing on this input; assign
58                                     immediately
59                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
60                                     after critical timing expires
61                             end
62                         endcase
63                     end
64             end
65     end
66 endmodule

```

Listing 3.4: RSFQ SPLITT verilog model.

The digital simulation results for the RSFQ SPLITT is shown in Fig. 3.9 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 3.10.

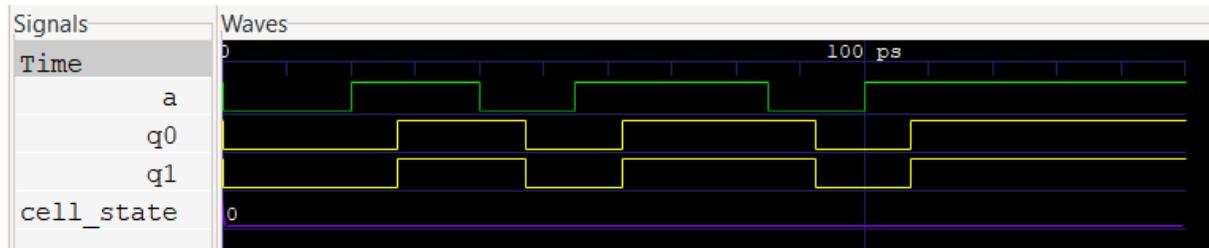


Figure 3.9: RSFQ SPLITT digital simulation results.

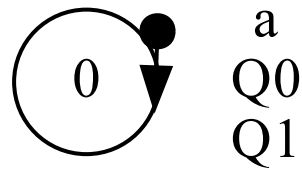


Figure 3.10: RSFQ SPLITT Mealy finite state diagram.

Power consumption

Table 3.4: RSFQ SPLITT power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	1510	1.65
2	1510	3.29
5	1510	8.23
10	1510	16.5
20	1510	32.9
50	1510	82.3

3.1.3 MERGET

The MERGET joins two input pulse signal lines and provides a single output pulse signal line. If there is a pulse on either input lines, the MERGET will generate a pulse on the output signal line. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

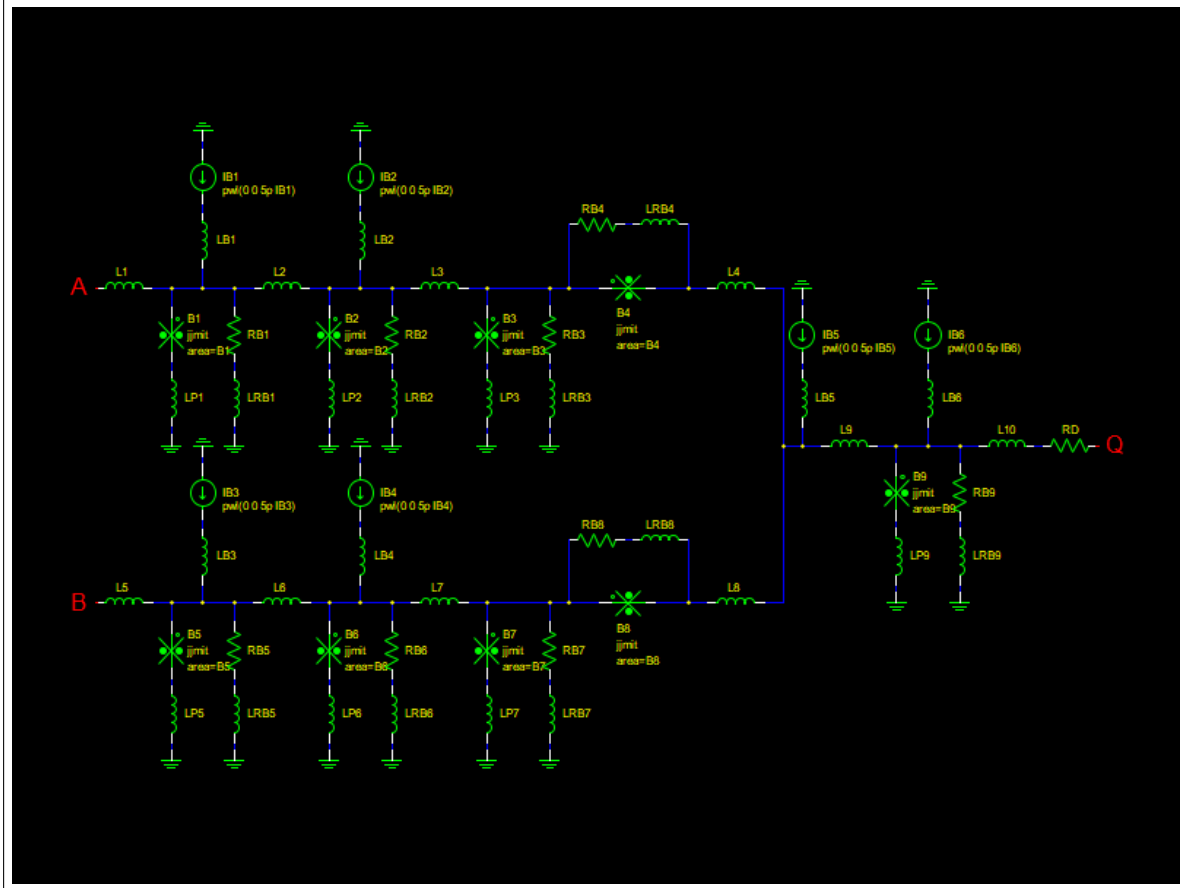


Figure 3.11: Schematic of RSFQ MERGET.

Layout

The physical layout of the RSFQ MERGET is shown in Fig. 3.12. The height of the layout is $70\ \mu\text{m}$ and the width is $50\ \mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

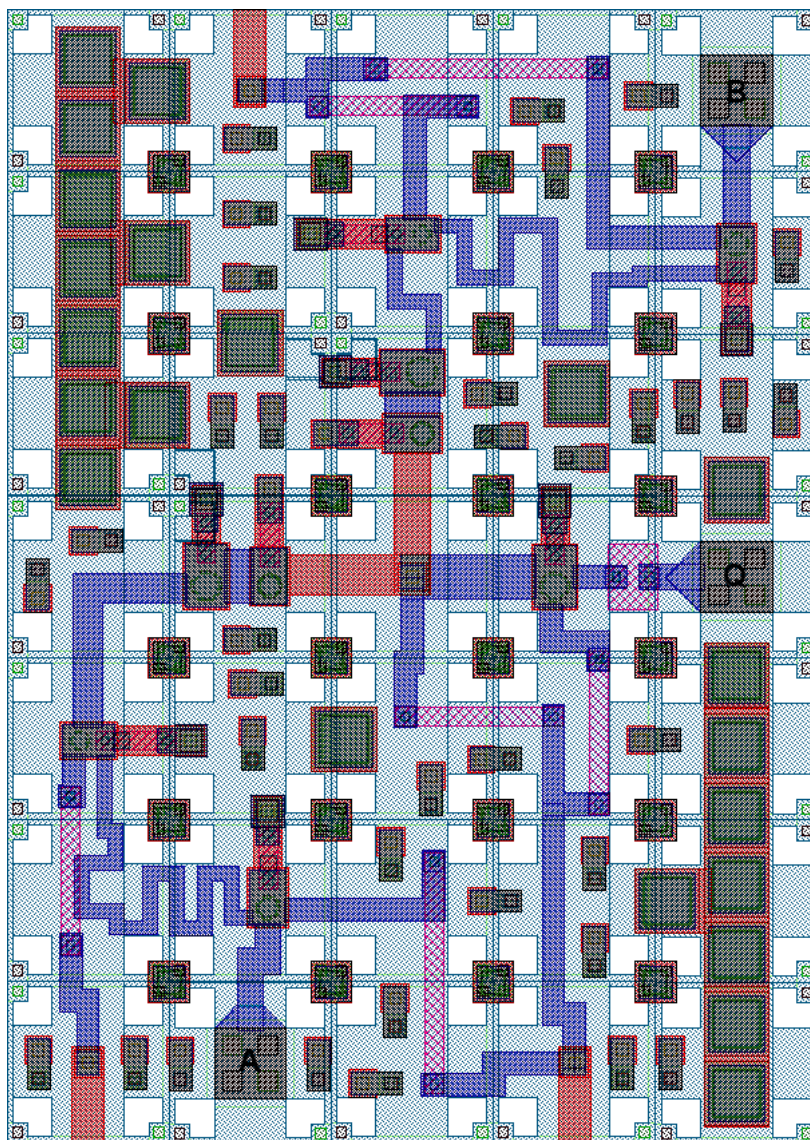


Figure 3.12: RSFQ MERGET layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/11.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 10 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      b      q
8  .subckt THmit11_MERGET a b q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=1.6
27 .param B2=1.10
28 .param B3=2.61
29 .param B4=1.56
30 .param B5=1.6
31 .param B6=1.10
32 .param B7=2.61
33 .param B8=1.56
34 .param B9=2.5
35
36 .param IB1=112u
37 .param IB2=170u
38 .param IB3=112u
39 .param IB4=170u
40 .param IB5=171u
41 .param IB6=175u
42
43 .param LB1=LB
44 .param LB2=LB
45 .param LB3=LB
46 .param LB4=LB
47 .param LB5=LB
48 .param LB6=LB
49
50 .param L1=Lptl
51 .param L2=8.8600p
52 .param L3=2.4587p
53 .param L4=1.5077p
54 .param L5=Lptl
55 .param L6=8.8600p
56 .param L7=2.4587p
57 .param L8=1.5077p
58 .param L9=1.0056p
59 .param L10=Lptl
60
61 .param LP1=LP
62 .param LP2=LP
63 .param LP3=LP
64 .param LP5=LP
65 .param LP6=LP
66 .param LP7=LP
67 .param LP9=LP
68
69 .param RB1=B0Rs/B1
70 .param RB2=B0Rs/B2
71 .param RB3=B0Rs/B3
72 .param RB4=B0Rs/B4
73 .param RB5=B0Rs/B5
74 .param RB6=B0Rs/B6
75 .param RB7=B0Rs/B7
76 .param RB8=B0Rs/B8
77 .param RB9=B0Rs/B9
78
79 .param LRB1=(RB1/Rsheet)*Lsheet+LP
80 .param LRB2=(RB2/Rsheet)*Lsheet+LP
81 .param LRB3=(RB3/Rsheet)*Lsheet+LP
82 .param LRB4=(RB4/Rsheet)*Lsheet
83 .param LRB5=(RB5/Rsheet)*Lsheet+LP
84 .param LRB6=(RB6/Rsheet)*Lsheet+LP
85 .param LRB7=(RB7/Rsheet)*Lsheet+LP
86 .param LRB8=(RB8/Rsheet)*Lsheet
87 .param LRB9=(RB9/Rsheet)*Lsheet+LP
88
89 B1 1 2 jjmit area=B1
90 B2 4 5 jjmit area=B2
91 B3 7 8 jjmit area=B3
92 B4 7 9 jjmit area=B4
93 B5 11 12 jjmit area=B5
94 B6 14 15 jjmit area=B6
95 B7 17 18 jjmit area=B7
96 B8 17 19 jjmit area=B8
97 B9 21 22 jjmit area=B9
98
99 IB1 0 3 pwl(0 0 5p IB1)
100 IB2 0 6 pwl(0 0 5p IB2)
101 IB3 0 13 pwl(0 0 5p IB3)
102 IB4 0 16 pwl(0 0 5p IB4)
103 IB5 0 20 pwl(0 0 5p IB5)
104 IB6 0 23 pwl(0 0 5p IB6)
105
106 LB1 3 1 2.504E-012
107 LB2 6 4 7.477E-013
108 LB3 13 11 3.72E-012
109 LB4 16 14 2.115E-012
110 LB5 20 10 1.701E-012
111 LB6 23 21 1.272E-012
112
113 L1 a 1 1.299E-012
114 L2 1 4 8.804E-012
115 L3 4 7 2.465E-012
116 L4 9 10 1.51E-012
117 L5 b 11 1.373E-012
118 L6 11 14 8.866E-012
119 L7 14 17 2.459E-012
120 L8 19 10 1.516E-012
121 L9 10 21 1.002E-012
122 L10 21 24 7.408E-013
123
124 RD 24 q RD
125
126 LP1 2 0 4.76E-013
127 LP2 5 0 4.103E-013
128 LP3 8 0 4.056E-013
129 LP5 12 0 3.991E-013

```

```

130 | LP6 15 0 5.058E-013
131 | LP7 18 0 4.187E-013
132 | LP9 22 0 3.717E-013
133 |
134 | RB1 1 101 RB1
135 | LRB1 101 0 LRB1
136 | RB2 4 104 RB2
137 | LRB2 104 0 LRB2
138 | RB3 7 107 RB3
139 | LRB3 107 0 LRB3
140 | RB4 7 109 RB4
141 | LRB4 109 9 LRB4
142 | RB5 11 111 RB5
143 | LRB5 111 0 LRB5
144 | RB6 14 114 RB6
145 | LRB6 114 0 LRB6
146 | RB7 17 117 RB7
147 | LRB7 117 0 LRB7
148 | RB8 17 119 RB8
149 | LRB8 119 19 LRB8
150 | RB9 21 121 RB9
151 | LRB9 121 0 LRB9
152 |
153 | .ends

```

Listing 3.5: RSFQ MERGET JoSIM netlist.**Table 3.5:** RSFQ MERGET pin list.

Pin	Description
a	Data input
b	Data input
q	Data output

The JoSIM simulation results for the RSFQ MERGET are shown in Fig. 3.13. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b**,
- (d) the phase over the input JJ of pin **b**,
- (e) the phase over the output JJ of pin **q**,
- (f) the phase over the input JJ of the load cell connected to pin **q** via a PTL.

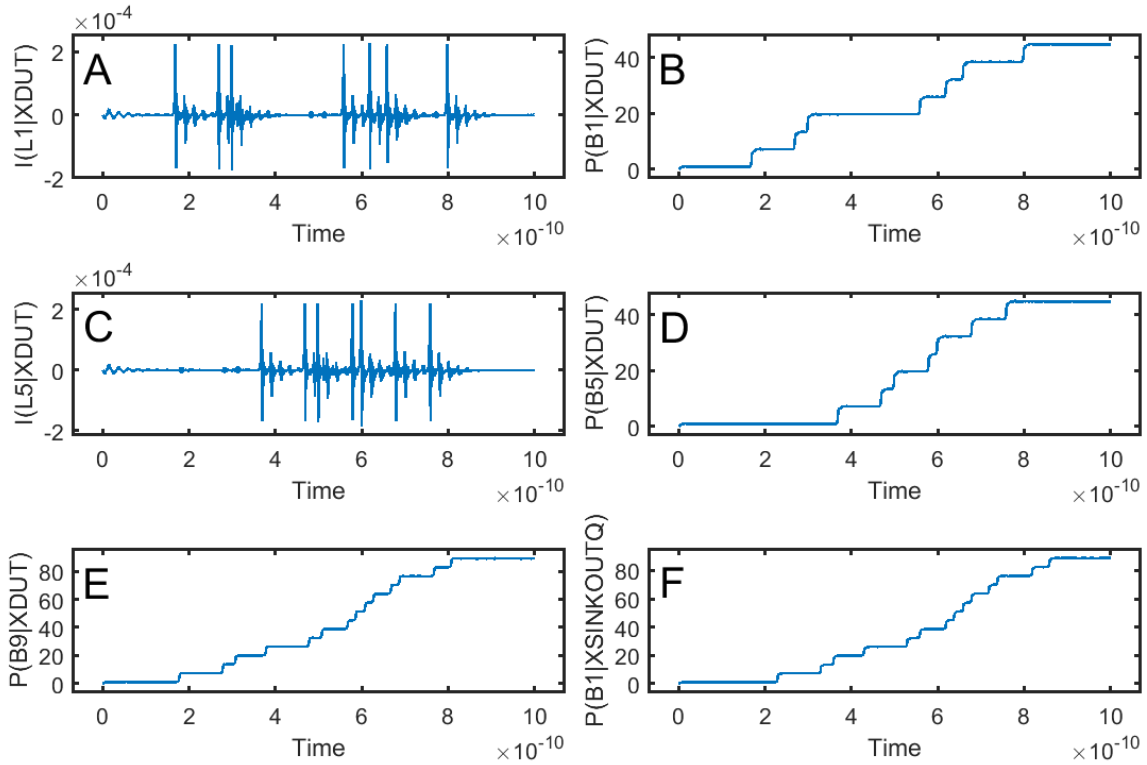


Figure 3.13: RSFQ MERGET analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 11 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_MERGET_v3p0_extracted (a, b, q);
9
10 input
11     a, b;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_a_q = 9.3,
21     delay_state0_b_q = 9.5,
22     ct_state0_a_a = 9.3,
23     ct_state0_a_b = 3.3,
24     ct_state0_b_a = 3.9,
25     ct_state0_b_b = 9.3;
26
27 reg
28     errorsignal_a,
29     errorsignal_b;
30
31 integer
32     outfile,
33     cell_state; // internal state of the cell
34
35 initial
36     begin
37         errorsignal_a = 0;
38         errorsignal_b = 0;
39         cell_state = 0; // Startup state
40         q = 0; // All outputs start at 0
41     end
42
43 always @(posedge a or negedge a) // execute at positive and negative edges of input
44     begin
45         if ($time>4) // arbitrary steady-state time)
46             begin
47                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
48                     begin
49                         outfile = $fopen("errors.txt", "a");
50                         $fdisplay(outfile, "Violation of critical timing in module %m; %0d ps.\n
51                             ↳ ", $stime);
52                         $fclose(outfile);
53                         q <= 1'bX; // Set all outputs to unknown
54                     end
55                 if (errorsignal_a == 0)
56                     begin
57                         case (cell_state)
58                             0: begin
59                                 q <= #(delay_state0_a_q) !q;
60                                 errorsignal_a = 1; // Critical timing on this input; assign
61                                     ↳ immediately
62                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
63                                     ↳ after critical timing expires
64                                 errorsignal_b = 1; // Critical timing on this input; assign
65                                     ↳ immediately
66                                 errorsignal_b <= #(ct_state0_a_b) 0; // Clear error signal
67                                     ↳ after critical timing expires

```

```

63         end
64     endcase
65 end
66 end
67 end
68
69 always @(posedge b or negedge b) // execute at positive and negative edges of input
70 begin
71     if ($time>4) // arbitrary steady-state time)
72     begin
73         if (errorsignal_b == 1'b1) // A critical timing is active for this input
74         begin
75             outfile = $fopen("errors.txt", "a");
76             $fdisplay(outfile, "Violation of critical timing in module %m; %0d ps.\n
77                 ↪ ", $stime);
78             $fclose(outfile);
79             q <= 1'bX; // Set all outputs to unknown
80         end
81         if (errorsignal_b == 0)
82         begin
83             case (cell_state)
84             0: begin
85                 q <= #(delay_state0_b_q) !q;
86                 errorsignal_a = 1; // Critical timing on this input; assign
87                     ↪ immediately
88                 errorsignal_a <= #(ct_state0_b_a) 0; // Clear error signal
89                     ↪ after critical timing expires
90                 errorsignal_b = 1; // Critical timing on this input; assign
91                     ↪ immediately
92                 errorsignal_b <= #(ct_state0_b_b) 0; // Clear error signal
93                     ↪ after critical timing expires
94             end
95             endcase
96         end
97     end
98 end
99 endmodule

```

Listing 3.6: RSFQ MERGET verilog model.

The digital simulation results for the RSFQ MERGET is shown in Fig. 3.14 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 3.15.

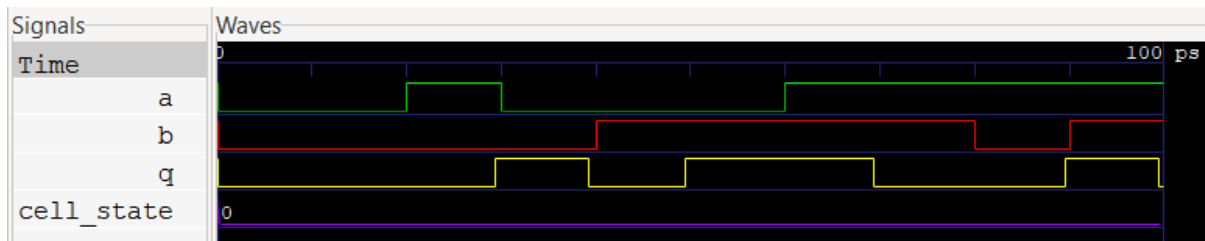


Figure 3.14: RSFQ MERGET digital simulation results.

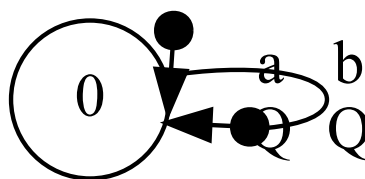


Figure 3.15: RSFQ MERGET Mealy finite state diagram.

Power consumption

Table 3.6: RSFQ MERGET power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	2370	3.36
2	2370	6.72
5	2370	16.8
10	2370	33.6
20	2370	67.2
50	2370	168

3.1.4 Always0T Asynchronous

The Always0T Asynchronous cell provides an output which is always zero. Two versions of the cell exist: One with an **a** input port, as seen in Fig. 3.16, and one without an **a** input port, as seen in Fig. 3.17. The cell is designed to connect directly to a PTL.

Schematic

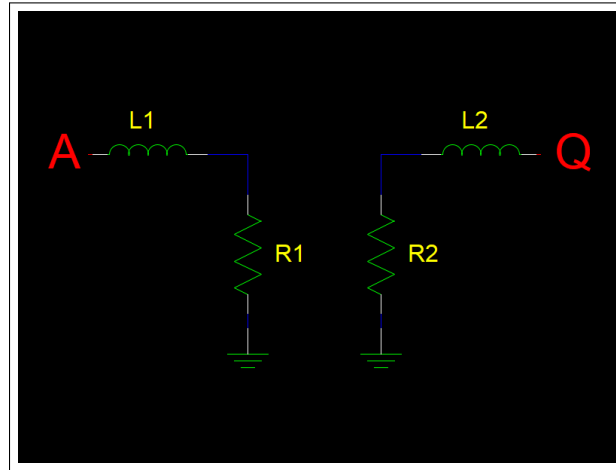


Figure 3.16: Schematic of RSFQ Always0T Asynchronous.

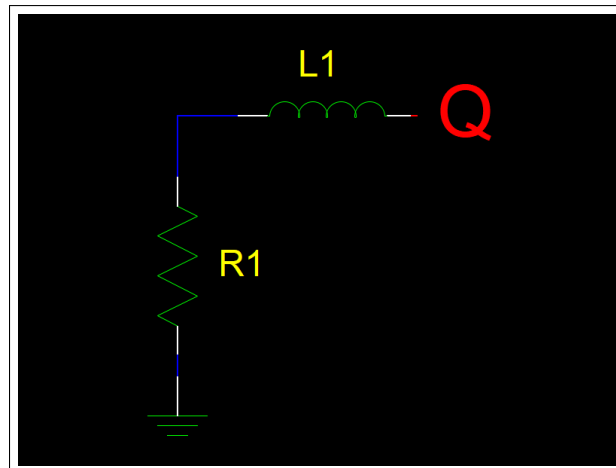


Figure 3.17: Schematic of RSFQ Always0T Asynchronous without an **a** input port.

Layout

The physical layouts of the two RSFQ Always0T Asynchronous cell versions are shown in Fig. 3.18a and 3.18b respectively. The height of the layout for both versions of the cell is $70\ \mu\text{m}$ and the width is $10\ \mu\text{m}$. The PTL connection extends to connect to a PTL on either M1 or M3.

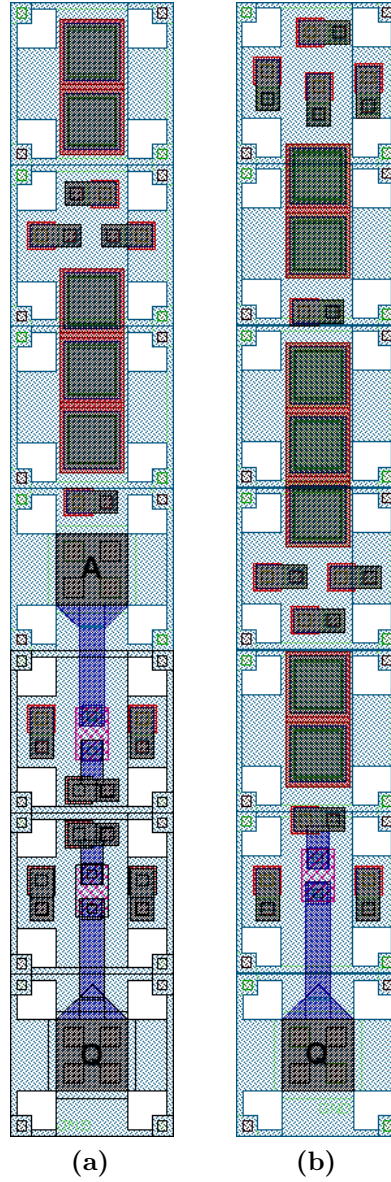


Figure 3.18: The physical layouts for (a) the RSFQ Always0T Asynchronous and (b) the RSFQ Always0T Asynchronous without an **a** input port.

Analog model

```

1  * Back-annotated simulation file written by InductEx v.6.1.52 on 2022/08/26.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 26 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports          a          q
8  .subckt THmit11_ALWAYS0T_ASYNC a q
9  .param Lpt1=2p
10
11 .param L1=Lpt1
12 .param L2=Lpt1
13
14 .param R1=2
15 .param R2=2
16
17 L1 a 1 1.386E-012
18 L2 2 q 1.441E-012
19
20 R1 1 0 R1
21 R2 2 0 R2
22
23 .ends

```

Listing 3.7: RSFQ Always0T Asynchronous JoSIM netlist.

Table 3.7: RSFQ Always0T Asynchronous pin list.

Pin	Description
a	Data input
q	Data output

```

1  * Back-annotated simulation file written by InductEx v.6.1.52 on 2022/08/26.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 26 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports          a          q
8  .subckt THmit11_ALWAYS0T_ASYNC_NOA a q
9
10 .param Lpt1=2p
11
12 .param L1=Lpt1
13
14 .param RQ=2
15
16 L1 1 q 1.67E-012
17
18 R1 1 0 R1
19
20 .ends

```

Listing 3.8: RSFQ Always0T Asynchronous, without an **a** input port, JoSIM netlist.

Table 3.8: RSFQ Always0T Asynchronous, without an **a** input port, pin list.

Pin	Description
q	Data output

Digital model

```

1 // -----
2 // Author: L. Schindler
3 // Version: 3.0
4 // Last modification date: 26 August 2022
5 // Last modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_ALWAYS0T_ASYNC (a, q);
9
10 input
11     a;
12
13 output
14     q;
15
16 reg
17     q;
18
19 initial
20     begin
21         q = 0; // Output always 0
22     end
23
24 always
25     begin
26         #10 q = 0; // Output always 0
27     end
28 endmodule

```

Listing 3.9: RSFQ Always0T Asynchronous verilog model.

```

1 // -----
2 // Author: L. Schindler
3 // Version: 3.0
4 // Last modification date: 26 August 2022
5 // Last modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_ALWAYS0T_ASYNC_NOA (q);
9
10
11 output
12     q;
13
14 reg
15     q;
16
17 initial
18     begin
19         q = 0; // Output always 0
20     end
21
22 always
23     begin
24         #10 q = 0; // Output always 0
25     end
26 endmodule

```

Listing 3.10: RSFQ Always0T Asynchronous, without an **a** input port, verilog model.

3.1.5 Always0T Synchronous

The Always0T Synchronous cell provides an output which is always zero synchronous to a clock signal. Two versions of the cell exist: One with an **a** input port, as seen in Fig. 3.19, and one without an **a** input port, as seen in Fig. 3.20. The cell is designed to connect directly to a PTL.

Schematic

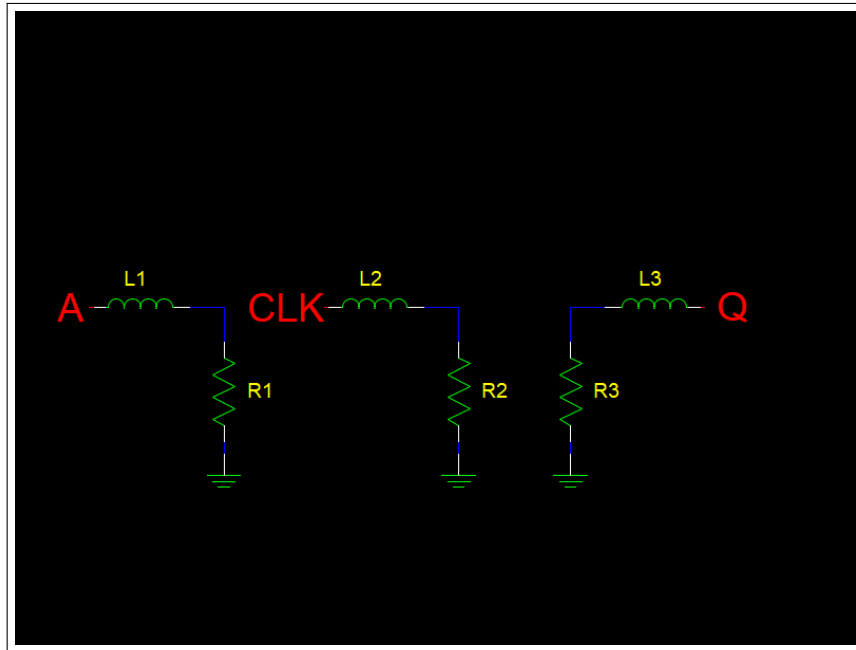


Figure 3.19: Schematic of RSFQ Always0T Synchronous.

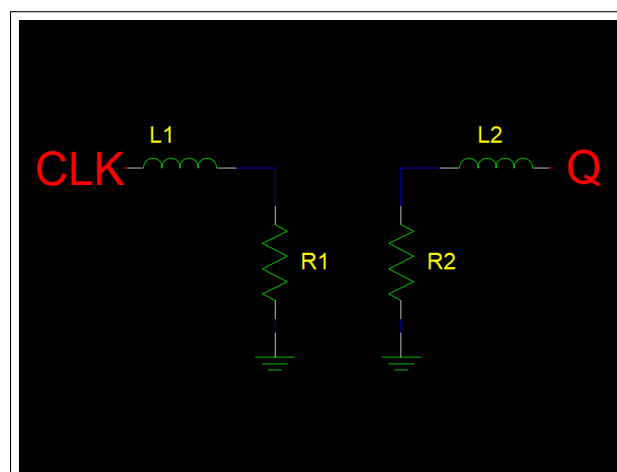


Figure 3.20: Schematic of RSFQ Always0T Synchronous without an **a** input port.

Layout

The physical layouts of the two RSFQ Always0T Synchronous cell versions are shown in Fig. 3.21a and 3.21b respectively. The height for both versions of the Always0T Synchronous layout is $70\ \mu m$ and the width is $10\ \mu m$. The PTL connection extends to connect to a PTL on either M1 or M3.

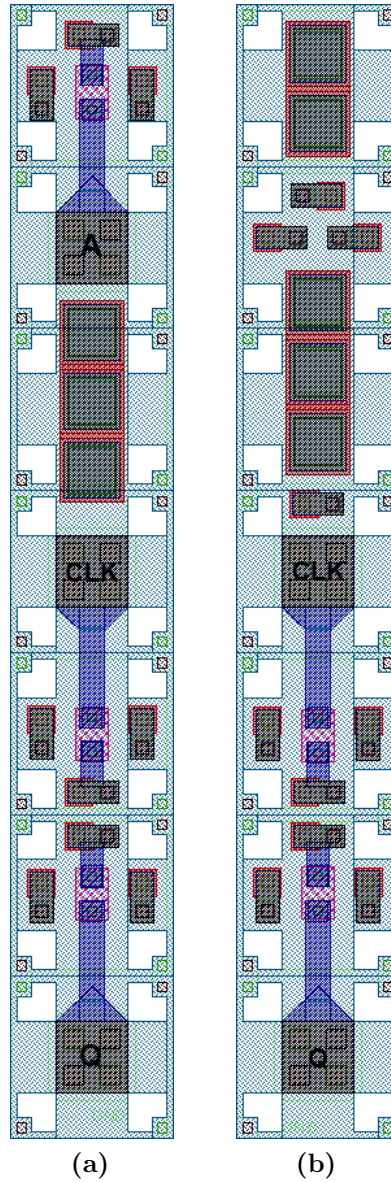


Figure 3.21: The physical layouts for (a) the RSFQ Always0T Synchronous and (b) the RSFQ Always0T Synchronous without an **a** input port.

Analog model

```

1 | * Back-annotated simulation file written by InductEx v.6.1.52 on 2022/08/26.
2 | * Author: L. Schindler
3 | * Version: 3.0
4 | * Last modification date: 26 August 2022
5 | * Last modification by: T. Hall
6 |
7 | *$Ports          a          q
8 | .subckt THmit11_ALWAYS0T_SYNC a q
9 | .param Lpt1=2p
10 | .param L1=Lpt1
11 | .param L2=Lpt1
12 | .param L3=Lpt1
13 | .param R1=2
14 | .param R2=2
15 | .param R3=2
16 | L1 a 1 1.282E-012
17 | L2 clk 2 1.401E-012
18 | L3 3 q 1.428E-012
19 | R1 1 0 R1
20 | R2 2 0 R2
21 | R3 3 0 R3
22 | .ends

```

Listing 3.11: RSFQ Always0T Synchronous JoSIM netlist.

Table 3.9: RSFQ Always0T Synchronous pin list.

Pin	Description
a	Data input
clk	Clock input
q	Data output

```

1 | * Back-annotated simulation file written by InductEx v.6.1.52 on 2022/08/26.
2 | * Author: L. Schindler
3 | * Version: 3.0
4 | * Last modification date: 26 August 2022
5 | * Last modification by: T. Hall
6 |
7 | *$Ports          a          q
8 | .subckt THmit11_ALWAYS0T_SYNC_NOA a q
9 | .param Lpt1=2p
10 | .param L1=Lpt1
11 | .param L2=Lpt1
12 | .param R1=2
13 | .param R2=2
14 | L1 clk 1 1.386E-012
15 | L2 2 q 1.441E-012
16 | R1 1 0 R1
17 | R2 2 0 R2
18 | .ends

```

Listing 3.12: RSFQ Always0T Synchronous, without an **a** input port, JoSIM netlist.

Table 3.10: RSFQ Always0T Synchronous, without an **a** input port, pin list.

Pin	Description
clk	Clock input
q	Data output

Digital model

```

1 // -----
2 // Author: L. Schindler
3 // Version: 3.0
4 // Last modification date: 26 August 2022
5 // Last modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_ALWAYS0T_SYNC (a, clk, q);
9
10 input
11     a, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 initial
20     begin
21         q = 0; // Output always 0
22     end
23
24 always @(posedge clk or negedge clk)
25     begin
26         #10 q = 0; // Output always 0
27     end
28 endmodule

```

Listing 3.13: RSFQ Always0T Synchronous verilog model.

```

1 // -----
2 // Author: L. Schindler
3 // Version: 3.0
4 // Last modification date: 26 August 2022
5 // Last modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_ALWAYS0T_SYNC_NOA ( clk, q);
9
10 input
11     clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 initial
20     begin
21         q = 0; // Output always 0
22     end
23
24 always @(posedge clk or negedge clk)
25     begin
26         #10 q = 0; // Output always 0
27     end
28 endmodule

```

Listing 3.14: RSFQ Always0T Synchronous, without an **a** input port, verilog model.

3.2 Logic Cells

3.2.1 AND2T

The RSFQ AND2T cell generates an output pulse if pulses from both input signal lines were received before the clock signal. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

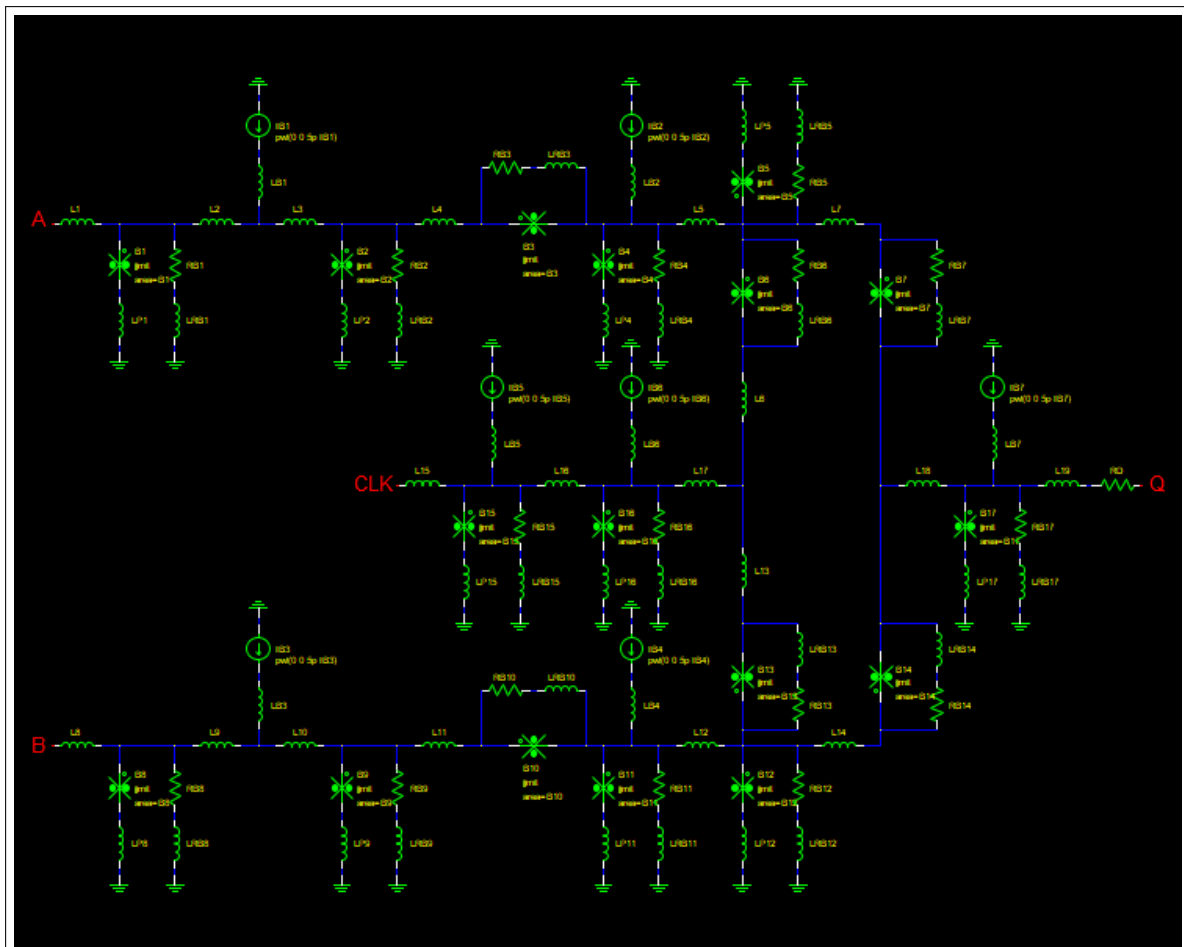


Figure 3.22: Schematic of RSFQ AND2T.

Layout

The physical layout of the RSFQ AND2T is shown in Fig. 3.23. The layout height is $70\ \mu\text{m}$ and the width is $50\ \mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

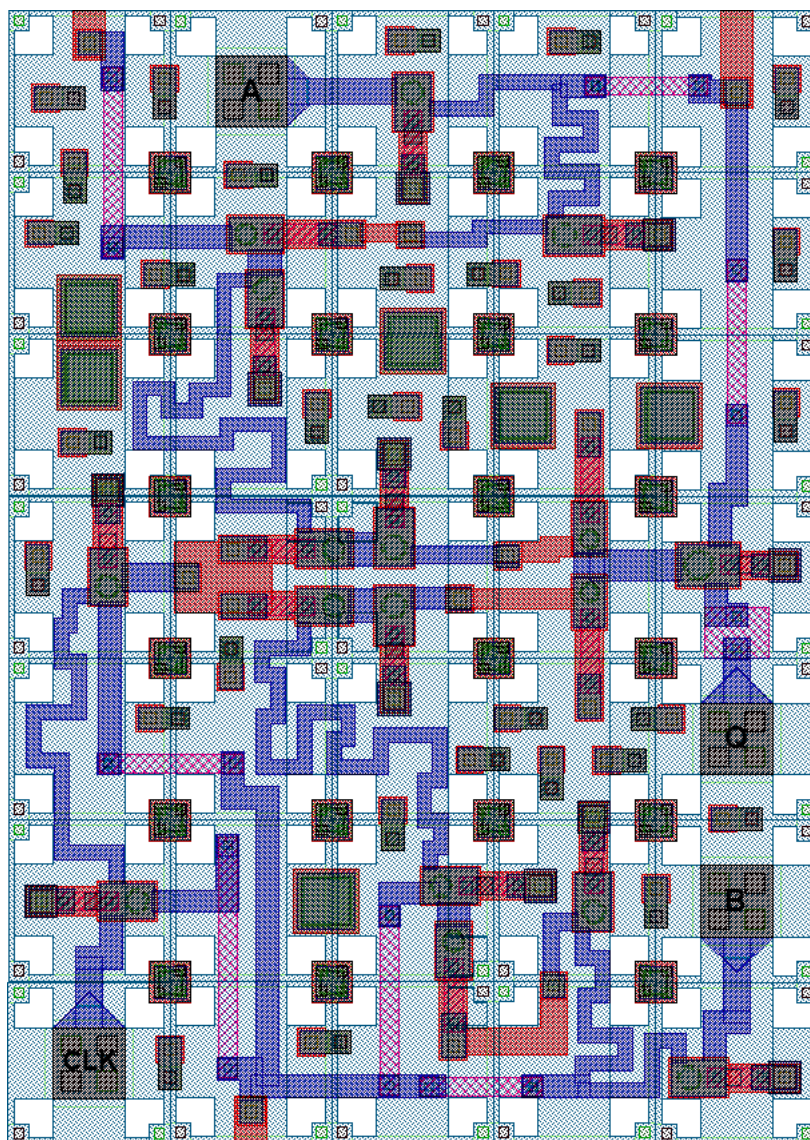


Figure 3.23: RSFQ AND2T Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/23.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 23 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      b      clk      q
8  .subckt THmit11_AND2T a b clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=1.6
27 .param B2=1.66
28 .param B3=1.26
29 .param B4=1.44
30 .param B5=1.79
31 .param B6=1.41
32 .param B7=1.03
33 .param B8=1.6
34 .param B9=1.66
35 .param B10=1.26
36 .param B11=1.44
37 .param B12=1.79
38 .param B13=1.41
39 .param B14=1.03
40 .param B15=1.6
41 .param B16=1.56
42 .param B17=2.5
43
44 .param IB1=249u
45 .param IB2=148u
46 .param IB3=249u
47 .param IB4=148u
48 .param IB5=112u
49 .param IB6=203u
50 .param IB7=175u
51
52 .param LB1=LB
53 .param LB2=LB
54 .param LB3=LB
55 .param LB4=LB
56 .param LB5=LB
57 .param LB6=LB
58 .param LB7=LB
59
60 .param L1=Lptl
61 .param L2=2.7125p
62 .param L3=3.6844p
63 .param L4=4.4946p
64 .param L5=10.8125p
65 .param L6=0.8299p
66 .param L7=2.6546p
67 .param L8=Lptl
68 .param L9=2.7125p
69 .param L10=3.6844p
70 .param L11=4.4946p
71 .param L12=10.8125p
72 .param L13=0.8299p
73 .param L14=2.6546p
74 .param L15=Lptl
75 .param L16=7.3588p
76 .param L17=1.0183p
77 .param L18=0.9951p
78 .param L19=Lptl
79
80 .param LP1=LP
81 .param LP2=LP
82 .param LP4=LP
83 .param LP5=LP
84 .param LP8=LP
85 .param LP9=LP
86 .param LP11=LP
87 .param LP12=LP
88 .param LP15=LP
89 .param LP16=LP
90 .param LP17=LP
91
92 .param RB1=B0Rs/B1
93 .param RB2=B0Rs/B2
94 .param RB3=B0Rs/B3
95 .param RB4=B0Rs/B4
96 .param RB5=B0Rs/B5
97 .param RB6=B0Rs/B6
98 .param RB7=B0Rs/B7
99 .param RB8=B0Rs/B8
100 .param RB9=B0Rs/B9
101 .param RB10=B0Rs/B10
102 .param RB11=B0Rs/B11
103 .param RB12=B0Rs/B12
104 .param RB13=B0Rs/B13
105 .param RB14=B0Rs/B14
106 .param RB15=B0Rs/B15
107 .param RB16=B0Rs/B16
108 .param RB17=B0Rs/B17
109
110 .param LRB1=(RB1/Rsheet)*Lsheet+LP
111 .param LRB2=(RB2/Rsheet)*Lsheet+LP
112 .param LRB3=(RB3/Rsheet)*Lsheet
113 .param LRB4=(RB4/Rsheet)*Lsheet+LP
114 .param LRB5=(RB5/Rsheet)*Lsheet+LP
115 .param LRB6=(RB6/Rsheet)*Lsheet
116 .param LRB7=(RB7/Rsheet)*Lsheet
117 .param LRB8=(RB8/Rsheet)*Lsheet+LP
118 .param LRB9=(RB9/Rsheet)*Lsheet+LP
119 .param LRB10=(RB10/Rsheet)*Lsheet
120 .param LRB11=(RB11/Rsheet)*Lsheet+LP
121 .param LRB12=(RB12/Rsheet)*Lsheet+LP
122 .param LRB13=(RB13/Rsheet)*Lsheet
123 .param LRB14=(RB14/Rsheet)*Lsheet
124 .param LRB15=(RB15/Rsheet)*Lsheet+LP
125 .param LRB16=(RB16/Rsheet)*Lsheet+LP
126 .param LRB17=(RB17/Rsheet)*Lsheet+LP
127
128 B1 1 2 jjmit area=B1
129 B2 5 6 jjmit area=B2

```

```

130 B3 7 8 jjmit area=B3
131 B4 8 9 jjmit area=B4
132 B5 11 12 jjmit area=B5
133 B6 11 13 jjmit area=B6
134 B7 15 16 jjmit area=B7
135 B8 17 18 jjmit area=B8
136 B9 21 22 jjmit area=B9
137 B10 23 24 jjmit area=B10
138 B11 24 25 jjmit area=B11
139 B12 27 28 jjmit area=B12
140 B13 27 29 jjmit area=B13
141 B14 30 16 jjmit area=B14
142 B15 31 32 jjmit area=B15
143 B16 34 35 jjmit area=B16
144 B17 37 38 jjmit area=B17
145
146 IB1 0 4 pwl(0 0 5p IB1)
147 IB2 0 10 pwl(0 0 5p IB2)
148 IB3 0 20 pwl(0 0 5p IB3)
149 IB4 0 26 pwl(0 0 5p IB4)
150 IB5 0 33 pwl(0 0 5p IB5)
151 IB6 0 36 pwl(0 0 5p IB6)
152 IB7 0 39 pwl(0 0 5p IB7)
153
154 LB1 4 3 5.738E-013
155 LB2 10 8 1.925E-012
156 LB3 20 19 8.945E-013
157 LB4 26 24 8.334E-013
158 LB5 33 31 1.882E-012
159 LB6 36 34 2.156E-012
160 LB7 39 37 1.99E-012
161
162 L1 a 1 1.421E-012
163 L2 1 3 2.715E-012
164 L3 3 5 3.674E-012
165 L4 5 7 4.505E-012
166 L5 8 11 1.08E-011
167 L6 13 14 8.262E-013
168 L7 11 15 2.657E-012
169 L8 b 17 1.744E-012
170 L9 17 19 2.706E-012
171 L10 19 21 3.675E-012
172 L11 21 23 4.485E-012
173 L12 24 27 1.081E-011
174 L13 14 29 8.334E-013
175 L14 27 30 2.652E-012
176 L15 clk 31 1.606E-012
177 L16 31 34 7.368E-012
178 L17 34 14 1.014E-012
179 L18 16 37 9.896E-013
180 L19 37 40 7.389E-013
181
182 RD 40 q RD
183
184 LP1 2 0 3.818E-013
185 LP2 6 0 5.451E-013
186 LP4 9 0 7.058E-013
187 LP5 0 12 7.873E-013
188 LP8 18 0 4.185E-013
189 LP9 22 0 5.92E-013
190 LP11 25 0 4.503E-013
191 LP12 28 0 7.626E-013
192 LP15 32 0 3.764E-013
193 LP16 35 0 5.096E-013
194 LP17 38 0 3.396E-013
195
196 RB1 1 101 RB1
197 LRB1 101 0 LRB1
198 RB2 5 105 RB2
199 LRB2 105 0 LRB2
200 RB3 7 107 RB3
201 LRB3 107 8 LRB3
202 RB4 8 108 RB4
203 LRB4 108 0 LRB4
204 RB5 111 11 RB5
205 LRB5 0 111 LRB5
206 RB6 11 113 RB6
207 LRB6 113 13 LRB6
208 RB7 15 115 RB7
209 LRB7 115 16 LRB7
210 RB8 17 117 RB8
211 LRB8 117 0 LRB8
212 RB9 21 121 RB9
213 LRB9 121 0 LRB9
214 RB10 23 123 RB10
215 LRB10 123 24 LRB10
216 RB11 24 124 RB11
217 LRB11 124 0 LRB11
218 RB12 27 127 RB12
219 LRB12 127 0 LRB12
220 RB13 129 27 RB13
221 LRB13 29 129 LRB13
222 RB14 130 30 RB14
223 LRB14 16 130 LRB14
224 RB15 31 131 RB15
225 LRB15 131 0 LRB15
226 RB16 34 134 RB16
227 LRB16 134 0 LRB16
228 RB17 37 137 RB17
229 LRB17 137 0 LRB17
230
231 .ends

```

Listing 3.15: RSFQ AND2T JoSIM netlist.**Table 3.11:** RSFQ AND2T pin list.

Pin	Description
a	Data input
b	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ AND2T using JoSIM is shown in Fig. 3.24. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b**,
- (d) the phase over the input JJ of pin **b**,
- (e) the current through the input inductor connected to pin **clk**,
- (f) the phase over the input JJ of pin **clk**,
- (g) the phase over the output JJ of pin **q**, and
- (h) the phase over the input JJ of the load circuit connected via a PTL to pin **q**.

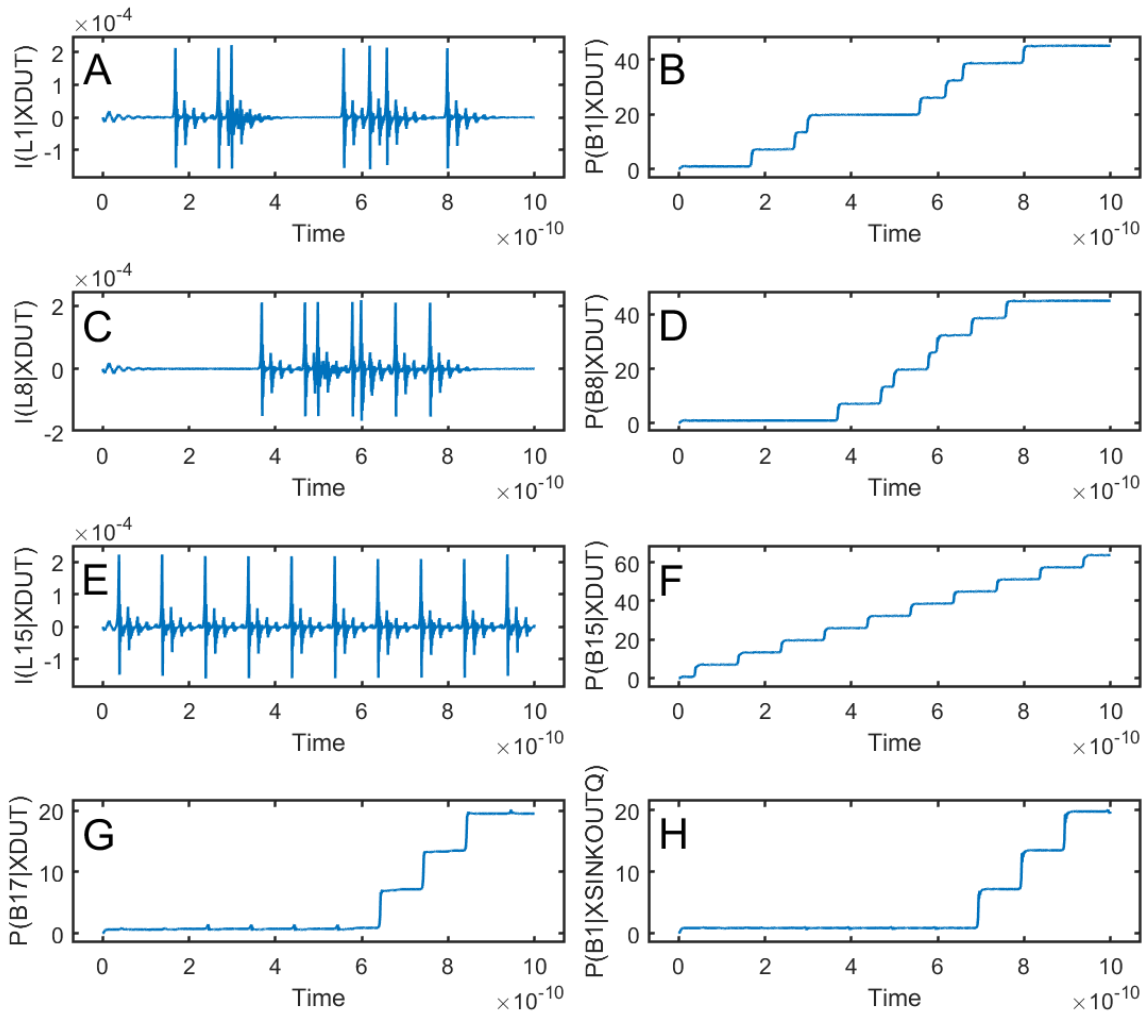


Figure 3.24: RSFQ AND2T analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 23 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_AND2T_v3p0_extracted (a, b, clk, q);
9
10 input
11     a, b, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state3_clk_q = 5.7,
21     ct_state0_clk_a = 2.7,
22     ct_state0_clk_b = 2.7,
23     ct_state1_clk_a = 0.3,
24     ct_state1_clk_b = 1.0,
25     ct_state2_clk_a = 1.0,
26     ct_state3_a_clk = 1.4,
27     ct_state3_b_clk = 1.5;
28
29 reg
30     errorsignal_a,
31     errorsignal_b,
32     errorsignal_clk;
33
34 integer
35     outfile,
36     cell_state; // internal state of the cell
37
38 initial
39     begin
40         errorsignal_a = 0;
41         errorsignal_b = 0;
42         errorsignal_clk = 0;
43         cell_state = 0; // Startup state
44         q = 0; // All outputs start at 0
45     end
46
47 always @(posedge a or negedge a) // execute at positive and negative edges of input
48     begin
49         if ($time>4) // arbitrary steady-state time)
50             begin
51                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
52                     begin
53                         outfile = $fopen("errors.txt", "a");
54                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0dps.\n
55                             ↪", $time);
56                         $fclose(outfile);
57                         q <= 1'bX; // Set all outputs to unknown
58                     end
59                 if (errorsignal_a == 0)
60                     begin
61                         case (cell_state)
62                             0: begin
63                                 cell_state = 1; // Blocking statement -- immediately
64                                 end
65                             1: begin
66                                 end
67                             2: begin

```

```

67         cell_state = 3; // Blocking statement -- immediately
68     end
69     3: begin
70         errorsignal_clk = 1; // Critical timing on this input; assign
71             ↪ immediately
72         errorsignal_clk <= #(ct_state3_a_clk) 0; // Clear error signal
73             ↪ after critical timing expires
74     end
75 endcase
76 end
77 end
78 always @(posedge b or negedge b) // execute at positive and negative edges of input
79 begin
80     if ($time>4) // arbitrary steady-state time)
81     begin
82         if (errorsignal_b == 1'b1) // A critical timing is active for this input
83         begin
84             outfile = $fopen("errors.txt", "a");
85             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
86             $fclose(outfile);
87             q <= 1'bX; // Set all outputs to unknown
88         end
89         if (errorsignal_b == 0)
90         begin
91             case (cell_state)
92             0: begin
93                 cell_state = 2; // Blocking statement -- immediately
94             end
95             1: begin
96                 cell_state = 3; // Blocking statement -- immediately
97             end
98             2: begin
99             end
100            3: begin
101                errorsignal_clk = 1; // Critical timing on this input; assign
102                    ↪ immediately
103                errorsignal_clk <= #(ct_state3_b_clk) 0; // Clear error signal
104                    ↪ after critical timing expires
105            end
106        endcase
107    end
108 end
109 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
110 begin
111     if ($time>4) // arbitrary steady-state time)
112     begin
113         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
114         begin
115             outfile = $fopen("errors.txt", "a");
116             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
117             $fclose(outfile);
118             q <= 1'bX; // Set all outputs to unknown
119         end
120         if (errorsignal_clk == 0)
121         begin
122             case (cell_state)
123             0: begin
124                 errorsignal_a = 1; // Critical timing on this input; assign
125                     ↪ immediately
126                 errorsignal_a <= #(ct_state0_clk_a) 0; // Clear error signal
127                     ↪ after critical timing expires
128                 errorsignal_b = 1; // Critical timing on this input; assign
129                     ↪ immediately
130                 errorsignal_b <= #(ct_state0_clk_b) 0; // Clear error signal

```

```

128                                     ↪ after critical timing expires
129     end
130 1: begin
131     cell_state = 0; // Blocking statement -- immediately
132     errorsignal_a = 1; // Critical timing on this input; assign
133         ↪ immediately
134     errorsignal_a <= #(ct_state1_clk_a) 0; // Clear error signal
135         ↪ after critical timing expires
136     errorsignal_b = 1; // Critical timing on this input; assign
137         ↪ immediately
138     errorsignal_b <= #(ct_state1_clk_b) 0; // Clear error signal
139         ↪ after critical timing expires
140     end
141 2: begin
142     cell_state = 0; // Blocking statement -- immediately
143     errorsignal_a = 1; // Critical timing on this input; assign
144         ↪ immediately
145     errorsignal_a <= #(ct_state2_clk_a) 0; // Clear error signal
146         ↪ after critical timing expires
147     end
148 3: begin
149     q <= #(delay_state3_clk_q) !q;
150     cell_state = 0; // Blocking statement -- immediately
151     end
152 endcase
153 end
154 end
155 endmodule

```

Listing 3.16: RSFQ AND2T verilog model.

The digital simulation results for the RSFQ AND2T is shown in Fig. 3.25 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 3.26.

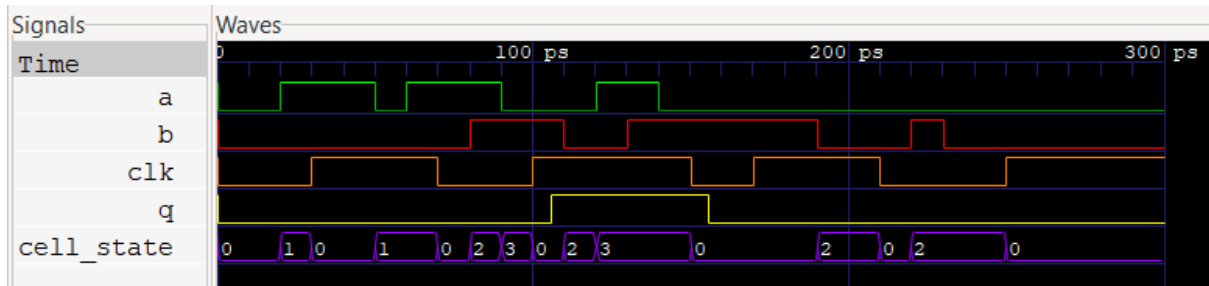


Figure 3.25: RSFQ AND2T digital simulation results.

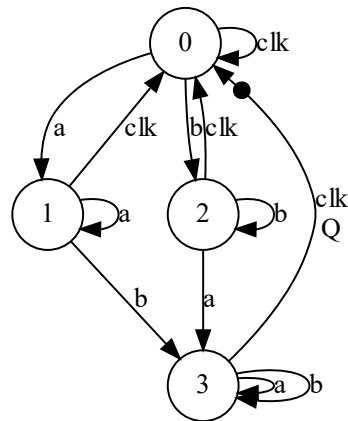


Figure 3.26: RSFQ AND2T Mealy finite state machine diagram.

Power Consumption

Table 3.12: RSFQ AND2T power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	3340	5.38
2	3340	10.8
5	3340	26.9
10	3340	53.8
20	3340	108
50	3340	269

3.2.2 OR2T

The RSFQ OR2T cell generates an output pulse if an input pulse from either input lines was received before the clock signal. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

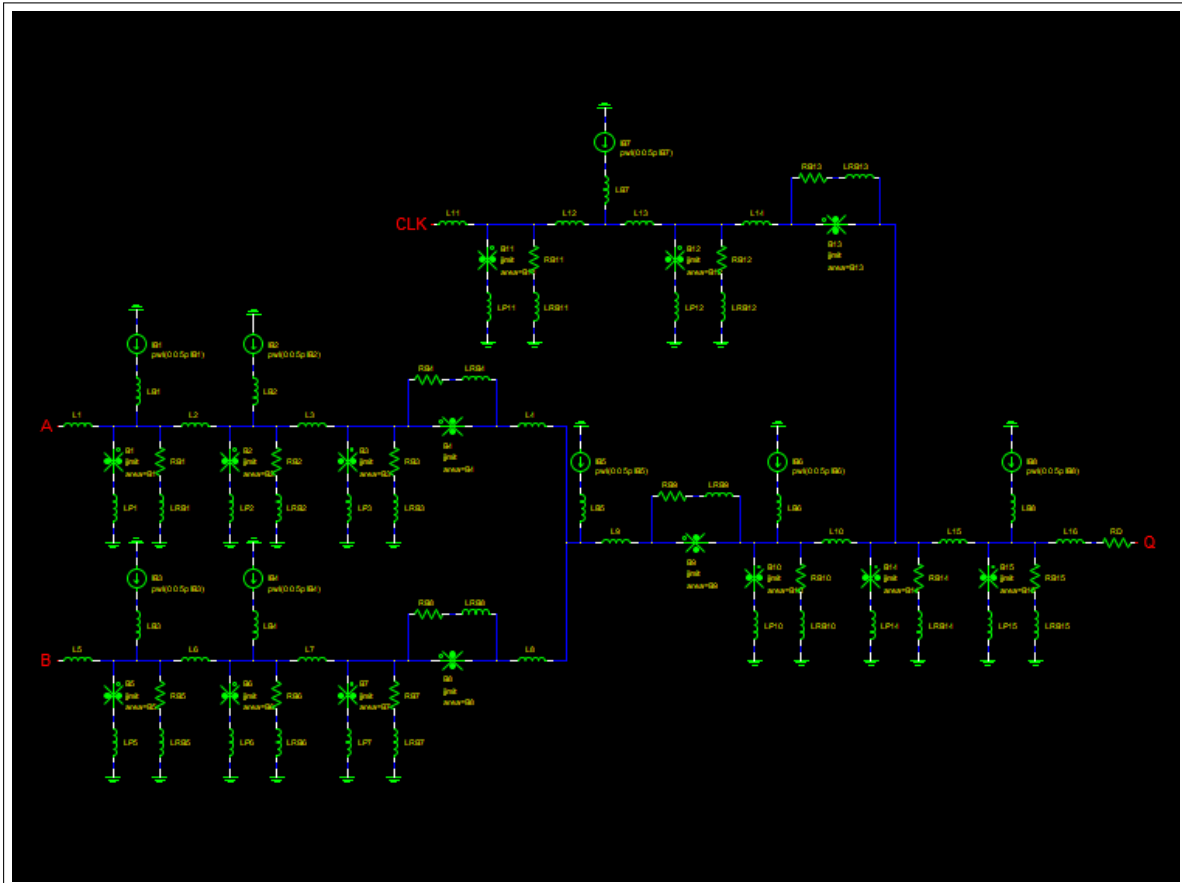


Figure 3.27: Schematic of RSFQ OR2T.

Layout

The physical layout for the RSFQ OR2T is shown in Fig. 3.28. The layout height is $70\ \mu\text{m}$ and the width is $40\ \mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

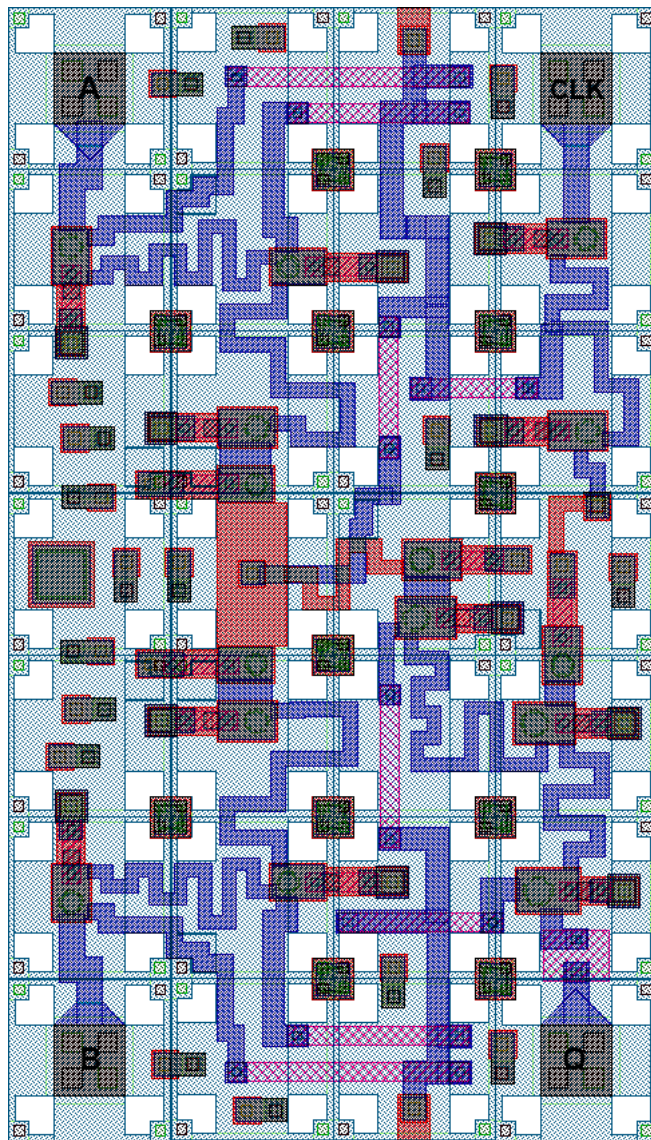


Figure 3.28: RSFQ OR2T Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/18.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 16 August 2022
5  * Last modification by: T. Hall
6
7  *$ports      a      b      clk      q
8  .subckt THmit11_OR2T a b clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=1.6
27 .param B2=1.28
28 .param B3=1.65
29 .param B4=1.25
30 .param B5=1.6
31 .param B6=1.28
32 .param B7=1.65
33 .param B8=1.25
34 .param B9=1.67
35 .param B10=1.95
36 .param B11=1.6
37 .param B12=1.59
38 .param B13=1.44
39 .param B14=2.09
40 .param B15=2.5
41
42 .param IB1=112u
43 .param IB2=156u
44 .param IB3=112u
45 .param IB4=156u
46 .param IB5=210u
47 .param IB6=176u
48 .param IB7=242u
49 .param IB8=175u
50
51 .param LB1=LB
52 .param LB2=LB
53 .param LB3=LB
54 .param LB4=LB
55 .param LB5=LB
56 .param LB6=LB
57 .param LB7=LB
58 .param LB8=LB
59
60 .param L1=Lptl
61 .param L2=6.7019p
62 .param L3=6.2519p
63 .param L4=0.7723p
64 .param L5=Lptl
65 .param L6=6.7019p
66 .param L7=6.2519p
67 .param L8=0.7723p
68 .param L9=4.6089p
69 .param L10=7.7141p
70 .param L11=Lptl
71 .param L12=2.7407p
72 .param L13=3.4070p
73 .param L14=4.5212p
74 .param L15=4.5560p
75 .param L16=Lptl
76
77 .param LP1=LP
78 .param LP2=LP
79 .param LP3=LP
80 .param LP5=LP
81 .param LP6=LP
82 .param LP7=LP
83 .param LP10=LP
84 .param LP11=LP
85 .param LP12=LP
86 .param LP14=LP
87 .param LP15=LP
88
89 .param RB1=B0Rs/B1
90 .param RB2=B0Rs/B2
91 .param RB3=B0Rs/B3
92 .param RB4=B0Rs/B4
93 .param RB5=B0Rs/B5
94 .param RB6=B0Rs/B6
95 .param RB7=B0Rs/B7
96 .param RB8=B0Rs/B8
97 .param RB9=B0Rs/B9
98 .param RB10=B0Rs/B10
99 .param RB11=B0Rs/B11
100 .param RB12=B0Rs/B12
101 .param RB13=B0Rs/B13
102 .param RB14=B0Rs/B14
103 .param RB15=B0Rs/B15
104
105 .param LRB1=(RB1/Rsheet)*Lsheet+LP
106 .param LRB2=(RB2/Rsheet)*Lsheet+LP
107 .param LRB3=(RB3/Rsheet)*Lsheet+LP
108 .param LRB4=(RB4/Rsheet)*Lsheet
109 .param LRB5=(RB5/Rsheet)*Lsheet+LP
110 .param LRB6=(RB6/Rsheet)*Lsheet+LP
111 .param LRB7=(RB7/Rsheet)*Lsheet+LP
112 .param LRB8=(RB8/Rsheet)*Lsheet
113 .param LRB9=(RB9/Rsheet)*Lsheet
114 .param LRB10=(RB10/Rsheet)*Lsheet+LP
115 .param LRB11=(RB11/Rsheet)*Lsheet+LP
116 .param LRB12=(RB12/Rsheet)*Lsheet+LP
117 .param LRB13=(RB13/Rsheet)*Lsheet
118 .param LRB14=(RB14/Rsheet)*Lsheet+LP
119 .param LRB15=(RB15/Rsheet)*Lsheet+LP
120
121 B1 1 2 jjmit area=B1
122 B2 4 5 jjmit area=B2
123 B3 7 8 jjmit area=B3
124 B4 7 9 jjmit area=B4
125 B5 11 12 jjmit area=B5
126 B6 14 15 jjmit area=B6
127 B7 17 18 jjmit area=B7
128 B8 17 19 jjmit area=B8
129 B9 21 22 jjmit area=B9

```

```

130 B10 22 23 jjmit area=B10
131 B11 26 27 jjmit area=B11
132 B12 30 31 jjmit area=B12
133 B13 32 25 jjmit area=B13
134 B14 25 33 jjmit area=B14
135 B15 34 35 jjmit area=B15
136
137 IB1 0 3 pwl(0 0 5p IB1)
138 IB2 0 6 pwl(0 0 5p IB2)
139 IB3 0 13 pwl(0 0 5p IB3)
140 IB4 0 16 pwl(0 0 5p IB4)
141 IB5 0 20 pwl(0 0 5p IB5)
142 IB6 0 24 pwl(0 0 5p IB6)
143 IB7 0 29 pwl(0 0 5p IB7)
144 IB8 0 36 pwl(0 0 5p IB8)
145
146 LB1 3 1 4.448E-012
147 LB2 6 4 2.203E-012
148 LB3 13 11 4.442E-012
149 LB4 16 14 2.187E-012
150 LB5 20 10 3.603E-012
151 LB6 24 22 1.443E-012
152 LB7 29 28 1.069E-012
153 LB8 36 34 8.669E-013
154
155 L1 a 1 1.444E-012
156 L2 1 4 6.665E-012
157 L3 4 7 6.254E-012
158 L4 9 10 7.702E-013
159 L5 b 11 1.443E-012
160 L6 11 14 6.656E-012
161 L7 14 17 6.244E-012
162 L8 19 10 7.773E-013
163 L9 10 21 4.573E-012
164 L10 22 25 7.702E-012
165 L11 clk 26 1.282E-012
166 L12 26 28 2.735E-012
167 L13 28 30 3.418E-012
168 L14 30 32 4.48E-012
169 L15 25 34 4.553E-012
170 L16 34 37 8.104E-013
171
172 RD 37 q RD
173
174 LP1 2 0 4.195E-013
175 LP2 5 0 5.813E-013
176 LP3 8 0 4.455E-013
177 LP5 12 0 4.193E-013
178 LP6 15 0 5.831E-013
179 LP7 18 0 4.774E-013
180 LP10 23 0 4.819E-013
181 LP11 27 0 3.541E-013
182 LP12 31 0 4.657E-013
183 LP14 33 0 4.19E-013
184 LP15 35 0 3.333E-013
185
186 RB1 1 101 RB1
187 LRB1 101 0 LRB1
188 RB2 4 104 RB2
189 LRB2 104 0 LRB2
190 RB3 7 107 RB3
191 LRB3 107 0 LRB3
192 RB4 7 109 RB4
193 LRB4 109 9 LRB4
194 RB5 11 111 RB5
195 LRB5 111 0 LRB5
196 RB6 14 114 RB6
197 LRB6 114 0 LRB6
198 RB7 17 117 RB7
199 LRB7 117 0 LRB7
200 RB8 17 119 RB8
201 LRB8 119 19 LRB8
202 RB9 21 121 RB9
203 LRB9 121 22 LRB9
204 RB10 22 122 RB10
205 LRB10 122 0 LRB10
206 RB11 26 126 RB11
207 LRB11 126 0 LRB11
208 RB12 30 130 RB12
209 LRB12 130 0 LRB12
210 RB13 32 132 RB13
211 LRB13 132 25 LRB13
212 RB14 25 125 RB14
213 LRB14 125 0 LRB14
214 RB15 34 134 RB15
215 LRB15 134 0 LRB15
216
217 .ends

```

Listing 3.17: RSFQ OR2T JoSIM netlist.**Table 3.13:** RSFQ OR2T pin list.

Pin	Description
a	Data input
b	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ OR2T using JoSIM is shown in Fig. 3.29. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b**,
- (d) the phase over the input JJ of pin **b**,
- (e) the current through the input inductor connected to pin **clk**,
- (f) the phase over the input JJ of pin **clk**,
- (g) the phase over the output JJ of pin **q**, and
- (h) the phase over the input JJ of the load circuit connected via a PTL to pin **q**.

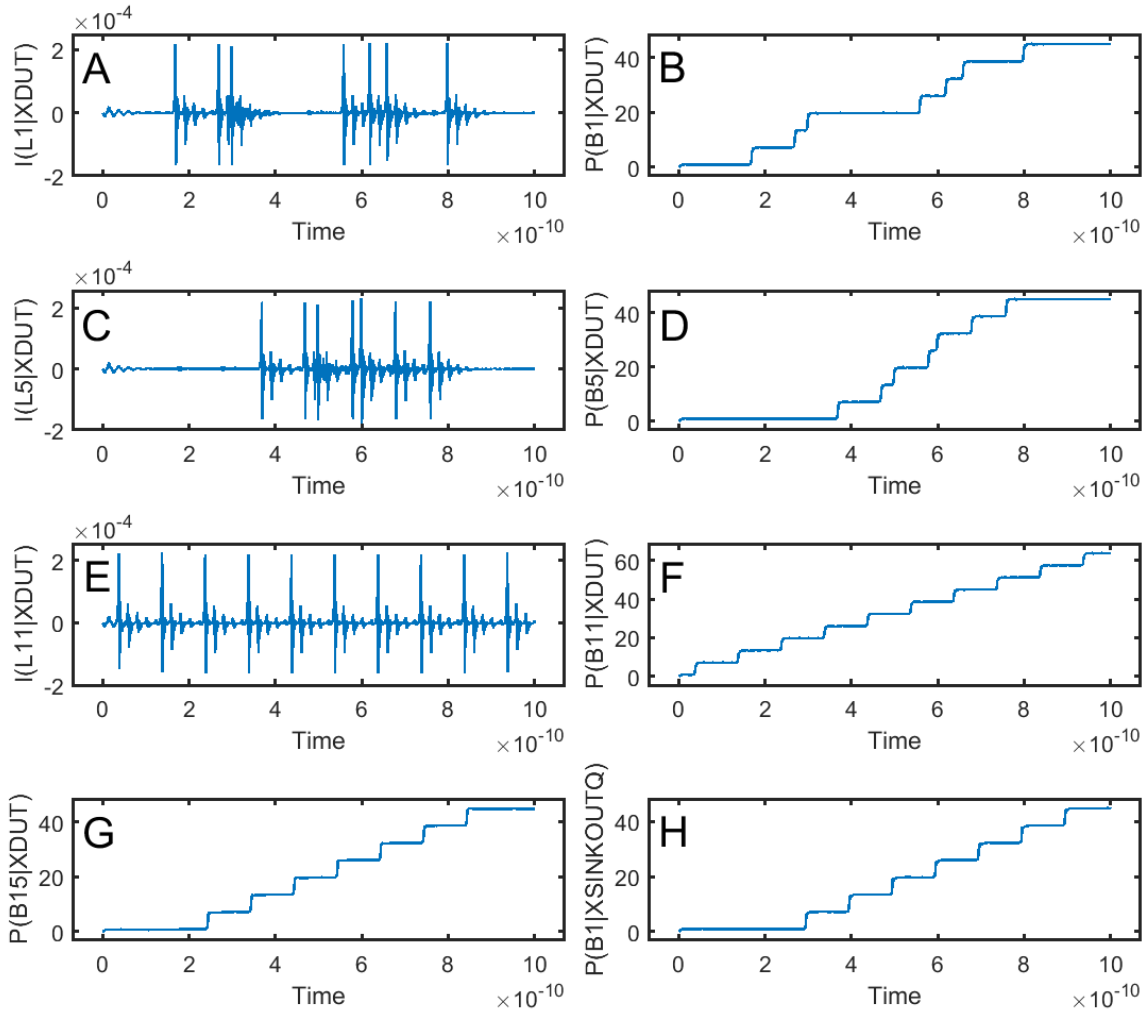


Figure 3.29: RSFQ OR2T analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 18 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit1l_OR2T_v3p0_extracted (a, b, clk, q);
9
10 input
11     a, b, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state1_clk_q = 6.5,
21     ct_state0_a_clk = 1.6,
22     ct_state0_b_clk = 1.6,
23     ct_state1_a_clk = 3.6,
24     ct_state1_b_clk = 3.7;
25
26 reg
27     errorsignal_a,
28     errorsignal_b,
29     errorsignal_clk;
30
31 integer
32     outfile,
33     cell_state; // internal state of the cell
34
35 initial
36     begin
37         errorsignal_a = 0;
38         errorsignal_b = 0;
39         errorsignal_clk = 0;
40         cell_state = 0; // Startup state
41         q = 0; // All outputs start at 0
42     end
43
44 always @(posedge a or negedge a) // execute at positive and negative edges of input
45     begin
46         if ($time>4) // arbitrary steady-state time)
47             begin
48                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
49                     begin
50                         outfile = $fopen("errors.txt", "a");
51                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n",
52                             ↪ ", $time);
53                         $fclose(outfile);
54                         q <= 1'bX; // Set all outputs to unknown
55                     end
56                 if (errorsignal_a == 0)
57                     begin
58                         case (cell_state)
59                             0: begin
60                                 cell_state = 1; // Blocking statement -- immediately
61                                 errorsignal_clk = 1; // Critical timing on this input; assign
62                                     ↪ immediately
63                                 errorsignal_clk <= #(ct_state0_a_clk) 0; // Clear error signal
64                                     ↪ after critical timing expires
65                             end
66                             1: begin
67                                 errorsignal_clk = 1; // Critical timing on this input; assign

```

```

65         ↪ immediately
        errorsignal_clk <= #(ct_state1_a_clk) 0; // Clear error signal
        ↪ after critical timing expires
66     end
67     endcase
68 end
69 end
70 end
71
72 always @(posedge b or negedge b) // execute at positive and negative edges of input
73 begin
74     if ($time>4) // arbitrary steady-state time)
75     begin
76         if (errorsignal_b == 1'b1) // A critical timing is active for this input
77         begin
78             outfile = $fopen("errors.txt", "a");
79             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
            ↪ ", $time);
80             $fclose(outfile);
81             q <= 1'bX; // Set all outputs to unknown
82         end
83         if (errorsignal_b == 0)
84         begin
85             case (cell_state)
86             0: begin
87                 cell_state = 1; // Blocking statement -- immediately
88                 errorsignal_clk = 1; // Critical timing on this input; assign
89                 ↪ immediately
89                 errorsignal_clk <= #(ct_state0_b_clk) 0; // Clear error signal
90                 ↪ after critical timing expires
90             end
91             1: begin
92                 errorsignal_clk = 1; // Critical timing on this input; assign
93                 ↪ immediately
93                 errorsignal_clk <= #(ct_state1_b_clk) 0; // Clear error signal
94                 ↪ after critical timing expires
94             end
95             endcase
96         end
97     end
98 end
99
100 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
101 begin
102     if ($time>4) // arbitrary steady-state time)
103     begin
104         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
105         begin
106             outfile = $fopen("errors.txt", "a");
107             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
            ↪ ", $time);
108             $fclose(outfile);
109             q <= 1'bX; // Set all outputs to unknown
110         end
111         if (errorsignal_clk == 0)
112         begin
113             case (cell_state)
114             0: begin
115                 end
116             1: begin
117                 q <= #(delay_state1_clk_q) !q;
118                 cell_state = 0; // Blocking statement -- immediately
119             end
120             endcase
121         end
122     end
123 end
124 endmodule

```

Listing 3.18: RSFQ OR2T verilog model.

The digital simulation results for the RSFQ OR2T is shown in Fig. 3.30 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 3.31.

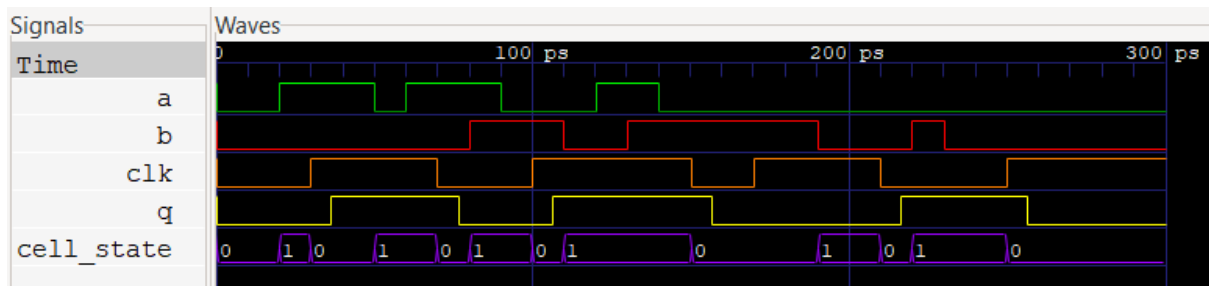


Figure 3.30: RSFQ OR2T digital simulation results.

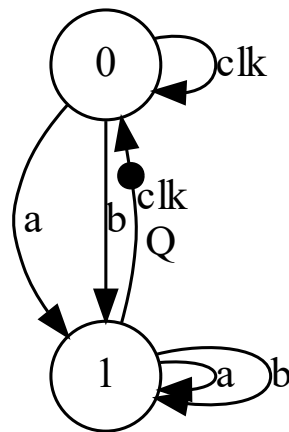


Figure 3.31: RSFQ OR2T Mealy finite state machine diagram.

Power Consumption

Table 3.14: RSFQ OR2T power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	3480	5.05
2	3480	10.1
5	3480	25.2
10	3480	50.5
20	3480	101
50	3480	252

3.2.3 XORT

The RSFQ XORT cell generates an output pulse exclusively if a pulse from a single input line was received before the clock signal. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

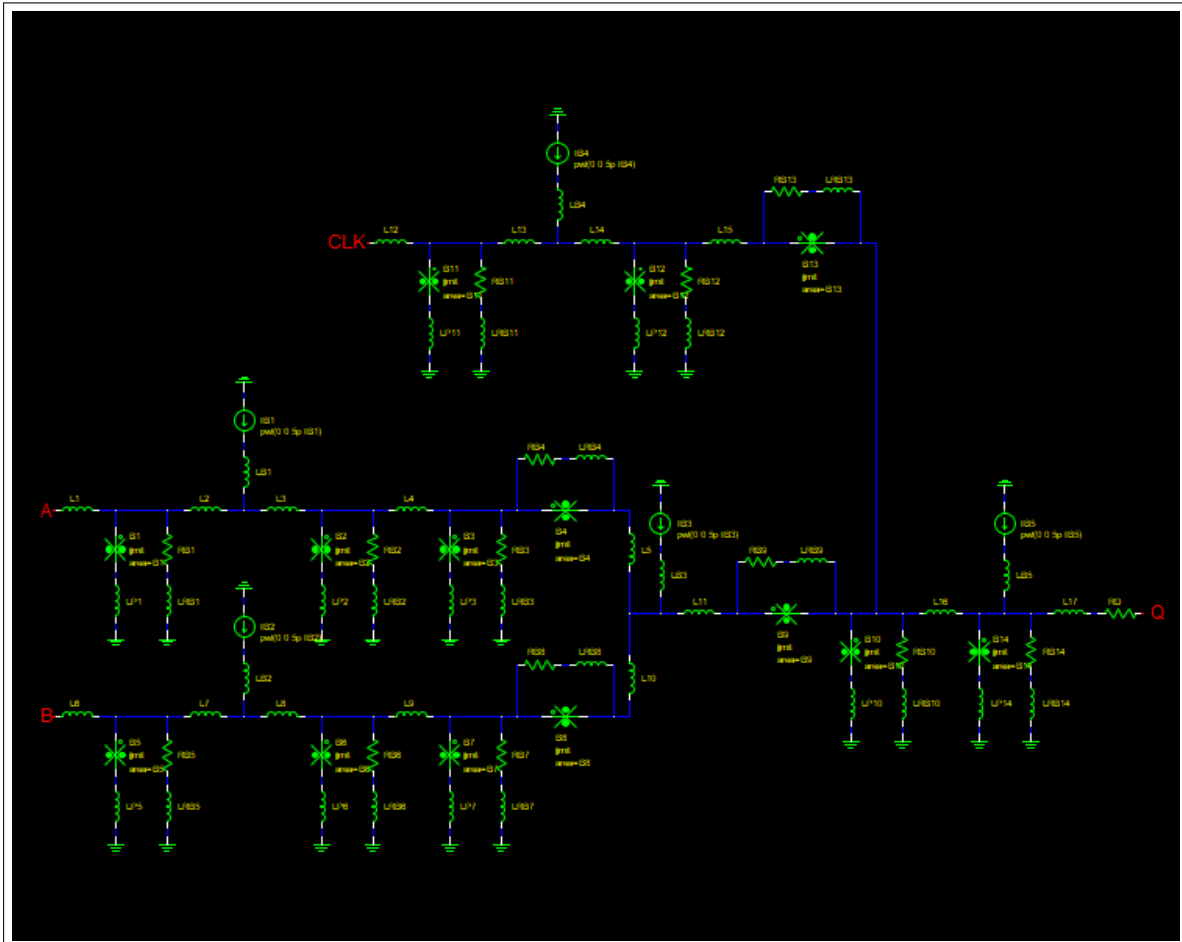


Figure 3.32: Schematic of RSFQ XORT.

Layout

The physical layout of the RSFQ XORT is shown in Fig. 3.33. The layout height is $70\ \mu\text{m}$ and the width is $50\ \mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

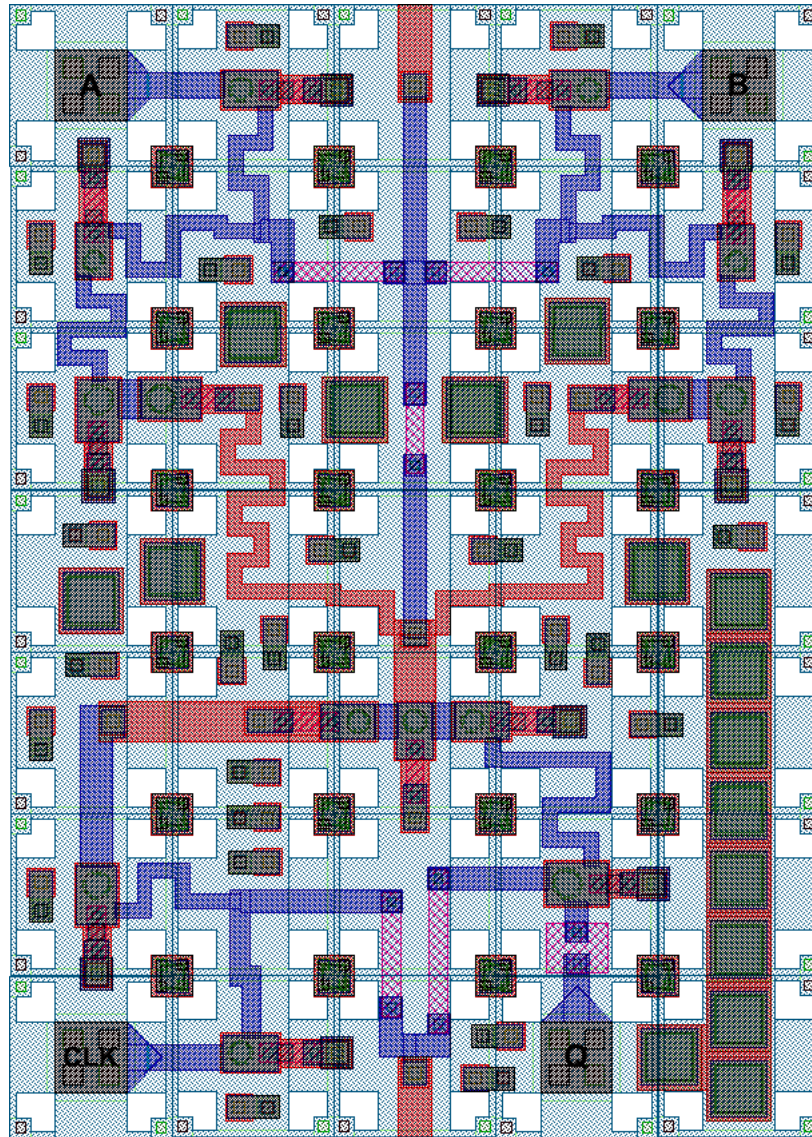


Figure 3.33: RSFQ XORT Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/09/06.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 6 September 2022
5  * Last modification by: T. Hall
6
7  *$ports      a      b      clk      q
8  .subckt THmit11_XORT a b clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=1.6
27 .param B2=1.25
28 .param B3=2.45
29 .param B4=2.22
30 .param B5=1.6
31 .param B6=1.25
32 .param B7=2.45
33 .param B8=2.22
34 .param B9=1.56
35 .param B10=1.52
36 .param B11=1.6
37 .param B12=2.16
38 .param B13=1.47
39 .param B14=2.5
40
41 .param IB1=230u
42 .param IB2=230u
43 .param IB3=373u
44 .param IB4=245u
45 .param IB5=175u
46
47 .param LB1=LB
48 .param LB2=LB
49 .param LB3=LB
50 .param LB4=LB
51 .param LB5=LB
52
53 .param LP1=LP
54 .param LP2=LP
55 .param LP3=LP
56 .param LP5=LP
57 .param LP6=LP
58 .param LP7=LP
59 .param LP10=LP
60 .param LP11=LP
61 .param LP12=LP
62 .param LP14=LP
63
64 .param L1=Lptl
65 .param L2=2.7271p
66 .param L3=3.7490p
67 .param L4=3.4856p
68 .param L5=11.3746p
69 .param L6=Lptl
70 .param L7=2.7271p
71 .param L8=3.7490p
72 .param L9=3.4856p
73 .param L10=11.3746p
74 .param L11=1.0849p
75 .param L12=Lptl
76 .param L13=1.8920p
77 .param L14=2.9216p
78 .param L15=4.4126p
79 .param L16=5.1623p
80 .param L17=Lptl
81
82 .param RB1=B0Rs/B1
83 .param RB2=B0Rs/B2
84 .param RB3=B0Rs/B3
85 .param RB4=B0Rs/B4
86 .param RB5=B0Rs/B5
87 .param RB6=B0Rs/B6
88 .param RB7=B0Rs/B7
89 .param RB8=B0Rs/B8
90 .param RB9=B0Rs/B9
91 .param RB10=B0Rs/B10
92 .param RB11=B0Rs/B11
93 .param RB12=B0Rs/B12
94 .param RB13=B0Rs/B13
95 .param RB14=B0Rs/B14
96
97 .param LRB1=(RB1/Rsheet)*Lsheet+LP
98 .param LRB2=(RB2/Rsheet)*Lsheet+LP
99 .param LRB3=(RB3/Rsheet)*Lsheet+LP
100 .param LRB4=(RB4/Rsheet)*Lsheet
101 .param LRB5=(RB5/Rsheet)*Lsheet+LP
102 .param LRB6=(RB6/Rsheet)*Lsheet+LP
103 .param LRB7=(RB7/Rsheet)*Lsheet+LP
104 .param LRB8=(RB8/Rsheet)*Lsheet
105 .param LRB9=(RB9/Rsheet)*Lsheet
106 .param LRB10=(RB10/Rsheet)*Lsheet+LP
107 .param LRB11=(RB11/Rsheet)*Lsheet+LP
108 .param LRB12=(RB12/Rsheet)*Lsheet+LP
109 .param LRB13=(RB13/Rsheet)*Lsheet
110 .param LRB14=(RB14/Rsheet)*Lsheet+LP
111
112 B1 1 2 jjmit area=B1
113 B2 5 6 jjmit area=B2
114 B3 7 8 jjmit area=B3
115 B4 7 9 jjmit area=B4
116 B5 11 12 jjmit area=B5
117 B6 15 16 jjmit area=B6
118 B7 17 18 jjmit area=B7
119 B8 17 19 jjmit area=B8
120 B9 21 22 jjmit area=B9
121 B10 22 23 jjmit area=B10
122 B11 24 25 jjmit area=B11
123 B12 28 29 jjmit area=B12
124 B13 30 22 jjmit area=B13
125 B14 31 32 jjmit area=B14
126
127 IB1 0 4 pw1(0 0 5p IB1)
128 IB2 0 14 pw1(0 0 5p IB2)
129 IB3 0 20 pw1(0 0 5p IB3)

```

```

130 IB4 0 27 pwl(0 0 5p IB4)
131 IB5 0 33 pwl(0 0 5p IB5)
132
133 LB1 4 3 8.211E-013
134 LB2 14 13 8.097E-013
135 LB3 20 10 2.305E-012
136 LB4 27 26 2.017E-012
137 LB5 33 31 1.71E-012
138
139 L1 a 1 1.358E-012
140 L2 1 3 2.704E-012
141 L3 3 5 3.77E-012
142 L4 5 7 3.482E-012
143 L5 9 10 1.13E-011
144 L6 b 11 1.36E-012
145 L7 11 13 2.741E-012
146 L8 13 15 3.753E-012
147 L9 15 17 3.499E-012
148 L10 10 19 1.147E-011
149 L11 10 21 1.081E-012
150 L12 clk 24 1.322E-012
151 L13 24 26 1.894E-012
152 L14 26 28 2.925E-012
153 L15 28 30 4.408E-012
154 L16 22 31 5.166E-012
155 L17 31 34 5.652E-013
156
157 RD 34 q RD
158
159 LP1 2 0 3.691E-013
160 LP2 6 0 4.306E-013
161 LP3 8 0 4.459E-013
162 LP5 12 0 3.668E-013
163 LP6 16 0 4.344E-013
164 LP7 18 0 4.281E-013
165 LP10 23 0 4.896E-013
166 LP11 25 0 4.012E-013
167 LP12 29 0 4.248E-013
168 LP14 32 0 3.342E-013
169
170 RB1 1 101 RB1
171 LRB1 101 0 LRB1
172 RB2 5 105 RB2
173 LRB2 105 0 LRB2
174 RB3 7 107 RB3
175 LRB3 107 0 LRB3
176 RB4 7 109 RB4
177 LRB4 109 9 LRB4
178 RB5 11 111 RB5
179 LRB5 111 0 LRB5
180 RB6 15 115 RB6
181 LRB6 115 0 LRB6
182 RB7 17 117 RB7
183 LRB7 117 0 LRB7
184 RB8 17 119 RB8
185 LRB8 119 19 LRB8
186 RB9 21 121 RB9
187 LRB9 121 22 LRB9
188 RB10 22 122 RB10
189 LRB10 122 0 LRB10
190 RB11 24 124 RB11
191 LRB11 124 0 LRB11
192 RB12 28 128 RB12
193 LRB12 128 0 LRB12
194 RB13 30 130 RB13
195 LRB13 130 22 LRB13
196 RB14 31 131 RB14
197 LRB14 131 0 LRB14
198
199 .ends

```

Listing 3.19: RSFQ XORT JoSIM netlist.**Table 3.15:** RSFQ XORT pin list.

Pin	Description
a	Data input
b	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ XORT using JoSIM is shown in Fig. 3.34. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b**,
- (d) the phase over the input JJ of pin **b**,
- (e) the current through the input inductor connected to pin **clk**,
- (f) the phase over the input JJ of pin **clk**,
- (g) the phase over the output JJ of pin **q**, and
- (h) the phase over the input JJ of the load circuit connected via a PTL to pin **q**.

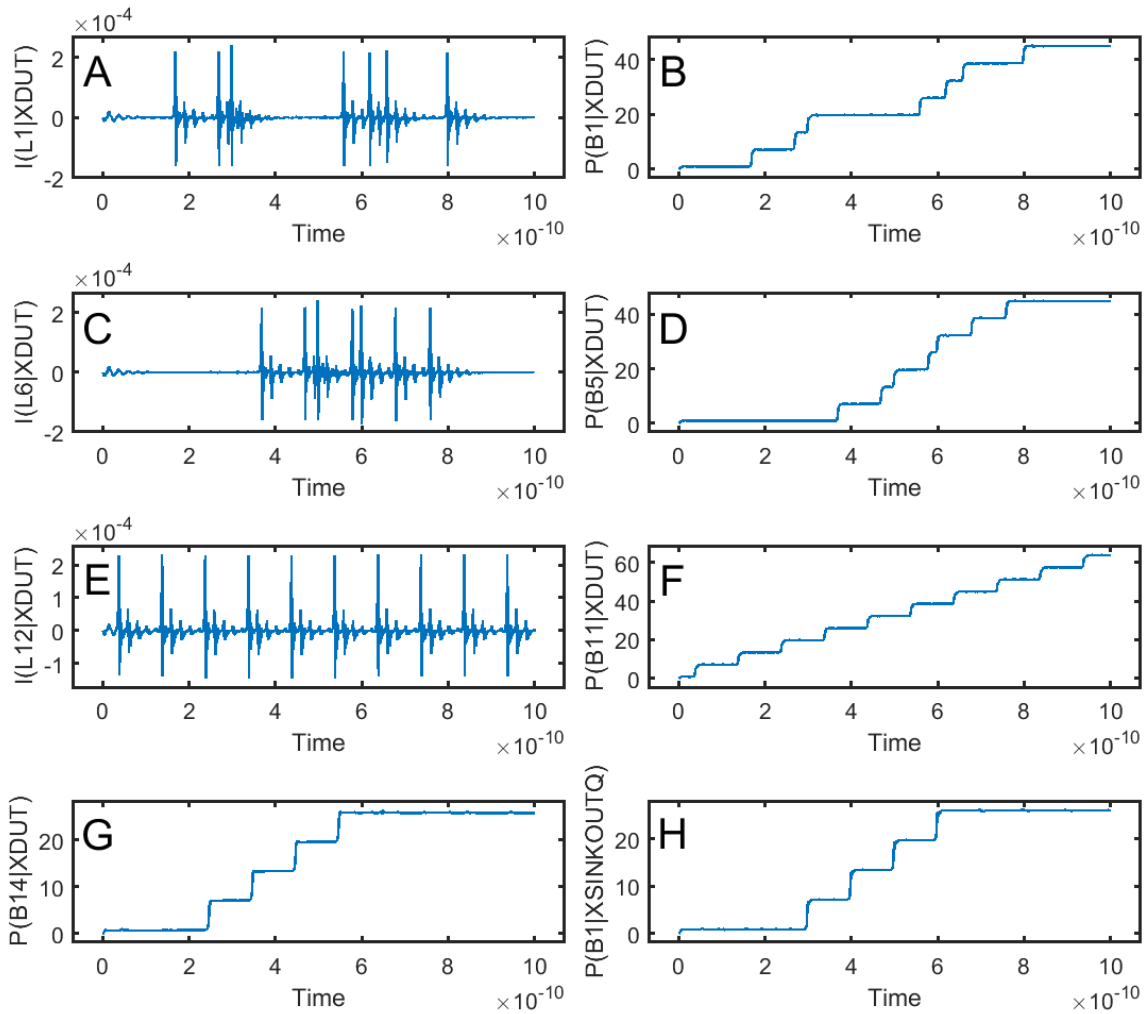


Figure 3.34: RSFQ XORT analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 6 September 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit1l_XORT_v3p0_extracted (a, b, clk, q);
9
10 input
11     a, b, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state1_clk_q = 8.8,
21     delay_state2_clk_q = 8.8,
22     ct_state0_clk_a = 3.2,
23     ct_state0_clk_b = 3.1,
24     ct_state1_a_b = 9.5,
25     ct_state1_a_clk = 6.6,
26     ct_state1_b_b = 5.2,
27     ct_state1_clk_b = 7.8,
28     ct_state2_a_a = 5.2,
29     ct_state2_b_a = 9.5,
30     ct_state2_b_clk = 6.7,
31     ct_state2_clk_a = 7.6;
32
33 reg
34     errorsignal_a,
35     errorsignal_b,
36     errorsignal_clk;
37
38 integer
39     outfile,
40     cell_state; // internal state of the cell
41
42 initial
43     begin
44         errorsignal_a = 0;
45         errorsignal_b = 0;
46         errorsignal_clk = 0;
47         cell_state = 0; // Startup state
48         q = 0; // All outputs start at 0
49     end
50
51 always @(posedge a or negedge a) // execute at positive and negative edges of input
52     begin
53         if ($time>4) // arbitrary steady-state time)
54             begin
55                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
56                     begin
57                         outfile = $fopen("errors.txt", "a");
58                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
59                         $fclose(outfile);
60                         q <= 1'bX; // Set all outputs to unknown
61                     end
62                 if (errorsignal_a == 0)
63                     begin
64                         case (cell_state)
65                             0: begin
66                                 cell_state = 1; // Blocking statement -- immediately

```

```

67         end
68     1: begin
69         errorsignal_b = 1; // Critical timing on this input; assign
70             ↪ immediately
71         errorsignal_b <= #(ct_state1_a_b) 0; // Clear error signal
72             ↪ after critical timing expires
73         errorsignal_clk = 1; // Critical timing on this input; assign
74             ↪ immediately
75         errorsignal_clk <= #(ct_state1_a_clk) 0; // Clear error signal
76             ↪ after critical timing expires
77     end
78     2: begin
79         cell_state = 0; // Blocking statement -- immediately
80         errorsignal_a = 1; // Critical timing on this input; assign
81             ↪ immediately
82         errorsignal_a <= #(ct_state2_a_a) 0; // Clear error signal
83             ↪ after critical timing expires
84     end
85 endcase
86 end
87 end
88
89 always @(posedge b or negedge b) // execute at positive and negative edges of input
90 begin
91     if ($time>4) // arbitrary steady-state time)
92     begin
93         if (errorsignal_b == 1'b1) // A critical timing is active for this input
94         begin
95             outfile = $fopen("errors.txt", "a");
96             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d.ps.\n", $time);
97             ↪ ", $time);
98             $fclose(outfile);
99             q <= 1'bX; // Set all outputs to unknown
100         end
101         if (errorsignal_b == 0)
102         begin
103             case (cell_state)
104             0: begin
105                 cell_state = 2; // Blocking statement -- immediately
106             end
107             1: begin
108                 cell_state = 0; // Blocking statement -- immediately
109                 errorsignal_b = 1; // Critical timing on this input; assign
110                     ↪ immediately
111                 errorsignal_b <= #(ct_state1_b_b) 0; // Clear error signal
112                     ↪ after critical timing expires
113             end
114             2: begin
115                 errorsignal_a = 1; // Critical timing on this input; assign
116                     ↪ immediately
117                 errorsignal_a <= #(ct_state2_b_a) 0; // Clear error signal
118                     ↪ after critical timing expires
119                 errorsignal_clk = 1; // Critical timing on this input; assign
120                     ↪ immediately
121                 errorsignal_clk <= #(ct_state2_b_clk) 0; // Clear error signal
122                     ↪ after critical timing expires
123             end
124         endcase
125     end
126 end
127
128 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
129 begin
130     if ($time>4) // arbitrary steady-state time)
131     begin
132         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
133         begin
134             outfile = $fopen("errors.txt", "a");

```

```

124         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d.ps.\n
           ↪ ", $stime);
125     $fclose(outfile);
126     q <= 1'bX; // Set all outputs to unknown
127 end
128 if (errorsignal_clk == 0)
129     begin
130         case (cell_state)
131             0: begin
132                 errorsignal_a = 1; // Critical timing on this input; assign
133                     ↪ immediately
134                 errorsignal_a <= #(ct_state0_clk_a) 0; // Clear error signal
135                     ↪ after critical timing expires
136                 errorsignal_b = 1; // Critical timing on this input; assign
137                     ↪ immediately
138                 errorsignal_b <= #(ct_state0_clk_b) 0; // Clear error signal
139                     ↪ after critical timing expires
140             end
141             1: begin
142                 q <= #(delay_state1_clk_q) !q;
143                 cell_state = 0; // Blocking statement -- immediately
144                 errorsignal_b = 1; // Critical timing on this input; assign
145                     ↪ immediately
146                 errorsignal_b <= #(ct_state1_clk_b) 0; // Clear error signal
147                     ↪ after critical timing expires
148             end
149             2: begin
150                 q <= #(delay_state2_clk_q) !q;
151                 cell_state = 0; // Blocking statement -- immediately
152                 errorsignal_a = 1; // Critical timing on this input; assign
153                     ↪ immediately
154                 errorsignal_a <= #(ct_state2_clk_a) 0; // Clear error signal
155                     ↪ after critical timing expires
156             end
157         endcase
158     end
159 end
160 end
161 endmodule

```

Listing 3.20: RSFQ XORT verilog model.

The digital simulation results for the RSFQ XORT is shown in Fig. 3.35 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 3.36.

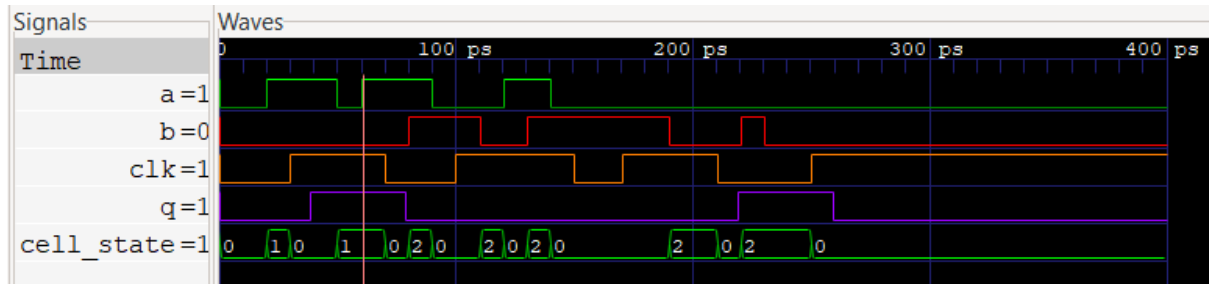


Figure 3.35: RSFQ XORT digital simulation results.

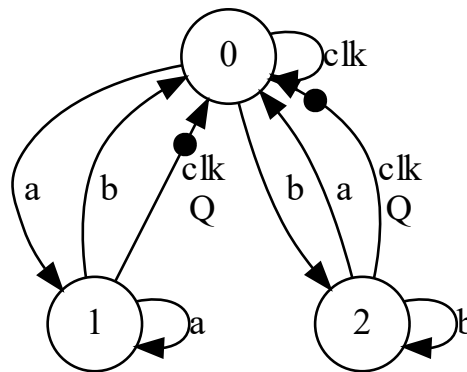


Figure 3.36: RSFQ XORT Mealy finite state machine diagram.

Power Consumption

Table 3.16: RSFQ XORT power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	3260	5.35
2	3260	10.7
5	3260	26.7
10	3260	53.5
20	3260	107
50	3260	267

3.2.4 NOTT

The RSFQ NOTT cell is a signal inverting cell driven by a clock pulse signal line. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

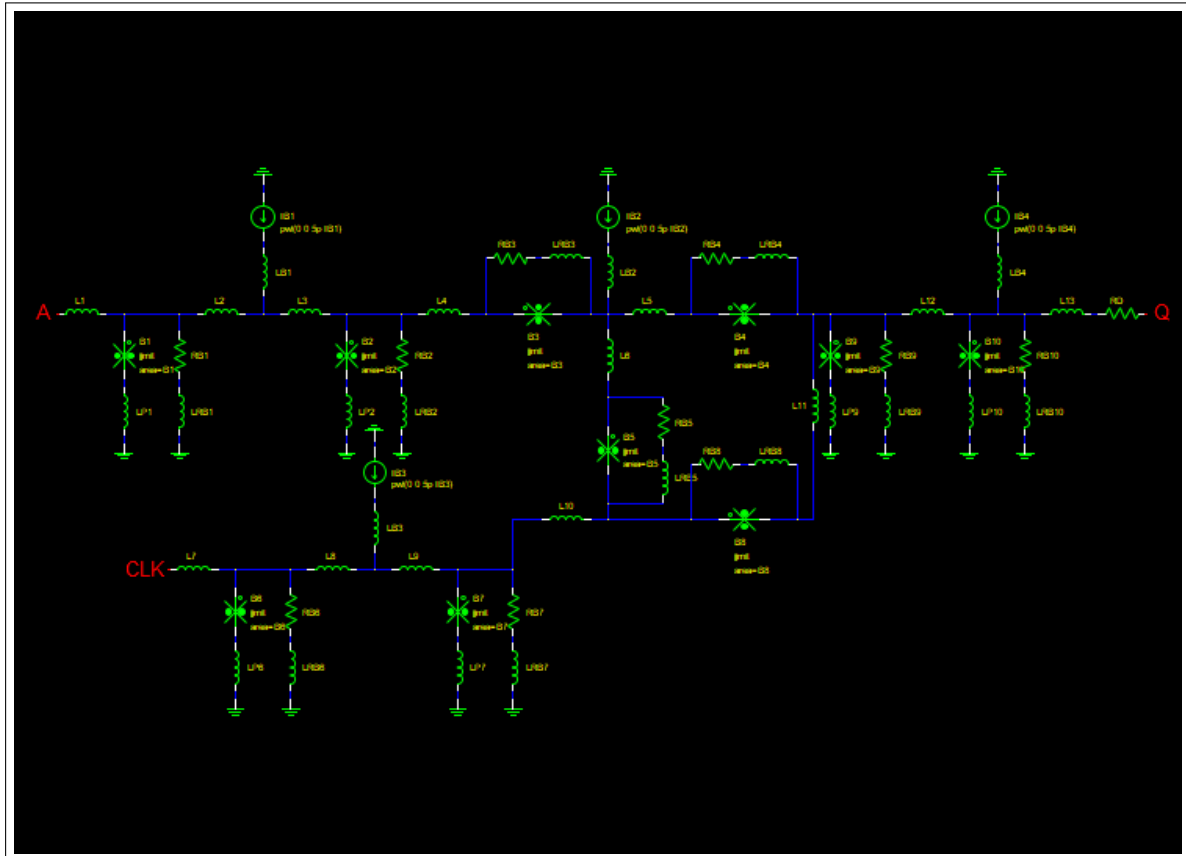


Figure 3.37: Schematic of RSFQ NOTT.

Layout

The physical layout of the RSFQ NOTT is shown in Fig. 3.38. The layout height is $70\ \mu\text{m}$ and the width is $40\ \mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

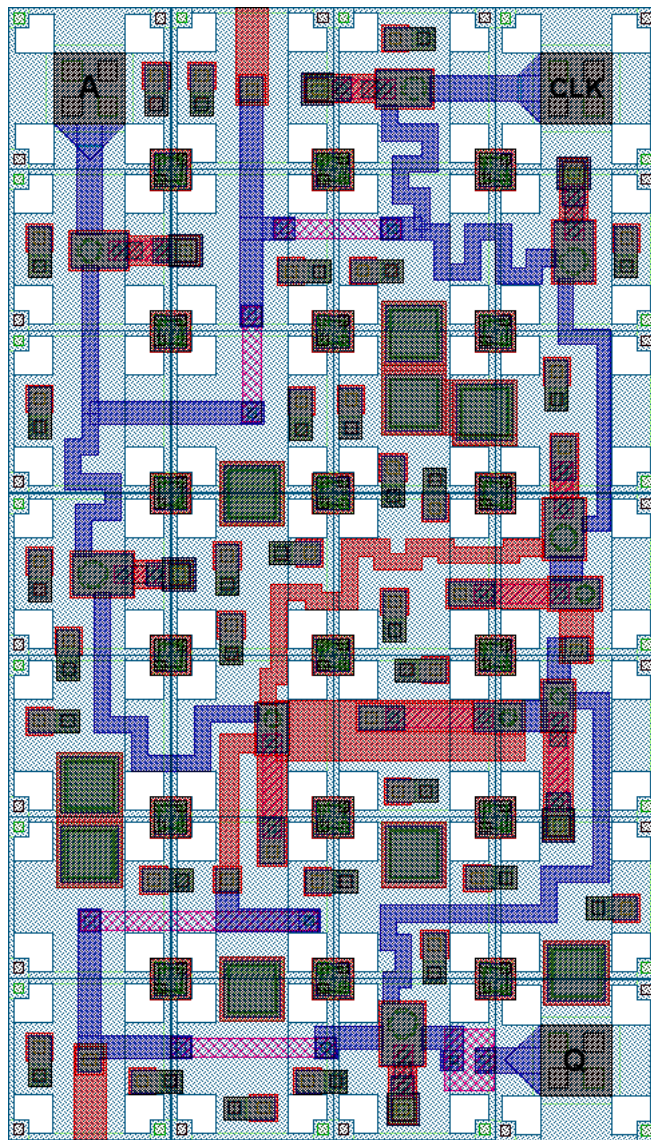


Figure 3.38: RSFQ NOTT Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/25.
2  * Author: T. Hall
3  * Version: 3.0
4  * Last modification date: 24 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      clk      q
8  .subckt THmit11_NOTT a clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=1.6
27 .param B2=2.38
28 .param B3=0.77
29 .param B4=0.73
30 .param B5=1.98
31 .param B6=1.6
32 .param B7=2.18
33 .param B8=0.87
34 .param B9=0.82
35 .param B10=2.5
36
37 .param IB1=267u
38 .param IB2=114u
39 .param IB3=241u
40 .param IB4=175u
41
42 .param LB1=LB
43 .param LB2=LB
44 .param LB3=LB
45 .param LB4=LB
46
47 .param L1=Lptl
48 .param L2=2.3963p
49 .param L3=2.9984p
50 .param L4=5.5602p
51 .param L5=1.8403p
52 .param L6=8.5410p
53 .param L7=Lptl
54 .param L8=2.8313p
55 .param L9=3.4874p
56 .param L10=5.2166p
57 .param L11=1.5349p
58 .param L12=8.5298p
59 .param L13=Lptl
60
61 .param LP1=LP
62 .param LP2=LP
63 .param LP6=LP
64 .param LP7=LP
65 .param LP9=LP
66 .param LP10=LP
67
68 .param RB1=B0Rs/B1
69 .param RB2=B0Rs/B2
70 .param RB3=B0Rs/B3
71 .param RB4=B0Rs/B4
72 .param RB5=B0Rs/B5
73 .param RB6=B0Rs/B6
74 .param RB7=B0Rs/B7
75 .param RB8=B0Rs/B8
76 .param RB9=B0Rs/B9
77 .param RB10=B0Rs/B10
78
79 .param LRB1=(RB1/Rsheet)*Lsheet+LP
80 .param LRB2=(RB2/Rsheet)*Lsheet+LP
81 .param LRB3=(RB3/Rsheet)*Lsheet
82 .param LRB4=(RB4/Rsheet)*Lsheet
83 .param LRB5=(RB5/Rsheet)*Lsheet
84 .param LRB6=(RB6/Rsheet)*Lsheet+LP
85 .param LRB7=(RB7/Rsheet)*Lsheet+LP
86 .param LRB8=(RB8/Rsheet)*Lsheet
87 .param LRB9=(RB9/Rsheet)*Lsheet+LP
88 .param LRB10=(RB10/Rsheet)*Lsheet+LP
89
90 B1 1 2 jjmit area=B1
91 B2 5 6 jjmit area=B2
92 B3 7 8 jjmit area=B3
93 B4 10 11 jjmit area=B4
94 B5 12 13 jjmit area=B5
95 B6 14 15 jjmit area=B6
96 B7 18 19 jjmit area=B7
97 B8 13 20 jjmit area=B8
98 B9 11 21 jjmit area=B9
99 B10 22 23 jjmit area=B10
100
101 IB1 0 4 pw1(0 0 5p IB1)
102 IB2 0 9 pw1(0 0 5p IB2)
103 IB3 0 17 pw1(0 0 5p IB3)
104 IB4 0 24 pw1(0 0 5p IB4)
105
106 LB1 4 3 2.006E-012
107 LB2 9 8 4.261E-012
108 LB3 17 16 4.788E-013
109 LB4 24 22 1.011E-012
110
111 L1 a 1 1.41E-012
112 L2 1 3 2.393E-012
113 L3 3 5 3E-012
114 L4 5 7 5.514E-012
115 L5 8 10 1.851E-012
116 L6 8 12 8.457E-012
117 L7 clk 14 1.492E-012
118 L8 14 16 2.817E-012
119 L9 16 18 3.474E-012
120 L10 18 13 5.206E-012
121 L11 11 20 1.537E-012
122 L12 11 22 8.509E-012
123 L13 22 25 7.694E-013
124
125 RD 25 q RD
126
127 LP1 2 0 4.024E-013
128 LP2 6 0 4.485E-013
129 LP6 15 0 3.866E-013

```

```

130 | LP7 19 0 4.773E-013
131 | LP9 21 0 5.266E-013
132 | LP10 23 0 3.249E-013
133 |
134 | RB1 1 101 RB1
135 | LRB1 101 0 LRB1
136 | RB2 5 105 RB2
137 | LRB2 105 0 LRB2
138 | RB3 7 107 RB3
139 | LRB3 107 8 LRB3
140 | RB4 10 110 RB4
141 | LRB4 110 11 LRB4
142 | RB5 12 112 RB5
143 | LRB5 112 13 LRB5

144 | RB6 14 114 RB6
145 | LRB6 114 0 LRB6
146 | RB7 18 118 RB7
147 | LRB7 118 0 LRB7
148 | RB8 13 113 RB8
149 | LRB8 113 20 LRB8
150 | RB9 11 111 RB9
151 | LRB9 111 0 LRB9
152 | RB10 22 122 RB10
153 | LRB10 122 0 LRB10
154 |
155 | .ends

```

Listing 3.21: RSFQ NOTT JoSIM netlist.**Table 3.17:** RSFQ NOTT pin list.

Pin	Description
a	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ NOTT using JoSIM is shown in Fig. 3.39. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **clk**,
- (d) the phase over the input JJ of pin **clk**,
- (e) the phase over the output JJ of pin **q**, and
- (f) the phase over the input JJ of the load circuit connected via a PTL to pin **q**.

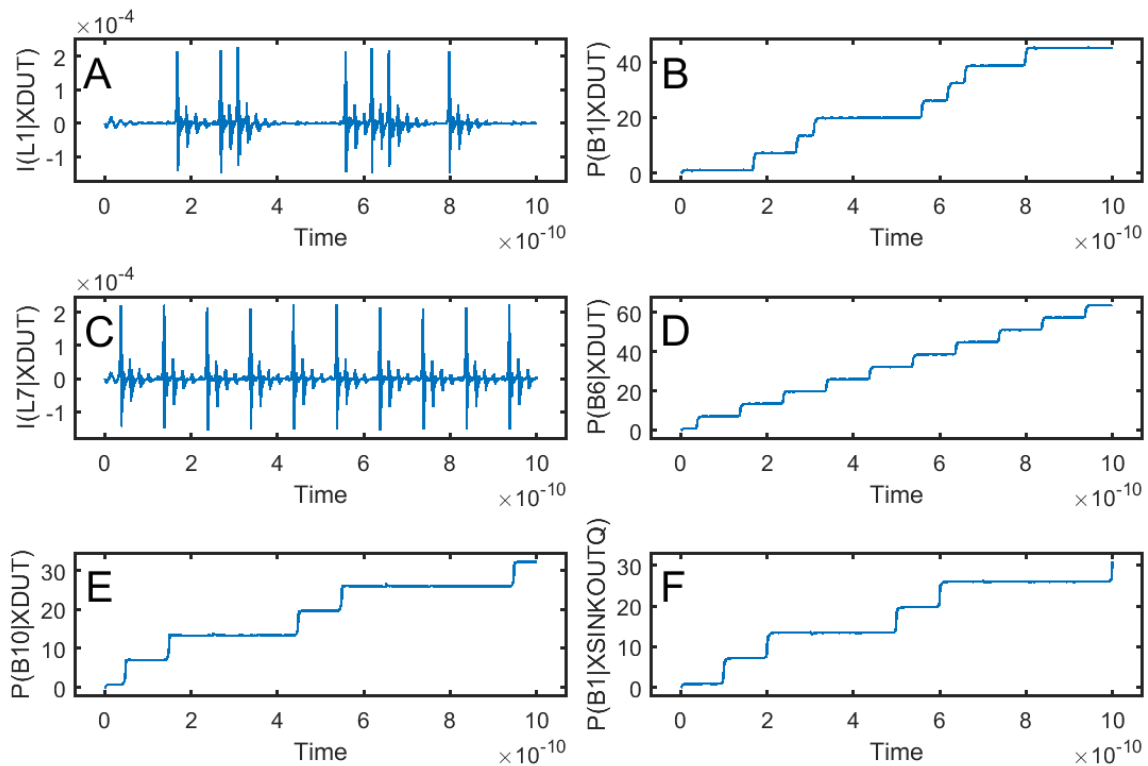


Figure 3.39: RSFQ NOTT analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 25 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_NOTT_v3p0_extracted (a, clk, q);
9
10 input
11     a, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_clk_q = 10.5,
21     ct_state0_a_clk = 1.6,
22     ct_state0_clk_a = 6.9,
23     ct_state0_clk_clk = 8.4,
24     ct_state1_a_clk = 0.6;
25
26 reg
27     errorsignal_a,
28     errorsignal_clk;
29
30 integer
31     outfile,
32     cell_state; // internal state of the cell
33
34 initial
35     begin
36         errorsignal_a = 0;
37         errorsignal_clk = 0;
38         cell_state = 0; // Startup state
39         q = 0; // All outputs start at 0
40     end
41
42 always @(posedge a or negedge a) // execute at positive and negative edges of input
43     begin
44         if ($time>4) // arbitrary steady-state time)
45             begin
46                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
47                     begin
48                         outfile = $fopen("errors.txt", "a");
49                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
50                             ↪ ", $stime);
51                         $fclose(outfile);
52                         q <= 1'bX; // Set all outputs to unknown
53                     end
54                 if (errorsignal_a == 0)
55                     begin
56                         case (cell_state)
57                             0: begin
58                                 cell_state = 1; // Blocking statement -- immediately
59                                 errorsignal_clk = 1; // Critical timing on this input; assign
60                                     ↪ immediately
61                                 errorsignal_clk <= #(ct_state0_a_clk) 0; // Clear error signal
62                                     ↪ after critical timing expires
63                             end
64                             1: begin
65                                 errorsignal_clk = 1; // Critical timing on this input; assign
66                                     ↪ immediately
67                                 errorsignal_clk <= #(ct_state1_a_clk) 0; // Clear error signal

```

```

64                                     ↪ after critical timing expires
65                                     end
66                                     endcase
67                                     end
68                                     end
69
70 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
71 begin
72     if ($time>4) // arbitrary steady-state time)
73     begin
74         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
75         begin
76             outfile = $fopen("errors.txt", "a");
77             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
78                 ↪ ", $time);
79             $fclose(outfile);
80             q <= 1'bX; // Set all outputs to unknown
81         end
82         if (errorsignal_clk == 0)
83         begin
84             case (cell_state)
85             0: begin
86                 q <= #(delay_state0_clk_q) !q;
87                 errorsignal_a = 1; // Critical timing on this input; assign
88                     ↪ immediately
89                 errorsignal_a <= #(ct_state0_clk_a) 0; // Clear error signal
90                     ↪ after critical timing expires
91                 errorsignal_clk = 1; // Critical timing on this input; assign
92                     ↪ immediately
93                 errorsignal_clk <= #(ct_state0_clk_clk) 0; // Clear error
94                     ↪ signal after critical timing expires
95             end
96             1: begin
97                 cell_state = 0; // Blocking statement -- immediately
98             end
99         endcase
100     end
101 end
102 end
103 endmodule

```

Listing 3.22: RSFQ NOTT verilog model.

The digital simulation results for the RSFQ NOTT is shown in Fig. 3.40 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 3.41.

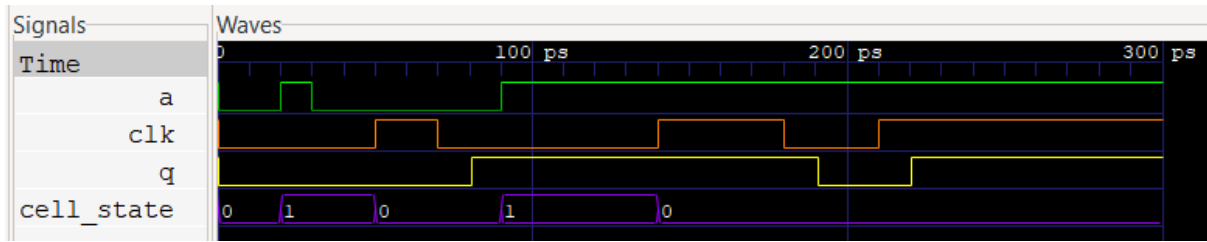


Figure 3.40: RSFQ NOTT digital simulation results.

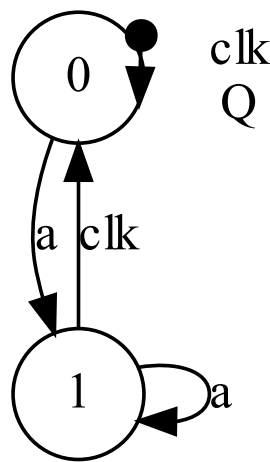


Figure 3.41: RSFQ NOTT Mealy finite state machine diagram.

Power Consumption

Table 3.18: RSFQ NOTT power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	2070	3.19
2	2070	6.38
5	2070	16.0
10	2070	31.9
20	2070	63.8
50	2070	160

3.3 Buffers

3.3.1 DFFT

The RSFQ DFFT, D flip-flop, is a multi-state device used to transmit an input set pulse synchronised with a reset (typically clock) signal. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

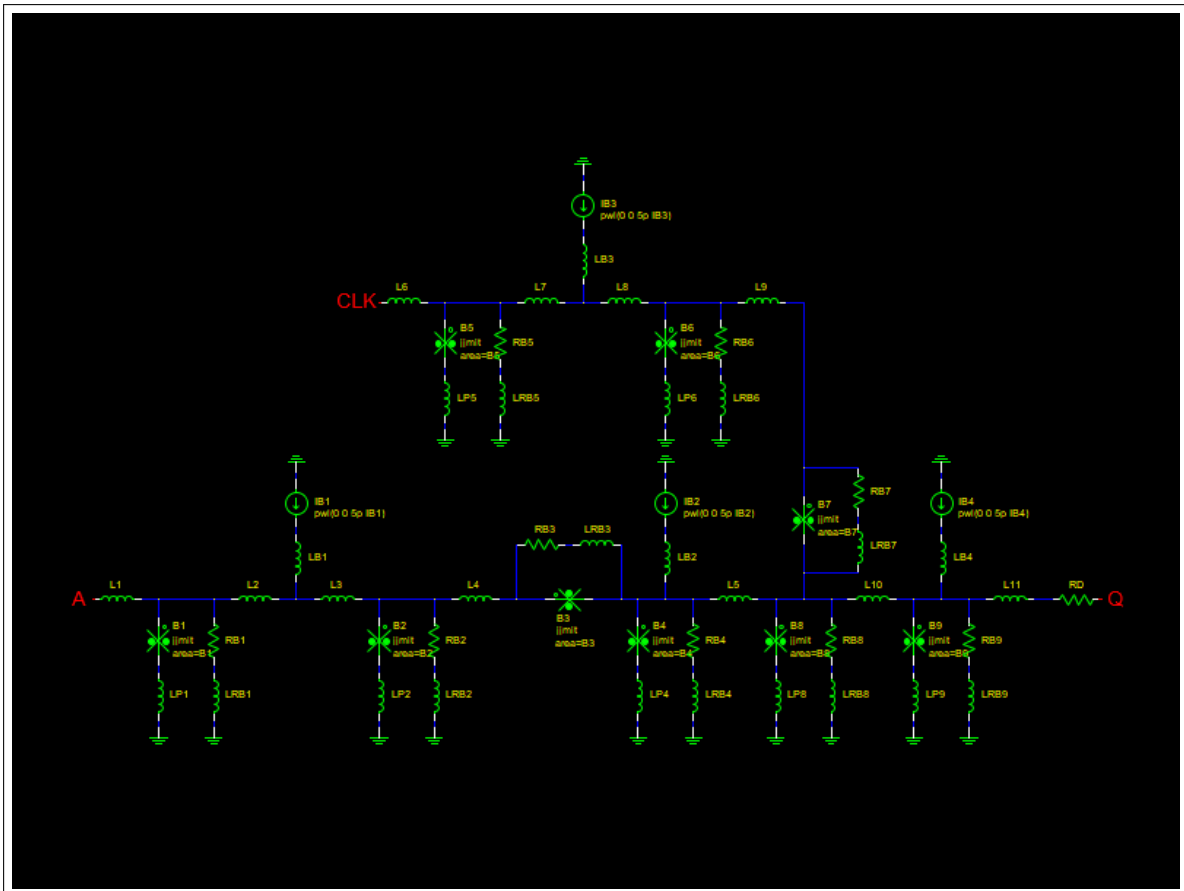


Figure 3.42: Schematic of RSFQ DFFT.

Layout

The physical layout of the RSFQ DFFT is shown in Fig. 3.43. The layout height is $70\text{ }\mu\text{m}$ and the width is $30\text{ }\mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

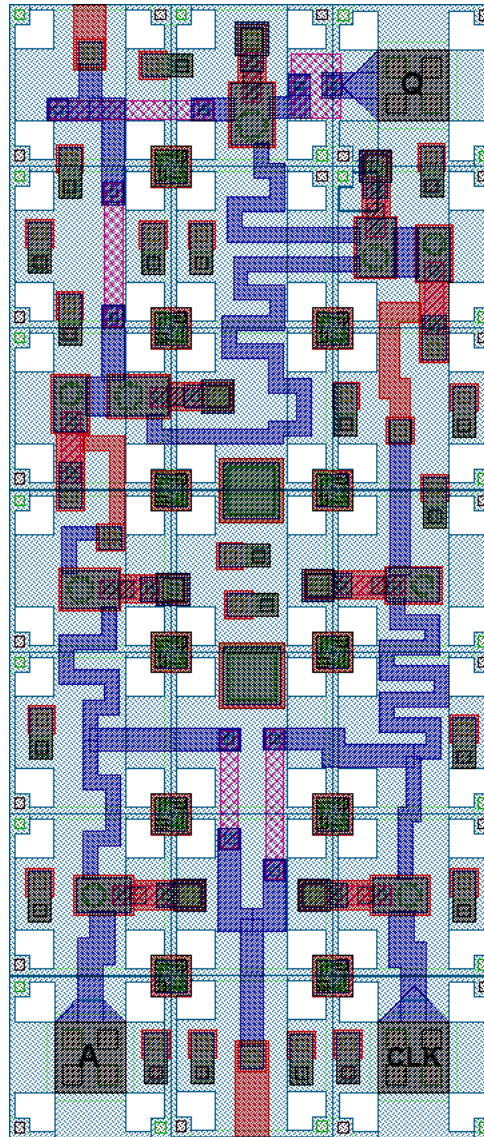


Figure 3.43: RSFQ DFFT Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/17.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 16 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      clk      q
8  .subckt THmit11_DFFT a clk q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=1.6
27 .param B2=1.82
28 .param B3=1.36
29 .param B4=2.07
30 .param B5=1.6
31 .param B6=1.27
32 .param B7=1.28
33 .param B8=1.89
34 .param B9=2.5
35
36 .param IB1=262u
37 .param IB2=206u
38 .param IB3=194u
39 .param IB4=175u
40
41 .param LB1=LB
42 .param LB2=LB
43 .param LB3=LB
44 .param LB4=LB
45
46 .param L1=Lptl
47 .param L2=2.8908p
48 .param L3=3.6865p
49 .param L4=4.7771p
50 .param L5=9.5023p
51 .param L6=Lptl
52 .param L7=2.2014p
53 .param L8=5.6199p
54 .param L9=5.3129p
55 .param L10=4.5588p
56 .param L11=Lptl
57
58 .param LP1=LP
59 .param LP2=LP
60 .param LP4=LP
61 .param LP5=LP
62 .param LP6=LP
63 .param LP8=LP
64 .param LP9=LP
65
66 .param RB1=B0Rs/B1
67 .param RB2=B0Rs/B2
68 .param RB3=B0Rs/B3
69 .param RB4=B0Rs/B4
70 .param RB5=B0Rs/B5
71 .param RB6=B0Rs/B6
72 .param RB7=B0Rs/B7
73 .param RB8=B0Rs/B8
74 .param RB9=B0Rs/B9
75
76 .param LRB1=(RB1/Rsheet)*Lsheet+LP
77 .param LRB2=(RB2/Rsheet)*Lsheet+LP
78 .param LRB3=(RB3/Rsheet)*Lsheet
79 .param LRB4=(RB4/Rsheet)*Lsheet+LP
80 .param LRB5=(RB5/Rsheet)*Lsheet+LP
81 .param LRB6=(RB6/Rsheet)*Lsheet+LP
82 .param LRB7=(RB7/Rsheet)*Lsheet
83 .param LRB8=(RB8/Rsheet)*Lsheet+LP
84 .param LRB9=(RB9/Rsheet)*Lsheet+LP
85
86 B1 1 2 jjmit area=B1
87 B2 5 6 jjmit area=B2
88 B3 7 8 jjmit area=B3
89 B4 8 9 jjmit area=B4
90 B5 12 13 jjmit area=B5
91 B6 16 17 jjmit area=B6
92 B7 18 11 jjmit area=B7
93 B8 11 19 jjmit area=B8
94 B9 20 21 jjmit area=B9
95
96 IB1 0 4 pwl(0 0 5p IB1)
97 IB2 0 10 pwl(0 0 5p IB2)
98 IB3 0 15 pwl(0 0 5p IB3)
99 IB4 0 22 pwl(0 0 5p IB4)
100
101 LB1 4 3 1.76E-012
102 LB2 10 8 1.06E-012
103 LB3 15 14 1.853E-012
104 LB4 22 20 6.105E-013
105
106 L1 a 1 1.401E-012
107 L2 1 3 2.877E-012
108 L3 3 5 3.673E-012
109 L4 5 7 4.739E-012
110 L5 8 11 9.527E-012
111 L6 clk 12 1.422E-012
112 L7 12 14 2.188E-012
113 L8 14 16 5.588E-012
114 L9 16 18 5.337E-012
115 L10 11 20 4.566E-012
116 L11 20 23 7.347E-013
117 RD 23 q RD
118
119 LP1 2 0 3.976E-013
120 LP2 6 0 4.101E-013
121 LP4 9 0 4.172E-013
122 LP5 13 0 3.949E-013
123 LP6 17 0 5.178E-013
124 LP8 19 0 4.435E-013
125 LP9 21 0 3.068E-013
126
127 RB1 1 101 RB1
128 LRB1 101 0 LRB1
129 RB2 5 105 RB2

```

```

130 | LRB2 105 0 LRB2
131 | RB3 7 107 RB3
132 | LRB3 107 8 LRB3
133 | RB4 8 108 RB4
134 | LRB4 108 0 LRB4
135 | RB5 12 112 RB5
136 | LRB5 112 0 LRB5
137 | RB6 16 116 RB6
138 | LRB6 116 0 LRB6
139 | RB7 18 118 RB7

140 | LRB7 118 11 LRB7
141 | RB8 11 111 RB8
142 | LRB8 111 0 LRB8
143 | RB9 20 120 RB9
144 | LRB9 120 0 LRB9
145 |
146 | .ends

```

Listing 3.23: RSFQ DFFT JoSIM netlist.**Table 3.19:** RSFQ DFFT pin list.

Pin	Description
a	Data input
clk	Clock input
q	Data output

The simulation results for the RSFQ DFFT using JoSIM is shown in Fig. 3.44. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **clk**,
- (d) the phase over the input JJ of pin **clk**,
- (e) the phase over the output JJ of pin **q**, and
- (f) the phase over the input JJ of the load circuit connected via a PTL to pin **q**.

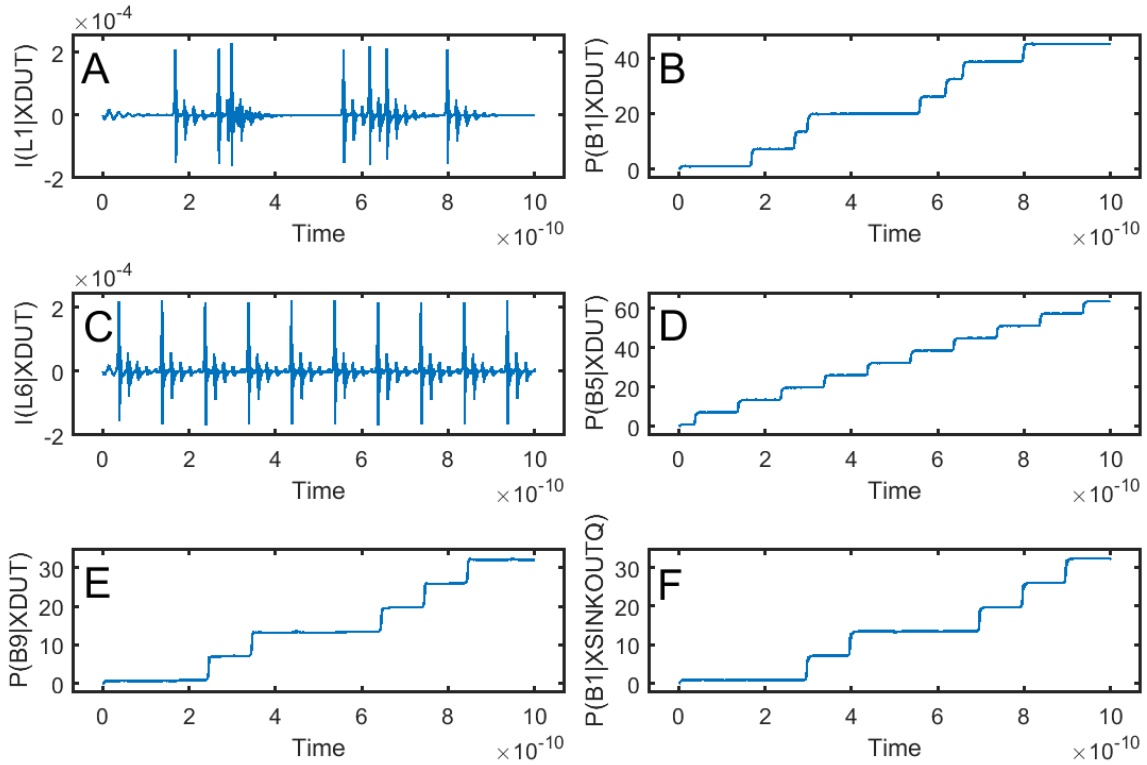


Figure 3.44: RSFQ DFFT analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 17 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_DFFT_v3p0_extracted (a, clk, q);
9
10 input
11     a, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state1_clk_q = 8.0,
21     ct_state0_clk_a = 2.3,
22     ct_state1_clk_a = 0.7;
23
24 reg
25     errorsignal_a,
26     errorsignal_clk;
27
28 integer
29     outfile,
30     cell_state; // internal state of the cell
31
32 initial
33     begin
34         errorsignal_a = 0;
35         errorsignal_clk = 0;
36         cell_state = 0; // Startup state
37         q = 0; // All outputs start at 0
38     end
39
40 always @(posedge a or negedge a) // execute at positive and negative edges of input
41     begin
42         if ($time>4) // arbitrary steady-state time)
43             begin
44                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
45                     begin
46                         outfile = $fopen("errors.txt", "a");
47                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0dps.\n
48                             ↪ ", $time);
49                         $fclose(outfile);
50                         q <= 1'bX; // Set all outputs to unknown
51                     end
52                 if (errorsignal_a == 0)
53                     begin
54                         case (cell_state)
55                             0: begin
56                                 cell_state = 1; // Blocking statement -- immediately
57                             end
58                             1: begin
59                                 end
60                         endcase
61                     end
62                 end
63             end
64         end
65
66 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
67     begin
68         if ($time>4) // arbitrary steady-state time)

```

```

67     begin
68         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
69             begin
70                 outfile = $fopen("errors.txt", "a");
71                 $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
72                 $fclose(outfile);
73                 q <= 1'bX; // Set all outputs to unknown
74             end
75         if (errorsignal_clk == 0)
76             begin
77                 case (cell_state)
78                     0: begin
79                         errorsignal_a = 1; // Critical timing on this input; assign
80                             ↪ immediately
81                         errorsignal_a <= #(ct_state0_clk_a) 0; // Clear error signal
82                             ↪ after critical timing expires
83                     end
84                     1: begin
85                         q <= #(delay_state1_clk_q) !q;
86                         cell_state = 0; // Blocking statement -- immediately
87                         errorsignal_a = 1; // Critical timing on this input; assign
88                             ↪ immediately
89                         errorsignal_a <= #(ct_state1_clk_a) 0; // Clear error signal
90                             ↪ after critical timing expires
91                     end
92                 endcase
93             end
94         end
95     end
96 endmodule

```

Listing 3.24: RSFQ DFFT verilog model.

The digital simulation results for the RSFQ DFFT is shown in Fig. 3.45 and the Mealy finite state machine diagram, extracted using *TimEx*, is shown in Fig. 3.46.

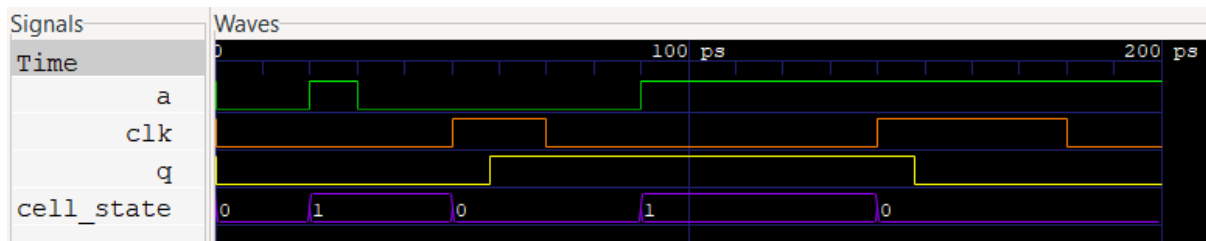


Figure 3.45: RSFQ DFFT digital simulation results.

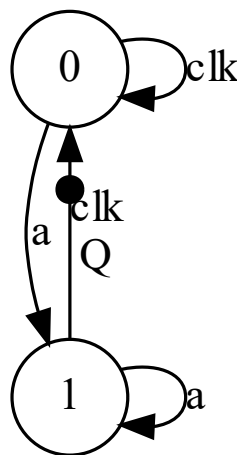


Figure 3.46: RSFQ DFFT Mealy finite state machine diagram.

Power Consumption

Table 3.20: RSFQ DFFT power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	2180	3.18
2	2180	6.36
5	2180	15.9
10	2180	31.8
20	2180	63.6
50	2180	159

3.3.2 NDROT

The NDROT, non-destructive readout, cell is a memory device controlled by a set, reset and clock input signal. When an input set signal is received, the NDROT will generate an output pulse after each clock signal until an input reset signal is received. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

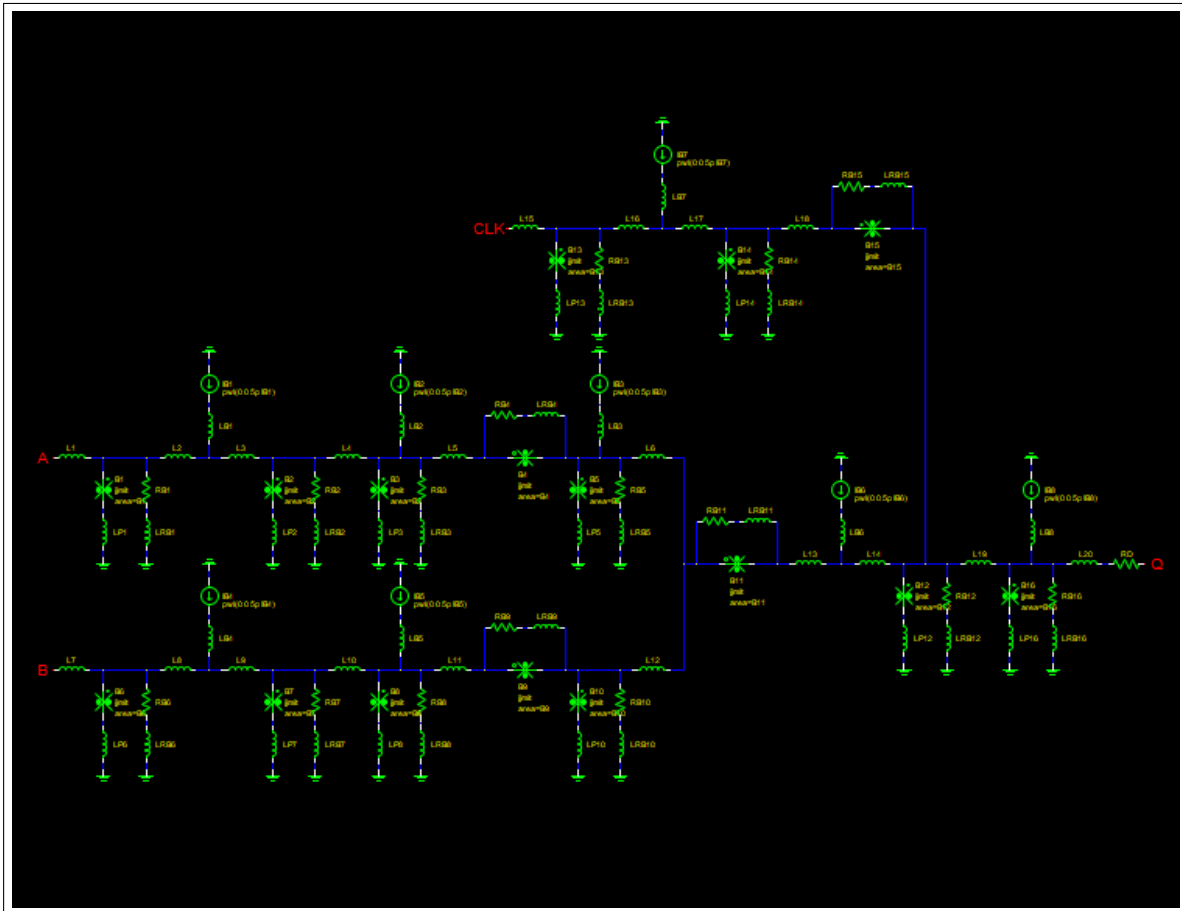


Figure 3.47: Schematic of RSFQ NDROT.

Layout

The physical layout of the RSFQ NDROT is shown in Fig. 3.48. The layout height is $70\ \mu\text{m}$ and the width is $50\ \mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

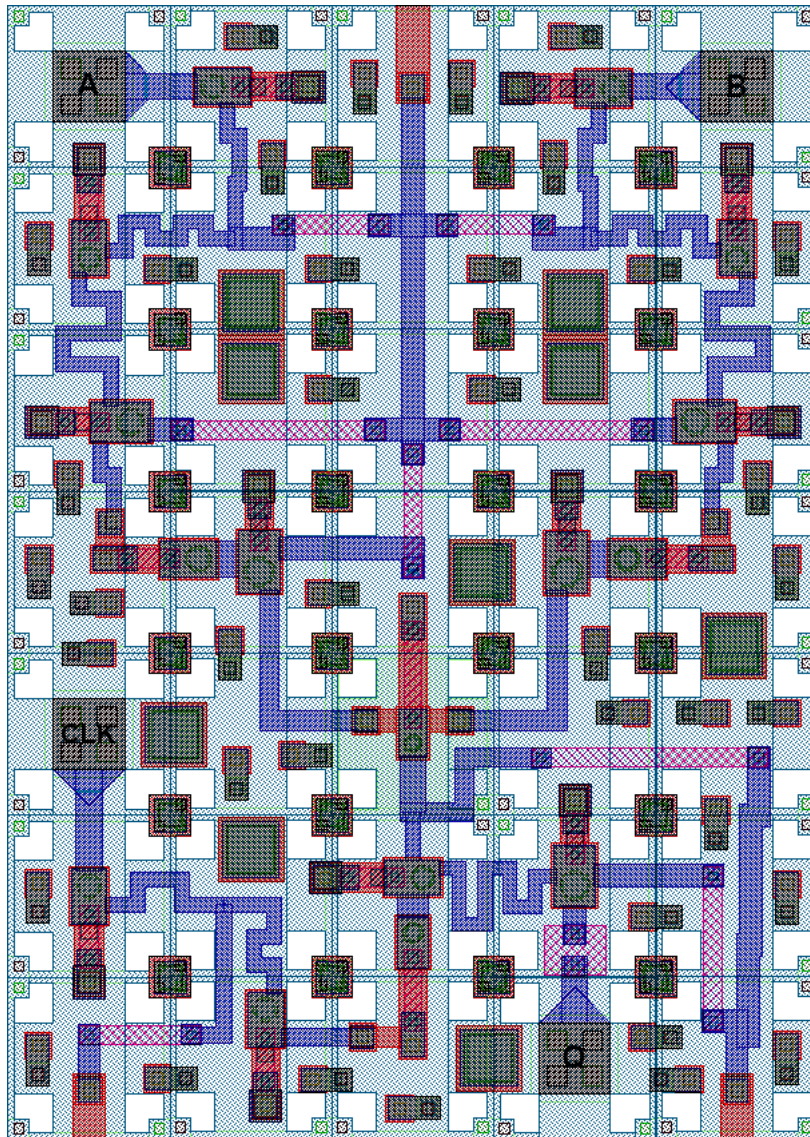


Figure 3.48: RSFQ NDROT Layout.

Analog model

```

1  * Back-annotated simulation file written      63  .param L3=2.7660p
    ↪ by InductEx v.6.1.52 on 2022/09/01.      64  .param L4=4.1774p
2  * Author: L. Schindler                      65  .param L5=3.2486p
3  * Version: 3.0                             66  .param L6=3.7241p
4  * Last modification date: 30 August 2022    67  .param L7=Lpt1
5  * Last modification by: T. Hall             68  .param L8=2.3873p
6                                              69  .param L9=2.7660p
7  *$Ports          a          b          clk    q  70  .param L10=4.1774p
    ↪                                              71  .param L11=3.2486p
8  .subckt THmit11_NDR0T a b clk q             72  .param L12=3.7241p
9  .model jjmit jj(rtype=1, vg=2.8mV, cap      73  .param L13=0.8049p
    ↪ =0.07pF, r0=160, rn=16, icrit=0.1mA      74  .param L14=0.8860p
    ↪ )                                          75  .param L15=Lpt1
10 .param Phi0=2.067833848E-15                 76  .param L16=2.6122p
11 .param B0=1                                 77  .param L17=2.6435p
12 .param Ic0=0.0001                          78  .param L18=3.5452p
13 .param IcRs=100u*6.859904418               79  .param L19=4.6837p
14 .param B0Rs=IcRs/Ic0*B0                    80  .param L20=Lpt1
15 .param Rsheet=2                            81
16 .param Lsheet=1.13e-12                     82  .param LP1=LP
17 .param LP=0.5p                             83  .param LP2=LP
18 .param IC=2.5                              84  .param LP3=LP
19 .param ICreceive=1.6                       85  .param LP5=LP
20 .param ICtrans=2.5                         86  .param LP6=LP
21 .param LB=2p                               87  .param LP7=LP
22 .param BiasCoef=0.7                       88  .param LP8=LP
23 .param Lpt1=2p                             89  .param LP10=LP
24 .param RD=1.36                             90  .param LP12=LP
25                                              91  .param LP13=LP
26 .param B1=1.6                              92  .param LP14=LP
27 .param B2=1.45                             93  .param LP16=LP
28 .param B3=2.15                             94
29 .param B4=1.73                             95  .param RB1=B0Rs/B1
30 .param B5=2.54                             96  .param RB2=B0Rs/B2
31 .param B6=1.6                              97  .param RB3=B0Rs/B3
32 .param B7=1.45                             98  .param RB4=B0Rs/B4
33 .param B8=2.15                             99  .param RB5=B0Rs/B5
34 .param B9=1.73                            100  .param RB6=B0Rs/B6
35 .param B10=2.54                            101  .param RB7=B0Rs/B7
36 .param B11=0.74                           102  .param RB8=B0Rs/B8
37 .param B12=1.57                           103  .param RB9=B0Rs/B9
38 .param B13=1.6                             104  .param RB10=B0Rs/B10
39 .param B14=1.6                             105  .param RB11=B0Rs/B11
40 .param B15=1.08                           106  .param RB12=B0Rs/B12
41 .param B16=2.5                             107  .param RB13=B0Rs/B13
42                                              108  .param RB14=B0Rs/B14
43 .param IB1=268u                            109  .param RB15=B0Rs/B15
44 .param IB2=129u                            110  .param RB16=B0Rs/B16
45 .param IB3=225u                            111
46 .param IB4=268u                            112  .param LRB1=(RB1/Rsheet)*Lsheet+LP
47 .param IB5=129u                            113  .param LRB2=(RB2/Rsheet)*Lsheet+LP
48 .param IB6=115u                            114  .param LRB3=(RB3/Rsheet)*Lsheet+LP
49 .param IB7=254u                            115  .param LRB4=(RB4/Rsheet)*Lsheet
50 .param IB8=175u                            116  .param LRB5=(RB5/Rsheet)*Lsheet+LP
51                                              117  .param LRB6=(RB6/Rsheet)*Lsheet+LP
52 .param LB1=LB                              118  .param LRB7=(RB7/Rsheet)*Lsheet+LP
53 .param LB2=LB                              119  .param LRB8=(RB8/Rsheet)*Lsheet+LP
54 .param LB3=LB                              120  .param LRB9=(RB9/Rsheet)*Lsheet
55 .param LB4=LB                              121  .param LRB10=(RB10/Rsheet)*Lsheet+LP
56 .param LB5=LB                              122  .param LRB11=(RB11/Rsheet)*Lsheet
57 .param LB6=LB                              123  .param LRB12=(RB12/Rsheet)*Lsheet+LP
58 .param LB7=LB                              124  .param LRB13=(RB13/Rsheet)*Lsheet+LP
59 .param LB8=LB                              125  .param LRB14=(RB14/Rsheet)*Lsheet+LP
60                                              126  .param LRB15=(RB15/Rsheet)*Lsheet
61 .param L1=Lpt1                             127  .param LRB16=(RB16/Rsheet)*Lsheet+LP
62 .param L2=2.3873p                         128

```

```

129 B1 1 2 jjmit area=B1
130 B2 5 6 jjmit area=B2
131 B3 7 8 jjmit area=B3
132 B4 10 11 jjmit area=B4
133 B5 11 12 jjmit area=B5
134 B6 15 16 jjmit area=B6
135 B7 19 20 jjmit area=B7
136 B8 21 22 jjmit area=B8
137 B9 24 25 jjmit area=B9
138 B10 25 26 jjmit area=B10
139 B11 14 27 jjmit area=B11
140 B12 30 31 jjmit area=B12
141 B13 32 33 jjmit area=B13
142 B14 36 37 jjmit area=B14
143 B15 38 30 jjmit area=B15
144 B16 39 40 jjmit area=B16
145
146 IB1 0 4 pwl(0 0 5p IB1)
147 IB2 0 9 pwl(0 0 5p IB2)
148 IB3 0 13 pwl(0 0 5p IB3)
149 IB4 0 18 pwl(0 0 5p IB4)
150 IB5 0 23 pwl(0 0 5p IB5)
151 IB6 0 29 pwl(0 0 5p IB6)
152 IB7 0 35 pwl(0 0 5p IB7)
153 IB8 0 41 pwl(0 0 5p IB8)
154
155 LB1 4 3 8.139E-013
156 LB2 9 7 6.905E-013
157 LB3 13 11 2.05E-012
158 LB4 18 17 7.93E-013
159 LB5 23 21 6.92E-013
160 LB6 29 28 2.339E-012
161 LB7 35 34 2.399E-012
162 LB8 41 39 1.726E-012
163
164 L1 a 1 1.001E-012
165 L2 1 3 2.395E-012
166 L3 3 5 2.766E-012
167 L4 5 7 4.17E-012
168 L5 7 10 3.25E-012
169 L6 11 14 3.695E-012
170 L7 b 15 1.003E-012
171 L8 15 17 2.401E-012
172 L9 17 19 2.779E-012
173 L10 19 21 4.183E-012
174 L11 21 24 3.255E-012
175 L12 25 14 3.713E-012
176 L13 27 28 8.093E-013
177 L14 28 30 8.804E-013
178 L15 clk 32 1.351E-012
179 L16 32 34 2.624E-012
180 L17 34 36 2.63E-012
181 L18 36 38 3.565E-012
182 L19 30 39 4.674E-012
183 L20 39 42 6.06E-013
184
185 RD 42 q RD
186
187 LP1 2 0 3.626E-013
188 LP2 6 0 4.608E-013
189 LP3 8 0 3.732E-013
190 LP5 12 0 4.072E-013
191 LP6 16 0 3.63E-013
192 LP7 20 0 4.592E-013
193 LP8 22 0 3.742E-013
194 LP10 26 0 3.912E-013
195 LP12 31 0 4.701E-013
196 LP13 33 0 3.832E-013
197 LP14 37 0 4.271E-013
198 LP16 40 0 3.764E-013
199
200 RB1 1 101 RB1
201 LRB1 101 0 LRB1
202 RB2 5 105 RB2
203 LRB2 105 0 LRB2
204 RB3 7 107 RB3
205 LRB3 107 0 LRB3
206 RB4 10 110 RB4
207 LRB4 110 11 LRB4
208 RB5 11 111 RB5
209 LRB5 111 0 LRB5
210 RB6 15 115 RB6
211 LRB6 115 0 LRB6
212 RB7 19 119 RB7
213 LRB7 119 0 LRB7
214 RB8 21 121 RB8
215 LRB8 121 0 LRB8
216 RB9 24 124 RB9
217 LRB9 124 25 LRB9
218 RB10 25 125 RB10
219 LRB10 125 0 LRB10
220 RB11 14 127 RB11
221 LRB11 127 27 LRB11
222 RB12 30 130 RB12
223 LRB12 130 0 LRB12
224 RB13 32 132 RB13
225 LRB13 132 0 LRB13
226 RB14 36 136 RB14
227 LRB14 136 0 LRB14
228 RB15 38 138 RB15
229 LRB15 138 30 LRB15
230 RB16 39 139 RB16
231 LRB16 139 0 LRB16
232
233 .ends

```

Listing 3.25: RSFQ NDROT JoSIM netlist.

Table 3.21: RSFQ NDROT pin list.

Pin	Description
a	Data input (set signal)
b	Data input (reset signal)
clk	Clock input
q	Data output

The simulation results for the RSFQ NDROT using JoSIM is shown in Fig. 3.49. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a** (set signal),
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the input inductor connected to pin **b** (reset signal),
- (d) the phase over the input JJ of pin **b**,
- (e) the current through the input inductor connected to pin **clk**,
- (f) the phase over the input JJ of pin **clk**,
- (g) the phase over the output JJ of pin **q**, and
- (h) the phase over the input JJ of the load circuit connected via a PTL to pin **q**.

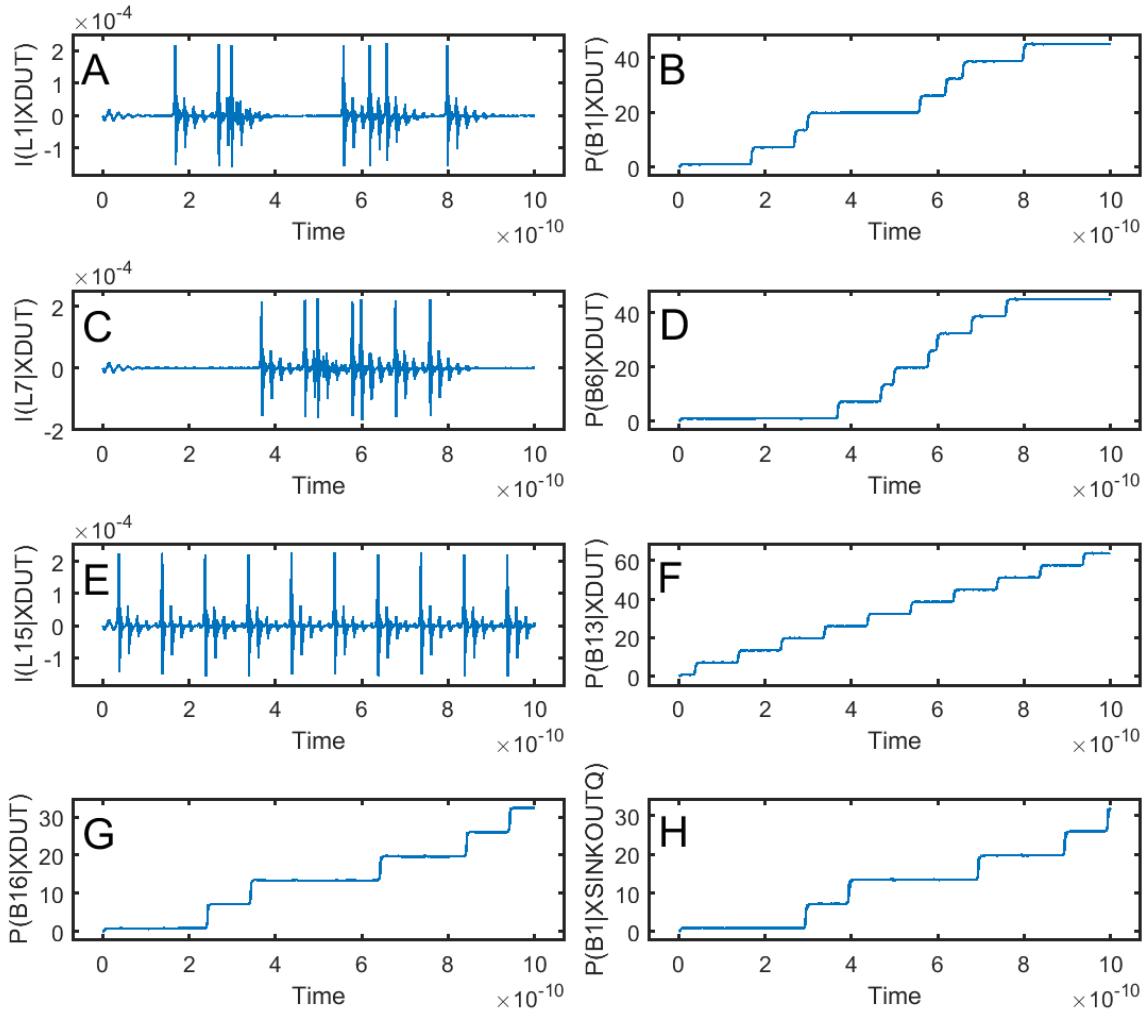


Figure 3.49: RSFQ NDROT analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 1 September 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_NDROT_v3p0_extracted (a, b, clk, q);
9
10 input
11     a, b, clk;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state1_clk_q = 5.8,
21     ct_state0_a_b = 0.2,
22     ct_state1_b_a = 0.3,
23     ct_state1_clk_clk = 8.4;
24
25 reg
26     errorsignal_a,
27     errorsignal_b,
28     errorsignal_clk;
29
30 integer
31     outfile,
32     cell_state; // internal state of the cell
33
34 initial
35     begin
36         errorsignal_a = 0;
37         errorsignal_b = 0;
38         errorsignal_clk = 0;
39         cell_state = 0; // Startup state
40         q = 0; // All outputs start at 0
41     end
42
43 always @(posedge a or negedge a) // execute at positive and negative edges of input
44     begin
45         if ($time>4) // arbitrary steady-state time)
46             begin
47                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
48                     begin
49                         outfile = $fopen("errors.txt", "a");
50                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n", $time);
51                         $fclose(outfile);
52                         q <= 1'bX; // Set all outputs to unknown
53                     end
54                 if (errorsignal_a == 0)
55                     begin
56                         case (cell_state)
57                             0: begin
58                                 cell_state = 1; // Blocking statement -- immediately
59                                 errorsignal_b = 1; // Critical timing on this input; assign
60                                     immediately
61                                 errorsignal_b <= #(ct_state0_a_b) 0; // Clear error signal
62                                     after critical timing expires
63                             end
64                             1: begin
65                                 end
66                         endcase
67                     end
68             end
69     end

```

```

65         end
66     end
67 end
68
69 always @(posedge b or negedge b) // execute at positive and negative edges of input
70 begin
71     if ($time>4) // arbitrary steady-state time)
72     begin
73         if (errorsignal_b == 1'b1) // A critical timing is active for this input
74         begin
75             outfile = $fopen("errors.txt", "a");
76             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
77                 ↳ ", $time);
78             $fclose(outfile);
79             q <= 1'bX; // Set all outputs to unknown
80         end
81         if (errorsignal_b == 0)
82         begin
83             case (cell_state)
84             0: begin
85                 end
86             1: begin
87                 cell_state = 0; // Blocking statement -- immediately
88                 errorsignal_a = 1; // Critical timing on this input; assign
89                 ↳ immediately
90                 errorsignal_a <= #(ct_state1_b_a) 0; // Clear error signal
91                 ↳ after critical timing expires
92             end
93             endcase
94         end
95     end
96 end
97
98 always @(posedge clk or negedge clk) // execute at positive and negative edges of input
99 begin
100     if ($time>4) // arbitrary steady-state time)
101     begin
102         if (errorsignal_clk == 1'b1) // A critical timing is active for this input
103         begin
104             outfile = $fopen("errors.txt", "a");
105             $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
106                 ↳ ", $time);
107             $fclose(outfile);
108             q <= 1'bX; // Set all outputs to unknown
109         end
110         if (errorsignal_clk == 0)
111         begin
112             case (cell_state)
113             0: begin
114                 end
115             1: begin
116                 q <= #(delay_state1_clk_q) !q;
117                 errorsignal_clk = 1; // Critical timing on this input; assign
118                 ↳ immediately
119                 errorsignal_clk <= #(ct_state1_clk_clk) 0; // Clear error
120                 ↳ signal after critical timing expires
121             end
122             endcase
123         end
124     end
125 end
126 end
127 endmodule

```

Listing 3.26: RSFQ NDROT verilog model.

The digital simulation results for the RSFQ NDROT is shown in Fig. 3.50 and the Mealy finite state diagram, extracted using *TimEx*, is shown in Fig. 3.51.

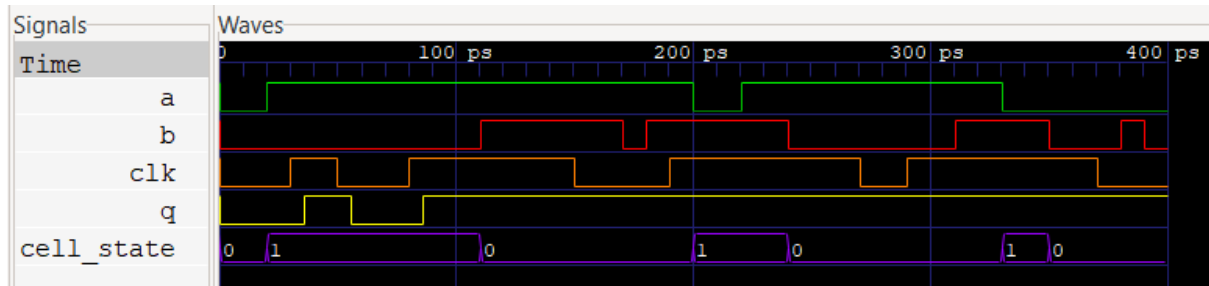


Figure 3.50: RSFQ NDROT digital simulation results.

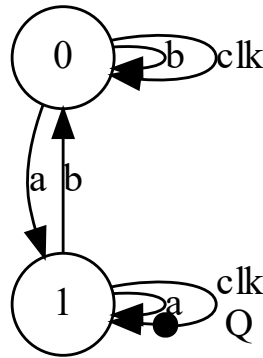


Figure 3.51: RSFQ NDROT Mealy finite state machine diagram.

Power Consumption

Table 3.22: RSFQ NDROT power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	4060	5.80
2	4060	11.6
5	4060	29.0
10	4060	58.0
20	4060	116
50	4060	290

3.3.3 BUFFT

The RSFQ BUFFT cell is a buffer cell intended for clock balancing. It is designed to have the same a-to-q delay as the SPLITT cell. The cell has integrated PTL transmitters and receivers and is designed to connect directly to a PTL.

Schematic

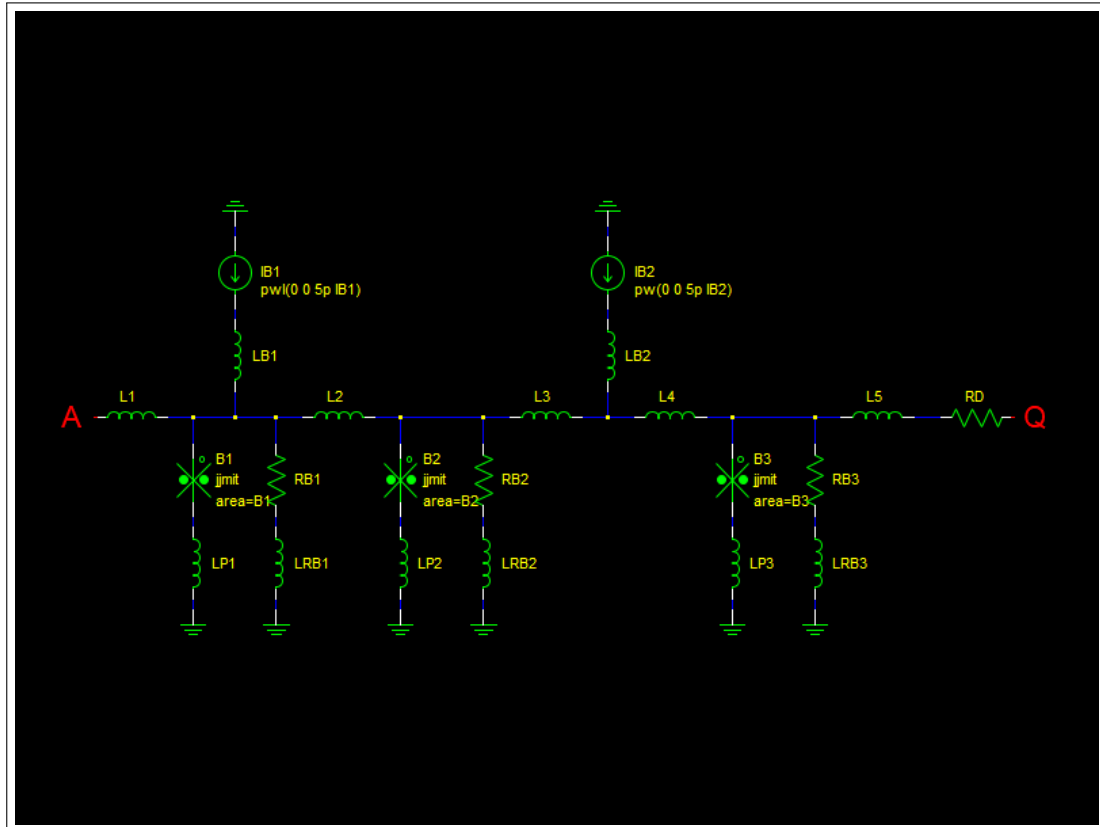


Figure 3.52: Schematic of RSFQ BUFFT.

Layout

The physical layout for the RSFQ BUFFT is shown in Fig. 3.53. The layout height is $70\ \mu\text{m}$ and the width is $20\ \mu\text{m}$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

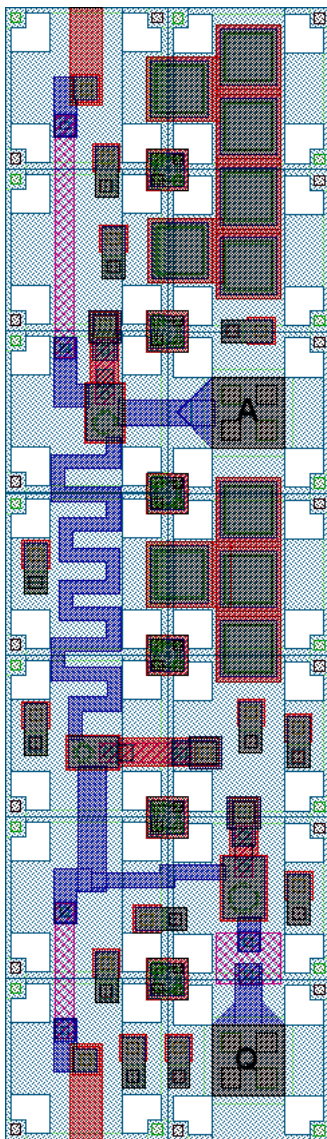


Figure 3.53: RSFQ BUFFT Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/11.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 11 August 2022
5  * Last modification by: T. Hall
6
7  *$Ports          a          q
8  .subckt THmit11_BUFFT a q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=1.6
27 .param B2=0.92
28 .param B3=2.5
29
30 .param IB1=112u
31 .param IB2=218u
32
33 .param LB1=LB
34 .param LB2=LB
35
36 .param L1=Lptl
37 .param L2=12.2588p
38 .param L3=1.1532p
39 .param L4=2.4523p
40 .param L5=Lptl
41
42 .param LP1=LP
43 .param LP2=LP
44 .param LP3=LP
45
46 .param RB1=B0Rs/B1
47 .param RB2=B0Rs/B2
48 .param RB3=B0Rs/B3
49
50 .param LRB1=(RB1/Rsheet)*Lsheet+LP
51 .param LRB2=(RB2/Rsheet)*Lsheet+LP
52 .param LRB3=(RB3/Rsheet)*Lsheet+LP
53
54 B1 1 2 jjmit area=B1
55 B2 4 5 jjmit area=B2
56 B3 8 9 jjmit area=B3
57
58 IB1 0 3 pwl(0 0 5p IB1)
59 IB2 0 7 pwl(0 0 5p IB2)
60
61 LB1 3 1 1.087E-012
62 LB2 7 6 8.18E-013
63
64 L1 a 1 1.138E-012
65 L2 1 4 1.216E-011
66 L3 4 6 1.152E-012
67 L4 6 8 2.432E-012
68 L5 8 10 6.041E-013
69
70 RD 10 q RD
71
72 LP1 2 0 3.595E-013
73 LP2 5 0 4.661E-013
74 LP3 9 0 3.254E-013
75
76 RB1 1 101 RB1
77 LRB1 101 0 LRB1
78 RB2 4 104 RB2
79 LRB2 104 0 LRB2
80 RB3 8 108 RB3
81 LRB3 108 0 LRB3
82
83 .ends

```

Listing 3.27: RSFQ BUFFT JoSIM netlist.

Table 3.23: RSFQ BUFFT pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ BUFFT using JoSIM is shown in Fig. 3.54. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the phase over the output JJ of pin **q**, and
- (d) the phase over the input JJ of the load circuit connected via a PTL to pin **q**.

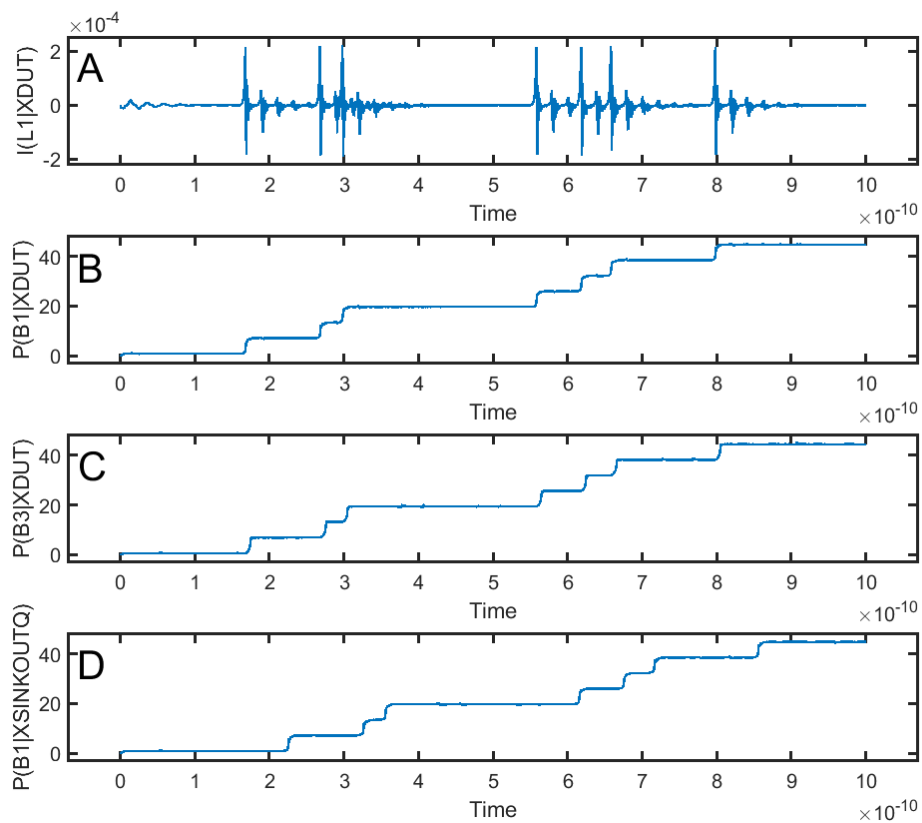


Figure 3.54: RSFQ BUFFT analog simulation results.

Digital model

```

1 // -----
2 // Author L. Schindler
3 // Version: 3.0
4 // Last Modification date: 11 August 2022
5 // Last Modification by: T. Hall
6 // -----
7 `timescale 1ps/100fs
8 module THmit11_BUFFT_v3p0_extracted (a, q);
9
10 input
11     a;
12
13 output
14     q;
15
16 reg
17     q;
18
19 real
20     delay_state0_a_q = 6.0,
21     ct_state0_a_a = 20.2;
22
23 reg
24     errorsignal_a;
25
26 integer
27     outfile,
28     cell_state; // internal state of the cell
29
30 initial
31     begin
32         errorsignal_a = 0;
33         cell_state = 0; // Startup state
34         q = 0; // All outputs start at 0
35     end
36
37 always @(posedge a or negedge a) // execute at positive and negative edges of input
38     begin
39         if ($time>4) // arbitrary steady-state time)
40             begin
41                 if (errorsignal_a == 1'b1) // A critical timing is active for this input
42                     begin
43                         outfile = $fopen("errors.txt", "a");
44                         $fdisplay(outfile, "Violation_of_critical_timing_in_module_%m;_%0d_ps.\n
45                             ↪ ", $stime);
46                         $fclose(outfile);
47                         q <= 1'bX; // Set all outputs to unknown
48                     end
49                 if (errorsignal_a == 0)
50                     begin
51                         case (cell_state)
52                             0: begin
53                                 q <= #(delay_state0_a_q) !q;
54                                 errorsignal_a = 1; // Critical timing on this input; assign
55                                     ↪ immediately
56                                 errorsignal_a <= #(ct_state0_a_a) 0; // Clear error signal
57                                     ↪ after critical timing expires
58                             end
59                         endcase
60                     end
61             end
62     end
63 endmodule

```

Listing 3.28: RSFQ BUFFT verilog model.

The digital simulation results for the RSFQ BUFFT is shown in Fig. 3.55 and the Mealy finite state machine diagram, extracted using *TimEx*, is shown in Fig. 3.56.

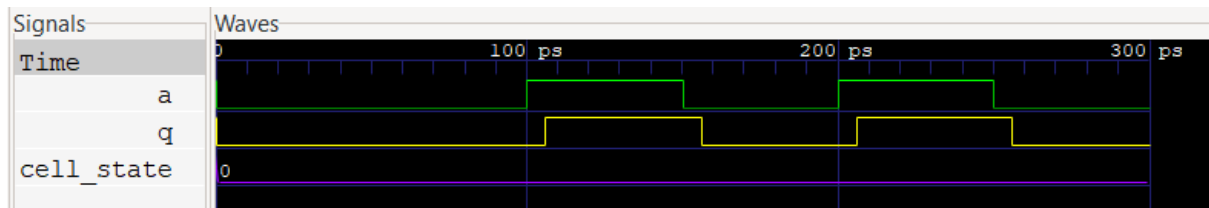


Figure 3.55: RSFQ BUFFT digital simulation results.

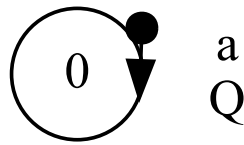


Figure 3.56: RSFQ BUFFT Mealy finite state machine diagram.

Power Consumption

Table 3.24: RSFQ BUFFT power consumption.

Clock rate (GHz)	Static power (nW)	Dynamic power (nW)
1	858	1.04
2	858	2.08
5	858	5.19
10	858	10.4
20	858	20.8
50	858	51.9

3.4 Interface Cells

3.4.1 DCSFQ-PTLTX

The RSFQ DCSFQ-PTLTX is an interface cell designed to convert input voltage pulses into SFQ pulses. The DCSFQ-PTLTX has an integrated PTL transmitter and is intended to connect directly to a PTL output.

Schematic

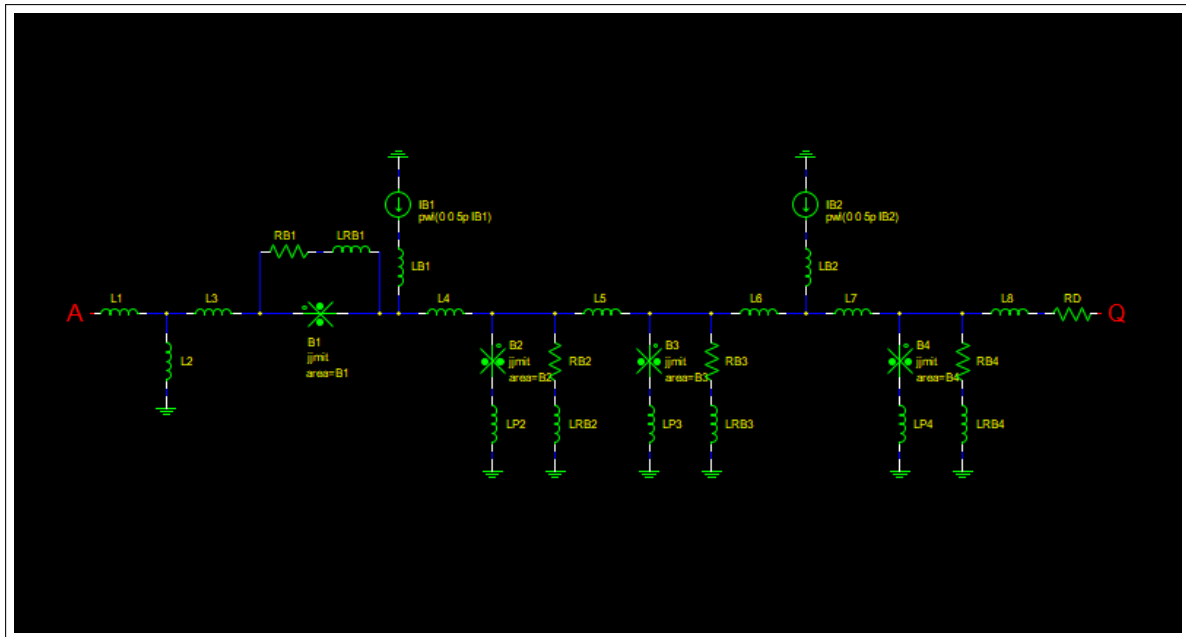


Figure 3.57: Schematic of RSFQ DCSFQ-PTLTX.

Layout

The physical layout for the RSFQ DCSFQ-PTLTX is shown in Fig. 3.58. The layout height is $70\ \mu m$ and the width is $20\ \mu m$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

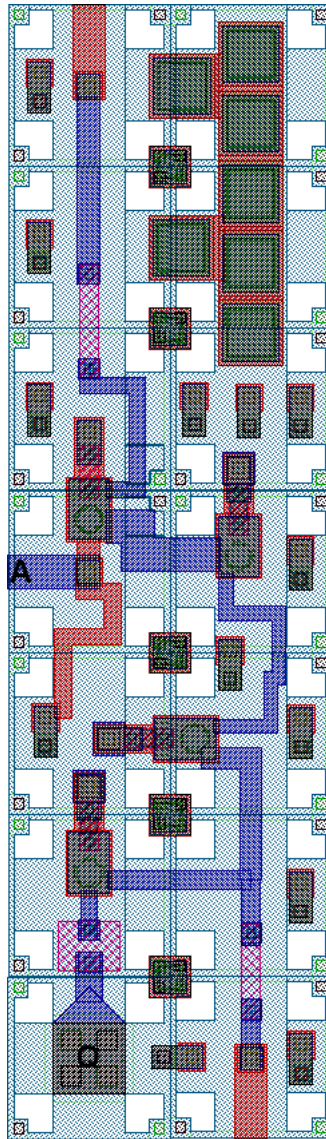


Figure 3.58: RSFQ DCSFQ-PTLTX Layout.

Analog model

```

1  * Back-annotated simulation file written
    ↳ by InductEx v.6.1.52 on 2022/08/09.
2  * Author: L. Schindler
3  * Version: 3.0
4  * Last modification date: 14 April 2022
5  * Last modification by: T. Hall
6
7  *$Ports      a      q
8  .subckt THmit11_DCSFQ-PTLTx a q
9  .model jjmit jj(rtype=1, vg=2.8mV, cap
    ↳ =0.07pF, r0=160, rn=16, icrit=0.1mA
    ↳ )
10 .param Phi0=2.067833848E-15
11 .param B0=1
12 .param Ic0=0.0001
13 .param IcRs=100u*6.859904418
14 .param B0Rs=IcRs/Ic0*B0
15 .param Rsheet=2
16 .param Lsheet=1.13e-12
17 .param LP=0.5p
18 .param IC=2.5
19 .param ICreceive=1.6
20 .param ICtrans=2.5
21 .param LB=2p
22 .param BiasCoef=0.7
23 .param Lptl=2p
24 .param RD=1.36
25
26 .param B1=0.9*Ic
27 .param B2=0.9*Ic
28 .param B3=IC
29 .param B4=ICtrans
30
31 .param IB1=(11/9)*Ic0*B2
32 .param IB2=BiasCoef*Ic0*(B3+B4)
33
34 .param LB1=LB
35 .param LB2=LB
36
37 .param LP2=LP
38 .param LP3=LP
39 .param LP4=LP
40
41 .param L1=1p
42 .param L2=3.9p
43 .param L3=0.6p
44 .param L4=1.1p
45 .param L5=Phi0/(2*B2*Ic0)
46 .param L6=Phi0/(4*B3*Ic0)
47 .param L7=Phi0/(4*B3*Ic0)
48 .param L8=Lptl
49
50 .param RB1=B0Rs/B1
51 .param RB2=B0Rs/B2
52 .param RB3=B0Rs/B3
53 .param RB4=B0Rs/B4
54
55 .param LRB1=(RB1/Rsheet)*Lsheet
56 .param LRB2=(RB2/Rsheet)*Lsheet+LP
57 .param LRB3=(RB3/Rsheet)*Lsheet+LP
58 .param LRB4=(RB4/Rsheet)*Lsheet+LP
59
60 B1 2 3 jjmit area=B1
61 B2 5 6 jjmit area=B2
62 B3 7 8 jjmit area=B3
63 B4 11 12 jjmit area=B4
64
65 IB1 0 4 pw1(0 0 5p IB1)
66 IB2 0 10 pw1(0 0 5p IB2)
67
68 LB1 4 3 3.12E-012
69 LB2 10 9 8.782E-013
70
71 LP2 6 0 5.209E-013
72 LP3 8 0 5.058E-013
73 LP4 12 0 4.053E-013
74
75 L1 a 1 1.322E-012
76 L2 1 0 3.879E-012
77 L3 1 2 6.03E-013
78 L4 3 5 1.104E-012
79 L5 5 7 4.572E-012
80 L6 7 9 2.078E-012
81 L7 9 11 2.073E-012
82 L8 11 13 8.436E-013
83
84 RD 13 q RD
85
86 RB1 2 102 RB1
87 LRB1 102 3 LRB1
88 RB2 5 105 RB2
89 LRB2 105 0 LRB2
90 RB3 7 107 RB3
91 LRB3 107 0 LRB3
92 RB4 11 111 RB4
93 LRB4 111 0 LRB4
94
95 .ends

```

Listing 3.29: RSFQ DCSFQ-PTLTx JoSIM netlist.

Table 3.25: RSFQ DCSFQ-PTLTx pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ DCSFQ-PTLTx using JoSIM is shown in Fig. 3.59. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the current through the output inductor connected to pin **q**,
- (c) the phase over the output JJ of pin **q**, and
- (d) the phase over the input JJ of the load circuit connected through a PTL to pin **q**.

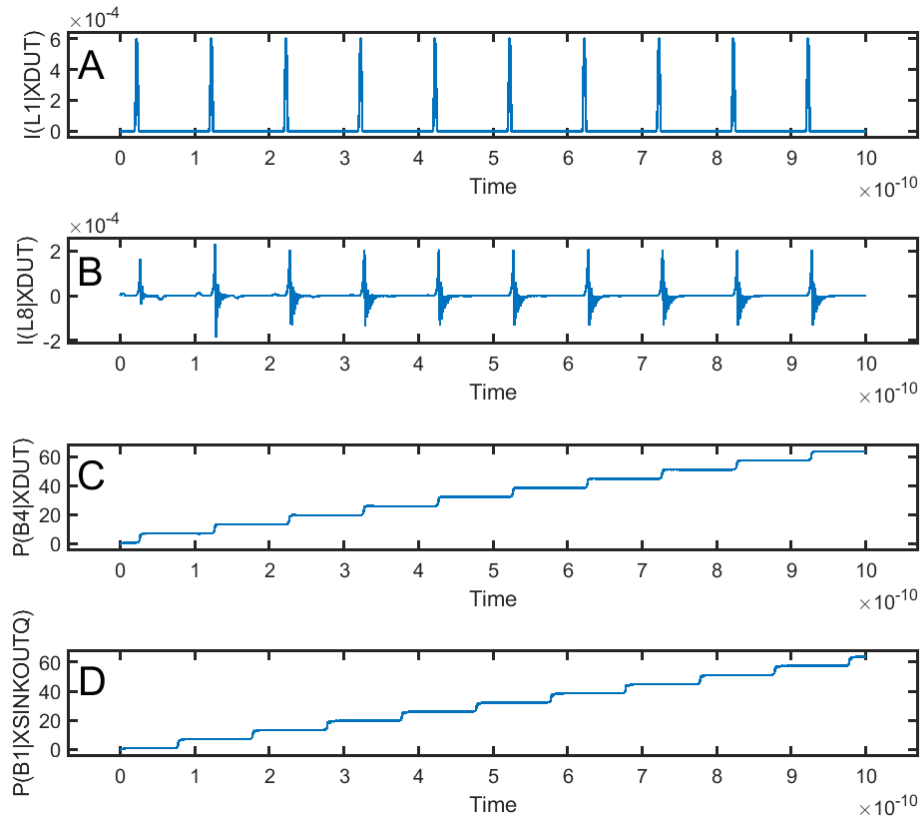


Figure 3.59: RSFQ DCSFQ-PTLTx analog simulation results.

3.4.2 PTLRX-SFQDC

The RSFQ PTLRX-SFQDC is an interface cell designed to convert SFQ pulses to an output voltage level which can be measured by standard equipment. The PTLRX-SFQDC has an integrated PTL receiver and is intended to be connected directly to a PTL input.

Schematic

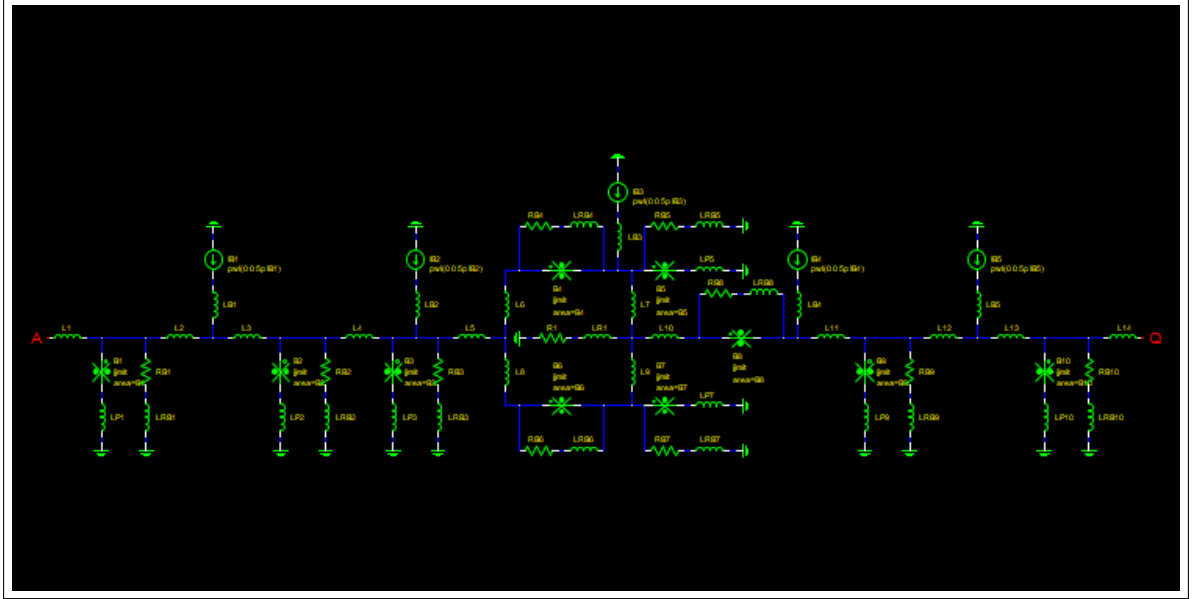


Figure 3.60: Schematic of RSFQ PTLRX-SFQDC.

Layout

The physical layout for the RSFQ PTLRX-SFQDC is shown in Fig. 3.61. The layout height is $70\ \mu m$ and the width is $40\ \mu m$. The biasing is brought in on M5 from external bias lines at the top and bottom of the cell. The PTL connection extends to connect to a PTL on either M1 or M3. All bias resistors are designed to be connected to a 2.6 mV voltage source.

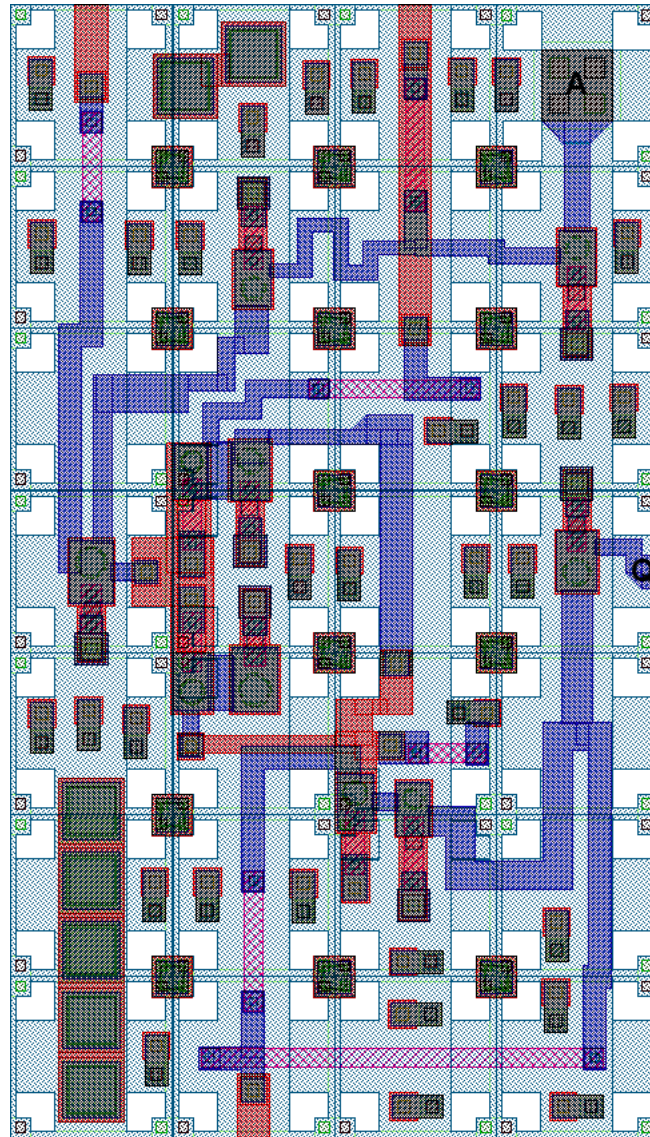


Figure 3.61: RSFQ PTLRX-SFQDC Layout.

Analog model

```

1  * Back-annotated simulation file written      64
    ↪ by InductEx v.6.1.52 on 2022/08/14.      65
2  * Adapted from Fluxonics SDQDC_v5          66
3  * Author: L. Schindler                      67
4  * Version: 3.0                             68
5  * Last modification date: 13 August 2022    69
6  * Last modification by: T. Hall             70
7                                              71
8  *$Ports          a          q              72
9  .subckt THmit11_PTLRX-SFQDC a q            73
10 .model jjmit jj(rtype=1, vg=2.8mV, cap      74
    ↪ =0.07pF, r0=160, rn=16, icrit=0.1mA     75
    ↪ )                                         76
11 .param Phi0=2.067833848E-15                77
12 .param B0=1                                78
13 .param Ic0=0.0001                          79
14 .param IcRs=100u*6.859904418               80
15 .param B0Rs=IcRs/Ic0*B0                    81
16 .param Rsheet=2                            82
17 .param Lsheet=1.13e-12                     83
18 .param LP=0.5p                             84
19 .param IC=2.5                               85
20 .param ICreceive=1.6                       86
21 .param ICtrans=2.5                         87
22 .param LB=2p                               88
23 .param BiasCoef=0.7                       89
24 .param Lptl=2p                             90
25 .param RD=1.36                            91
26                                              92
27 .param B1=1.6                              93
28 .param B2=1.62                            94
29 .param B3=2.76                            95
30 .param B4=1.34                            96
31 .param B5=1.98                            97
32 .param B6=1.82                            98
33 .param B7=3.15                            99
34 .param B8=1.65                            100
35 .param B9=1.22                            101
36 .param B10=2.18                           102
37                                              103
38 .param IB1=229u                             104
39 .param IB2=292u                             105
40 .param IB3=167u                             106
41 .param IB4=212u                             107
42 .param IB5=64u                             108
43                                              109
44 .param LB1=LB                               110
45 .param LB2=LB                               111
46 .param LB3=LB                               112
47 .param LB4=LB                               113
48 .param LB5=LB                               114
49                                              115
50 .param L1=Lptl                             116
51 .param L2=2.3453p                           117
52 .param L3=3.9596p                           118
53 .param L4=4.5142p                           119
54 .param L5=0.7362p                           120
55 .param L6=0.9663p                           121
56 .param L7=5.6790p                           122
57 .param L8=0.8679p                           123
58 .param L9=3.2739p                           124
59 .param L10=0.2245p                          125
60 .param L11=0.695p                           126
61 .param L12=3.036p                           127
62 .param L13=1.6165p                          128
63 .param L14=1.510p                          129

.param LR1=0.7066p
.param R1=5.74

.param LP1=LP
.param LP2=LP
.param LP3=LP
.param LP5=LP
.param LP7=LP
.param LP9=LP
.param LP10=LP

.param RB1=B0Rs/B1
.param RB2=B0Rs/B2
.param RB3=B0Rs/B3
.param RB4=B0Rs/B4
.param RB5=B0Rs/B5
.param RB6=B0Rs/B6
.param RB7=B0Rs/B7
.param RB8=B0Rs/B8
.param RB9=B0Rs/B9
.param RB10=B0Rs/B10

.param LRB1=(RB1/Rsheet)*Lsheet+LP
.param LRB2=(RB2/Rsheet)*Lsheet+LP
.param LRB3=(RB3/Rsheet)*Lsheet+LP
.param LRB4=(RB4/Rsheet)*Lsheet
.param LRB5=(RB5/Rsheet)*Lsheet+LP
.param LRB6=(RB6/Rsheet)*Lsheet
.param LRB7=(RB7/Rsheet)*Lsheet+LP
.param LRB8=(RB8/Rsheet)*Lsheet
.param LRB9=(RB9/Rsheet)*Lsheet+LP
.param LRB10=(RB10/Rsheet)*Lsheet+LP

B1 1 2 jjmit area=B1
B2 5 6 jjmit area=B2
B3 7 8 jjmit area=B3
B4 11 12 jjmit area=B4
B5 12 13 jjmit area=B5
B6 16 17 jjmit area=B6
B7 17 18 jjmit area=B7
B8 20 21 jjmit area=B8
B9 23 24 jjmit area=B9
B10 27 28 jjmit area=B10

IB1 0 4 pwl(0 0 5p IB1)
IB2 0 9 pwl(0 0 5p IB2)
IB3 0 14 pwl(0 0 5p IB3)
IB4 0 22 pwl(0 0 5p IB4)
IB5 0 26 pwl(0 0 5p IB5)

LB1 4 3 6.848E-013
LB2 9 7 4.296E-012
LB3 14 12 2.614E-012
LB4 22 21 2.738E-012
LB5 26 25 4.096E-012

L1 a 1 1.455E-012
L2 1 3 2.346E-012
L3 3 5 3.959E-012
L4 5 7 4.522E-012
L5 7 10 7.386E-013
L6 11 10 9.705E-013
L7 12 15 5.626E-012
L8 10 16 8.718E-013
L9 15 17 3.248E-012

```

```

130 | L10 15 20 2.239E-013
131 | L11 21 23 6.903E-013
132 | L12 23 25 3.044E-012
133 | L13 25 27 1.603E-012
134 | L14 27 q 1.635E-012
135 |
136 | LR1 19 15 7.057E-013
137 | R1 0 19 R1
138 |
139 | LP1 2 0 4.229E-013
140 | LP2 6 0 4.418E-013
141 | LP3 8 0 4.758E-013
142 | LP5 13 0 4.434E-013
143 | LP7 18 0 6.22E-013
144 | LP9 24 0 3.836E-013
145 | LP10 28 0 3.18E-013
146 |
147 | RB1 1 101 RB1
148 | LRB1 101 0 LRB1
149 | RB2 5 105 RB2
150 | LRB2 105 0 LRB2

```

```

151 | RB3 7 107 RB3
152 | LRB3 107 0 LRB3
153 | RB4 11 111 RB4
154 | LRB4 111 12 LRB4
155 | RB5 12 112 RB5
156 | LRB5 112 0 LRB5
157 | RB6 16 116 RB6
158 | LRB6 116 17 LRB6
159 | RB7 17 117 RB7
160 | LRB7 117 0 LRB7
161 | RB8 20 120 RB8
162 | LRB8 120 21 LRB8
163 | RB9 23 123 RB9
164 | LRB9 123 0 LRB9
165 | RB10 27 127 RB10
166 | LRB10 127 0 LRB10
167 |
168 | .ends

```

Listing 3.30: RSFQ PTLRX-SFQDC JoSIM netlist.

Table 3.26: RSFQ PTLRX-SFQDC pin list.

Pin	Description
a	Data input
q	Data output

The simulation results for the RSFQ PTLRX-SFQDC using JoSIM is shown in Fig. 3.62. The figure shows the graph for:

- (a) the current through the input inductor connected to pin **a**,
- (b) the phase over the input JJ of pin **a**,
- (c) the current through the output inductor connected to pin **q**, and
- (c) the voltage over the load resistor connected to pin **q**.

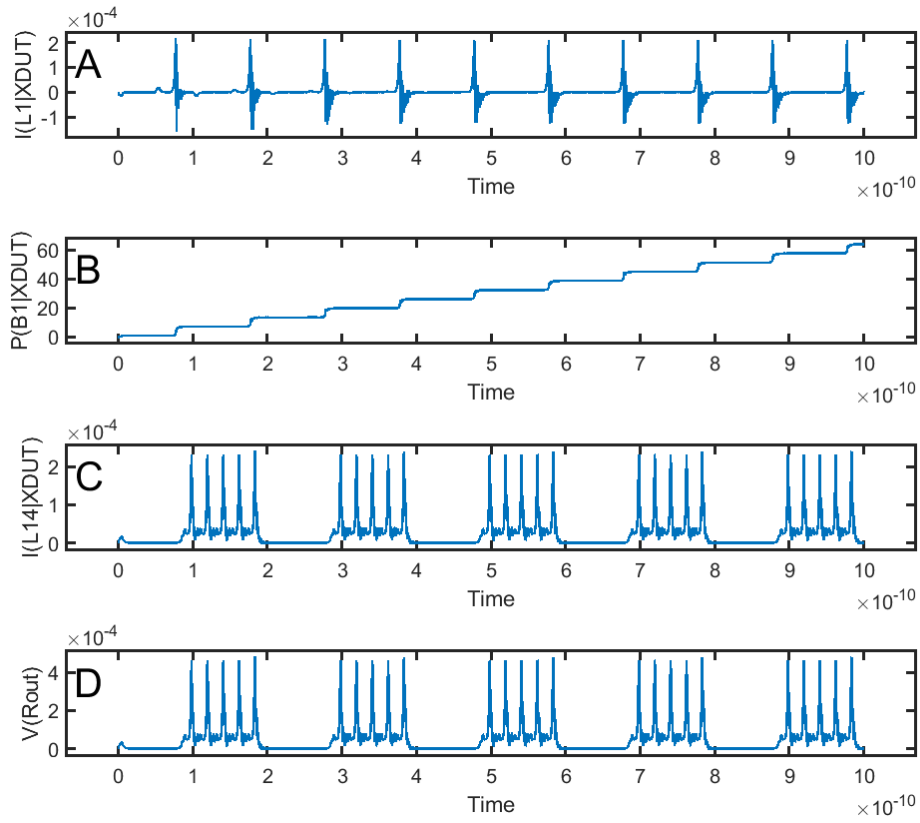


Figure 3.62: RSFQ PTLRX-SFQDC analog simulation results.

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